

SEO Agent v4.2 on n8n

How the agent works, every workflow and node explained, how to build, test and deploy it, and the results of the full system test run on techand.ai on 2026-10-01 (keyword ladder, discovery, audits, full report, rank tracker, form path, e-mail, API).

Instance: n8n 2.41.4 (Docker Compose, SQLite, Gutenberg PDF service), project `w01WpMiHjdduGvf1`. Source of truth: [n8n/seo-agent/](#) (build scripts, prompts, Code nodes, harness, workflows JSON).

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1. What the agent does

The SEO Agent is an enterprise SEO automation built as n8n workflows. One public form (or one API call) gives a customer:

- **I know my keyword** — competitor research, a GO / GO_WITH_CHANGES / AVOID verdict, a keyword report and a finished, QA-checked page (Word + PDF).
- **Suggest keywords for my business** — multi-source keyword research with an opportunity score, AI relevance screen, topic clusters, AI search demand and a live SERP check; form users then pick the keyword to write for.
- **Just describe my website** — a structured description of the site from its homepage.
- **Audit my website** — a technical Site Audit (crawl, probes, scoring) or a Full SEO Report (plus competitors, keywords, backlinks, PageSpeed, E-E-A-T content review, AI visibility).
- **Check if my keyword is right** — the verdict alone.
- **Rank my site for a keyword** (new, v4.2) — the **keyword ladder**: feasibility of the destination keyword, a ladder of 8-12 pages from winnable long-tail terms up to the head term with an internal link map and timeline, the first page written now, weekly position tracking with the recommended next rung.

Data comes from DataForSEO (SERP, Labs, On-Page, Backlinks, WHOIS, AI Optimization), Jina Reader (pages → markdown), Google PageSpeed Insights and RDAP; reasoning from Claude (Opus 5.5 for brief, copy and editing; Sonnet 5.5 for analysis, verdicts, relevance and critique; Haiku 4.5 for the site description). Reports are rendered to PDF by Gotenberg and delivered by e-mail (Gmail SMTP), form download or API callback.

2. Architecture and workflows

Seven workflows live in the n8n project. The main workflow holds every mode; the others are small and single-purpose. Sub-runs (ladder pages, discovery follow-ups, form audits) are started through *API Entry* as separate executions, which keeps form flows short and lets every stage deliver on its own.

Workflow	Id	Role	Nodes	State
SEO Agent v4 — Audit, Keywords, Verdict & Content	SE0agentV4Full01	main workflow: every mode, the keyword ladder, delivery	331	active
SEO Agent — API	SE0agentAPIentry	webhook front door: validates, answers 202/400, starts the main workflow	7	active
SEO Agent — Rank Tracker	SE0agentTracker1	weekly position checks for every ladder, progress e-mail / callback	18	active
SEO Agent — Site Tracker	SE0agentSiteTrk1	weekly per-site performance: Search Console, GA4, Google Trends, live checks, actions; e-mail / callback	78	inactive (on purpose)
SEO Agent — Console Alerts	SE0agentConsole1	IMAP watcher: Search Console notification mails (manual actions, security, coverage, CWV) → alerts	16	inactive (on purpose)
SEO Agent — Content Cadence	SE0agentCadence1	weekly blog pages per site (ladder → striking distance → trends), delivered as HTML + Markdown + meta.json	27	inactive (on purpose)
SEO Agent — Site Admin	SE0agentSiteAdm1	delete / pause / resume / cadence / unpublish for a tracked site (published sub-workflow)	25	inactive (on purpose)
SEO Agent — AI Visibility Tracker	SE0agentAIVisib1	weekly buyer questions × ChatGPT / Perplexity / Gemini / Claude / Google AI Mode + AI Overviews: mentions, citations, share of voice, sources AI trusts, lost questions	55	inactive (on purpose)
SEO Agent — Backlink Monitor	SE0agentBacklnk1	weekly watch (lost / spammy links) and monthly full report: broken-link reclaim, link gap, unlinked mentions, prospects with outreach drafts	45	inactive (on purpose)
SEO Agent — Audit Scheduler	SE0agentAuditSc1	monthly technical re-audit per site; each audit stores its findings, diffs against the last one and hands over a fix pack	16	inactive (on purpose)
SEO Agent — Publish to WordPress	SE0agentWordPres	optional: generated page → WordPress draft (inactive sub-workflow)	4	inactive (on purpose)
SEO Agent — Error Handler	SE0agentErrHandl	error workflow: e-mails ops when a run fails	3	active
SEO Agent — Test Callback Receiver	SE0agentCallback	testing: stores API callbacks as executions	2	active
SEO Agent — Test Runner	SE0agentTestRun1	testing only: starts any SEO Agent workflow with a JSON body (kept unpublished)	5	inactive (on purpose)

How a request flows

1. **Entry:** form page (basic auth) or POST /webhook/seo-keyword-check (header auth) → the API workflow validates, answers 202 and starts the main workflow through *API Entry*.

2. **Intake:** *Normalize Input* → *Rate Limit* (per-address limit, estimated Claude budget) → *Input OK?* (errors become a form page or a callback rejection).
3. **Site read** when needed (Jina + Claude Haiku) → *Route Mode*.
4. **Mode pipeline** (discovery · keyword/verdict/ladder · audit · describe) → report builder → Gutenberg PDF.
5. **Delivery:** e-mail (SMTP), form completion page (download) or callback JSON with PDF/Word bytes; the run ledger (DataForSEO spend per node, AI calls, duration) rides along.
6. **Follow-ups as separate executions:** ladder pages, discovery report/content/audit, form audits (a form completion page ends its execution, so nothing may be queued behind it).

Design rules learned the hard way (all enforced in the build): every model node and every DataForSEO/Jina/PageSpeed HTTP node carries a credential slot; Code nodes read binaries with `getBinaryDataBuffer`; thinking tokens count against max tokens (brief 24k, copy/edit 32k, streaming on above ~21k); Form completion nodes are always last; error workflows must be published; script-created Webhook nodes need a `webhookId`.

3. Repository, build, test and deploy

Path	Purpose
build_v4.py	Builds the main workflow from the v3 export with asserted, anchored patches (sections 1-11); writes workflows/SEO_Agent_v4.json, harness/code_v4/, CHANGES_v3_to_v4.md; then runs build_api.py and build_ladder_extras.py.
build_api.py · build_ladder_extras.py · ladder_common.py	API front door; Rank Tracker, WordPress publisher, Test Runner; shared Data Table columns.
v5/prompts/*.txt · v5/code/*.js	Agent prompts and new Code nodes (read by the build). Edit here, never in the JSON.
harness/	Offline test harness: runs every Code node in the n8n container's Node against fixtures (no paid calls). Must end "0 failed".
workflows/*.json	Generated workflow JSON (import with the ids inside). SEO_Agent_v3_original_export.json is the untouched source.
REVIEW.md · LADDER_FEATURE_SPEC.md · docs/	Findings, every change, live test log, roadmap; the ladder spec with decisions; this guide and the exported run traces.

Build

```
cd n8n/seo-agent && python3 build_v4.py          # check the exit code: patches assert the exact text they replace
```

Test (offline)

```
cd harness
docker cp . n8n-n8n-1:/tmp/harness && docker exec -u root n8n-n8n-1 chown -R node:node /tmp/harness
docker exec n8n-n8n-1 sh -c 'cd /tmp/harness && CODE_DIR=/tmp/harness/code_v4
OUT_DIR=/tmp/harness/out node scenarios_v5.js'    # expect "161 passed, 0 failed"
```

Deploy

```
docker exec n8n-n8n-1 n8n import:workflow --input=/tmp/SEO_Agent_v4.json --
projectId=wolWpMiHjdduGvfl    # upsert by id
# activation: POST /api/v1/workflows/<id>/activate with an n8n API key, or without a key:
docker exec n8n-n8n-1 n8n publish:workflow --id=SEOagentV4Full01 && docker restart n8n-n8n-1
# CLI publish takes effect after a restart
# verify without logging in: GET /form/<Start Form webhookId> → 401 (registered, auth
required); POST /webhook/seo-keyword-check → 403
```

Credentials are created/updated without the UI through `n8n import:credentials --input=<json> --projectId=...` (upsert by id, encrypted on import). Never export or print credentials.

4. Configuration, guards and credentials

CONFIG (top of *Normalize Input*)

```
const CONFIG = {
  brand: "Dev SEO",    // from n8n/.env (SEO_BRAND) at build time; shown on reports and e-mails
  crawl_max_pages: 200,           // DataForSEO on-page crawl size (also used when reading
  crawled pages)
  crawl_js: false,                // true = render JavaScript during the crawl (slower,
  costs more)
  qa_max_rounds: 2,    // one editor pass: live tests showed a second pass inflates length
  without raising the score           // how many times the Editor may revise content that
  fails QA
  default_country: 'United States',
  ai_budget_usd: 10,             // estimated Claude spend allowed per budget period (see
  Rate Limit); raised from 3 on 2026-10-01 at the user's request to finish the system test – the
  Anthropic Console workspace limit is the hard cap
  ai_budget_period: 'day',       // 'day' | 'month' – daily window while testing; the
  Anthropic Console workspace limit is the hard cap
  ladder_pages_default: 1,       // ladder mode: pages written immediately (1-3; each page
  is its own content run)
  ladder_tracker_cadence: 'weekly', // the Rank Tracker workflow runs weekly; 'monthly' is
  documentation only until its schedule is changed
  ladder_auto_next_rung: false,   // false = the tracker recommends the next rung (one-click
  form link + API body); true is not wired
  wordpress_publish: false       // true + wordpress_url in the request = every generated
  page is also created as a WordPress DRAFT
};
```

Rate Limit

```
const LIMITS = { per_key_per_day: 6, global_per_day: 150 };
const EST = { content: 1.2, discover: 0.35, verdict: 0.2, full: 0.25, site_audit: 0, describe:
0.03, ladder: 0.55, report: 0.3, track: 0, published: 0, checkin: 0, profile: 0, case_study:
0, ai_visibility: 0.06, backlinks: 0.04 };
```

Keys: e-mail, else client IP (API) or the ladder/discover tag, else domain or keyword. The AI budget is an estimate-based guard (streaming model calls report no token usage); the Anthropic Console workspace limit is the real cap. The cap was raised 3 → 10 at the user's request on 2026-10-01 to finish the system test.

Credentials (fixed ids, pre-wired in the JSON)

Credential	Type	Id	Used by
Anthropic (SEO Agent)	anthropicApi	SEOcredAnthropic	Anthropic Chat Model, Claude — AI Brief, Claude — Competitor Analyzer, Claude — Content Reviewer, Claude — Copywriter, Claude — Critic, Claude — Editor, Claude — Keyword Relevance, Claude — Ladder Relevance, Claude — Outreach Writer, Claude — Prompt Writer, Claude — Site Brief, Claude — Site Describer, Claude — Strategy Brief, Verdict Chat Model

DataForSEO	httpBasicAuth	SE0credDataForSE	AI Demand, AI SERP, Ask LLMs, Backlink Gap, Backlink Summary, Candidate SERP, DataForSEO Whois, Domain Overview, Facts SERP, Fallback SERP, Fetch GBP, Find Competitors, Get Crawl Summary, Get Crawled Pages, Google Trends, Keyword Data, Keyword Ideas, Ranked Keywords, Run AI Requests, Run BL Requests, Run Competitor Keywords, Run Crawl Extras, Run Gap Requests, Run Ladder Research, Run Research, SERP Check, SERP Check (Site), SERP Top 10, Site Authority, Start Crawl
Gmail IMAP (SEO Agent)	imap	SE0credImapGmail	Mailbox Trigger
Gmail SMTP (SEO Agent)	smtp	SE0credSmtgGmail	Email Ops, Email Owner (Alert), Send AI Report, Send Audit Report, Send BL Report, Send Cadence Note, Send Ladder Report, Send Progress Email, Send Report, Send Site Report
Google PageSpeed	httpHeaderAuth	SE0credPageSpeed	PageSpeed
Google Service Account (SEO Agent)	googleApi	SE0credGoogleSvc	GA4 Accounts, GA4 Report, GA4 Streams, GSC Audit Query, GSC Query, GSC Sitemaps (Audit), GSC Sites, GSC Sites (Audit), Inspect URL
Jina Reader	httpHeaderAuth	SE0credJinaReade	Read Client Pages, Read Competitor Homes, Read Competitor Pages, Read Site (AI), Read Website, Retry Blocked Pages
SEO API Key	httpHeaderAuth	SE0credApiHeader	Webhook Trigger
SEO API Key (test)	httpHeaderAuth	SE0credApiTest01	Test Webhook
SEO Form Login	httpBasicAuth	SE0credFormLogin	Start Form
WordPress (SEO Agent)	wordpressApi	SE0credWordPress	Create WP Draft

Also: the API caller secret is the header credential "SEO API Key" (X-API-Key) on the API workflow; the form login is the basic-auth credential "SEO Form Login". The SMTP credential needs a Google App Password.

5. Data model and API contract

Data Tables (created on first use by the Data Table node, referenced by name)

Table	Columns	Written by	Read by
seo_ladders	ladder_id (string), domain (string), head_keyword (string), rung (number), page_no (number), keyword (string), supporting (string), page_type (string), target_url (string), page_exists (boolean), status (string), months (string), start_date (string), country (string), location_code (number), language_code (string), email (string), callback_url (string), request_id (string)	Ladder Rows → Save Ladder Rows (one row per ladder page; status planned / writing)	Rank Tracker (Load Ladders)
seo_rank_history	ladder_id (string), keyword (string), checked_at (string), position (number), url (string), serp_features (string), domain (string), rung (number)	Rank Tracker (Parse Positions → Save History; position 0 = not in top 50, -1 = check failed)	Rank Tracker (deltas, stop rule)

API

POST /webhook/seo-keyword-check, header X-API-Key. Body: mode (keyword | verdict | audit | discover | describe | ladder), keyword, country (name or ISO-2), domain, business, customers, business_facts, cta, tone, goal, page_type, competitors, receive ["Keyword Report", "Page Content"], report_type, pages_now (ladder, 1-3), wordpress_url, email, callback_url (https; http only for localhost while testing). Answers **202** {status: accepted, request_id, estimated_minutes} or **400**.

Callbacks (POST to callback_url, same request_id): stage = site_description · keyword_strategy · content (verdict, keyword data, QA summary, ladder_id/rung/head when part of a ladder, run_ledger, pdf + file as base64) · site_audit · full_report · ladder_plan (head, feasibility, rungs, top_page, link_map, timeline, write_now, requirements, stats, pdf/file) · rank_tracker (progress; positions, rungs, gains, drops, next_step with a ready-to-send api_body). Rejections: {status: rejected, request_id, error}.

Ladder page runs carry force_content, ladder_id, ladder_rung, ladder_head, ladder_page_no, ladder_links (top page + sibling; planned pages are allowed link targets).

6. Main workflow, stage by stage

Every node of **SEO Agent v4** grouped by stage, in pipeline order. Descriptions come from the node itself (leading comment of the Code, HTTP target, IF condition, agent prompt, form fields) so they stay true to the build.

Entry points & forms

A six-option public form (basic auth) or the Execute Workflow Trigger "API Entry", which the API front door, the ladder page runs, the discovery follow-ups and the audit-form spawns all use. Each form option opens its own page of fields.

Node	Type	What it does
Start Form	Form trigger	Form page with fields: What do you want? credential: SEO Form Login
Choose Path	Switch	Switch: outputs keyword, discover, describe, audit, verdict_check, ladder, track, published, checkin, case_study, profile, ai_visibility, backlinks + fallback
Page Keyword	Form page	Form page with fields: Target Keyword, Target Country, Page Type, Website Domain, Existing page URL (optional), City or area (local pages only), Video URL for this page (optional), Video transcript (optional), Business facts to include (optional), Main call to action (optional), Tone of voice, Content goal, Extra Features, What do you want to receive?, Email me the report
Page Discover	Form page	Form page with fields: What does your business do?, Who are your customers?, Website Domain, Content goal, Business facts to include (optional), Target Country, Extra Features, What do you want to receive?, Email me the reports
Page Describe	Form page	Form page with fields: Website Domain, Email me the report
Page Audit	Form page	Form page with fields: Website Domain, Target Country, Report type, Pages to crawl, Render JavaScript, Main competitors, Email me the report
Page Verdict	Form page	Form page with fields: Target Keyword, Target Country, What does your business do?, Website Domain, Email me the result
Page Ladder	Form page	Form page with fields: Keyword you want to rank for, Website Domain, Target Country, What does your business do?, Who are your customers?, Business facts to include (optional), Content goal, Tone of voice, Main call to action (optional), Pages to write now, Email me the plan and the pages
Unmatched Choice	Code	Guard: unknown form option.
API Entry	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.

Intake, validation & guards

Normalize Input turns any form page or API body into one normalised object (mode, keyword, domain, country codes, deliverables, ladder fields, CONFIG values). Validation problems are returned as data, not thrown; Rate Limit enforces the

per-address daily limit and the estimated Claude budget; Input OK? routes errors to a form page or to the caller's callback.

Node	Type	What it does
Normalize Input	Code	v4: validation errors no longer crash the run. They are returned as { validation_error } and shown to the user (form completion page) or returned as HTTP 400 (API). See "Input OK?" right after Rate Limit.
Rate Limit	Code	Simple abuse guard: limits runs per requester and per day (state lives in workflow static data; it persists only for active/production executions, not manual test runs).
Input OK?	IF	Routes on: {{ !\$json.validation_error }}
Error Via API?	IF	Routes on: {{ !!\$json.via_webhook }}
API Reject	Code	An API request failed validation inside the main run. Tell the caller through callback_url; with no callback_url, fail loudly so the error workflow alerts ops.
POST Rejection	HTTP Request	HTTP POST {{ \$json.callback_url }}
Show Input Error	Form page	Form completion page: "Something needs fixing". Ends the form flow (the execution stays in waiting).

Site read & description

When the business is not described, the homepage is read through Jina Reader and summarised by Claude Haiku into a structured site description (services, audience, USPs, seed keywords); unreadable sites get an honest fallback.

Node	Type	What it does
Need Site Read?	IF	Routes on: {{ \$json.need_site_description }}
Read Website	HTTP Request	HTTP GET =https://r.jina.ai/https://{{ \$json.domain }} — Jina Reader (page → markdown) (credential: Jina Reader)
Clean Site Text	Code	Strips the Jina Reader markdown of navigation, images and boilerplate; sets read_ok and the page title.
Site Read OK?	IF	Routes on: {{ !!\$json.read_ok }}
Site Describer	AI Agent	AI agent with structured output. Prompt starts: "You are a senior business analyst and SEO strategist."
Claude — Site Describer	Anthropic model	Claude model node: claude-haiku-4-5-20251001; max tokens 4096 credential: Anthropic (SEO Agent)
Parser — Site Describer	Output parser	Structured output parser; fields: business_name, one_line_summary, business_description, industry, products_or_services, target_audience, unique_selling_points, location_served, tone_of_brand, suggested_meta_title, suggested_meta_description, seed_keywords
Parse Description	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences or surrounding text
Describe Fallback	Code	The homepage could not be read (blocked, timeout, empty). Do not call the LLM with nothing; carry on with what the user typed.
Route Mode	Switch	Switch: outputs describe, discover, keyword, audit, ladder, track, published, checkin, profile, case_study, ai_visibility, backlinks + fallback

Unmatched Mode	Code	Guard: unknown mode after routing.
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Describe mode

Renders the site description page (form) or posts it to the callback (API).

Node	Type	What it does
Build Description Page	Code	Renders the site-description result page (HTML) for the form, with a warning when the site could not be read.
Via Webhook (Describe)?	IF	Routes on: {{ !!\$json.via_webhook }}
Build Describe Response	Code	API delivery for "describe my website": the description goes to the callback instead of a form page
Show Description	Form page	Form completion page: "showText". Ends the form flow (the execution stays in waiting).

Keyword discovery

Seeds by buyer motive → seven DataForSEO Labs pulls (ideas, long-tail, related, live SERP, competitor rankings) → one pool with an opportunity score → Claude relevance screen → clusters, quick wins, questions → AI search demand → live SERP check of the priority keywords → Keyword Strategy report (Word + PDF). Form users then pick the keyword on "Choose your keyword"; the follow-ups they ticked (report / content / audit) start as separate executions.

Node	Type	What it does
Keyword Seeds	AI Agent	AI agent with structured output. Prompt starts: "You are an SEO keyword strategist."
Anthropic Chat Model	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (medium) credential: Anthropic (SEO Agent)
Parser — Keyword Seeds	Output parser	Structured output parser; fields: primary_seed, seed_groups, services, seeds
Seed List	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences
Research Requests	Code	Discover mode: one item per DataForSEO Labs request (executed by "Run Research"). Sources: ideas - keyword_ideas for up to 20 seeds (relevance order, SERP features included) suggestions - keyword_suggestions (long-tail phrases that contain the seed) for the 3 main seeds related - related_keywords ("searches related to", depth 2) for the 2 main seeds serp - live SERP for the primary seed -> who ranks (competitor keywo
Run Research	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Competitor Keyword Requests	Code	From the primary-seed SERP, pick the 3 closest competitors and fetch the keywords they rank for (top 20 positions). Always emits at least one request so the chain never stalls.
Run Competitor Keywords	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Collect Research	Code	Merges every research response into one keyword pool with metrics, SERP features and competitor positions, then scores each keyword's OPPORTUNITY before the AI relevance pass: traffic potential (volume x CTR after SERP features) x winnability (difficulty, top-10 link strength) x commercial value (CPC) x intent weight for the client's goal x trend

Relevance Chunks	Code	Splits the top candidates into chunks of 80 for the AI relevance pass (the agent runs once per item).
Keyword Relevance	AI Agent	AI agent with structured output. Prompt starts: "You are an SEO strategist screening keyword ideas for one specific business."
Claude — Keyword Relevance	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — Keyword Relevance	Output parser	Structured output parser; fields: items
Rank Keywords	Code	Applies the AI relevance verdicts, computes the final opportunity score, builds topic clusters and the content plan, and chooses the priority keywords (live SERP check next) and the keyword for the content pipeline.
AI Demand	HTTP Request	HTTP POST https://api.dataforseo.com/v3/ai_optimization/ai_keyword_data/keywords_search_volume/live — AI search demand (credential: DataForSEO)
Priority Keywords	Code	One item per priority keyword for the live SERP check ("Candidate SERP"). Always at least one item.
Candidate SERP	HTTP Request	HTTP POST https://api.dataforseo.com/v3/serp/google/organic/live/advanced — live Google SERP (credential: DataForSEO)
Build Keyword Strategy	Code	Final keyword strategy: merges AI search demand and the live SERP checks into the ranked keyword set and renders the "Keyword Strategy" report. Also exposes keyword_suggestions for the downstream file/email/content steps.
Choice Step?	IF	Routes on: <code>{{ !\$json.via_webhook && !(\$json.include_content \$json.include_seo_report) && ((\$json.choice_options []).length > 1) }}</code>
Choose Keyword	Form page	Form page defined from JSON at runtime: <code>{{ JSON.stringify([{ fieldLabel: 'Which keyword should we write for?', fieldType: 'dropdown', fieldOptions: { values: (\$json.choice_options</code>
Apply Choice	Code	Discovery choice step (form only): the user's pick replaces the system's pipeline keyword.
Build Keyword File	Code	Wraps the Keyword Strategy HTML into the Word-compatible file with a dated file name; computes the run ledger.
Prepare PDF Ideas	Code	Copy of the report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name)
Render PDF Ideas	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF Ideas	Code	Merge the PDF (when the renderer succeeded) with the original Word-compatible file. Never fails the run.
Spawn Follow-ups	Code	Discover mode: the extras the user ticked (keyword report / page content for the chosen keyword, site audit or full report) run as SEPARATE executions through API Entry. A form completion page ends the execution it is in, so the download page can only be the last step; API callers still get one callback per stage with the same request_id.
Start Follow-up Runs	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode each, wait=False) — starts a separate execution
Via Webhook (Ideas)?	IF	Routes on: <code>{{ !\$json.via_webhook }}</code>

Build Ideas Response	Code	API delivery for "suggest keywords": the keyword strategy (JSON + PDF + Word) goes to the callback. If content or an audit was also requested, those arrive later as separate callbacks (stage: content / site_audit).
Download Keyword Ideas	Form page	Form completion page: "Your keyword ideas are ready" (returns the file for download). Ends the form flow (the execution stays in waiting).

Keyword pipeline: research

Used by keyword, verdict and ladder runs. Sitemaps give real internal-link targets and detect an existing page; the client's own pages are read for facts; the live SERP (top 30, PAA, AI Overview) picks competitors whose pages are read (with a browser-engine retry for bot-protected pages) and distilled into a term model, questions and statistics; a second search collects verified facts with sources; Claude analyses the competition; keyword metrics and site authority complete the picture.

Node	Type	What it does
Prepare Keyword Run	Code	Single entry point for the keyword pipeline (SERP -> competitors -> brief -> content). Reached from Route Mode (keyword mode) or from the Discover fork (top suggested keyword).
Has Domain?	IF	Routes on: {{ \$json.domain }} notEmpty
Fetch Sitemap Index	HTTP Request	HTTP GET =https://{ \$json.domain }/sitemap.xml
Parse Sitemap Index	Code	Input: the fetched /sitemap.xml (text). Output: the sitemap files to read (max 5).
Fetch Sitemaps	HTTP Request	HTTP GET {{ \$json.url }}
Collect Site URLs	Code	Builds the list of REAL pages on the user's site (from sitemap files) so the brief and the copywriter only link to URLs that exist, and detects whether a page for this keyword already exists.
Read Client Pages?	IF	Routes on: {{ !!\$json.domain }}
Client Page Requests	Code	Read the client's own pages (existing page on the topic, homepage, one related service page) so the brief and the copy are grounded in real facts instead of placeholders. One item per URL for "Read Client Pages".
Read Client Pages	HTTP Request	HTTP GET =https://r.jina.ai/{{ \$json.url }} — Jina Reader (page → markdown) (credential: Jina Reader)
Collect Client Pages	Code	Cleans the client's pages and extracts cheap trust signals (phone, email, years, certifications, counts).
SERP Top 10	HTTP Request	HTTP POST https://api.dataforseo.com/v3/serp/google/organic/live/advanced — live Google SERP (credential: DataForSEO)
Pick Top 6	Code	Picks up to 8 organic competitor URLs from the live SERP, skipping the client's own domain, social sites, marketplaces, job boards, courses and software vendors.
Read Competitor Pages	HTTP Request	HTTP GET =https://r.jina.ai/{{ \$json.url }} — Jina Reader (page → markdown) (credential: Jina Reader)
Find Blocked Pages	Code	Which competitor pages came back blocked, empty or too thin? Those get a second attempt through Jina's browser engine + proxy pool.
Any Blocked?	IF	Routes on: {{ (\$json.blocked_count 0) > 0 }}
Blocked Page Items	Code	One item per blocked page for the retry (this node only runs when at least one page was blocked)

Retry Blocked Pages	HTTP Request	HTTP GET =https://r.jina.ai/{{ \$json.url }} — Jina Reader (page → markdown) (credential: Jina Reader)
Analyze Competitor Pages	Code	Turns the raw competitor pages into (1) a per-competitor digest for the analyst, (2) a TERM MODEL — the words and phrases most top pages share (what the topic is "made of"), (3) the questions competitors answer, (4) the statistics they cite, plus the SERP features from the same search. Longer excerpts than v3 (9,000 chars kept per page).
Facts SERP	HTTP Request	HTTP POST https://api.dataforseo.com/v3/serp/google/organic/live/advanced — live Google SERP (credential: DataForSEO)
Collect Facts	Code	Verified facts: statistics from a second live search ("<keyword> statistics <year>"), kept only with their source URL, so the writer can cite real numbers instead of inventing them or hiding behind [Statistic].
Competitor Analyzer	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO content strategist."
Claude — Competitor Analyzer	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (medium) credential: Anthropic (SEO Agent)
Parser — Competitor Analyzer	Output parser	Structured output parser; fields: search_intent, intent_explanation, ranking_page_types, recommended_word_count, common_topics, common_heading_patterns, content_gaps, competitor_strengths, competitor_weaknesses, unique_angle, must_include_entities, faq_opportunities, answer_block_question, serp_feature_strategy...
Parse Analysis	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences or surrounding text
Keyword Data	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/keyword_overview/live — keyword metrics (credential: DataForSEO)
Has Domain (Authority)?	IF	Routes on: {{ !!\$json.domain }}
Site Authority	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/domain_rank_overview/live — site authority / domain overview (credential: DataForSEO)
Merge Keyword Data	Code	Merges keyword metrics (volume, CPC, difficulty, intent, trend, competitor link strength) and the site authority into the context for the verdict.

Verdict

Claude Sonnet weighs demand, difficulty, link strength, intent, SERP features and the site's authority into GO / GO_WITH_CHANGES / AVOID with reasons, what must change, expected visits and time to rank. Ladder runs branch here; AVOID stops content unless the run is a forced ladder page.

Node	Type	What it does
Verdict Agent	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO strategist deciding whether a business should target a specific keyword."
Verdict Chat Model	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (medium) credential: Anthropic (SEO Agent)

Parser — Verdict	Output parser	Structured output parser; fields: verdict, score, reasons, risks, recommended_page, secondary_keywords, what_must_change, expected_monthly_visits_top3, time_to_rank_months
Parse Verdict	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences or surrounding text
Ladder Mode?	IF	Routes on: {{ \$json.mode === 'ladder' }}
Verdict Router	IF	Routes on: {{ \$json.verdict === 'AVOID' && !\$json.force_content }}
Need Content?	IF	Routes on: {{ \$json.include_content }}

Content context: profile, proof, page role, video (v4.4)

Before the brief, every content run loads the site's business profile (author, expert reviewer, address / NAP) and its recorded case studies from the Data Tables seo_profiles / seo_case_studies, reads the video page when the request carries a video (YouTube watch page or Vimeo oEmbed: title, upload date, duration), and Brief Context derives the page role (hub for the ladder's top page or a pillar page, case study, local page, standard), the cluster pages a hub links down to, and the prompt blocks for the brief and the copy (author and experience, proof, role rules, video).

Node	Type	What it does
Load Profile (Brief)	Data Table	Data Table: row get — table seo_profiles
Load Case Studies (Brief)	Data Table	Data Table: row get — table seo_case_studies
Has Video?	IF	Routes on: {{ !!(('\$('Parse Verdict').first().json.video {}).fetch_url) }}
Fetch Video Details	HTTP Request	HTTP GET {{ \$('Parse Verdict').first().json.video.fetch_url }}
Brief Context	Code	Content runs (v4.4): what the brief, the copy, the schema and the blog package need beyond the keyword research — the site's business profile (author, reviewer, address / NAP), proof from its case studies, the page role (hub, case study, local, standard), the cluster pages a hub links down to, and the video details (YouTube page / Vimeo oEmbed, read by Fetch Video Details).

On-demand monitors and audit upgrades (v4.5)

Check my AI visibility / Check my backlinks (form + API) start the monitor workflows at once for one domain; Track my site stores the monitor settings (seo_monitors). Every audit: brand & entity check against the Google Business Profile (a listing counts only with the site's website or phone) and the homepage Organization schema; audit history with the since-the-last-audit diff; internal links to add; the fix pack (robots.txt, llms.txt, redirect map, schema, internal links; zipped for the e-mail).

Node	Type	What it does
Page AI Visibility	Form page	Form page with fields: Website Domain, Target Country, Main services or products (optional), Competitors (optional), Email me the report
Page Backlinks	Form page	Form page with fields: Website Domain, Target Country, Competitors (optional), Email me the report
Monitor Start	Code	"Check my AI visibility" / "Check my backlinks" (form or API mode ai_visibility / backlinks): the monitor runs at once for this one domain, in its own execution (Manual Run of SEO Agent — AI Visibility Tracker / Backlink Monitor), and delivers to this requester. Tracked sites keep their settings.

AI Visibility Run?	IF	Routes on: {{ \$('Normalize Input').first().json.mode === 'ai_visibility' }}
Start AI Visibility	Execute Workflow	Execute Workflow → SEOagentAIVisib1 (mode once, wait=False) — starts a separate execution
Start Backlink Monitor	Execute Workflow	Execute Workflow → SEOagentBacklnk1 (mode once, wait=False) — starts a separate execution
Monitor Started	Code	Confirms the on-demand monitor run (form completion page / API callback stage ai_visibility_started or backlinks_started).
Via Webhook (Monitor)?	IF	Routes on: {{ !\$('Normalize Input').first().json.via_webhook }}
Build Monitor Response	Code	API answer for modes ai_visibility / backlinks: the run started; the report follows as its own callback (stage ai_visibility / backlinks).
Monitor Check Started	Form page	Form completion page: "={{ \$json.mode === 'ai_visibility' ? 'Your AI visibility check has started' : 'Your backlink check has started' }}". Ends the form flow (the execution stays in waiting).
Ensure Monitors Table (Track)	Data Table	Data Table: table create — table seo_monitors
Load Monitors (Track)	Data Table	Data Table: row get — table seo_monitors
Monitor Row (Track)	Code	"Track my site": the site's monitor settings (seo_monitors, upsert by site_id; exact columns). Values given now (competitors from the form, the API's `monitors` object) replace the stored ones; everything else is kept, and a new site starts with every monitor on.
Save Monitors (Track)	Data Table	Data Table: row upsert — table seo_monitors
Load Profile (Audit)	Data Table	Data Table: row get — table seo_profiles
GBP Request	Code	Brand / entity check in the audit: the business name to look up on Google Business Profile (DataForSEO Business Data, \$0.0054): from the business profile, else the homepage's Organization / LocalBusiness schema, else og:site_name. No name = no lookup.
Look Up GBP?	IF	Routes on: {{ !\$json.skip }}
Fetch GBP	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Entity Check	Code	Brand & entity consistency (category AI Search Readiness): the same name, phone and website on the Google Business Profile, the site's own schema and the business profile; Organization schema with sameAs links; a claimed listing. AI assistants and Google's local results rely on one consistent entity. Findings are appended to the Search Console step's extra_findings so Build Site Issues scores them.
Ensure Audits Table	Data Table	Data Table: table create — table seo_audits
Ensure Findings Table	Data Table	Data Table: table create — table seo_audit_findings
Load Audit History	Data Table	Data Table: row get — table seo_audits
Load Audit Findings	Data Table	Data Table: row get — table seo_audit_findings
Audit Diff	Code	Code node.
Audit Row	Code	The audit's row in seo_audits (exact columns).
Save Audit	Data Table	Data Table: row insert — table seo_audits

Finding Rows	Code	One seo_audit_findings row per scored finding (exact columns); the key ignores numbers so "7 pages ..." and "5 pages ..." are the same issue.
Any Finding Rows?	IF	Routes on: {{ !\$json.skip }}
Save Findings	Data Table	Data Table: row insert — table seo_audit_findings
Fix Pack	Code	Fix pack for every audit: ready-to-deploy files built from the audit's own data — robots.txt (current rules kept, sitemap line, AI search crawlers allowed, blocking "Disallow: /" removed for search engines), llms.txt (site summary + key pages by section), a redirect map for broken pages (CSV + nginx + Apache), Organization / LocalBusiness / WebSite schema from the business profile, Google Business Profile and homepage
Zip Fix Pack	compression	compression
Restore Audit Item	Code	The audit item with the fix-pack files and, when zipping worked, fix-pack.zip for the e-mail (the Compression node replaces the item's binaries).

Business profile & case studies (v4.4)

Two free-standing modes. "Set up my business profile" / API mode profile merges the submitted fields over the stored row (empty keeps, "-" clears), upserts seo_profiles and reports the E-E-A-T and local readiness with a Person / LocalBusiness preview; free and never rate-limited. "Write a case study" / API mode case_study stores the intake in seo_case_studies (proof for later pages), logs the page in seo_content_log and starts the case-study page as its own content run (page type Case Study, force_content, internal key casestudy:); "I published a page" later marks the case study published so other pages may link it.

Node	Type	What it does
Page Profile	Form page	Form page with fields: Website Domain, Author name, Author job title, Author credentials and experience, Author bio, Author profile links (one per line), Author photo URL (optional), Topics the author knows best, Expert reviewer (optional), Company name, Category (for local search), Street address, City, Region / state / emirate, Postal code, Country, Phone, Public e-mail on the website, Opening hours, Areas you serve, Google Maps link (optional), Logo URL (optional), Email (optional)
Ensure Profile Table	Data Table	Data Table: table create — table seo_profiles
Load Profile (Profile)	Data Table	Data Table: row get — table seo_profiles
Profile Row	Code	"Set up my business profile" (v4.4): merges the submitted fields over the stored profile — an empty field keeps the stored value, "-" clears it — into one row for the Data Table seo_profiles (upsert by site_id). Every content run for the domain then carries the author, reviewer and address.
Save Profile	Data Table	Data Table: row upsert — table seo_profiles
Profile Delivery	Code	Restores the profile after the Data Table step and reports what it gives every page: the author box and Person schema (E-E-A-T), the LocalBusiness / NAP block for local pages, and what is still missing.
Via Webhook (Profile)?	IF	Routes on: {{ !\$('Normalize Input').first().json.via_webhook }}
Build Profile Response	Code	API answer for mode "profile" (POSTed to callback_url): the stored profile, what it adds to every page and what is still missing.

Profile Saved	Form page	Form completion page: "Business profile saved for {{ \$json.domain }}" . Ends the form flow (the execution stays in waiting).
Page Case Study	Form page	Form page with fields: Website Domain, Target Country, Service you delivered, Client name or description, May we name the client?, Client industry, Client location, The challenge, What you did, Timeline, Results (with numbers), Client quote (optional), Quote by (name, role), Keyword for this page (optional), Email me the case study
Ensure Case Table	Data Table	Data Table: table create — table seo_case_studies
Ensure Log Table (Case)	Data Table	Data Table: table create — table seo_content_log
Case Study Row	Code	"Write a case study" (v4.4): the intake becomes one row of seo_case_studies — the proof library that every later page for the site may cite — and the case-study page itself is written as its own content run (Spawn Case Study Run -> API Entry), logged in seo_content_log.
Save Case Study	Data Table	Data Table: row upsert — table seo_case_studies
Case Study Log Row	Code	The content-log row for the case-study page (status started), so "I published a page" finds it by keyword or request id.
Log Case Study	Data Table	Data Table: row upsert — table seo_content_log
Spawn Case Study Run	Code	Starts the case-study page as a content run (mode keyword, page type Case Study, force_content) in its own execution through API Entry: a form completion page ends the execution it is in, and the run takes 10-15 minutes. Internal key "casestudy:" = not charged to the owner's daily runs.
Start Case Study Run	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode once, wait=False) — starts a separate execution
Case Study Delivery	Code	Restores the case-study intake after the Data Table steps and the spawn, and tells the owner what happens next.
Via Webhook (Case Study)?	IF	Routes on: {{ !\$('Normalize Input').first().json.via_webhook }}
Build Case Study Response	Code	API answer for mode "case_study" (POSTed to callback_url): the intake is stored and the page run started; the page itself follows as a "content" callback.
Case Study Started	Form page	Form completion page: "Your case study is being written". Ends the form flow (the execution stays in waiting).
Mark Case Study Published	Data Table	Data Table: row update — table seo_case_studies

Content generation & QA

Opus writes the brief and the page from the research; Sonnet critiques; the deterministic QA computes a content score (structure, topic coverage, readability, specificity, clean language, length), fixes what it can and asks the Opus editor for one revision when needed.

Node	Type	What it does
Strategy Brief	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO content strategist."

Claude — Strategy Brief	Anthropic model	Claude model node: claude-opus-5-5; max tokens 24576; adaptive thinking (medium); streaming credential: Anthropic (SEO Agent)
Parser — Strategy Brief	Output parser	Structured output parser; fields: meta_title, meta_description, slug, h1, primary_keyword, secondary_keywords, target_audience, search_intent, tone, target_word_count, answer_block, outline, gaps_to_cover, unique_angle...
Parse Brief	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences or surrounding text
Copywriter	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO copywriter and subject-matter editor writing for"
Claude — Copywriter	Anthropic model	Claude model node: claude-opus-5-5; max tokens 32768; adaptive thinking (medium); streaming credential: Anthropic (SEO Agent)
Critic	AI Agent	AI agent with structured output. Prompt starts: "You are a demanding editor-in-chief and SEO lead reviewing a draft before it goes to a client."
Claude — Critic	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (medium) credential: Anthropic (SEO Agent)
Parser — Critic	Output parser	Structured output parser; fields: score, verdict, strengths, problems, banned_phrases_found, unsupported_claims, missing_brief_items, revision_instructions
Parse Critique	Code	Accepts an object (from the Structured Output Parser) or a string; attaches the critique to the draft for Content QA.
Content QA	Code	Deterministic quality gate for the generated page. v5 adds: banned-phrase (AI cliché) detection, readability (Flesch), term coverage against the competitor term model / brief, specificity, section-length checks, the editor's critique, and a 0-100 CONTENT SCORE. Schema is generated here (page-type aware), never by the model.
Needs Revision?	IF	Routes on: <code>{{ \$json.content_qa.passed === false && \$json.qa_round < (\$json.qa_max_rounds 2) }}</code>
Editor	AI Agent	AI agent with structured output. Prompt starts: "You are a meticulous SEO editor."
Claude — Editor	Anthropic model	Claude model node: claude-opus-5-5; max tokens 32768; adaptive thinking (medium); streaming credential: Anthropic (SEO Agent)

Report & delivery (keyword / content)

The Word-compatible report (verdict, keyword overview, SERP features, competitor analysis, page content, QA, schema) is rendered to PDF by Gotenberg and delivered by e-mail, form download or API callback; the optional WordPress gate runs first.

Node	Type	What it does
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Build Word File	Code	Builds the keyword report / page content document (Word-compatible HTML): verdict, keyword overview, SERP features, competitor analysis, the page content with meta tags, QA results, topic coverage, sources, title/meta alternatives, editorial review and JSON-LD; computes the run ledger (DataForSEO spend per node, AI calls, duration).
Prepare PDF Report	Code	Copy of the report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name)
Render PDF Report	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF Report	Code	Merge the PDF (when the renderer succeeded) with the original Word-compatible file. Never fails the run.
Publish To WordPress?	IF	Routes on: <code>{{ !!\$json.publish_wordpress && !!\$json.wordpress_url && !!\$json.page_markdown }}</code>
Publish WordPress Draft	Execute Workflow	Execute Workflow → SEOagentWordPres (mode once, wait=False) — starts a separate execution
Has Email?	IF	Routes on: <code>{{ \$json.email }}</code> notEmpty
Send Report	Send Email	Sends e-mail via SMTP to <code>{{ \$json.email }}</code> ; subject: =Your <code>{{ \$json.report_type }}</code> for " <code>{{ \$json.keyword }}</code> " — <code>{{ \$json.brand }}</code> credential: Gmail SMTP (SEO Agent)
Deliver Via Download?	IF	Routes on: <code>{{ !\$json.via_webhook && \$json.mode !== 'discover' }}</code>
Download Report	Form page	Form completion page: "Your report is ready" (returns the file for download). Ends the form flow (the execution stays in waiting).
Is Webhook?	IF	Routes on: <code>{{ \$json.via_webhook }}</code>
Build Webhook Response	Code	API callback payload for content runs: status, stage, ladder fields, verdict, keyword data, QA summary, run ledger, PDF and Word bytes.
Has Callback?	IF	Routes on: <code>{{ \$json.callback_url }}</code> notEmpty
POST Callback	HTTP Request	HTTP POST <code>{{ \$json.callback_url }}</code>

Keyword ladder

Four DataForSEO pulls around the head term → topic-filtered pool with opportunity score → Claude relevance anchored on the head term → rungs by difficulty (1: ≤25 long-tail, 2: 26-45, 3: 46-60, top = head), one page per topic cluster, existing pages matched, internal link map, timeline, feasibility → Keyword Ladder Plan (Word + PDF) → rows into the Data Table `seo_ladders` → delivery → the first page(s) start as separate content runs.

Node	Type	What it does
Ladder Requests	Code	Ladder mode: research pulls around the head term (executed by "Run Ladder Research"). suggestions - long-tail phrases that contain the head term related - "searches related to" the head term, depth 2 ideas - keyword ideas for the head term + the site's main services (topic neighbours) ranked - keywords the client's domain already ranks for (current positions for the rungs)
Run Ladder Research	HTTP Request	HTTP POST <code>{{ \$json.endpoint }}</code> — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)

Ladder Pool	Code	Ladder mode: merges the research pulls into one pool around the head term, keeps only keywords that share its topic, scores opportunity (same formula as discovery) and records where the client's site ranks today.
Ladder Relevance Chunks	Code	Splits the top candidates into chunks of 80 for the AI relevance pass (the agent runs once per item).
Ladder Keyword Relevance	AI Agent	AI agent with structured output. Prompt starts: "You are an SEO strategist screening keyword ideas for one specific business."
Claude — Ladder Relevance	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — Ladder Relevance	Output parser	Structured output parser; fields: items
Ladder Plan	Code	Ladder mode: applies the AI relevance screen, assigns keywords to rungs by difficulty, builds one page per topic cluster, maps internal links, matches existing pages, estimates the timeline and picks the pages to write now.
Build Ladder Report	Code	Builds the Keyword Ladder Plan document: one-page summary, feasibility with alternatives, the ladder per rung (pages, supporting keywords, difficulty, your position, existing or planned URL), internal link map, pages written now, requirements, tracking, later keywords, method; slims the item and computes the run ledger.
Prepare PDF Ladder	Code	Copy of the report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name)
Render PDF Ladder	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF Ladder	Code	Merge the PDF (when the renderer succeeded) with the original Word-compatible file. Never fails the run.
Ensure Ladder Table	Data Table	Data Table: table create — table seo_ladders
Ensure History Table	Data Table	Data Table: table create — table seo_rank_history
Ladder Rows	Code	One row per ladder page for the "seo_ladders" Data Table (the rank tracker reads these every week).
Save Ladder Rows	Data Table	Data Table: row insert — table seo_ladders
Ladder Delivery	Code	Restores the ladder report item (json + PDF/Word binaries) after the Data Table steps and records whether the rows were stored.
Spawn Page Runs	Code	Starts one content run (mode "keyword") per page in write_now, as separate executions through API Entry. Each run delivers its own PDF/Word by e-mail and/or callback, tagged with the ladder id and rung.
Start Page Runs	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode each, wait=False) — starts a separate execution
Has Email (Ladder)?	IF	Routes on: {{ !!\$json.email }}
Send Ladder Report	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: =Your keyword ladder plan for "{{ \$json.keyword }}" — {{ \$json.brand }} credential: Gmail SMTP (SEO Agent)
Via Webhook (Ladder)?	IF	Routes on: {{ !!\$json.via_webhook }}

Build Ladder Response	Code	API delivery for ladder mode: the plan (JSON + PDF + Word) goes to the callback. The pages written now arrive later as separate callbacks (stage: content) carrying the same request_id and ladder_id.
Download Ladder Plan	Form page	Form completion page: "Your keyword ladder plan is ready" (returns the file for download). Ends the form flow (the execution stays in waiting).

Technical audit: crawl & probes

Form audits are spawned as their own execution so the "started" page can show at once. DataForSEO On-Page crawls up to 200 pages (polled every minute, 25 minutes max); extra reports list duplicate tags, non-indexable pages and redirect chains; host probes (no redirects) and content probes (robots, sitemap, llms.txt, 11 crawler user-agents) measure what browsers and crawlers really get.

Node	Type	What it does
Audit Via Form?	IF	Routes on: <code>{{ !\$json.via_webhook }}</code>
Spawn Audit Run	Code	Audit requested through the form: the audit itself runs as a separate execution (API Entry, e-mail delivery) so the "Your audit has started" page can be shown at once — a form completion page ends the execution it is in.
Start Audit Run	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode once, wait=False) — starts a separate execution
Audit Started	Form page	Form completion page: "Your audit has started". Ends the form flow (the execution stays in waiting).
Start Crawl	HTTP Request	HTTP POST https://api.dataforseo.com/v3/on_page/task_post — start On-Page crawl (credential: DataForSEO)
Save Task ID	Code	Stores the DataForSEO On-Page task id; throws a clear error when the crawl could not be started.
Wait 1 min	Wait	Waits 60 (execution is persisted while waiting).
Get Crawl Summary	HTTP Request	HTTP GET https://api.dataforseo.com/v3/on_page/summary/{{ \$json.crawl_task_id }} — crawl summary (credential: DataForSEO)
Check Crawl	Code	Reads the crawl summary; sets crawl_finished / progress; aborts after 25 polls with "took too long".
Crawl Done?	IF	Routes on: <code>{{ \$json.crawl_finished }}</code>
Get Crawled Pages	HTTP Request	HTTP POST https://api.dataforseo.com/v3/on_page/pages — crawled pages (credential: DataForSEO)
Crawl Extra Requests	Code	Extra DataForSEO On-Page reports for the finished crawl (each becomes one HTTP request)
Run Crawl Extras	HTTP Request	HTTP POST <code>{{ \$json.endpoint }}</code> — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Collect Crawl Extras	Code	Consolidates the extra On-Page reports into one object. Every report is optional.
Build Probes	Code	Host probes: raw responses WITHOUT following redirects, to see how apex/www/http resolve. Content probes (robots, sitemap, llms.txt, crawler access) live in "Content Probes" and DO follow redirects.

Run Probes	HTTP Request	HTTP GET {{ \$json.url }}
Content Probes	Code	Content probes: what browsers, search crawlers and AI crawlers actually receive (redirects followed).
Run Content Probes	HTTP Request	HTTP GET {{ \$json.url }}
Analyze Probes	Code	Interprets host and content probes: canonical host and redirects, caching and TTFB, security headers, robots.txt, sitemap, llms.txt, crawler access.
Full Report?	IF	Routes on: {{ \$json.audit_level === 'full' }}

Full report modules

Key-page HTML and schema checks, competitor discovery (Labs + live SERPs), domain overview and age (WHOIS + RDAP), ranked keywords and gaps, backlinks and link gap, PageSpeed for you and competitors, an E-E-A-T content review by Claude, and AI visibility (brand SERP, AI Overviews, live ChatGPT/Perplexity answers).

Node	Type	What it does
Pick Key Pages	Code	Chooses the homepage, service pages and articles whose HTML and schema get inspected (hub pages excluded).
Fetch Page HTML	HTTP Request	HTTP GET {{ \$json.url }}
Analyze HTML	Code	Checks titles, descriptions, headings, canonical, Open Graph, schema presence and sameAs links on the key pages.
URLs To Check	Code	Collects image and profile URLs from JSON-LD for a HEAD check (public http(s) hosts only — SSRF guard).
Check Schema URLs	HTTP Request	HTTP HEAD {{ \$json.url }}
Schema Findings	Code	Turns broken schema URLs into findings (Organization logo Critical, publisher logo High).
Find Competitors	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/competitors_domain/live — Labs competitors (credential: DataForSEO)
Fallback Queries	Code	Queries used to discover competitors on Google: what this site's key pages are about, taken from H1s/titles of shallow pages.
Fallback SERP	HTTP Request	HTTP POST https://api.dataforseo.com/v3/serp/google/organic/live/advanced — live Google SERP (credential: DataForSEO)
Pick Competitors	Code	Combines Labs competitors with live-SERP competitors into up to 3 domains plus the client.
Domain Overview	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/domain_rank_overview/live — site authority / domain overview (credential: DataForSEO)
DataForSEO Whois	HTTP Request	HTTP POST =https://api.dataforseo.com/v3/domain_analytics/whois/overview/live — WHOIS (credential: DataForSEO)
RDAP Lookup	HTTP Request	HTTP GET =https://rdap.org/domain/{{ \$('Pick Competitors').item.json.domain }} — RDAP domain registration
Competitor Analysis	Code	Benchmarks organic keywords, top-10 rankings, traffic and domain age against the competitors (RDAP fills WHOIS gaps).

KW Targets	Code	One item per domain for the ranked-keywords pull.
Ranked Keywords	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/ranked_keywords/live — keywords a domain ranks for (credential: DataForSEO)
Keyword Ideas	HTTP Request	HTTP POST https://api.dataforseo.com/v3/dataforseo_labs/google/keyword_ideas/live — keyword ideas (credential: DataForSEO)
Analyze Keywords	Code	Keyword gap, easy wins and striking-distance keywords from ranked keywords and keyword ideas.
BL Targets	Code	One item per domain for the backlink summary.
Backlink Summary	HTTP Request	HTTP POST https://api.dataforseo.com/v3/backlinks/summary/live — backlink summary (credential: DataForSEO)
Backlink Gap	HTTP Request	HTTP POST https://api.dataforseo.com/v3/backlinks/domain_intersection/live — backlink gap (credential: DataForSEO)
Analyze Backlinks	Code	Referring domains and link gap per competitor; marks data unavailable when the Backlinks API is off.
PS Targets	Code	One item per domain for PageSpeed Insights.
PageSpeed	HTTP Request	HTTP GET =https://www.googleapis.com/pagespeedonline/v5/runPagespeed?url{{ encodeURIComponent(\$json.url) }}&strateg — PageSpeed Insights (credential: Google PageSpeed)
Analyze PageSpeed	Code	Core Web Vitals and performance scores for you and the competitors → findings.
Competitor Homes	Code	Competitor homepages to read for the content review.
Read Competitor Homes	HTTP Request	HTTP GET =https://r.jina.ai/{{ \$json.url }} — Jina Reader (page → markdown) (credential: Jina Reader)
Build Content Prompt	Code	Assembles the E-E-A-T content-review prompt from your pages and the competitors' homepages.
Content Reviewer	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO and E-E-A-T auditor."
Claude — Content Reviewer	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 16384; adaptive thinking (medium) credential: Anthropic (SEO Agent)
Parser — Content Reviewer	Output parser	Structured output parser; fields: skipped, business_context, eeat_signals, findings, strengths, competitor_comparison, binding_constraint
Parse Content Review	Code	Accepts an object (from the Structured Output Parser) or a string with/without code fences or surrounding text
AI Queries	Code	Live Google queries for the AI-visibility and brand checks. Each item becomes one SERP request.
AI SERP	HTTP Request	HTTP POST https://api.dataforseo.com/v3/serp/google/organic/live/advanced — live Google SERP (credential: DataForSEO)
LLM Prompts	Code	Real AI-assistant answers via DataForSEO's LLM Responses API (ChatGPT with web search, Perplexity). Each item becomes one request to the endpoint in `endpoint` with body `body`.

Ask LLMs	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
AI Visibility	Code	Brand SERP, site: index estimate, AI Overview citations and live LLM answers → findings and what works.

Scoring & audit delivery

Build Site Issues scores every finding (Critical 30 / High 15 / Medium 7 / Low 3 / Info 1, scaled by share of pages, 70% diminishing returns, caps) into a health score and grade; the Site Audit or Full SEO Report is rendered to PDF and e-mailed / posted to the callback.

Node	Type	What it does
Build Site Issues	Code	Turns crawled pages, probes and every module's extra findings into scored findings, category scores, caps, health score and grade.
Is Full Report?	IF	Routes on: {{ \$json.site_audit.audit_level === 'full' }}
Build Full Report	Code	Builds the Full SEO Report document from the scored audit: health score and grade, top issues, every module's findings and what works, competitor benchmark, keywords, backlinks, PageSpeed, content review, AI visibility, methodology; computes the run ledger.
Build Audit Report	Code	Builds the technical Site Audit document (scored findings by category, what works, measurements, methodology); computes the run ledger.
Prepare PDF Audit	Code	Copy of the report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name)
Render PDF Audit	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF Audit	Code	Merge the PDF (when the renderer succeeded) with the original Word-compatible file. Never fails the run.
Via Webhook (Audit)?	IF	Routes on: {{ !!\$json.via_webhook }}
Build Audit Response	Code	API delivery for audits: scores, counts and the report files go to the callback (the e-mail is still sent).
Has Email (Audit)?	IF	Routes on: {{ !!\$json.email }}
Send Audit Report	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.report_title 'Site Audit' }} for {{ \$json.domain }}; {{ \$json.health_score }}/ credential: Gmail SMTP (SEO Agent)

Other nodes

Node	Type	What it does
Page Track	Form page	Form page with fields: Website Domain, Target Country, Search terms to track (optional), GA4 property ID (optional), Blog posts per week, Competitors (optional), Email me the weekly report
Ensure Sites Table (Track)	Data Table	Data Table: table create — table seo_sites

Site Row (Track)	Code	"Track my site": one row for the Data Table "seo_sites" (upserted by site_id); the Site Tracker then runs at once for this site. Live finding (2026-10-02): the Data Table "create" node before this one outputs the table object, not the request, so read Normalize Input explicitly.
Save Site (Track)	Data Table	Data Table: row upsert — table seo_sites
Tracker Payload (Track)	Code	What the Site Tracker receives through its "Manual Run" trigger: run this one site now (first report in a few minutes).
Start Site Tracker	Execute Workflow	Execute Workflow → SEOagentSiteTrk1 (mode once, wait=False) — starts a separate execution
Via Webhook (Track)?	IF	Routes on: {{ !\$('#Normalize Input').first().json.via_webhook }}
Build Track Response	Code	API answer for mode "track" (POSTed to callback_url): registration confirmed; the first report follows as stage "site_tracker".
Tracking Started	Form page	Form completion page: "=Tracking is set up for {{ \$('#Normalize Input').first().json.domain }}" . Ends the form flow (the execution stays in waiting).
Blog Package	Code	Framework-agnostic blog package for every content run: the article as clean HTML (with the JSON-LD blocks), as Markdown, and a meta.json (title, slug, meta description, H1, keywords, headings, links, schema, word count, content score, suggested URL, publish checklist). Attached to the e-mail and returned in the API callback next to the PDF / Word report. Runs without content pass through untouched. v4.4: byline + aut
Page Published	Form page	Form page with fields: Website Domain, Keyword of the page, Published page URL
Ensure Log Table (Published)	Data Table	Data Table: table create — table seo_content_log
Fetch Published Page	HTTP Request	HTTP GET {{ \$('#Normalize Input').first().json.published_url }}
Load Content Log (Published)	Data Table	Data Table: row get — table seo_content_log
Load Ladders (Published)	Data Table	Data Table: row get — table seo_ladders
Load Query History (Published)	Data Table	Data Table: row get — table seo_query_history
Publish Check	Code	"I published a page": verifies the live page (HTTP status, noindex, title, meta description, H1, canonical, JSON-LD, length, keyword use, images; v4.4: author byline + Person schema and date for articles, TOC and cluster links for hubs, LocalBusiness + visible phone, VideoObject + transcript), resolves it to the content-log row and/or the ladder row for that keyword, prepares the rows to store, and suggests pages to
Published Log Row	Code	The content-log row for the published page (upserted by site_id + keyword).
Save Published (Log)	Data Table	Data Table: row upsert — table seo_content_log
Published Ladder Row	Code	The ladder row update when the published keyword belongs to a ladder (status published, real URL).
Ladder Page?	IF	Routes on: {{ !\$json.skip }}
Mark Ladder Published	Data Table	Data Table: row update — table seo_ladders
Published Delivery	Code	Restores the publish-check item after the Data Table steps and records whether the rows were stored.

Via Webhook (Published)?	IF	Routes on: {{ !\$('#Normalize Input').first().json.via_webhook }}
Build Published Response	Code	API answer for mode "published" (POSTed to callback_url): the publish check, the link suggestions and the stored state.
Published Recorded	Form page	Form completion page: "=Recorded: {{ \$json.keyword }}" . Ends the form flow (the execution stays in waiting).
Ensure Cadence Table (Track)	Data Table	Data Table: table create — table seo_cadence
Cadence Row (Track)	Code	"Track my site": the content cadence for this site (blog posts per week, 0 = off), upserted into seo_cadence.
Save Cadence (Track)	Data Table	Data Table: row upsert — table seo_cadence
GSC Audit Plan	Code	Search Console + site-structure checks for the audit: keeps the incoming audit item, picks the pages to inspect in Search Console (home, shallow and well-linked indexable pages, max 25), lists the crawled URLs for the sitemap comparison, and the 28-day window.
Fetch Sitemap (Audit)	HTTP Request	HTTP GET {{ \$json.sitemap_url }}
Sitemap Children (Audit)	Code	The site's /sitemap.xml: if it is an index, read up to 3 child sitemaps (the rest is counted); otherwise nothing more to fetch.
Any Sitemap Children?	IF	Routes on: {{ !\$json.skip }}
Fetch Sitemap Children (Audit)	HTTP Request	HTTP GET {{ \$json.url }}
GSC Sites (Audit)	HTTP Request	HTTP GET https://www.googleapis.com/webmasters/v3/sites (credential: Google Service Account (SEO Agent))
GSC Audit Requests	Code	Resolves the Search Console property for the audited domain and builds the requests: search performance for the last 28 days (totals, top pages, top queries) and one URL inspection per sampled page. Skips everything when the service account has no access.
GSC Audit?	IF	Routes on: {{ !\$json.skip }}
GSC Audit Query	HTTP Request	HTTP POST {{ \$json.url }} (credential: Google Service Account (SEO Agent))
GSC Sitemaps (Audit)	HTTP Request	HTTP GET {{ \$('GSC Audit Requests').first().json.sitemaps_url }} (credential: Google Service Account (SEO Agent))
GSC Audit Findings	Code	Turns the Search Console answers (sitemaps, search performance, URL inspections) and the sitemap-vs-crawl comparison into audit findings, a "search_console" block and a "site_structure" block for the report; then hands the audit item on to Build Site Issues.
Page Check-in	Form page	Form page with fields: Website Domain, Any manual action in Search Console?, Any security issue in Search Console?, Notes (optional), Pages report export (optional CSV), Email (optional)
Check-in Intake	Code	"Search Console check-in" form page: reads the optional Pages-report CSV upload into text (the file is only readable here, on the form item) and hands every field on to Normalize Input.
Ensure Checkin Table	Data Table	Data Table: table create — table seo_console_checkins
Ensure Alerts Table (Check-in)	Data Table	Data Table: table create — table seo_console_alerts

Parse Check-in	Code	Monthly Search Console check-in: the two yes/no answers plus the optional "Pages" export (Table.csv: Reason, Source, Validation, Trend, Pages; a per-reason URL list: URL, Last crawled; or Chart.csv: Date, Not indexed, Indexed) -> one row per site and month, and the summary shown back.
Checkin Row	Code	The seo_console_checkins row (upserted by site_id + month).
Save Check-in	Data Table	Data Table: row upsert — table seo_console_checkins
Checkin Alert?	IF	Routes on: {{ !!(('\$('Parse Check-in').first().json.checkin_row {}).manual_action (\$('Parse Check-in').first().json.checkin_row {}).security_issue) }}
Alert From Checkin	Code	A reported manual action / security issue also becomes a console alert (so the Monday report and the brief carry it).
Save Check-in Alert	Data Table	Data Table: row upsert — table seo_console_alerts
Check-in Delivery	Code	Restores the check-in item after the Data Table steps and records whether the row was stored.
Via Webhook (Check-in)?	IF	Routes on: {{ !!(('\$('Normalize Input').first().json.via_webhook) }}
Build Check-in Response	Code	API answer for mode "checkin" (POSTed to callback_url).
Check-in Recorded	Form page	Form completion page: "=Check-in recorded for {{ \$json.domain }}" . Ends the form flow (the execution stays in waiting).

7. The other workflows

SEO Agent — API (SEOagentAPIentry)

webhook front door: validates, answers 202/400, starts the main workflow.

Node	Type	What it does
Webhook Trigger	Webhook	Webhook POST /webhook/seo-keyword-check (headerAuth auth) credential: SEO API Key
Quick Validate	Code	Fast validation so callers get an immediate, useful 400. The full validation happens again in the main workflow.
Valid?	IF	Routes on: {{ \$json.ok }}
Respond Accepted	Respond to Webhook	Responds to the webhook: {{ JSON.stringify({ status: 'accepted', request_id: \$json.request_id, mode: \$json.mode, keyword: \$json.keyword, domain: (HTTP 202) }) }}
Prepare Payload	Code	Code node.
Start SEO Run	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode once, wait=False) — starts a separate execution
Respond Bad Request	Respond to Webhook	Responds to the webhook: {{ JSON.stringify({ status: 'rejected', request_id: \$json.request_id, error: \$json.error }) }} (HTTP 400)

SEO Agent — Rank Tracker (SEOagentTracker1)

weekly position checks for every ladder, progress e-mail / callback.

Node	Type	What it does
Weekly Schedule	Schedule Trigger	Schedule trigger: [{"field": "weeks", "weeksInterval": 1, "triggerAtDay": 1, "triggerAtHour": 8, "triggerAtMinute": 0}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Ladder Table	Data Table	Data Table: table create — table seo_ladders
Ensure History Table	Data Table	Data Table: table create — table seo_rank_history
Load Ladders	Data Table	Data Table: row get — table seo_ladders
Load History	Data Table	Data Table: row get — table seo_rank_history
Tracker Plan	Code	Rank tracker: one SERP check (Google top 50) per tracked keyword: the primary keyword of every ladder page, plus the head term. Reads the ladders and the history from the Data Tables, skips finished ladders and caps the spend per run.
Any Checks?	IF	Routes on: {{ !\$json.nothing_to_do }}
SERP Check	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)

Parse Positions	Code	Locates the client's domain in each SERP (top 50): position, ranking URL and the features present. One history row per keyword; position 0 = not in the top 50, -1 = the check failed.
Save History	Data Table	Data Table: row insert — table seo_rank_history
Tracker Report	Code	Rank tracker report: per ladder, positions vs the previous check, rungs reached, drops and the recommended next step.
Has Report?	IF	Routes on: {{ !\$json.nothing_to_do }}
Has Email (Tracker)?	IF	Routes on: {{ !\$json.email }}
Send Progress Email	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)
Has Callback (Tracker)?	IF	Routes on: {{ !\$json.callback_url }}
POST Progress	HTTP Request	HTTP POST {{ \$json.callback_url }}
Nothing To Track	No Op	No operation (branch end).

SEO Agent — Site Tracker (SEOagentSiteTrk1)

weekly per-site performance: Search Console, GA4, Google Trends, live checks, actions; e-mail / callback.

Node	Type	What it does
Weekly Schedule	Schedule Trigger	Schedule trigger: [{"field": "weeks", "weeksInterval": 1, "triggerAtDay": 1, "triggerAtHour": 9, "triggerAtMinute": 0}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Sites Table	Data Table	Data Table: table create — table seo_sites
Ensure Metrics Table	Data Table	Data Table: table create — table seo_site_metrics
Ensure Query Table	Data Table	Data Table: table create — table seo_query_history
Ensure History Table (Site)	Data Table	Data Table: table create — table seo_rank_history
Ensure Trends Table	Data Table	Data Table: table create — table seo_trends
Ensure Log Table (Site)	Data Table	Data Table: table create — table seo_content_log
Load Content Log	Data Table	Data Table: row get — table seo_content_log
Ensure Alerts Table (Site)	Data Table	Data Table: table create — table seo_console_alerts
Ensure Checkin Table (Site)	Data Table	Data Table: table create — table seo_console_checkins
Load Console Alerts	Data Table	Data Table: row get — table seo_console_alerts
Load Check-ins	Data Table	Data Table: row get — table seo_console_checkins
Ensure AI Visibility Table (Site)	Data Table	Data Table: table create — table seo_ai_visibility
Ensure Snapshots Table (Site)	Data Table	Data Table: table create — table seo_backlink_snapshots
Ensure Audits Table (Site)	Data Table	Data Table: table create — table seo_audits
Load AI Visibility (Site)	Data Table	Data Table: row get — table seo_ai_visibility
Load Backlink Snapshots (Site)	Data Table	Data Table: row get — table seo_backlink_snapshots

Load Audits (Site)	Data Table	Data Table: row get — table seo_audits
Load Sites	Data Table	Data Table: row get — table seo_sites
Load Ladders	Data Table	Data Table: row get — table seo_ladders
Load Rank History	Data Table	Data Table: row get — table seo_rank_history
Load Site Metrics	Data Table	Data Table: row get — table seo_site_metrics
Load Query History	Data Table	Data Table: row get — table seo_query_history
Site Plan	Code	Site tracker plan: one job per tracked site. Sites come from the Data Table "seo_sites" (registered through the form option "Track my site" / API mode "track") plus every domain that has a keyword ladder (registered automatically, e-mail from the ladder). Google data is read for the last 28 complete days ending 3 days ago (Search Console data is final after about 3 days), against the 28 days before and the same 28 da
Any Sites?	IF	Routes on: {{ !\$json.nothing_to_do }}
Nothing To Track (Site)	No Op	No operation (branch end).
GSC Sites	HTTP Request	HTTP GET https://www.googleapis.com/webmasters/v3/sites (credential: Google Service Account (SEO Agent))
Resolve Properties	Code	Matches each site to a Search Console property the service account can read: "sc-domain:" (domain property) first, then a URL-prefix property for the host (https preferred), or the property named at registration.
GSC Requests	Code	Search Analytics requests per connected site: clicks by day for 56 days, the same 28 days a year ago, top queries (with page) and pages for both periods, and every tracked keyword as an exact query (with the ranking page) for both periods. All free (Google quota).
Any GSC?	IF	Routes on: {{ !\$json.skip }}
GSC Query	HTTP Request	HTTP POST {{ \$json.url }} (credential: Google Service Account (SEO Agent))
Parse GSC	Code	Consolidates the Search Analytics responses per site: totals for both periods and a year ago, daily clicks, query movers (winners, losers, new, lost), striking-distance queries, CTR gaps, page movers and decaying pages, tracked keyword positions.
Need GA4 Detect?	IF	Routes on: {{ \$('Resolve Properties').all().some(i => !i.json.ga4_property_id) }}
GA4 Accounts	HTTP Request	HTTP GET https://analyticsadmin.googleapis.com/v1beta/accountSummaries?pageSize=200 (credential: Google Service Account (SEO Agent))
Stream Requests	Code	GA4 auto-detection, step 1: one dataStreams request per GA4 property the service account can see, so the property whose web stream URL matches the site can be found without the owner looking up the property id. Skipped when every site has an id.
Any Streams?	IF	Routes on: {{ !\$json.skip }}
GA4 Streams	HTTP Request	HTTP GET {{ \$json.url }} (credential: Google Service Account (SEO Agent))

GA4 Requests	Code	GA4 Data API reports per site with a property id (given at registration, or detected from the web data stream whose URL matches the domain): organic-channel totals for both periods, organic landing pages, organic sessions by day. Free (Google quota).
Any GA4?	IF	Routes on: {{ !\$json.skip }}
GA4 Report	HTTP Request	HTTP POST {{ \$json.url }} (credential: Google Service Account (SEO Agent))
Parse GA4	Code	Consolidates the GA4 reports per site: organic vs all-channel totals for both periods, organic share, landing pages with deltas, daily organic sessions.
Inspect Requests	Code	URL Inspection (Search Console) for the ladder pages: is each planned/written page indexed? Only on connected properties, max 20 per site.
Any Inspections?	IF	Routes on: {{ !\$json.skip }}
Inspect URL	HTTP Request	HTTP POST https://searchconsole.googleapis.com/v1/urlInspection/index:inspect (credential: Google Service Account (SEO Agent))
Parse Inspection	Code	Index status per inspected URL: verdict PASS = indexed; coverage state, last crawl, canonical mismatch, robots.txt state.
Trends Requests	Code	Google Trends (through DataForSEO) for the head terms: ladder heads first, then the site's own keywords; one keyword per request, past 12 months, interest graph + related queries (top / rising). Max 4 per site; \$0.011 per request (measured 2026-10-02).
Any Trends?	IF	Routes on: {{ !\$json.skip }}
Google Trends	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Parse Trends	Code	Trend per keyword: direction (last 4 weeks vs the 12-month average), latest and peak, related rising / top searches, DataForSEO cost.
Site SERP Requests	Code	Live Google check (DataForSEO SERP, top 50, ~\$0.015 each) for the site's own tracked keywords. Ladder keywords are checked by the Rank Tracker workflow and only read here; keywords without a market (location) are skipped. Max 20 per site.
Any SERP?	IF	Routes on: {{ !\$json.skip }}
SERP Check (Site)	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Parse Site SERP	Code	Locates the site in each SERP (top 50): one seo_rank_history row per keyword (ladder_id = site id, rung 0). position 0 = not in the top 50, -1 = check failed.
Save Site Rank History	Data Table	Data Table: row insert — table seo_rank_history
Site Metrics	Code	Joins everything per site: Search Console, GA4, index status, trends and the live SERP checks -> KPIs, alerts, ranked actions (with ready-to-send API bodies), and the exact rows for the Data Tables seo_site_metrics / seo_query_history / seo_sites.
Save Site Metrics	Data Table	Data Table: row insert — table seo_site_metrics

Query Rows	Code	Rows for the Data Table "seo_query_history": every tracked keyword plus the top 100 queries of the period, per site.
Any Query Rows?	IF	Routes on: {{ !\$json.skip }}
Save Query Rows	Data Table	Data Table: row insert — table seo_query_history
Trend Rows	Code	Rows for the Data Table "seo_trends" (the Content Cadence reads the rising related searches from here).
Any Trend Rows?	IF	Routes on: {{ !\$json.skip }}
Save Trends	Data Table	Data Table: row insert — table seo_trends
Site Rows	Code	One "seo_sites" row per site (upserted by site_id): resolved Search Console property, GA4 property id, last run and status.
Save Sites	Data Table	Data Table: row upsert — table seo_sites
Brief Input	Code	Compact facts for the AI brief (one item per site): totals, movers, opportunities, tracked keywords, pages, traffic, trends, alerts, actions.
Site Brief	AI Agent	AI agent with structured output. Prompt starts: "You are a senior SEO analyst writing the weekly performance brief for one website."
Claude — Site Brief	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 8192; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — Site Brief	Output parser	Structured output parser; fields: headline, summary, highlights, actions, watch
Site Report	Code	Weekly performance report per site: e-mail HTML (KPI tiles, actions, tracked keywords, ladder pages, movers, opportunities, traffic, trends), the same document as a file for the PDF renderer, and the fields for the callback. Uses the AI brief when present, the facts otherwise.
Prepare PDF Site	Code	Copy of each report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name); one item per site.
Render PDF Site	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF Site	Code	Attach the PDF (when the renderer succeeded) next to the HTML file, per site; a renderer failure never fails the run.
Has Email (Site)?	IF	Routes on: {{ !!\$json.email }}
Send Site Report	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)
Build Site Callback	Code	Callback payload per site for API callers: the report data plus the PDF (bytes read from n8n's binary store); heavy internals dropped.
Has Callback (Site)?	IF	Routes on: {{ !!\$json.callback_url }}
POST Site Report	HTTP Request	HTTP POST {{ \$json.callback_url }}

SEO Agent — Console Alerts (SEOagentConsole1)

IMAP watcher: Search Console notification mails (manual actions, security, coverage, CWV) → alerts.

Node	Type	What it does
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Mailbox Trigger	emailReadImap	emailReadImap credential: Gmail IMAP (SEO Agent)
Classify Notification	Code	Google Search Console notification e-mails (the agent's mailbox is a user of the property) -> typed alerts; anything else is ignored. Covers what the API cannot see: manual actions, security issues, site-wide indexing / Core Web Vitals / rich-result / sitemap notices.
Relevant?	IF	Routes on: {{ !!\$json.relevant }}
Ignore	No Op	No operation (branch end).
Ensure Alerts Table	Data Table	Data Table: table create — table seo_console_alerts
Load Sites (Alerts)	Data Table	Data Table: row get — table seo_sites
Load Ladders (Alerts)	Data Table	Data Table: row get — table seo_ladders
Alert Row	Code	Row for the Data Table seo_console_alerts (upserted by message id, so a re-read mail never duplicates).
Save Alert	Data Table	Data Table: row upsert — table seo_console_alerts
Urgent?	IF	Routes on: {{ \$('Classify Notification').first().json.severity === 'critical' }}
Alert Target	Code	Who to tell about an urgent notice: the site's e-mail / callback from seo_sites, the ladder's as fallback.
Has Email (Alert)?	IF	Routes on: {{ !!\$json.email }}
Email Owner (Alert)	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject_line }} credential: Gmail SMTP (SEO Agent)
Has Callback (Alert)?	IF	Routes on: {{ !!\$json.callback_url }}
POST Alert	HTTP Request	HTTP POST {{ \$json.callback_url }}
Stored	No Op	No operation (branch end).

SEO Agent — Content Cadence (SEOagentCadence1)

weekly blog pages per site (ladder → striking distance → trends), delivered as HTML + Markdown + meta.json.

Node	Type	What it does
Weekly Schedule	Schedule Trigger	Schedule trigger: [{"field": "weeks", "weeksInterval": 1, "triggerAtDay": 1, "triggerAtHour": 10, "triggerAtMinute": 0}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Log Table	Data Table	Data Table: table create — table seo_content_log
Ensure Cadence Table	Data Table	Data Table: table create — table seo_cadence
Ensure Trends Table (Cadence)	Data Table	Data Table: table create — table seo_trends
Load Sites	Data Table	Data Table: row get — table seo_sites
Load Cadence	Data Table	Data Table: row get — table seo_cadence
Load Ladders	Data Table	Data Table: row get — table seo_ladders

Load Query History	Data Table	Data Table: row get — table seo_query_history
Load Trends	Data Table	Data Table: row get — table seo_trends
Load Content Log	Data Table	Data Table: row get — table seo_content_log
Cadence Plan	Code	Content cadence: picks this week's blog topics per site (pages per week from seo_cadence, max 3), in this order: the keyword ladder's next planned pages (rung order, with the ladder's internal links), striking-distance queries from Search Console (positions 4-20 with impressions: strengthen the page that already ranks), rising related searches from Google Trends. Skips keywords written in the last 90 days, pages already published.
Any Pages?	IF	Routes on: {{ !\$json.nothing_to_do }}
Nothing To Write	No Op	No operation (branch end).
Dry Run?	IF	Routes on: {{ !\$json.dry_run }}
Planned Only	No Op	No operation (branch end).
Start Content Runs	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode each, wait=False) — starts a separate execution
Log Rows	Code	One seo_content_log row per page started this week (status "started"; "I published a page" later sets published_url).
Save Content Log	Data Table	Data Table: row insert — table seo_content_log
Ladder Marks	Code	Ladder rows of the pages started this week -> status "writing" (so the Rank Tracker and the next cadence know).
Any Ladder Marks?	IF	Routes on: {{ !\$json.skip }}
Mark Ladder Writing	Data Table	Data Table: row update — table seo_ladders
Cadence Note	Code	One note per site: this week's page(s) and why, what comes next, and pages still waiting for a publish link.
Has Email (Cadence)?	IF	Routes on: {{ !\$json.email }}
Send Cadence Note	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)
Has Callback (Cadence)?	IF	Routes on: {{ !\$json.callback_url }}
POST Cadence Note	HTTP Request	HTTP POST {{ \$json.callback_url }}

SEO Agent — Site Admin (SEOagentSiteAdm1)

delete / pause / resume / cadence / unpublish for a tracked site (published sub-workflow).

Node	Type	What it does
Admin Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Cadence Table (Admin)	Data Table	Data Table: table create — table seo_cadence
Ensure Log Table (Admin)	Data Table	Data Table: table create — table seo_content_log
Ensure Sites Table (Admin)	Data Table	Data Table: table create — table seo_sites
Ensure Monitors Table (Admin)	Data Table	Data Table: table create — table seo_monitors

Ensure Prospects Table (Admin)	Data Table	Data Table: table create — table seo_link_prospects
Ensure AI Prompts Table (Admin)	Data Table	Data Table: table create — table seo_ai_prompts
Admin Action	Code	Site admin: delete pause resume cadence (pages_per_week 0-3) unpublish (keyword) monitors (AI visibility / backlinks / monthly audit settings) prospect (prospect_domain + status contacted / won / rejected / ignored / new, note) ai_prompts (add custom questions, switch some off) — by site_id or domain. The table-ensure nodes before this one output table objects, so the request is read from the trigger (l
Load Monitors (Admin)	Data Table	Data Table: row get — table seo_monitors
Monitor Row (Admin)	Code	Site Admin "monitors": merges the given settings over the stored seo_monitors row (exact columns). Settings: ai_visibility, ai_engines (chatgpt, perplexity, gemini, claude, ai_overview, ai_mode), ai_prompts_max (3-15), backlinks, audit_monthly, audit_pages (50-1000), audit_js, competitors (domains), brand_names.
Set Monitors	Data Table	Data Table: row upsert — table seo_monitors
Load Link Prospects (Admin)	Data Table	Data Table: row get — table seo_link_prospects
Prospect Rows (Admin)	Code	Site Admin "prospect": sets the status (and note) of a link prospect for the site — every type of that domain unless `type` is given. Rows keep all other fields (exact seo_link_prospects columns). Status won also stamps won_at.
Set Prospects	Data Table	Data Table: row upsert — table seo_link_prospects
Load AI Prompts (Admin)	Data Table	Data Table: row get — table seo_ai_prompts
AI Prompt Rows (Admin)	Code	Site Admin "ai_prompts": adds custom buyer questions (source custom, tracked weekly before the automatic ones) and switches questions off (by prompt id or exact text). Exact seo_ai_prompts columns; upsert by prompt_id.
Set AI Prompts	Data Table	Data Table: row upsert — table seo_ai_prompts
Admin Route	Switch	Switch: outputs delete, status, cadence, unpublish, monitors, prospect, ai_prompts
Delete Site	Data Table	Data Table: row deleteRows — table seo_sites
Set Site Status	Data Table	Data Table: row update — table seo_sites
Cadence Row (Admin)	Code	seo_cadence row for the admin action "cadence" (upserted by site_id).
Set Cadence	Data Table	Data Table: row upsert — table seo_cadence
Unpublish Log	Data Table	Data Table: row update — table seo_content_log
Unpublish Ladder	Data Table	Data Table: row update — table seo_ladders
Admin Result	Code	Rows affected by the admin action (0 = no site / row with that id or keyword).

SEO Agent — AI Visibility Tracker (SEOagentAIVisib1)

weekly buyer questions × ChatGPT / Perplexity / Gemini / Claude / Google AI Mode + AI Overviews: mentions, citations, share of voice, sources AI trusts, lost questions.

Node	Type	What it does
Weekly Schedule	Schedule Trigger	Schedule trigger: [{"field": "weeks", "weeksInterval": 1, "triggerAtDay": [1], "triggerAtHour": 7, "triggerAtMinute": 0}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Monitors Table	Data Table	Data Table: table create — table seo_monitors
Ensure AI Prompts Table	Data Table	Data Table: table create — table seo_ai_prompts
Ensure AI Answers Table	Data Table	Data Table: table create — table seo_ai_answers
Ensure AI Visibility Table	Data Table	Data Table: table create — table seo_ai_visibility
Ensure Prospects Table	Data Table	Data Table: table create — table seo_link_prospects
Ensure Sites Table	Data Table	Data Table: table create — table seo_sites
Ensure Profiles Table	Data Table	Data Table: table create — table seo_profiles
Load Sites	Data Table	Data Table: row get — table seo_sites
Load Ladders	Data Table	Data Table: row get — table seo_ladders
Load Monitors	Data Table	Data Table: row get — table seo_monitors
Load Profiles	Data Table	Data Table: row get — table seo_profiles
Load AI Prompts	Data Table	Data Table: row get — table seo_ai_prompts
Load AI Visibility	Data Table	Data Table: row get — table seo_ai_visibility
Load Link Prospects	Data Table	Data Table: row get — table seo_link_prospects
AI Plan	Code	AI visibility plan (weekly, Monday 07:00, and on demand): one job per site — the brand names to look for, the competitors, the topics, the stored prompt set (seo_ai_prompts: stable week to week so the trend means something), whether new prompts must be written, the engines, the previous week's numbers and whether the monthly market view (DataForSEO LLM Mentions) is due. ---- shared by the monitors (AI Visibility Trac
Any Sites (AI)?	IF	Routes on: {{ !\$json.nothing_to_do }}
Nothing To Do (AI)	No Op	No operation (branch end).
Site Text Jobs	Code	Sites that need questions but have no topics and no business name on file (e.g. an ad-hoc check of a new domain): the homepage is read (Jina Reader) so the Prompt Writer knows what the business sells.
Need Site Text?	IF	Routes on: {{ !\$json.skip }}
Read Site (AI)	HTTP Request	HTTP GET =https://r.jina.ai/https://{{ \$json.domain }} — Jina Reader (page → markdown) (credential: Jina Reader)
Prompt Jobs	Code	Sites whose prompt set is short get one Prompt Writer job each (Claude writes the buyer questions once; they are stored and reused weekly).
Any Prompt Jobs?	IF	Routes on: {{ !\$json.skip }}
Prompt Writer	AI Agent	AI agent with structured output. Prompt starts: "You write the questions real buyers type into ChatGPT, Perplexity, Gemini and Google's AI Mode before they choose a provider."

Claude — Prompt Writer	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 8192; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — Prompt Writer	Output parser	Structured output parser; fields: prompts
Prompt Rows	Code	New prompt rows for seo_ai_prompts: the Prompt Writer's questions (Claude), or template questions when the writer failed or did not run. Exact table columns. Existing prompts are never rewritten, so the weekly trend compares the same questions.
Any New Prompts?	IF	Routes on: {{ !\$json.skip }}
Save Prompts	Data Table	Data Table: row insert — table seo_ai_prompts
AI Requests	Code	One DataForSEO request per question x engine (ChatGPT = the real chatgpt.com answer through the LLM scraper; Perplexity, Gemini and Claude = LLM Responses with web search; Google AI Mode with the question; the Google AI Overview on the matching search phrase), plus the monthly market view (LLM Mentions: which domains AI answers cite for the main topic). Costs measured live on 2026-10-02.
Any AI Requests?	IF	Routes on: {{ !\$json.skip }}
Run AI Requests	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Parse AI Answers	Code	One row per question x engine for seo_ai_answers (exact columns): did the answer mention the brand, cite one of your pages, at which rank among the businesses it names, which competitors / businesses appeared and which sources it cited. The market view (LLM Mentions) is read by AI Metrics, not stored here. Engines: chatgpt (llm_scraper), perplexity / gemini / claude (llm_responses), ai_overview / ai_mode (SERP).
Any Answer Rows?	IF	Routes on: {{ !\$json.skip }}
Save AI Answers	Data Table	Data Table: row insert — table seo_ai_answers
AI Metrics	Code	Per site: how often AI answers mention / cite the business, share of voice against the competitors, rank in recommendation lists, the per-engine picture, the sources AI trusts (PR and listing targets), your pages it cites, the questions where competitors win (content gaps), the monthly market view (LLM Mentions), week-over-week changes, alerts and ranked actions. Also the seo_ai_visibility row and 'ai_source' rows fo
AI Vis Rows	Code	The seo_ai_visibility rows (one per site per run; exact columns).
Save AI Visibility	Data Table	Data Table: row insert — table seo_ai_visibility
AI Prospect Rows	Code	Sites AI cites become link / PR prospects (type ai_source) in seo_link_prospects, next to the Backlink Monitor's prospects; statuses are kept.
Any AI Prospects?	IF	Routes on: {{ !\$json.skip }}
Save AI Prospects	Data Table	Data Table: row upsert — table seo_link_prospects
AI Brief Input	Code	Compact facts per site for the AI-visibility brief (Claude Sonnet, low effort): numbers, per engine, competitors, sources, gaps, market, alerts.
AI Brief	AI Agent	AI agent with structured output. Prompt starts: "You are an AI-search (GEO) strategist."

Claude — AI Brief	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 8192; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — AI Brief	Output parser	Structured output parser; fields: headline, summary, actions, watch
AI Report	Code	Weekly AI-visibility report per site (e-mail HTML + the same document for the PDF): headline, KPI tiles with last week's change, alerts, actions (with ready-to-send API bodies for content gaps), the answer grid (question x engine), competitors' share, the sources AI trusts, your cited pages, the monthly market view and the method. Uses the Claude brief when present, the computed facts otherwise. ---- shared report he
Prepare PDF AI	Code	Copy of each report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name); one item per site.
Render PDF AI	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF AI	Code	Attach the PDF (when the renderer succeeded) next to the report's other files, per site; a renderer failure never fails the run.
Build AI Callback	Code	Callback payload per site for API callers: the AI-visibility data plus the PDF (bytes from n8n's binary store); internals dropped.
Has Email (AI)?	IF	Routes on: {{ !!\$json.email }}
Send AI Report	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)
Has Callback (AI)?	IF	Routes on: {{ !!\$json.callback_url }}
POST AI Report	HTTP Request	HTTP POST {{ \$json.callback_url }}

SEO Agent — Backlink Monitor (SEOagentBacklnk1)

weekly watch (lost / spammy links) and monthly full report: broken-link reclaim, link gap, unlinked mentions, prospects with outreach drafts.

Node	Type	What it does
Weekly Schedule	Schedule Trigger	Schedule trigger: [{"field": "weeks", "weeksInterval": 1, "triggerAtDay": 1, "triggerAtHour": 7, "triggerAtMinute": 30}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Monitors Table	Data Table	Data Table: table create — table seo_monitors
Ensure Snapshots Table	Data Table	Data Table: table create — table seo_backlink_snapshots
Ensure Prospects Table	Data Table	Data Table: table create — table seo_link_prospects
Ensure Sites Table	Data Table	Data Table: table create — table seo_sites
Ensure Profiles Table	Data Table	Data Table: table create — table seo_profiles
Ensure Case Table	Data Table	Data Table: table create — table seo_case_studies
Ensure Log Table	Data Table	Data Table: table create — table seo_content_log
Load Sites	Data Table	Data Table: row get — table seo_sites

Load Ladders	Data Table	Data Table: row get — table seo_ladders
Load Monitors	Data Table	Data Table: row get — table seo_monitors
Load Profiles	Data Table	Data Table: row get — table seo_profiles
Load Backlink Snapshots	Data Table	Data Table: row get — table seo_backlink_snapshots
Load Link Prospects	Data Table	Data Table: row get — table seo_link_prospects
Load Content Log	Data Table	Data Table: row get — table seo_content_log
Load Case Studies	Data Table	Data Table: row get — table seo_case_studies
BL Plan	Code	Backlink monitor plan (weekly Monday 07:30; the first run of each month is the FULL run, the others are LIGHT): per site the window since the last check, the competitors (configured, else found once through DataForSEO Labs and kept), the brand names for mention search, the pages worth pitching (ladder top / published pages, case studies) and the existing prospects. Light runs only watch for lost and spammy links and
Any Sites (BL)?	IF	Routes on: {{ !\$json.nothing_to_do }}
Nothing To Do (BL)	No Op	No operation (branch end).
BL Requests	Code	Backlink requests per site (DataForSEO Backlinks API, \$0.024 per request + \$0.000036 per row, measured 2026-10-02). Every run: summary, links lost since the last check, new links since the last check. Full runs add: links pointing to broken pages (reclaim), the referring domains (to see which prospects linked), 12 months of history, unlinked brand mentions (Content Analysis) and, once, the competitors.
Run BL Requests	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Gap Requests	Code	Link gap (full runs): the domains that link to at least one competitor but not to you (backlinks/domain_intersection; the linking domain is the item's `target` field), one request per site. Competitors: configured, kept from earlier runs, or picked now from the Labs competitors (big platforms and the site itself excluded).
Any Gap?	IF	Routes on: {{ !\$json.skip }}
Run Gap Requests	HTTP Request	HTTP POST {{ \$json.endpoint }} — DataForSEO endpoint chosen per item (endpoint/body in the item) (credential: DataForSEO)
Parse Backlinks	Code	Per site: link profile now vs the last check, links lost (important ones = alert), new links (spammy ones flagged), links pointing to broken pages on the site (reclaim with a redirect), 12 months of history, unlinked brand mentions, the link gap (domains linking to competitors, not to you), alerts and the prospect candidates (gap / mention / lost / reclaim). Rank and spam score are DataForSEO's 0-1000 and 0-100 scale
Outreach Jobs	Code	Up to 8 new prospects per site get an outreach draft (Claude, low effort): lost links first (ask to restore), then unlinked mentions (ask for the link), then link-gap and AI-source domains (pitch a page worth linking to). Prospects that already have a draft are skipped.
Any Outreach?	IF	Routes on: {{ !\$json.skip }}
Outreach Writer	AI Agent	AI agent with structured output. Prompt starts: "You write short, honest link-outreach e-mails for ... (...)."

Claude — Outreach Writer	Anthropic model	Claude model node: claude-sonnet-5-5; max tokens 8192; adaptive thinking (low) credential: Anthropic (SEO Agent)
Parser — Outreach Writer	Output parser	Structured output parser; fields: drafts
Prospect Rows	Code	The link-prospect pipeline (seo_link_prospects, upsert by site + prospect domain + type; exact columns): this run's candidates merged with the stored rows (status, notes and drafts kept), outreach drafts attached, and prospects that now link to you marked "won" automatically.
Any Prospect Rows?	IF	Routes on: {{ !\$json.skip }}
Save Prospects	Data Table	Data Table: row upsert — table seo_link_prospects
BL Snapshot Rows	Code	One seo_backlink_snapshots row per site per run (exact columns); lists are kept short.
Save BL Snapshot	Data Table	Data Table: row insert — table seo_backlink_snapshots
BL Report	Code	Backlink report per site: monthly FULL report (profile vs last month, 12-month history, lost / new / spammy links, broken links to reclaim, link gap, unlinked mentions, the prospect pipeline with outreach drafts) or, on light weeks, a short ALERT e-mail when an important link was lost or spam arrived. Attachments: prospects.csv and, when needed, disavow-candidates.txt. Sites with nothing to report stay quiet. ---- sh
Prepare PDF BL	Code	Copy of each report HTML named index.html for the PDF renderer (Gotenberg requires that exact file name); one item per site.
Render PDF BL	HTTP Request	HTTP POST http://gotenberg:3000/forms/chromium/convert/html — Gotenberg HTML → PDF
Attach PDF BL	Code	Attach the PDF (when the renderer succeeded) next to the report's other files, per site; a renderer failure never fails the run.
Build BL Callback	Code	Callback payload per site for API callers: the backlink data, the PDF and the CSV / disavow files (bytes from n8n's binary store).
Has Email (BL)?	IF	Routes on: {{ !!\$json.email }}
Send BL Report	Send Email	Sends e-mail via SMTP to {{ \$json.email }}; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)
Has Callback (BL)?	IF	Routes on: {{ !!\$json.callback_url }}
POST BL Report	HTTP Request	HTTP POST {{ \$json.callback_url }}

SEO Agent — Audit Scheduler (SEOagentAuditSc1)

monthly technical re-audit per site; each audit stores its findings, diffs against the last one and hands over a fix pack.

Node	Type	What it does
Monthly Schedule	Schedule Trigger	Schedule trigger: [{"field": "cronExpression", "expression": "0 6 1 * *"}]
Manual Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Ensure Monitors Table	Data Table	Data Table: table create — table seo_monitors
Ensure Audits Table	Data Table	Data Table: table create — table seo_audits

Ensure Sites Table	Data Table	Data Table: table create — table seo_sites
Ensure Profiles Table	Data Table	Data Table: table create — table seo_profiles
Load Sites	Data Table	Data Table: row get — table seo_sites
Load Ladders	Data Table	Data Table: row get — table seo_ladders
Load Monitors	Data Table	Data Table: row get — table seo_monitors
Load Profiles	Data Table	Data Table: row get — table seo_profiles
Load Audits	Data Table	Data Table: row get — table seo_audits
Audit Schedule Plan	Code	Monthly technical re-audit (1st of the month 06:00, and on demand): one technical audit per site with audits switched on, unless the site was audited in the last 25 days. Each audit runs as its own execution through API Entry (internal key "audit:") and delivers itself: report + "since the last audit" diff + fix pack. The crawl size / JavaScript rendering come from the site's monitor settings. ---- shared by the moni
Any Audits Due?	IF	Routes on: {{ !\$json.nothing_to_do }}
Start Scheduled Audits	Execute Workflow	Execute Workflow → SEOagentV4Full01 (mode each, wait=False) — starts a separate execution
Audits Started	Code	What was started (each audit delivers its own report, diff and fix pack by e-mail / callback).
Nothing Due	No Op	No operation (branch end).

SEO Agent — Publish to WordPress (SEOagentWordPres)

optional: generated page → WordPress draft (inactive sub-workflow).

Node	Type	What it does
From Content Run	Execute Workflow Trigger	Execute Workflow Trigger (passthrough): entry for runs started by another workflow or the API front door.
Build WP Draft	Code	Turns the generated page into a WordPress REST payload: a DRAFT post with title, slug, excerpt (meta description), HTML content, JSON-LD blocks and SEO-plugin meta (Yoast / RankMath keys; they must be exposed to the REST API by the plugin).
Create WP Draft	HTTP Request	HTTP POST {{ \$json.wordpress_url + '/wp-json/wp/v2/posts' }} (credential: WordPress (SEO Agent))
WP Result	Code	Normalised result of the draft creation (inspect it in the execution; nothing else depends on it).

SEO Agent — Error Handler (SEOagentErrHandl)

error workflow: e-mails ops when a run fails.

Node	Type	What it does
Error Trigger	Error Trigger	Error Trigger: receives failed executions of workflows that name this one as their error workflow.

Format Alert	Code	Turns the Error Trigger payload into a readable ops alert.
Email Ops	Send Email	Sends e-mail via SMTP to kinnngahmed@gmail.com; subject: {{ \$json.subject }} credential: Gmail SMTP (SEO Agent)

SEO Agent — Test Callback Receiver (SEOagentCallback)

testing: stores API callbacks as executions.

Node	Type	What it does
Callback Webhook	Webhook	Webhook POST /webhook/seo-callback (none auth)
Summarise Callback	Code	Summarise the SEO Agent callback (the full payload stays in this execution for inspection)

SEO Agent — Test Runner (SEOagentTestRun1)

testing only: starts any SEO Agent workflow with a JSON body (kept unpublished).

Node	Type	What it does
Test Webhook	Webhook	Webhook POST /webhook/seo-test-run (headerAuth auth) credential: SEO API Key (test)
Route Test	Code	body.workflow picks the target (default: SEO Agent v4); the rest of the body is forwarded like the API does.
Respond Started	Respond to Webhook	Responds to the webhook: {{ JSON.stringify({ status: 'started', target: \$json.target, request_id: \$json.request_id }) }} (HTTP 202)
Prepare Forward	Code	Code node.
Run Target	Execute Workflow	Execute Workflow → ={{ \$('Route Test').first().json.target }} (mode once, wait=False) — starts a separate execution

8. Offline test harness

harness/scenarios_v5.js runs every Code node of the workflows against realistic fixtures (DataForSEO, Jina, PageSpeed, LLM outputs) in the n8n container's Node runtime, with a mini n8n executor (\$input, \$('Node'), \$runIndex, static data, getBinaryDataBuffer). Scenarios: validation and rate limit; keyword mode (full content pipeline, truncated output, AVOID path, empty SERP); verdict; describe; discovery (research pool, relevance, clusters, strategy report, choice step, spawned follow-ups); full report and technical audit (crawl, probes incl. apex→www, HTML/schema, competitors, keywords, backlinks, PageSpeed, content review, AI visibility, scoring); S7 ladder (validation, API and page-run normalisation, ladder links as candidates, requests, pool, relevance, rung bands, link map, timeline, report, PDF, Data Table rows with exact columns, delivery, spawned payload round-trip, callback, store failure, all-pulls-failed stop); S8 tracker (plan, positions, history columns, second check with gains/drops, stop rule, no ladders); S9 WordPress payload. Last run: **161 node runs, 0 failed**. Generated reports land in harness/sample-output/; live samples are prefixed LIVE - .

9. Live system test 2026-10-01

All runs used the user's real inputs: domain **techand.ai**, keyword **e invoicing in uae**, country United Arab Emirates, e-mail kinnngahmed@gmail.com. API-style runs were started through the Test Runner (same path as the API front door's *Start SEO Run*); form runs were driven with curl against the public form. Each section lists the nodes n8n executed, in order, from the execution data.

Execution 16: Test 1 — Technical site audit of techand.ai

Started through	Test Runner → API Entry; body: mode audit, report_type "Site Audit (technical issues only)", country AE, e-mail + callback
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 12:09:31.924 → 2026-10-01 12:15:04.655 · 5.5 min · last node: From Audit Form?
Result	Pass. Crawl of the site (DataForSEO On-Page, 5 polls), probes, scoring → health score 70/100 (1 Critical: broken pages; www not consolidated; HTML not cacheable). PDF + Word e-mailed (Gmail SMTP answered 250 OK, 208 KB) and posted to the callback.
Key output fields	<pre>{ "stage": "site_audit", "health_score": 70, "grade": "Needs Improvement", "issue_counts": { "Critical": 1, "High": 3, "Medium": 6, "Low": 10, "Info": 3 }, "top_issues": ["Broken pages (4xx, 5xx or timeout)", "www and non-www versions are not consolidated", "HTML cannot be cached by the CDN (no-store / private)", "Broken internal links"], "run_ledger": { "dataforseo_usd": 0.09, "dataforseo_calls": 7, "by_node": { "Start Crawl": 0.09 }, "ai_calls": 0, "ai_nodes": [], "started_at": "2026-10-01T12:09:32.319Z", "finished_at": "2026-10-01T12:14:58.933Z", "duration_min": 5.4 } }</pre>

Nodes executed (35)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Start Crawl → Save Task ID → Wait 1 min ×5 → Get Crawl Summary ×5 → Check Crawl ×5 → Crawl Done? ×5 → Get Crawled Pages → Crawl Extra Requests → Run Crawl Extras → Collect Crawl Extras → Build Probes → Run Probes → Content Probes → Run Content Probes → Analyze Probes → Full Report? → Build Site Issues → Is Full Report? → Build Audit Report → Prepare PDF Audit → Render PDF Audit → Attach PDF Audit → Via Webhook (Audit)? → Build Audit Response → Has Callback? → POST Callback → Has Email (Audit)? → Send Audit Report → From Audit Form?
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Execution 18: Test 2 — Keyword discovery for techand.ai

Started through	Test Runner → API Entry; body: mode discover, domain techand.ai, country AE, goal leads, e-mail + callback (no business text → site read + Claude Haiku description)
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 12:10:31.159 → 2026-10-01 12:12:35.867 · 2.1 min · last node: Discover Audit?
Result	Pass. 624 keywords researched → 240 AI-screened → 126 relevant → 46 clusters; start-with pick "e invoicing" (1,900 searches/mo, KD 9, CPC \$7.95; tax.gov.ae and mof.gov.ae rank; AI Overview present); AI search demand 64/mo. 88 KB PDF in the callback.

Key output fields

```
{
  "stage": "keyword_strategy",
  "summary": {
    "keywords_researched": 624,
    "ai_reviewed": 240,
    "relevant": 126,
    "clusters": 46,
    "competitors": [
      "installogic.ae",
      "craftinteractive.ae",
      "hamtinfotech.com"
    ]
  },
  "start_with": {
    "keyword": "e invoicing",
    "volume": 1900,
    "kd": 9,
    "cpc": 7.949999809265137,
    "intent": "commercial",
    "page_type": "Service Page",
    "topic": "e-invoicing uae",
    "traffic_potential": 371,
    "opportunity": 1070.8,
    "ai_search_volume": 64,
    "serp_features": [
      "ai_overview",
      "organic",
      "people_also_ask",
      "related_searches",
      "images"
    ],
    "why": "low competition; high commercial value (CPC 7.949999809265137); buyers comparing providers; 1 competitor(s) already rank for it; AI Overview present – write a quotable answer block.",
    "live": {
      "top_domains": [
        "tax.gov.ae",
        "mof.gov.ae",
        "edicomgroup.com",
        "tallysolutions.com",
        "europe.thomsonreuters.com"
      ],
      "your_position": null,
      "features": [
        "ai_overview",
        "people_also_ask",
        "images",
        "related_searches"
      ],
      "ai_overview": true,
      "paa": [
        "What is meant by e-invoicing?",
        "Is e-invoicing mandatory in the UAE?",
        "What is an e-invoice and how does it work?",
        "What is the rule for e-invoicing?"
      ]
    }
  }
}
```

	<pre>}, "run_ledger": { "dataforseo_usd": 0.2214, "dataforseo_calls": 17, "by_node": { "Candidate SERP": 0.012, "Run Research": 0.1512, "Run Competitor Keywords": 0.0381, "AI Demand": 0.02 }, "ai_calls": 2, "ai_nodes": ["Site Describer", "Keyword Seeds"], "started_at": "2026-10-01T12:10:31.289Z", "finished_at": "2026-10-01T12:12:33.976Z", "duration_min": 2 } }</pre>
Nodes executed (42)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Read Website → Clean Site Text → Site Read OK? → Claude — Site Describer → Parser — Site Describer → Site Describer → Parse Description → Route Mode → Anthropic Chat Model → Parser — Keyword Seeds → Keyword Seeds → Seed List → Research Requests → Run Research → Competitor Keyword Requests → Run Competitor Keywords → Collect Research → Relevance Chunks → Claude — Keyword Relevance ×3 → Parser — Keyword Relevance ×3 → Keyword Relevance → Rank Keywords → AI Demand → Priority Keywords → Candidate SERP → Build Keyword Strategy → Choice Step? → Build Keyword File → Prepare PDF Ideas → Render PDF Ideas → Attach PDF Ideas → Via Webhook (Ideas)? → Build Ideas Response → Has Callback? → POST Callback → Discover Extras? → Discover Audit?

Execution 21: Test 3 — Full SEO report for techand.ai

Started through	Test Runner → API Entry; body: mode audit, report_type "Full SEO Report", e-mail + callback
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 12:13:51.255 → 2026-10-01 12:22:05.494 · 8.2 min · last node: From Audit Form?
Result	Pass. Crawl + probes + key-page HTML/schema + competitor discovery + domain overview/WHOIS/RDAP + ranked keywords + backlinks + PageSpeed + Claude content review + AI visibility → health score 64/100 (1 Critical, 22 High). 443 KB e-mail (250 OK) and callback.

Key output fields

```
{
  "stage": "full_report",
  "health_score": 64,
  "grade": "Needs Improvement",
  "issue_counts": {
    "Critical": 1,
    "High": 22,
    "Medium": 18,
    "Low": 16,
    "Info": 3
  },
  "top_issues": [
    "Broken pages (4xx, 5xx or timeout)",
    "www and non-www versions are not consolidated",
    "HTML cannot be cached by the CDN (no-store / private)",
    "Malformed sameAs profile link",
    "Structured contact details are thinner than what the site shows",
    "Experience claim conflicts with the founding date",
    "Service pages have no Service schema"
  ],
  "run_ledger": {
    "dataforseo_usd": 1.02,
    "dataforseo_calls": 36,
    "by_node": {
      "Start Crawl": 0.09,
      "Find Competitors": 0.012,
      "Fallback SERP": 0.01,
      "Domain Overview": 0.0483,
      "DataForSEO Whois": 0.4848,
      "Ranked Keywords": 0.12,
      "Keyword Ideas": 0.048,
      "Backlink Summary": 0.096,
      "Backlink Gap": 0.024,
      "AI SERP": 0.022,
      "Ask LLMs": 0.0646
    },
    "ai_calls": 1,
    "ai_nodes": [
      "Content Reviewer"
    ],
    "started_at": "2026-10-01T12:13:51.371Z",
    "finished_at": "2026-10-01T12:21:58.374Z",
    "duration_min": 8.1
  }
}
```

Nodes executed (72)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Start Crawl → Save Task ID → Wait 1 min ×5 → Get Crawl Summary ×5 → Check Crawl ×5 → Crawl Done? ×5 → Get Crawled Pages → Crawl Extra Requests → Run Crawl Extras → Collect Crawl Extras → Build Probes → Run Probes → Content Probes → Run Content Probes → Analyze Probes → Full Report? → Pick Key Pages → Fetch Page HTML → Analyze HTML → URLs To Check → Check Schema URLs → Schema Findings → Find Competitors → Fallback Queries → Fallback SERP → Pick Competitors → Domain Overview → DataForSEO Whois → RDAP Lookup → Competitor Analysis → KW Targets → Ranked Keywords → Keyword Ideas → Analyze Keywords → BL Targets → Backlink Summary → Backlink Gap → Analyze Backlinks → PS Targets → PageSpeed → Analyze PageSpeed → Competitor Homes → Read Competitor Homes → Build Content Prompt → Claude — Content Reviewer → Parser — Content Reviewer → Content Reviewer → Parse Content Review → AI Queries → AI SERP → LLM Prompts → Ask LLMs → AI Visibility → Build Site Issues → Is Full Report? → Build Full Report → Prepare PDF Audit → Render PDF Audit → Attach PDF Audit → Via Webhook (Audit)? → Build Audit Response → Has Callback? → POST Callback → Has Email (Audit)? → Send Audit Report → From Audit Form?
----------------------------	--

Execution 23: Test 4a — Ladder for "e invoicing in uae" (first attempt)

Started through	Test Runner → API Entry; body: mode ladder, keyword "e invoicing in uae", domain techand.ai, pages_now 1, e-mail + callback
n8n execution	SEOagentV4Full01 · status error · 2026-10-01 12:14:49.732 → 2026-10-01 12:17:16.484 · 2.4 min · last node: Ladder Keyword Relevance
Result	Failed at "Ladder Keyword Relevance": the two nodes created by the ladder section had no credential slot (the build's credential pass ran before they existed). Verdict before the failure: GO_WITH_CHANGES 74/100. Fixed in the build (slot pass + assertions; "Ladder Pool" now stops with a clear message when every research pull fails).
Nodes executed (49)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Read Website → Clean Site Text → Site Read OK? → Claude — Site Describer → Parser — Site Describer → Site Describer → Parse Description → Route Mode → Prepare Keyword Run → Has Domain? → Fetch Sitemap Index → Parse Sitemap Index → Fetch Sitemaps → Collect Site URLs → Read Client Pages? → Client Page Requests → Read Client Pages → Collect Client Pages → SERP Top 10 → Pick Top 6 → Read Competitor Pages → Find Blocked Pages → Any Blocked? → Analyze Competitor Pages → Facts SERP → Collect Facts → Claude — Competitor Analyzer → Parser — Competitor Analyzer → Competitor Analyzer → Parse Analysis → Keyword Data → Has Domain (Authority)? → Merge Keyword Data → Verdict Chat Model → Parser — Verdict → Verdict Agent → Parse Verdict → Ladder Mode? → Ladder Requests → Run Ladder Research → Ladder Pool → Ladder Relevance Chunks → Claude — Ladder Relevance ✗ → Ladder Keyword Relevance ✗ Error: ERROR: Node does not have any credentials set

Execution 27: Test 4b — Ladder for "e invoicing in uae" (retry)

Started through	Same body as 4a after the fix
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 12:23:57.405 → 2026-10-01 12:29:40.053 · 5.7 min · last node: POST Callback
Result	Pass end to end in 5.6 min. Verdict GO_WITH_CHANGES 72/100 (KD 2, 880/mo, competitors average 5 referring domains). Ladder: 9 pages — rung 1 "uae e invoicing penalties", "e invoicing uae fta", "accredited service provider for e invoicing in uae", "uae e invoicing requirements"; rung 2 "e invoicing uae 2026", "list of asp for e invoicing in uae", "is e invoicing mandatory in uae", "e invoicing service providers in uae"; rung 3 empty; top page months 6-9. Data Tables created on first use, 9 rows stored; plan e-mailed (123 KB) and posted to the callback (72 KB PDF); first page run spawned.

Key output fields

```
{
  "stage": "ladder_plan",
  "ladder_id": "lad_mupia2l2c0tp",
  "head": {
    "keyword": "e invoicing in uae",
    "volume": 880,
    "kd": 2,
    "cpc": 6.85,
    "intent": "informational",
    "your_position": null
  },
  "feasibility": {
    "status": "winnable",
    "verdict": "GO_WITH_CHANGES",
    "score": 72,
    "reasons": [
      "Volume of 880/month with a CPC of $6.85 shows real commercial value. The mandate is close: Wave 1 must appoint an ASP by 30 Oct 2026 and go live on 1 Jan 2027, and Wave 1 is exactly Tech&'s enterprise audience (revenue ≥ AED 50m).",
      "Keyword difficulty is 2 (Easy) and competitors average only 5 referring domains and 10 backlinks. Link barriers are low, even though the site's authority is unknown.",
      "The content gaps are clear. All six ranking pages have 0 tables, and none offers a wave-finder, an ASP selection scorecard or a backwards-planned onboarding timeline. A structured page with a lead path can beat ClearTax (2,628 words) and EDICOM (1,888 words), and MoF (713 words, no lead path) is a weak competitor on conversion.",
      "The SERP is informational and mixed. It has government, Big-4 and vendor pages, an AI Overview and no featured snippet. A pure service page would mismatch that intent, so the page needs to be a guide-style service page."
    ],
    "what_must_change": [
      "Build a hybrid service page: open with a 40-70 word answer on what e-invoicing is and the mandatory dates, then give the detailed guide content, then the Dynamics 365 / Microsoft e-invoicing service offer.",
      "Add a Wave 1 vs Wave 2 wave-finder table, a backwards-planned timeline from 30 Oct 2026 and 31 Mar 2027, and an ASP selection scorecard labelled as Tech&'s own framework.",
      "Add question-style H2s and FAQ schema matching the People Also Ask questions, plus a sample invoice field table and a readiness checklist, with a free readiness assessment CTA.",
      "Add trust signals: a named expert author, a last-updated date, citations to MoF, Ministerial Decisions 243, 244 and 64, and Cabinet Decision 106, ASP partner or accreditation status, a UAE contact, and case studies.",
      "Support the page with spoke content on deadline, requirements, ASP list and ERP readiness (D365, SAP), and build internal links and a few UAE finance backlinks."
    ],
    "alternatives": [],
    "expected_visits_top3": 300,
    "time_to_rank_months": 4,
    "navigational": false
  },
}
```

```
"timeline": [
  {
    "rung": 1,
    "label": "Rung 1",
    "months": [
      1,
      2
    ],
    "pages": 4
  },
  {
    "rung": 2,
    "label": "Rung 2",
    "months": [
      3,
      5
    ],
    "pages": 4
  },
  {
    "rung": 4,
    "label": "Top – e invoicing in uae",
    "months": [
      6,
      9
    ],
    "pages": 1
  }
],
"write_now": [
  {
    "page_no": 1,
    "rung": 1,
    "keyword": "uae e invoicing penalties",
    "page_type": "Blog Post",
    "target_url": "https://techand.ai/uae-e-invoicing-penalties/",
    "exists": false,
    "supporting": [],
    "links_to": [
      {
        "url": "https://techand.ai/e-invoicing-in-uae/",
        "role": "top",
        "keyword": "e invoicing in uae"
      },
      {
        "url": "https://techand.ai/e-invoicing-uae-fta/",
        "role": "sibling",
        "keyword": "e invoicing uae fta"
      }
    ]
  }
]
```

Nodes executed (70)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Read Website → Clean Site Text → Site Read OK? → Claude — Site Describer → Parser — Site Describer → Site Describer → Parse Description → Route Mode → Prepare Keyword Run → Has Domain? → Fetch Sitemap Index → Parse Sitemap Index → Fetch Sitemaps → Collect Site URLs → Read Client Pages? → Client Page Requests → Read Client Pages → Collect Client Pages → SERP Top 10 → Pick Top 6 → Read Competitor Pages → Find Blocked Pages → Any Blocked? → Blocked Page Items → Retry Blocked Pages → Analyze Competitor Pages → Facts SERP → Collect Facts → Claude — Competitor Analyzer → Parser — Competitor Analyzer → Competitor Analyzer → Parse Analysis → Keyword Data → Has Domain (Authority)? → Merge Keyword Data → Verdict Chat Model → Parser — Verdict → Verdict Agent → Parse Verdict → Ladder Mode? → Ladder Requests → Run Ladder Research → Ladder Pool → Ladder Relevance Chunks → Claude — Ladder Relevance → Parser — Ladder Relevance → Ladder Keyword Relevance → Ladder Plan → Build Ladder Report → Prepare PDF Ladder → Render PDF Ladder → Attach PDF Ladder → Ensure Ladder Table → Ensure History Table → Ladder Rows → Save Ladder Rows → Ladder Delivery → Spawn Page Runs → Start Page Runs → Has Email (Ladder)? → Send Ladder Report → Via Webhook (Ladder)? → Build Ladder Response → Has Callback? → POST Callback
----------------------------	--

Execution 31: Test 4c (first attempt) — spawned rung-1 page run

Started through	Spawned by execution 27 through API Entry (mode keyword, force_content, ladder fields)
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 12:29:35.043 → 2026-10-01 12:29:35.494 · 0.0 min · last node: POST Rejection
Result	Rejected by the AI budget guard, by design: the day's estimate stood at \$1.83 and the content run adds \$1.20 (> \$3). The rejection was POSTed to the callback with the ladder's request_id.
Nodes executed (7)	API Entry → Normalize Input → Rate Limit → Input OK? → Error Via API? → API Reject → POST Rejection

Execution 29: Test 5 — Error handler check

Started through	Test Runner → API Entry; body: mode verdict without callback or e-mail (invalid on purpose)
n8n execution	SEOagentV4Full01 · status error · 2026-10-01 12:23:57.641 → 2026-10-01 12:23:57.928 · 0.0 min · last node: API Reject
Result	The run failed at "API Reject" as intended; the error workflow (execution 30) e-mailed ops (250 OK). Finding: in n8n 2.x an error workflow must be published; it had never fired before.
Nodes executed (6)	API Entry → Normalize Input → Rate Limit → Input OK? → Error Via API? → API Reject ✖ Error: ERROR: API request rejected (no callback_url to notify): API requests need a callback_url (https) or an email address so the result can be delivered. [line 4]

Execution 35: Test 6 — Rank Tracker manual run

Started through	Test Runner → Rank Tracker "Manual Run"
n8n execution	SEOagentTracker1 · status success · 2026-10-01 12:30:52.782 → 2026-10-01 12:31:51.481 · 1.0 min · last node: POST Progress
Result	Pass. 9 keywords checked (Google SERP live), 9 seo_rank_history rows written, progress e-mail (250 OK) and callback (stage rank_tracker); recommendation "work on rung 1". Measured \$0.028 per keyword at depth 100 → depth set to 50.

Key output fields

```
{
  "ladder_id": "lad_mupia2l2c0tp",
  "subject": "[Keyword ladder] e invoicing in uae – techand.ai: top
page not yet in the top 100",
  "rungs": [
    {
      "rung": 1,
      "pages": 4,
      "top10": 0,
      "top3": 0,
      "reached": false
    },
    {
      "rung": 2,
      "pages": 4,
      "top10": 0,
      "top3": 0,
      "reached": false
    }
  ],
  "next_step": {
    "action": "write_rung_1",
    "rung": 1,
    "text": "Work on rung 1: none of its pages is in the top 10 yet.
Write or improve: \"uae e invoicing penalties\" (already written –
publish it), \"e invoicing uae fta\", \"accredited service provider
for e invoicing in uae\", \"uae e invoicing requirements\".",
    "pages": [
      {
        "keyword": "uae e invoicing penalties",
        "page_type": "Blog Post",
        "target_url": "https://techand.ai/uae-e-invoicing-penalties/",
        "status": "writing"
      },
      {
        "keyword": "e invoicing uae fta",
        "page_type": "Guide",
        "target_url": "https://techand.ai/e-invoicing-uae-fta/",
        "status": "planned"
      },
      {
        "keyword": "accredited service provider for e invoicing in
uae",
        "page_type": "Service Page",
        "target_url": "https://techand.ai/accredited-service-provider-
for-e-invoicing-in-uae/",
        "status": "planned"
      },
      {
        "keyword": "uae e invoicing requirements",
        "page_type": "Guide",
        "target_url": "https://techand.ai/uae-e-invoicing-
requirements/",
        "status": "planned"
      }
    ]
  },
}
```

	<pre>"done": false }</pre>
Nodes executed (16)	Manual Run → Ensure Ladder Table → Ensure History Table → Load Ladders → Load History → Tracker Plan → Any Checks? → SERP Check → Parse Positions → Save History → Tracker Report → Has Report? → Has Email (Tracker)? → Send Progress Email → Has Callback (Tracker)? → POST Progress

Execution 38: Test 7 — Form path: "Check if my keyword is right"

Started through	Public form (basic auth) driven with curl: start page → Page Verdict → run → download
n8n execution	SEOagentV4Full01 · status waiting · 2026-10-01 13:12:48.718 → 2026-10-01 13:12:48.759 · 0.0 min · last node: Download Report
Result	Pass. 2.9 min; verdict GO_WITH_CHANGES 72/100 for "e invoicing in uae"; PDF download page served ("Your report is ready") and the same PDF e-mailed (76 KB, 250 OK).
Key output fields	<pre>{ "report_type": "Keyword Verdict", "verdict": "GO_WITH_CHANGES", "verdict_score": 72, "recommended_page": "Guide-style Service Page (e-invoicing readiness and implementation guide with lead CTAs)", "pdf_file_name": "keyword-verdict-e-invoicing-in-uae-2026-10-01.pdf", "run_ledger": { "dataforseo_usd": 0.0221, "dataforseo_calls": 3, "by_node": { "SERP Top 10": 0.006, "Keyword Data": 0.0121, "Facts SERP": 0.004 }, "ai_calls": 2, "ai_nodes": ["Competitor Analyzer", "Verdict Agent"], "started_at": "2026-10-01T13:21:45.723Z", "finished_at": "2026-10-01T13:24:37.788Z", "duration_min": 2.9 } }</pre>
Nodes executed (48)	Start Form → Choose Path → Page Verdict → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Prepare Keyword Run → Has Domain? → Fetch Sitemap Index → Parse Sitemap Index → Fetch Sitemaps → Collect Site URLs → Read Client Pages? → Client Page Requests → Read Client Pages → Collect Client Pages → SERP Top 10 → Pick Top 6 → Read Competitor Pages → Find Blocked Pages → Any Blocked? → Analyze Competitor Pages → Facts SERP → Collect Facts → Claude — Competitor Analyzer → Parser — Competitor Analyzer → Competitor Analyzer → Parse Analysis → Keyword Data → Has Domain (Authority)? → Merge Keyword Data → Verdict Chat Model → Parser — Verdict → Verdict Agent → Parse Verdict → Ladder Mode? → Verdict Router → Need Content? → Build Word File → Prepare PDF Report → Render PDF Report → Attach PDF Report → Has Email? → Send Report → Deliver Via Download? → Download Report

Execution 40: Form path: ladder page with an invalid domain

Started through	Public form: "Rank my site for a keyword" → Page Ladder with domain 10.0.0.1
n8n execution	SEOagentV4Full01 · status waiting · 2026-10-01 13:21:11.158 → 2026-10-01 13:21:11.191 · 0.0 min · last node: Show Input Error
Result	Pass (validation). Normalize Input rejected the domain; the "Something needs fixing" page showed "Please enter a public website address like example.com". No paid call.
Nodes executed (8)	Start Form → Choose Path → Page Ladder → Normalize Input → Rate Limit → Input OK? → Error Via API? → Show Input Error

Execution 41: Test 8 — Form path: discovery + "Choose your keyword"

Started through	Public form: "Suggest keywords for my business" with Keyword Report ticked
n8n execution	SEOagentV4Full01 · status waiting · 2026-10-01 13:26:11.033 → 2026-10-01 13:26:11.067 · 0.0 min · last node: Download Keyword Ideas
Result	Choice step passed: the JSON-defined dropdown rendered live (system suggestion + 7 keywords), option 4 "e invoicing implementation" was applied (chosen_by user) and the ideas download page was served. The chained keyword report never ran because the completion node ran first and ended the execution → redesigned (follow-ups are spawned, completion last).
Nodes executed (35)	Start Form → Choose Path → Page Discover → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Anthropic Chat Model → Parser — Keyword Seeds → Keyword Seeds → Seed List → Research Requests → Run Research → Competitor Keyword Requests → Run Competitor Keywords → Collect Research → Relevance Chunks → Claude — Keyword Relevance ×3 → Parser — Keyword Relevance ×3 → Keyword Relevance → Rank Keywords → AI Demand → Priority Keywords → Candidate SERP → Build Keyword Strategy → Choice Step? → Choose Keyword → Apply Choice → Build Keyword File → Prepare PDF Ideas → Render PDF Ideas → Attach PDF Ideas → Via Webhook (Ideas)? → Download Keyword Ideas

Execution 42: Test 9 — Form path: audit after the fix

Started through	Public form: "Audit my website" → Page Audit (technical)
n8n execution	SEOagentV4Full01 · status waiting · 2026-10-01 13:37:39.946 → 2026-10-01 13:37:39.979 · 0.0 min · last node: Audit Started
Result	Pass. "Your audit has started" page served within seconds; the audit itself ran as execution 43 (spawned).
Nodes executed (12)	Start Form → Choose Path → Page Audit → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Audit Via Form? → Spawn Audit Run → Start Audit Run → Audit Started

Execution 43: Test 9 — the spawned technical audit

Started through	Spawned by execution 42 through API Entry (mode audit, e-mail)
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 13:38:12.014 → 2026-10-01 13:42:42.645 · 4.5 min · last node: Send Audit Report
Result	Pass. 4.4 min, health score 70/100, PDF + Word e-mailed (208 KB, 250 OK).

Key output fields	<pre> { "stage": "site_audit", "health_score": 70, "grade": "Needs Improvement", "run_ledger": { "dataforseo_usd": 0.09, "dataforseo_calls": 7, "by_node": { "Start Crawl": 0.09 }, "ai_calls": 0, "ai_nodes": [], "started_at": "2026-10-01T13:38:12.121Z", "finished_at": "2026-10-01T13:42:37.751Z", "duration_min": 4.4 } } </pre>
Nodes executed (34)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Audit Via Form? → Start Crawl → Save Task ID → Wait 1 min ×4 → Get Crawl Summary ×4 → Check Crawl ×4 → Crawl Done? ×4 → Get Crawled Pages → Crawl Extra Requests → Run Crawl Extras → Collect Crawl Extras → Build Probes → Run Probes → Content Probes → Run Content Probes → Analyze Probes → Full Report? → Build Site Issues → Is Full Report? → Build Audit Report → Prepare PDF Audit → Render PDF Audit → Attach PDF Audit → Via Webhook (Audit)? → Build Audit Response → Has Callback? → Has Email (Audit)? → Send Audit Report

Execution 44: Test 10 — Form path: discovery with Site Audit ticked

Started through	Public form: "Suggest keywords for my business" + Extra Features "Site Audit"
n8n execution	SEOagentV4Full01 · status waiting · 2026-10-01 13:38:46.168 → 2026-10-01 13:38:46.200 · 0.0 min · last node: Show Input Error
Result	Stopped by the per-address rate limit (6 runs per e-mail per day); the validation page showed the limit message. Not bypassed.
Nodes executed (8)	Start Form → Choose Path → Page Discover → Normalize Input → Rate Limit → Input OK? → Error Via API? → Show Input Error

Execution 46: Test 4c — the ladder's first page, after the cap was raised

Started through	Test Runner → API Entry; the body Spawn Page Runs produces (mode keyword, keyword "uae e invoicing penalties", page_type Blog Post, force_content, ladder_id, rung 1, ladder_links; callback only)
n8n execution	SEOagentV4Full01 · status success · 2026-10-01 13:48:16.367 → 2026-10-01 13:58:33.798 · 10.3 min · last node: POST Callback
Result	Pass end to end in 10.3 min: verdict GO_WITH_CHANGES 62/100 for the rung keyword; first draft content score 83 (2,434 words vs 1,800, critic 58) → one Opus editor pass → final score 85/100, 12 H2s, 6 internal links incl. the ladder's planned top page and sibling; callback carries ladder_id / ladder_rung / ladder_head with the 187 KB PDF and 52 KB Word file.

Key output fields	<pre> { "stage": "content", "ladder_id": "lad_mupia2l2c0tp", "ladder_rung": 1, "ladder_head": "e invoicing in uae", "verdict": "GO_WITH_CHANGES", "score": 62, "qa_summary": "Quality check: 2 item(s) to review", "run_ledger": { "dataforseo_usd": 0.0221, "dataforseo_calls": 3, "by_node": { "SERP Top 10": 0.006, "Keyword Data": 0.0121, "Facts SERP": 0.004 }, "ai_calls": 5, "ai_nodes": ["Competitor Analyzer", "Verdict Agent", "Strategy Brief", "Copywriter", "Editor"], "started_at": "2026-10-01T13:48:16.462Z", "finished_at": "2026-10-01T13:58:32.899Z", "duration_min": 10.3 } } </pre>
Nodes executed (63)	API Entry → Normalize Input → Rate Limit → Input OK? → Need Site Read? → Route Mode → Prepare Keyword Run → Has Domain? → Fetch Sitemap Index → Parse Sitemap Index → Fetch Sitemaps → Collect Site URLs → Read Client Pages? → Client Page Requests → Read Client Pages → Collect Client Pages → SERP Top 10 → Pick Top 6 → Read Competitor Pages → Find Blocked Pages → Any Blocked? → Analyze Competitor Pages → Facts SERP → Collect Facts → Claude — Competitor Analyzer → Parser — Competitor Analyzer → Competitor Analyzer → Parse Analysis → Keyword Data → Has Domain (Authority)? → Merge Keyword Data → Verdict Chat Model → Parser — Verdict → Verdict Agent → Parse Verdict → Ladder Mode? → Verdict Router → Need Content? → Claude — Strategy Brief → Parser — Strategy Brief → Strategy Brief → Parse Brief → Claude — Copywriter → Copywriter → Claude — Critic → Parser — Critic → Critic → Parse Critique → Content QA ×2 → Needs Revision? ×2 → Claude — Editor → Editor → Build Word File → Prepare PDF Report → Render PDF Report → Attach PDF Report → Publish To WordPress? → Has Email? → Deliver Via Download? → Is Webhook? → Build Webhook Response → Has Callback? → POST Callback

Budget ledger of the session

#	Run	Claude (guard estimate)	DataForSEO (ledger)	Outcome
1	Technical audit	\$0.00	\$0.09	Pass, 5.4 min, score 70, e-mail 250 OK
2	Keyword discovery	\$0.38	\$0.22	Pass, 2.0 min, callback 88 KB PDF
3	Full SEO report	\$0.25	\$1.02	Pass, 8.1 min, score 64, e-mail 443 KB
4a	Ladder attempt 1	~\$0.23	~\$0.05	Failed (credential slot) → fixed

4b	Ladder retry	\$0.58	\$0.12	Pass, 5.6 min, 9 pages, 9 rows, e-mail + callback
4c	Ladder page (content)	\$1.20	\$0.02	First blocked by the \$3 cap (by design); passed after the cap was raised to \$10: 10.3 min, score 85
5	Error handler	\$0	\$0	Pass after publishing the error workflow
6	Rank tracker	\$0	\$0.25	Pass, 9 keywords, e-mail + callback
7	Form: verdict	\$0.20	\$0.02	Pass, 2.9 min, download + e-mail
8	Form: discovery + choice	\$0.35	\$0.22	Choice step pass; follow-up never ran → redesign
9	Form: audit (spawned)	\$0	\$0.09	Pass, started page instant, 4.4 min, e-mail
10	Form: discovery + audit	\$0	\$0	Stopped by the 6-runs-per-e-mail limit (by design)
Total		≈ \$3.20 (estimates)	≈ \$2.10	DataForSEO balance was \$32.6 before the day

Full test log (from REVIEW.md \$5c)

Date	Step	Result
2026-10-01	Credentials imported under the pre-wired IDs (DataForSEO, Jina, PageSpeed, Anthropic, form login, API key); SMTP slot created with a placeholder password pending a Google App Password	done
2026-10-01	Key validation: n8n API 200, DataForSEO balance \$32.93, Anthropic models list OK (Opus 5.5, Sonnet 5.5, Haiku 4.5 dated ID), Jina 200, PageSpeed 200	done
2026-10-01	Publish checks: n8n refused to activate while the Gmail OAuth nodes had no credential → replaced by SMTP Send Email nodes; webhook nodes created by script needed a webhookId to register → added	fixed
2026-10-01	Test 1, keyword verdict via API (no domain, local callback receiver SEO Agent – Test Callback Receiver)	Pass. 202 in 2 s; main run 2.3 min; verdict GO_WITH_CHANGES 52/100 with reasons quoting the real data (10 searches/mo, -50% trend, AI Overview citing 12 UAE providers, competitor word counts); callback received; DataForSEO \$0.02, 2 Claude calls
2026-10-01	Defects found in test 1	request_id dropped by the report builders (fixed); callback PDF was a filesystem reference, not bytes (fixed: bytes read through getBinaryDataBuffer); YouTube/LinkedIn accepted as fact sources (excluded); term model dominated by generic words (generic single words filtered, phrases ranked first); 3 of 8 competitor pages blocked by bot protection (new browser-engine + proxy retry pass)

2026-10-01	API completeness	describe / discover / audit runs started through the API used to reach form-only "completion" nodes; they now deliver through the callback with a s stage field (site_description, keyword_strategy, site_audit, full_report, content)
2026-10-01	Test 2a, discover → content via API	Rejected in <1 s by the form-era rule "extra reports are sent by email" although a callback was given, and the rejection could not be delivered because the error item dropped the callback. Fixed: a callback counts as delivery; rejections keep callback_url/request_id; audit e-mail only when an address exists
2026-10-01	Test 2b, discover stage	Pass. 2.2 min, 19 DataForSEO calls (\$0.22), 616 keywords → 240 AI-screened → 99 relevant → 41 clusters; AI search demand returned real numbers; priority picks "erp systems in uae" (260/mo, KD 6, CPC \$13.61) and "erp system dubai" (320/mo, KD 27, CPC \$12); 96 KB PDF in the callback with the right request_id. Tuned afterwards: goal-aware tiers, local competitor discovery, minimum volume for priority picks, "start with" card completeness
2026-10-01	Test 2b, content stage	Fail at Strategy Brief: both attempts stopped at exactly 8,192 completion tokens — adaptive thinking consumed the whole output budget and the structured parser got an empty response. Fixed: brief 24,576 tokens at medium effort; copywriter/editor 32,768; analysis models 16,384. Verdict before the failure was again specific (260/mo falling -35% QoQ, CPC \$13.61, KD 6 but 52 referring domains, niche mismatch)
2026-10-01	Test 3a, content run for "erp systems in uae" via API	Fail at Strategy Brief: "Streaming is required for operations that may take longer than 10 minutes" — the Anthropic SDK refuses non-streaming requests whose max_tokens exceed roughly 21k. Fixed: the node's "Stream Responses" option is on for the three Opus nodes (output shape unchanged)
2026-10-01	Spend guard	Conservative per-run estimates had reached \$2.95 of the \$3 guard while actual Anthropic spend was about \$0.80 (the two aborted 8,192-token briefs cost about \$0.43). Guard switched to a daily window; the Anthropic Console workspace limit remains the hard cap
2026-10-01	Test 3b, content run for "erp systems in uae" (streaming on)	Pass end to end: 26 min, callback with a 247 KB PDF and 69 KB Word file, request_id intact, verdict GO_WITH_CHANGES 58. Content quality is the step-change we wanted: distributor-specific comparison (landed cost, batch/expiry, PDCs, van sales), UAE compliance checklists (VAT 201, Corporate Tax, Peppol e-invoicing, WPS), a free-zone re-export worked example, an AED cost-driver worksheet, a migration risk register and a 10-question demo scorecard; government sources (tax.gov.ae, mof.gov.ae) cited. Critic 58/100 ("revise") with specific, fair objections

2026-10-01	Defects found in test 3b	Length ran from a 3,500-word target to 5,689 words over two editor rounds (sections "below target" warnings pushed expansion); a "Production Brief" note leaked above the H1; QA counted the keyword 2× because "ERP systems in the UAE" did not match "erp systems in uae" exactly, and term coverage ignored plurals; the editor took 11 + 5 minutes at high effort. Fixed: recommended length clamped by page type (900-2,400 service/landing, 1,200-3,000 guides) with 6-9 sections; copywriter and editor forbid notes and length overrun; QA strips pre-H1 notes, matches natural variants and plurals, trusts government/vendor sources, fails pages more than 25% off target; one editor pass; Opus at medium effort (expected run time 8-12 min)
2026-10-01	v4.2 keyword ladder built (see §3c): 224 nodes in the main workflow, Rank Tracker, WordPress publisher and Test Runner workflows; harness 155 node runs, 0 failed; imported through the CLI and published, n8n restarted (no API key available in this session, so activation went through <code>publish:workflow + restart</code>). Gmail SMTP credential updated with the Google App Password	done
2026-10-01	System test 1, technical audit for techand.ai (Test Runner → API Entry, e-mail + callback)	Pass. 5.4 min, 7 DataForSEO calls (\$0.09, crawl), health score 70/100 (1 critical: broken pages; www not consolidated; HTML not cacheable); PDF + Word e-mailed, Gmail SMTP answered 250 OK (208 KB message) — the e-mail path is verified
2026-10-01	System test 2, keyword discovery for techand.ai (site read + describe + discover, callback)	Pass. 2.0 min, 17 DataForSEO calls (\$0.22), 2 Claude calls (Haiku describer, Sonnet seeds + relevance); 624 keywords → 240 AI-screened → 126 relevant → 46 clusters; start-with pick "e invoicing" (1,900/mo, KD 9, CPC \$7.95, AI Overview present, tax.gov.ae/mof.gov.ae rank); AI search demand returned (e invoicing 64/mo in assistants); 88 KB PDF in the callback. Tuning note: the "e-invoicing uae" cluster landed in tier Later although it carries the highest opportunity, because the goal-aware tiering fills Now/Next with buyer-intent clusters first and the cluster was labelled a Guide — worth revisiting the tier rule (opportunity should be able to override the slot cap)
2026-10-01	System test 3, full SEO report for techand.ai (e-mail + callback)	Pass. 8.1 min (was 15-25 in the estimate), 36 DataForSEO calls (\$1.02 — WHOIS alone \$0.48 for 4 domains; candidate for the free RDAP-first order), 1 Claude call (Content Reviewer); health score 64/100 (1 critical, 22 high incl. malformed sameAs link, thin contact schema, no Service schema); PDF + Word e-mailed (443 KB, 250 OK) and posted to the callback
2026-10-01	System test 4a, ladder for "e invoicing in uae" on techand.ai (first attempt)	Fail at Ladder Keyword Relevance: the two nodes created by the ladder section had no credential slot (the build's credential pass ran before they existed), so all four research pulls returned nothing and the relevance model refused to run. Verdict before the failure: GO_WITH_CHANGES 74/100. Fixed: the slot pass now also runs after section 10, the build asserts that every model node and every DataForSEO/Jina/PageSpeed HTTP node carries a credential, and *Ladder Pool* stops with a clear message when every research pull fails (harness 156 node runs, 0 failed)

2026-10-01	Error handler check (deliberate API request without callback or e-mail → *API Reject* throws)	The error workflow had never fired for any of the earlier failed runs: in n8n 2.x an error workflow must be *published* to run, and SEO Agent — Error Handler had been imported unpublished. Published (n8n publish:workflow --id=SEOagentErrHandl) + restart; the deliberate failure then produced an Error Handler execution with *Email Ops* to kinnngahmed@gmail.com
2026-10-01	System test 4b, ladder for "e invoicing in uae" on techand.ai (retry; site read + describe, verdict, ladder chain, Data Tables, e-mail + callback)	Pass end to end. 5.6 min, 7 DataForSEO calls (\$0.12), 4 Claude calls (Haiku describer; Sonnet analyzer, verdict, ladder relevance). Verdict GO_WITH_CHANGES 72/100 with sharp, dated reasoning (Wave 1 ASP appointment by 30 Oct 2026, go-live 1 Jan 2027; KD 2; competitors average 5 referring domains; "hybrid guide-style service page"). Head term 880/mo, KD 2 → feasibility *winnable*. Ladder: 9 pages — rung 1 (months 1-2): "uae e invoicing penalties" (1,600/mo, KD 0), "e invoicing uae fta", "accredited service provider for e invoicing in uae", "uae e invoicing requirements"; rung 2 (months 3-5): "e invoicing uae 2026", "list of asp for e invoicing in uae", "is e invoicing mandatory in uae", "e invoicing service providers in uae"; rung 3 empty (every keyword around the term is easy); top page months 6-9. Data Tables seo_ladders / seo_rank_history were created on first use; 9 rows stored; plan e-mailed (123 KB, 250 OK) and posted to the callback (72 KB PDF); the first page run was spawned. Live samples: harness/sample-output/LIVE-ladder-plan-e-invoicing-in-uae.{pdf,doc.html,callback.json}
2026-10-01	System test 4c, the ladder's first page ("uae e invoicing penalties", rung 1, mode keyword with force_content and ladder links)	First attempt blocked by the AI budget guard by design (\$1.83 + \$1.20 > \$3; rejection POSTed to the callback). After the user raised the cap to \$10: Pass end to end — 10.3 min, 3 DataForSEO calls (\$0.02), 5 Claude calls (Sonnet analyzer + verdict, Opus brief + copywriter + editor, Sonnet critic); verdict for the rung keyword GO_WITH_CHANGES 62/100; first draft content score 83 (2,434 words vs 1,800, critic 58) → one editor pass; the copy links to the ladder's planned top page and sibling (both kept by the QA because ladder links count as internal-link candidates); final content score 85/100 after the editor (2,458 words — still over the 1,800 target, the same length tendency seen in the earlier content test; one snippet question left without a heading), 12 H2s, 6 internal links; callback carries ladder_id, ladder_rung, ladder_head with the 187 KB PDF and 52 KB Word file. Delivered through the callback (the e-mail address had reached its 6 runs/day). Live samples: harness/sample-output/LIVE-ladder-page-uae-e-invoicing-penalties.{pdf,doc.html,md}
2026-10-01	System test 6, Rank Tracker manual run (Test Runner → Manual Run)	Pass. 9 keywords checked, 9 seo_rank_history rows written, progress e-mail (250 OK) and callback (stage: rank_tracker) delivered; recommendation "work on rung 1" listing the written page as "publish it" and the three planned pages. Measured cost: \$0.028 per keyword at depth 100 (the spec assumed \$0.002, which is depth 10) → depth set to 50 (~\$0.015 per keyword, about \$0.60 per month for a 9-page ladder). Live sample: harness/sample-output/LIVE-tracker-email-e-invoicing-in-uae.html

2026-10-01	System test 7, form path (login supplied by the user; driven with curl: multipart POST with the page's auth token, then the per-page waiting URL)	Pass. Start page lists all 6 options; "Check if my keyword is right" → Page Verdict → run 2.9 min (3 DataForSEO calls \$0.02, 2 Claude calls) → verdict GO_WITH_CHANGES 72/100 for "e invoicing in uae" (880/mo +85% YoY, KD 2, competitors average 5 referring domains, intent informational → "guide-style service page") → *Download Report* completion served the PDF (saved as harness/sample-output/LIVE-form-verdict-e-invoicing-in-uae.pdf) and *Send Report* e-mailed it (76 KB, 250 OK). "Rank my site for a keyword" → Page Ladder renders all 11 fields; a submission with domain 10.0.0.1 ends on the "Something needs fixing" page with the exact validation message (no cost). One abandoned ladder form execution (39) is parked at its page and expires with the workflow timeout
2026-10-01	System test 8, discovery through the form with "Keyword Report" ticked → choice step	Pass for the choice step, fail for the follow-up. 2.8 min to the "Choose your keyword" page (JSON-defined dropdown rendered live: "Let the system choose" + 7 keywords with searches/mo, difficulty, intent, page type, and the system suggestion); option 4 "e invoicing implementation" submitted → *Apply Choice* recorded chosen_by: user; the keyword-ideas download page was served. But the chained keyword report never ran: *Via Webhook (Ideas)?* sat above *Discover Extras?* on the canvas, so the completion node ran first and ended the execution (n8n leaves it in *waiting*). Fixed by the spawn design above (harness 161 node runs, 0 failed). Observation: this run's priorities were low-volume (20-40/mo, difficulty unknown) because the business text drove the seeds; the API run that read the site found "e invoicing" at 1,900/mo — the seed step should blend both
2026-10-01	System test 9, audit through the form (technical) after the fix	Pass. "Your audit has started" page served within seconds; the audit itself ran as a spawned execution: 4.4 min, DataForSEO \$0.09, health score 70/100, PDF + Word e-mailed (208 KB, 250 OK). Note: executions that end on a form completion page stay in the *waiting* state until the workflow timeout (60 min) — cosmetic, n8n behaviour
2026-10-01	System test 10, discovery through the form with "Site Audit" ticked (to see the follow-up spawn live)	Stopped by the per-address rate limit (6 runs per e-mail per day — every test today used kinnngahmed@gmail.com); the "Something needs fixing" page showed the limit message. Not bypassed. The follow-up spawn is exercised by the harness and uses the same mechanism as the audit-form spawn verified in test 9. Live samples: harness/sample-output/LIVE-form-choose-keyword.html, LIVE-form-page-ladder.html
2026-10-01	Budget cap raised at the user's request	CONFIG.ai_budget_usd 3 → 10 in *Normalize Input* (build_v4.py section 10, commented) so the system test could finish the same day; the Anthropic Console workspace limit remains the hard cap. Lower it again when the testing phase is over. The per-address rate limit (6 runs/day) was left as is; the pending page run was started through the callback channel instead of the exhausted e-mail key

2026-10-01	Spend this session	DataForSEO about \$2.10 (full report \$1.02 of it, WHOIS \$0.48). Claude about \$3.20 by the guard's per-run estimates (audit 0, discovery 0.38, full report 0.25, ladder attempts 0.23 + 0.58, form verdict 0.20, form discovery 0.35, ladder page 1.20); the guard counter stands at \$3.58 for the day under the raised \$10 cap. The Test Runner is unpublished again; the test credential "SEO API Key (test)" remains without a known value (delete it or set a new one in the UI)
2026-10-02	v4.3 site tracking built (see §3d and TRACKING_FEATURE_SPEC.md): main workflow 235 nodes (mode "track"), Site Tracker (63 nodes), Site Admin (7 nodes); harness 212 node runs, 0 failed; imported through the CLI and published, n8n restarted. Credentials: SE0credGoogleSvc created as a placeholder (throwaway RSA key, e-mail replace-me@...) so the workflow runs until the real service-account key is pasted; SE0credApiTest01 (Test Runner key) reset to a fresh random value for the session	done
2026-10-02	System test A, Site Tracker on demand for techand.ai (virtual site from its ladder, Test Runner → Manual Run, e-mail + callback)	Pass end to end , 26 s: 9 ladder keywords tracked with their last Rank Tracker positions, Google Trends for the head term ("e invoicing in uae" rising, peak July 2026, \$0.011), Search Console and GA4 reported as not connected (placeholder credential), tables seo_sites / seo_site_metrics / seo_query_history created and 9 query rows stored, Claude brief written (headline "Connect Google Search Console and GA4...", 4 actions incl. publishing the rung-1 page), 62 KB PDF rendered, e-mail 250 OK, callback stage: site_tracker (80 KB) received. Live sample: harness/sample-output/LIVE-site-tracker-email-techand.ai.html
2026-10-02	Defects found in test A	Google's real error (invalid_grant: account not found, which pinpoints the placeholder credential) was hidden behind n8n's generic "Bad request" text → every parser now reads error.context.data first; the brief's facts showed unconnected sources as zeros ("0 impressions, live position 0") → facts now say "Search Console not connected" / "not in the top 50"; a registered site without an address now inherits the ladder's e-mail / callback; trend cost recorded (\$0.011 per keyword, not \$0.001)
2026-10-02	System test B, mode: track through the API with the test e-mail	Rejected by the per-address rate limit by design (the address had used its 6 runs on 2026-10-01 UTC; rejection POSTed to the callback). Re-run with callback only
2026-10-02	Defect found in test B (first run)	*Site Row (Track)* read its input item, which is the Data Table *create* node's table object, not the request → a junk row site_ was stored and the tracker found nothing to do. Fixed: the node reads *Normalize Input* explicitly; the harness now feeds the table object as input. The junk row was removed with the new Site Admin workflow({ action: delete, site_id }), which first failed with "Workflow is not active": sub-workflows must be *published* in n8n 2.x (noted for WordPress too)

2026-10-02	System test B3, mode: track for techand.ai with 3 search terms (callback only) → registration → tracker	Pass end to end. API Entry 202 → main run 1.3 s: row upserted into seo_sites (keywords, country, callback), stage: site_tracker_setup callback with the site id and keywords → Site Tracker 55 s: 12 tracked keywords (3 site terms + 9 ladder), 3 live SERP checks (\$0.045; techand.ai not in the top 50 for any), 4 trend requests (\$0.044), 3 seo_rank_history + 12 seo_query_history rows, brief, PDF (62.5 KB), e-mail to the ladder's address (inherited; 250 OK), callback received. Google errors now read invalid_grant: Invalid grant: account not found. Observation: low-volume terms gave "+1200% vs average" from sparse Trends series → such series are now reported as "too little search volume for a trend" (harness check added). Live sample: LIVE-site-tracker-email-techand.ai-registered.html
2026-10-02	v4.3 content cadence built (see §3e): blog package on every content run, "Blog posts per week" on Track my site, Content Cadence workflow (28 nodes), Site Admin with cadence / unpublish (14 nodes), "I published a page" mode (main workflow 256 nodes); harness 254 node runs, 0 failed; imported and published, n8n restarted	done
2026-10-02	Defect found live: Site Admin cadence before any workflow had created seo_cadence → "Data table not found"	Fixed: the admin workflow ensures its tables first and reads the request from its trigger (table-create nodes output the table object, same finding as *Site Row (Track)*)
2026-10-02	System test C1, Site Admin cadence for techand.ai (1 page per week)	Pass: seo_cadence row upserted (affected 1)
2026-10-02	System test C2, Content Cadence dry run for techand.ai	Pass: picks the ladder's next planned page "e invoicing uae fta" (rung 1, Guide) with links up to the top page and sideways to the written sibling; "coming next" lists the three remaining rung-1/2 pages; 8 candidates (striking-distance and trend sources empty because Search Console is not connected and the trends table was created in this run); nothing written, nothing logged
2026-10-02	System test C3, Content Cadence live for techand.ai (1 page)	Cadence run pass in 4 s: content run started as a separate execution (cadence: site_techand-ai key, not charged to the owner's daily limit), seo_content_log row (status started, request id cad_20261001_techand-ai_1), ladder row "e invoicing uae fta" → *writing*, cadence note e-mailed (250 OK, sample harness/sample-output/LIVE-cadence-note-techand.ai.html) and posted to the callback (stage: content_cadence). The spawned content run is logged in the row below
2026-10-02	System test P1, "I published a page" through the API (first attempt)	Rejected with "The published URL must be on techand.ai" for a correct URL: the n8n Code sandbox has no URL constructor , so new URL (. . .) inside try/catch silently produced an empty host. The same pattern was hiding in *Resolve Properties* (URL-prefix Search Console properties could never match), *GA4 Requests* (property detection by stream URL), *Blog Package* (internal-link classification), *URLs To Check* (the audit's schema-image probes were always skipped) and *Pick Top 6* (tracking parameters were never stripped). Fixed: regex host parsing everywhere; the harness now shadows URL so any new use fails in the tests (254 node runs, 0 failed)

2026-10-02	System test C3, the spawned content run for "e invoicing uae fta" (first attempt)	Failed at the Editor (terminated): the first Editor call errored after 33 s and the retry never returned until n8n terminated the execution. The host slept during the run and the container clock jumped by hours (durations of 32 minutes were recorded for a 20-second request in the next run), so the termination is most likely the execution timeout firing on the jumped clock rather than a model failure; the Anthropic node has no per-request timeout option to add. The Error Handler alerted ops. The run was restarted with the same request id and ladder fields (next row)
2026-10-02	System test C3 (restart), content run for "e invoicing uae fta" with the cadence's request id and ladder fields	Pass end to end (54 min wall-clock, of which the competitor-page stage with its bot-protection retry pass took the longest; Claude: strategy brief 2.4 min, copywriter 3.5 min, critic 23 s, one editor pass 1.8 min). Verdict for the rung keyword, then the page: 3,182 words, content score 89/100 , QA passed, 32 headings, Article + BreadcrumbList + FAQPage schema, internal links to the ladder's top page and sibling plus four real techand.ai pages, 24 external sources. Blog package delivered: e-invoicing-uae-fta.html, .md and .meta.json attached to the e-mail next to the PDF and Word file (364 KB, 250 OK) and returned in the callback (markdown, html, meta, PDF) with ladder_id / ladder_rung. Live samples: harness/sample-output/LIVE-blog-package-e-invoicing-uae-fta.{html,md,meta.json}
2026-10-02	System test P2, "I published a page" through the API (after the URL fix)	Functionally pass: the log row was found by keyword with its request id kept, upserted as *published* with the URL; the ladder row "e invoicing uae fta" was marked published with the real URL; stage: published callback received. Defect: *Publish Check* read its input item (the last Data Table load) instead of the fetch result, so the live check reported "HTTP 0" without the real reason (the fetch had timed out after 20 s on the site's slow 404 page). Fixed: the fetch result is read explicitly, the fetch timeout is 45 s, and the harness feeds the real input shape (254 node runs, 0 failed). Site Admin unpublish then reverted the test marking (affected 1)
2026-10-02	System test P3, "I published a page" by content_request_id (after the input fix, n8n restarted)	Pass: fetch in 2 s, the live check reports HTTP 404 honestly ("recorded, but the page could not be fetched; the Site Tracker checks it again on Monday"), the log row is resolved by the cadence's request id and the ladder row is found; both stored; stage: published callback received. Reverted with Site Admin unpublish so the ladder row is *writing* again until the owner really publishes the page
2026-10-02	Google service account installed (techand-site project, devai-seo@...), key pasted by the user in chat (rotate it after the testing phase); e-mail set in *Site Plan* CONFIG.service_account_email	done

2026-10-02	System test G1, Site Tracker on demand for techand.ai with real Google data	Pass end to end (2.5 min): Search Console property <code>sc-domain:techand.ai</code> resolved (full user), 30 Search Analytics requests in 5.5 s, 395 queries with impressions, 56 daily rows, year-ago empty (site younger than a year, handled); GA4 account 399185315 listed and property 543096312 detected from the web stream URL without being given; 3 GA4 reports (organic 46 sessions, 18% of 257, landing pages, daily); 9 URL inspections (planned pages "URL is unknown to Google", 3 transient "Internal error encountered." from Google); trends, 3 live checks, metrics and trend rows stored, brief with the real numbers ("impressions tripled to 4,226 ..."), e-mail 250 OK, callback. Every response shape matched the fixtures. Live sample: <code>harness/sample-output/LIVE-site-tracker-email-techand.ai-google.html</code>
2026-10-02	Tuning from the real data	Planned and written-but-unpublished ladder pages were inspected and reported as "not indexed" (they are not live yet) → only published or pre-existing pages are inspected; written pages get a "publish it" action, planned pages nothing; the brief is told not to call unpublished pages "not indexed". The brand query "tech and" was offered as a striking-distance win → brand queries (domain label, spaces ignored) are excluded from striking distance, CTR gaps, new queries and the cadence. Sparse trends no longer show a percentage
2026-10-02	System test G2, Site Tracker on demand after the tuning (real Google data)	Pass (1.7 min): only live pages inspected, planned pages reported as "planned, not written yet", the written page gets a "publish it" action, no brand query in the opportunities; brief headline built from the real numbers (impressions tripled to 4,226, 29 clicks). Site Admin unpublish now also resets <code>page_exists</code> so a reverted test publish no longer makes a page look live. Live sample: <code>harness/sample-output/LIVE-site-tracker-email-techand.ai-google.html</code>
2026-10-02	Search Console in the audit built (§3f): 11 nodes between the audit data collection and *Build Site Issues* (main workflow 268 nodes); harness 266 node runs, 0 failed; imported and published	done
2026-10-02	System test A1, technical audit of techand.ai with the Search Console step (live)	Ran end to end (4.5 min, e-mail + callback): property <code>sc-domain:techand.ai</code> , submitted sitemap found (43 URLs, read 2026-09-30, no errors), 28-day totals and top queries, "7 pages shown but never clicked" finding, structure summary (43 pages, 100% within 3 clicks, sections). Two defects: the HTTP node returns text responses under <code>data</code> , not <code>body</code> , so <code>/sitemap.xml</code> read as empty (and *Publish Check* would have read a live page as empty); and the crawl reports <code>www.techand.ai</code> URLs while canonicals use the bare domain, so no page counted as indexable and nothing was inspected. Fixed: text bodies read from <code>data</code> or <code>body</code> (harness fixtures now use <code>data</code>), host-agnostic URL matching for canonicals and for the sitemap-vs-crawl comparison (266 node runs, 0 failed). Re-run in the next row

2026-10-02	System test A2, technical audit of techand.ai (after the body/host fixes)	28 Search Console requests (3 analytics + 25 inspections) in 11.6 s; all 25 inspected URLs reported "unknown to Google" because the crawl's <code>www .</code> URLs were inspected while Google indexes the bare-host canonicals; the sitemap fetch timed out after 20 s on the slow host. Fixed: inspections use the page's canonical URL when it is on the domain (deduplicated), sitemap fetch 45 s with one retry
2026-10-02	System test A3, technical audit of techand.ai (final)	Pass (4.7 min): sitemap read (43 URLs, equal to the 43 Search Console reports), property <code>sc-domain: techand.ai</code> , submitted sitemap read 2026-09-30 without errors, 28-day totals, 25 canonical URLs inspected: 19 indexed, 4 "URL is unknown to Google", 2 "Discovered - currently not indexed" (the <code>/whatwedo/</code> section) → High finding with the URLs and the fix; "7 pages shown but never clicked" (Low); structure: 43 pages, 100% within 3 clicks, sections table; report section rendered, callback carries <code>search_console</code> and <code>site_structure</code> . Score unchanged at 70 (Crawlability 35)
2026-10-02	Console Alerts + monthly check-in built (§3g): Console Alerts workflow (17 nodes, imported, unpublished until the IMAP credential exists), check-in mode in the main workflow (283 nodes); harness 290 node runs, 0 failed	done
2026-10-02	System test K1, monthly check-in through the API for techand.ai (no manual action, no security issue, Pages chart export as text)	Pass (0.8 s): tables created, row stored for 2026-10 (chart: 43 indexed, 6 not indexed), no alert created, <code>stage: console_checkin</code> callback with the summary lines. The Site Tracker run that follows shows the check-in in the Monday report and drops the reminder for October
2026-10-02	System test T-K, Site Tracker on demand after the check-in (real Google data)	Pass (1 min): both new tables loaded, the October check-in appears in the report ("Last check-in (2026-10): manual action no, security issue no; Pages chart: 43 indexed, 6 not indexed"), the monthly reminder is gone for October, no notices (the Console Alerts watcher is not yet connected), brief headline unchanged. Test Runner unpublished again afterwards; final published set: main, API, Error Handler, Rank Tracker, Site Tracker, Site Admin, Content Cadence (Console Alerts and WordPress unpublished by design)
2026-10-02	Credentials and client settings moved to <code>n8n/.env</code> (§5b): <code>.env.seo.example</code> , <code>sync_credentials.py</code> , <code>apply_env.py</code> , <code>env_settings.py</code> ; brand / sender / ops address / service-account e-mail baked in at build time	done
2026-10-02	Live findings while moving to <code>.env</code>	(1) Exporting the stored credential values out of <code>n8n</code> was blocked by the auto-mode classifier, so only the values known from this session (Google service account, mail addresses, test key) are in <code>.env</code> ; the rest are left for the user to fill. (2) <code>{{ \$env.X }}</code> in a webhook trigger's header-auth credential → "No authentication data defined on node" (HTTP 500): trigger authentication never resolves expressions. (3) <code>\$env</code> in a Code node and in an e-mail node expression → "access to env vars denied": this <code>n8n</code> blocks env access in nodes by default. Design settled on literal values synced from <code>.env</code> by script (<code>\$env</code> expressions only as an opt-in mode for an instance with <code>N8N_BLOCK_ENV_ACCESS_IN_NODE=false</code>). (4) A container recreate empties <code>/tmp</code> , which held the execution-inspector scripts (recreated)

2026-10-02	System test E1, Site Tracker on demand after <code>apply_env.py</code> (Google credential and test key written from <code>.env</code> , sender from <code>.env</code>)	Pass (1 min): Search Console connected (<code>sc-domain:techand.ai</code>), 30 Search Analytics requests, 3 GA4 reports, service-account e-mail from <code>.env</code> in the connect instructions, e-mail sent (250 OK), callback. Test Runner unpublished afterwards
2026-10-02	All credential values placed in <code>n8n/.env</code> by the user (DataForSEO, Jina, PageSpeed, Anthropic, form login, Gmail App Password for SMTP + IMAP; <code>SEO_API_KEY</code> generated; <code>n8n</code> public API key stored as <code>N8N_API_KEY</code>) and applied with <code>apply_env.py</code> (10 credentials written, Console Alerts published)	Verified without spend : form login 401 → 200 with the <code>.env</code> login; API front door 403 with a wrong key, 400 (authenticated, body rejected) with <code>SEO_API_KEY</code> ; Jina 200; Anthropic 200; DataForSEO 200; SMTP 235 2.7.0 Accepted and IMAP OK authenticated with the App Password; no activation error for Console Alerts; PageSpeed 200 with a Lighthouse score returned (from the host and from inside the container; one earlier attempt failed to connect transiently). The Google account password the user pasted was not stored; all pasted values should be rotated after the testing phase
2026-10-02	State at the end of the session	techand.ai: tracked (site row + ladder), cadence 1 page per week (<code>seo_cadence</code>), first cadence page "e invoicing uae fta" delivered as a blog package and logged as <i>*started*</i> (waiting for the publish link). Test Runner unpublished again; its key value discarded. Spend for the cadence tests: about \$2.50 of Claude (two content attempts, one of them cut off at the Editor) and \$0.05 of DataForSEO
2026-10-02	Spend this session	DataForSEO about \$0.15 (10 trend requests, 3 SERP checks); Claude 2 brief calls (Sonnet 5.5, low effort, ~\$0.03 each by estimate; streaming off, so token usage is reported in the execution). Note: <code>seo_sites</code> now holds techand.ai with the test callback (<code>http://localhost:5678/webhook/seo-callback</code>) and no e-mail of its own (the ladder's e-mail is inherited); re-register through the form or API with your e-mail when the testing phase is over, or pause it with Site Admin
2026-10-02	v4.4 E-E-A-T and page types built (\$3h): business profile mode, case-study intake, hub / local / video page roles, Person / WebPage reviewedBy / LocalBusiness / VideoObject / ItemList schema, byline + author box + TOC + cluster block + snapshot + NAP + video embed in the package, new publish checks; critic prompt fixed to read the brief. Main workflow 312 nodes; harness 360 node runs, 0 failed; applied with <code>apply_env.py</code> (built, credentials synced, imported, published, restarted)	done

2026-10-02	Live checks of v4.4 (no AI spend)	API front door answers 400 with the new messages (profile without domain / content, case study without solution and results, video_url without https); the production form lists "Write a case study" and "Set up my business profile"; YouTube watch page read from the n8n container gives upload date, duration (PT3M34S) and title (no consent wall); Vimeo oEmbed answers JSON for public videos (404 HTML for a removed one, handled). Profile mode end to end (API, fictitious northwind-erp.com, local callback): seo_profiles created on first use, row upserted, readiness 88 % / 88 %, stage: profile callback posted (execution 132). The test row was deleted afterwards through the n8n Data Table API; seo_case_studies was created with its exact columns so content runs load an empty table rather than relying on the missing-table path. Not yet run live: a content run that uses a stored profile (hub / case study / local / video) — about \$1.20 of Claude each	pass
2026-10-01	Spend	DataForSEO total for all live tests about \$0.30. Anthropic spend is not metered by the workflow (streaming responses report no token usage); the two aborted briefs and two long editor passes make an estimate of \$2.5-3.5 for the day realistic. Set the Console workspace limit before more runs.	

10. Findings, fixes and roadmap

Found and fixed during this test

- **Credential slots for new nodes.** The two ladder nodes created after the build's credential pass had none; the build now re-runs the pass and asserts every model / data node carries one.
- **Error workflow never fired.** In n8n 2.x an error workflow must be published; *SEO Agent — Error Handler* had been imported unpublished. Published and verified with a deliberate failure.
- **Form completion pages end the execution.** Branches still queued (discovery report/content/audit, the audit "started" page order) never ran. Follow-ups are now spawned as separate executions; completion nodes are last; the WordPress gate runs before delivery.
- **Tracker cost.** \$0.028 per keyword at depth 100 → depth 50 (~\$0.015).
- **Ladder robustness.** Coarse AI topic labels are split per page type and capped at 6 keywords; an empty research pool stops with a clear message.

Observations for tuning

- Generated pages still overrun the length target (2,458 vs 1,800 words after one editor pass) — the same tendency as the earlier content test; consider a hard trim step in QA.
- Discovery priorities depend on the seed source: the API run that read the site found "e invoicing" (1,900/mo), the form run with a business text found only low-volume finance terms; blend both seed sources.
- The "e-invoicing uae" cluster landed in tier Later in the API discovery although it carried the highest opportunity; let opportunity override the goal-aware slot cap.
- WHOIS was \$0.48 of the \$1.02 full-report spend; query RDAP first.
- When a ladder page run is rejected by a guard and the user left an e-mail but no callback, only ops is told; e-mail the user too.

Roadmap (REVIEW.md §6)

1. **Parallelise the full report (M7).** Move Keywords, Backlinks, PageSpeed, Content Review and AI Visibility into sub-workflows started with Execute Workflow (no wait) and merge on completion; the wall-clock drops from ~20 to ~7 minutes and each module becomes independently testable and reusable through the API.
2. **Monthly cost roll-up:** ``run_ledger`` is per run; a Data Table or sheet that sums it per customer/month lets you price with margin.
3. **Hosted report link:** publish the PDF behind a signed URL so emails carry a link rather than a 200 KB attachment.
4. **Durable rate limiting and job tracking (M6):** a Data Table keyed by execution id with status (accepted / running / done / failed), which also powers a ``GET /status/:id`` endpoint for API customers and a "check my report" page for form users.
5. **Prompt re-tune on Sonnet 5.5:** the agents now run on Sonnet 5.5; after the first live runs, compare a few briefs/drafts against v3 outputs and adjust wording if needed.
6. **Quality evals for the content path:** keep 10–20 real briefs/drafts, score them with a rubric (keyword placement, no invented facts, FAQ format, table present), and run the set whenever a prompt changes. Content QA already produces most of the signals.
7. **Apply the production overlay** (``production/PRODUCTION.md``) before customer traffic, and move report email to a transactional sender on your domain.

8. **Competitor and cluster quality** (M11, L3): SERP-overlap clustering and service-based seed queries make the discovery report noticeably sharper.
9. **Ladder follow-ups**: e-mail the user (not only ops) when a ladder page run is rejected by the budget or rate guard; let the tracker start the next rung automatically when `ladder\_auto\_next\_rung` is on (today recommend-only); treat unknown difficulty (KD null) explicitly in the rung bands; let discovery's tiering promote a very high-opportunity cluster into "Now" even when it is informational; use RDAP before DataForSEO WHOIS in the full report (\$0.48 of the \$1.02 run).
10. **Tracking follow-ups** (see `TRACKING\_FEATURE\_SPEC.md`): Bing Webmaster Tools and Google Business Profile as further sources; charts in the PDF from the stored daily series and a hosted per-site dashboard reading `seo\_site\_metrics` / `seo\_query\_history`; feed Search Console striking-distance queries into discovery and ladder planning; auto-run the recommended improvement when `ladder\_auto\_next\_rung` is on; prune `seo\_query\_history` after 12 months.
11. **E-E-A-T follow-ups (v4.4)**: one live content run per page role with a real profile; several authors per site (pick by topic) instead of one default author; an author page generator (bio, credentials, list of articles, Person schema) from the profile; Site Admin actions to delete a profile or a case study; a case-study prompt in the Monday report for sites with none; transcripts fetched automatically (YouTube Data API captions need OAuth — today the owner pastes them); local pages planned by the ladder for geo head terms.
12. **Monitor follow-ups (v4.5)**: AI-visibility trend charts from `seo\_ai\_visibility`; brand-name disambiguation (a short domain label like "techand" can match another company — set `brand\_names` through Site Admin `monitors`); Google Business Profile through the official API (OAuth) for reviews and posts; a status page per prospect for outreach tracking; diff-aware fix pack (only what changed since the last audit); Bing / Copilot as an AI engine when DataForSEO offers it.

11. Appendix A: node inventory

SEO Agent v4 — Audit, Keywords, Verdict & Content

Node	Type	Version	Credential	Position
Spawn Follow-ups	Code	2		4552, -36
Start Follow-up Runs	Execute Workflow	1.2		4772, -36
Choose Path	Switch	3.4		0, 0
Page Keyword	Form page	2.5		448, 16
Start Form	Form trigger	2.6	SEO Form Login	-160, 32
Build Ideas Response	Code	2		4752, 64
Choose Keyword	Form page	2.5		5776, 80
Apply Choice	Code	2		5996, 80
Page Discover	Form page	2.5		272, 176
Download Keyword Ideas	Form page	2.5		3936, 224
Via Webhook (Ideas)?	IF	2.3		4552, 224
Keyword Seeds	AI Agent	3.1		2288, 240
Seed List	Code	2		2656, 240
Research Requests	Code	2		2876, 240
Run Research	HTTP Request	4.5	DataForSEO	3096, 240
Competitor Keyword Requests	Code	2		3316, 240
Run Competitor Keywords	HTTP Request	4.5	DataForSEO	3536, 240
Collect Research	Code	2		3756, 240
Relevance Chunks	Code	2		3976, 240
Keyword Relevance	AI Agent	3.1		4196, 240
Rank Keywords	Code	2		4456, 240
AI Demand	HTTP Request	4.5	DataForSEO	4676, 240
Priority Keywords	Code	2		4896, 240
Candidate SERP	HTTP Request	4.5	DataForSEO	5116, 240
Build Keyword Strategy	Code	2		5336, 240
Choice Step?	IF	2.3		5556, 240
Build Keyword File	Code	2		6216, 240
API Entry	Execute Workflow Trigger	1.2		-160, 288
Page Verdict	Form page	2.5		448, 288

Normalize Input	Code	2		656, 352
Page Audit	Form page	2.5		288, 384
Prepare PDF Ideas	Code	2		3912, 384
Render PDF Ideas	HTTP Request	4.5		4132, 384
Attach PDF Ideas	Code	2		4352, 384
Unmatched Choice	Code	2		160, 400
Claude — Keyword Relevance	Anthropic model	1.6	Anthropic (SEO Agent)	4096, 440
Parser — Keyword Relevance	Output parser	1.3		4316, 440
Build Full Report	Code	2		4464, 448
Anthropic Chat Model	Anthropic model	1.6	Anthropic (SEO Agent)	2368, 480
Parser — Keyword Seeds	Output parser	1.3		2480, 480
Route Mode	Switch	3.4		2064, 512
Read Website	HTTP Request	4.5	Jina Reader	1024, 528
Clean Site Text	Code	2		1248, 528
Site Read OK?	IF	2.3		1448, 528
Page Describe	Form page	2.5		112, 560
Page Ladder	Form page	2.5		448, 560
Rate Limit	Code	2		656, 576
Input OK?	IF	2.3		876, 576
Save Task ID	Code	2		2784, 608
Wait 1 min	Wait	1.1		3008, 608
Get Crawl Summary	HTTP Request	4.5	DataForSEO	3232, 608
Check Crawl	Code	2		3456, 608
Crawl Done?	IF	2.3		3680, 608
Get Crawled Pages	HTTP Request	4.5	DataForSEO	3904, 608
Build Site Issues	Code	2		4128, 608
Is Full Report?	IF	2.3		4352, 608
Prepare PDF Audit	Code	2		4664, 608
Render PDF Audit	HTTP Request	4.5		4884, 608
Attach PDF Audit	Code	2		5104, 608
Need Site Read?	IF	2.3		800, 624
Start Crawl	HTTP Request	4.5	DataForSEO	2496, 624
Build Describe Response	Code	2		3072, 624
Build Audit Report	Code	2		4576, 624
Send Audit Report	Send Email	2.1	Gmail SMTP (SEO Agent)	4800, 624

Site Describer	AI Agent	3.1		1536, 640
Parse Description	Code	2		1904, 640
Ranked Keywords	HTTP Request	4.5	DataForSEO	8160, 656
Analyze Keywords	Code	2		8608, 656
BL Targets	Code	2		8832, 656
Backlink Summary	HTTP Request	4.5	DataForSEO	9056, 656
Backlink Gap	HTTP Request	4.5	DataForSEO	9280, 656
Analyze Backlinks	Code	2		9504, 656
PS Targets	Code	2		9728, 656
PageSpeed	HTTP Request	4.5	Google PageSpeed	9952, 656
Analyze PageSpeed	Code	2		10176, 656
Competitor Homes	Code	2		10400, 656
Read Competitor Homes	HTTP Request	4.5	Jina Reader	10560, 672
Build Content Prompt	Code	2		10784, 672
Content Reviewer	AI Agent	3.1		11008, 672
Parse Content Review	Code	2		11376, 672
AI Queries	Code	2		11968, 672
AI SERP	HTTP Request	4.5	DataForSEO	12192, 672
AI Visibility	Code	2		12416, 672
KW Targets	Code	2		7904, 688
Keyword Ideas	HTTP Request	4.5	DataForSEO	8384, 704
Describe Fallback	Code	2		1668, 748
Pick Key Pages	Code	2		5152, 752
Page Track	Form page	2.5		448, 760
Via Webhook (Audit)?	IF	2.3		5304, 768
Build Audit Response	Code	2		5504, 768
API Reject	Code	2		1316, 776
POST Rejection	HTTP Request	4.5		1536, 776
Build Description Page	Code	2		2672, 784
Via Webhook (Describe)?	IF	2.3		2872, 784
Has Email (Audit)?	IF	2.3		4600, 784
Crawl Extra Requests	Code	2		3904, 800
Audit Via Form?	IF	2.3		2256, 824
Error Via API?	IF	2.3		1096, 836
Claude — Site Describer	Anthropic model	1.6	Anthropic (SEO Agent)	1536, 840

Run Probes	HTTP Request	4.5		4256, 848
Analyze Probes	Code	2		4480, 848
Full Report?	IF	2.3		4704, 848
Parser — Site Describer	Output parser	1.3		1696, 880
LLM Prompts	Code	2		12192, 880
Save Findings	Data Table	1.1		5928, 888
Analyze HTML	Code	2		5616, 896
URLs To Check	Code	2		5824, 896
Claude — Content Reviewer	Anthropic model	1.6	Anthropic (SEO Agent)	10896, 896
Parser — Content Reviewer	Output parser	1.3		11152, 896
Unmatched Mode	Code	2		2224, 912
Fetch Page HTML	HTTP Request	4.5		5376, 912
Schema Findings	Code	2		6288, 912
Show Description	Form page	2.5		2832, 928
Check Schema URLs	HTTP Request	4.5		6064, 928
Find Competitors	HTTP Request	4.5	DataForSEO	6528, 928
Fallback Queries	Code	2		6752, 928
Fallback SERP	HTTP Request	4.5	DataForSEO	6976, 928
Build Probes	Code	2		3936, 944
Spawn Audit Run	Code	2		2456, 954
Start Audit Run	Execute Workflow	1.2		2676, 954
Audit Started	Form page	2.5		2896, 954
Show Input Error	Form page	2.5		1316, 956
Page Published	Form page	2.5		448, 960
Run Crawl Extras	HTTP Request	4.5	DataForSEO	3904, 960
Pick Competitors	Code	2		7200, 960
Domain Overview	HTTP Request	4.5	DataForSEO	7424, 992
DataForSEO Whois	HTTP Request	4.5	DataForSEO	7632, 992
Fetch Sitemap Index	HTTP Request	4.5		1328, 1008
Parse Sitemap Index	Code	2		1520, 1008
Fetch Sitemaps	HTTP Request	4.5		1696, 1008
Client Page Requests	Code	2		2272, 1008
Read Client Pages	HTTP Request	4.5	Jina Reader	2472, 1008
Collect Client Pages	Code	2		2672, 1008
Ensure Audits Table	Data Table	1.1		4128, 1008

Ensure Findings Table	Data Table	1.1		4328, 1008
Load Audit History	Data Table	1.1		4528, 1008
Load Audit Findings	Data Table	1.1		4728, 1008
Audit Diff	Code	2		4928, 1008
Audit Row	Code	2		5128, 1008
Save Audit	Data Table	1.1		5328, 1008
Finding Rows	Code	2		5528, 1008
Any Finding Rows?	IF	2.3		5728, 1008
Fix Pack	Code	2		6128, 1008
Zip Fix Pack	n8n-nodes-base.compression	1.1		6328, 1008
Restore Audit Item	Code	2		6528, 1008
Competitor Analysis	Code	2		7856, 1008
Blog Package	Code	2		6814, 1028
Content Probes	Code	2		4456, 1048
Run Content Probes	HTTP Request	4.5		4656, 1048
Ask LLMs	HTTP Request	4.5	DataForSEO	12192, 1056
Editor	AI Agent	3.1		6048, 1072
Copywriter	AI Agent	3.1		5376, 1104
Collect Crawl Extras	Code	2		3904, 1120
Send Report	Send Email	2.1	Gmail SMTP (SEO Agent)	6464, 1152
RDAP Lookup	HTTP Request	4.5		7832, 1152
Page Check-in	Form page	2.5		448, 1160
Check-in Intake	Code	2		700, 1160
Prepare Keyword Run	Code	2		960, 1168
Has Domain?	IF	2.3		1152, 1168
Collect Site URLs	Code	2		1872, 1168
SERP Top 10	HTTP Request	4.5	DataForSEO	2048, 1168
Read Client Pages?	IF	2.3		2072, 1168
Pick Top 6	Code	2		2272, 1168
Read Competitor Pages	HTTP Request	4.5	Jina Reader	2496, 1168
Ensure Monitors Table (Track)	Data Table	1.1		3244, 1172
Load Monitors (Track)	Data Table	1.1		3464, 1172
Monitor Row (Track)	Code	2		3684, 1172
Save Monitors (Track)	Data Table	1.1		3904, 1172
Publish To WordPress?	IF	2.3		6924, 1268

Publish WordPress Draft	Execute Workflow	1.2		7144, 1268
Claude — Editor	Anthropic model	1.6	Anthropic (SEO Agent)	6224, 1280
Content QA	Code	2		5840, 1328
Build Word File	Code	2		6064, 1328
Has Email?	IF	2.3		6288, 1328
Find Blocked Pages	Code	2		2696, 1348
Any Blocked?	IF	2.3		2896, 1348
Ensure Cadence Table (Track)	Data Table	1.1		2474, 1352
Cadence Row (Track)	Code	2		2914, 1352
Save Cadence (Track)	Data Table	1.1		3134, 1352
Page Case Study	Form page	2.5		448, 1360
Critic	AI Agent	3.1		5616, 1364
Parse Critique	Code	2		5876, 1364
Claude — Copywriter	Anthropic model	1.6	Anthropic (SEO Agent)	5456, 1376
Build Track Response	Code	2		3684, 1392
Build Webhook Response	Code	2		6448, 1392
Has Callback?	IF	2.3		6896, 1392
POST Callback	HTTP Request	4.5		7120, 1392
Site Authority	HTTP Request	4.5	DataForSEO	4520, 1408
Is Webhook?	IF	2.3		6512, 1408
Strategy Brief	AI Agent	3.1		4768, 1424
Need Content?	IF	2.3		4480, 1472
Blocked Page Items	Code	2		3096, 1488
Retry Blocked Pages	HTTP Request	4.5	Jina Reader	3296, 1488
Prepare PDF Report	Code	2		6264, 1488
Render PDF Report	HTTP Request	4.5		6484, 1488
Attach PDF Report	Code	2		6704, 1488
Ensure Sites Table (Track)	Data Table	1.1		2364, 1512
Site Row (Track)	Code	2		2584, 1512
Save Site (Track)	Data Table	1.1		2804, 1512
Tracker Payload (Track)	Code	2		3024, 1512
Start Site Tracker	Execute Workflow	1.2		3244, 1512
Via Webhook (Track)?	IF	2.3		3464, 1512
Needs Revision?	IF	2.3		5888, 1520
Deliver Via Download?	IF	2.3		6288, 1536

Page Profile	Form page	2.5		448, 1560
Claude — Critic	Anthropic model	1.6	Anthropic (SEO Agent)	5516, 1564
Parser — Critic	Output parser	1.3		5736, 1564
Analyze Competitor Pages	Code	2		3280, 1568
Competitor Analyzer	AI Agent	3.1		3504, 1568
Parse Analysis	Code	2		3872, 1568
Keyword Data	HTTP Request	4.5	DataForSEO	4160, 1568
Has Domain (Authority)?	IF	2.3		4340, 1568
Parse Brief	Code	2		5120, 1568
Download Report	Form page	2.5		6512, 1600
Fetch Sitemap Children (Audit)	HTTP Request	4.5		2588, 1608
GSC Audit Query	HTTP Request	4.5	Google Service Account (SEO Agent)	3468, 1608
GSC Sitemaps (Audit)	HTTP Request	4.5	Google Service Account (SEO Agent)	3688, 1608
Parse Verdict	Code	2		4800, 1616
Tracking Started	Form page	2.5		3684, 1632
Claude — Strategy Brief	Anthropic model	1.6	Anthropic (SEO Agent)	4688, 1664
Parser — Strategy Brief	Output parser	1.3		4928, 1664
GSC Audit Plan	Code	2		1708, 1728
Fetch Sitemap (Audit)	HTTP Request	4.5		1928, 1728
Sitemap Children (Audit)	Code	2		2148, 1728
Any Sitemap Children?	IF	2.3		2368, 1728
GSC Sites (Audit)	HTTP Request	4.5	Google Service Account (SEO Agent)	2808, 1728
GSC Audit Requests	Code	2		3028, 1728
GSC Audit?	IF	2.3		3248, 1728
GSC Audit Findings	Code	2		3908, 1728
Facts SERP	HTTP Request	4.5	DataForSEO	3480, 1748
Collect Facts	Code	2		3680, 1748
Page AI Visibility	Form page	2.5		448, 1760
Merge Keyword Data	Code	2		4416, 1760
Fetch Video Details	HTTP Request	4.5		4860, 1772
Claude — Competitor Analyzer	Anthropic model	1.6	Anthropic (SEO Agent)	3392, 1792
Parser — Competitor Analyzer	Output parser	1.3		3648, 1792
Ladder Mode?	IF	2.3		4960, 1806
Verdict Router	IF	2.3		5120, 1808
Mark Ladder Published	Data Table	1.1		4564, 1812

Build Published Response	Code	2		5224, 1812
Parser — Verdict	Output parser	1.3		4736, 1856
Load Profile (Brief)	Data Table	1.1		4260, 1892
Load Case Studies (Brief)	Data Table	1.1		4460, 1892
Has Video?	IF	2.3		4660, 1892
Brief Context	Code	2		5060, 1892
Ensure Log Table (Published)	Data Table	1.1		2364, 1932
Fetch Published Page	HTTP Request	4.5		2584, 1932
Load Content Log (Published)	Data Table	1.1		2804, 1932
Load Ladders (Published)	Data Table	1.1		3024, 1932
Load Query History (Published)	Data Table	1.1		3244, 1932
Publish Check	Code	2		3464, 1932
Published Log Row	Code	2		3684, 1932
Save Published (Log)	Data Table	1.1		3904, 1932
Published Ladder Row	Code	2		4124, 1932
Ladder Page?	IF	2.3		4344, 1932
Published Delivery	Code	2		4784, 1932
Via Webhook (Published)?	IF	2.3		5004, 1932
Page Backlinks	Form page	2.5		448, 1960
Fetch GBP	HTTP Request	4.5	DataForSEO	4568, 1968
Verdict Agent	AI Agent	3.1		4816, 2000
Published Recorded	Form page	2.5		5224, 2052
Verdict Chat Model	Anthropic model	1.6	Anthropic (SEO Agent)	4576, 2064
Load Profile (Audit)	Data Table	1.1		3908, 2088
GBP Request	Code	2		4128, 2088
Look Up GBP?	IF	2.3		4348, 2088
Entity Check	Code	2		4788, 2088
Mark Case Study Published	Data Table	1.1		4014, 2132
Alert From Checkin	Code	2		3684, 2232
Save Check-in Alert	Data Table	1.1		3904, 2232
Build Check-in Response	Code	2		4564, 2232
Spawn Page Runs	Code	2		7720, 2336
Start Page Runs	Execute Workflow	1.2		7940, 2336
Ensure Checkin Table	Data Table	1.1		2364, 2352
Ensure Alerts Table (Check-in)	Data Table	1.1		2584, 2352

Parse Check-in	Code	2		2804, 2352
Checkin Row	Code	2		3024, 2352
Save Check-in	Data Table	1.1		3244, 2352
Checkin Alert?	IF	2.3		3464, 2352
Check-in Delivery	Code	2		4124, 2352
Via Webhook (Check-in)?	IF	2.3		4344, 2352
Ladder Requests	Code	2		4400, 2376
Run Ladder Research	HTTP Request	4.5	DataForSEO	4620, 2376
Ladder Pool	Code	2		4840, 2376
Ladder Relevance Chunks	Code	2		5060, 2376
Ladder Keyword Relevance	AI Agent	3.1		5280, 2376
Ladder Plan	Code	2		5540, 2376
Build Ladder Report	Code	2		5760, 2376
Check-in Recorded	Form page	2.5		4564, 2472
Prepare PDF Ladder	Code	2		5960, 2536
Render PDF Ladder	HTTP Request	4.5		6180, 2536
Attach PDF Ladder	Code	2		6400, 2536
Ensure Ladder Table	Data Table	1.1		6620, 2536
Ensure History Table	Data Table	1.1		6840, 2536
Ladder Rows	Code	2		7060, 2536
Save Ladder Rows	Data Table	1.1		7280, 2536
Ladder Delivery	Code	2		7500, 2536
Has Email (Ladder)?	IF	2.3		7720, 2536
Send Ladder Report	Send Email	2.1	Gmail SMTP (SEO Agent)	7940, 2536
Claude — Ladder Relevance	Anthropic model	1.6	Anthropic (SEO Agent)	5180, 2576
Parser — Ladder Relevance	Output parser	1.3		5400, 2576
Build Profile Response	Code	2		3684, 2652
Via Webhook (Ladder)?	IF	2.3		7720, 2736
Build Ladder Response	Code	2		7940, 2736
Ensure Profile Table	Data Table	1.1		2364, 2772
Load Profile (Profile)	Data Table	1.1		2584, 2772
Profile Row	Code	2		2804, 2772
Save Profile	Data Table	1.1		3024, 2772
Profile Delivery	Code	2		3244, 2772
Via Webhook (Profile)?	IF	2.3		3464, 2772

Profile Saved	Form page	2.5		3684, 2892
Download Ladder Plan	Form page	2.5		7940, 2916
Build Case Study Response	Code	2		4564, 3072
Ensure Case Table	Data Table	1.1		2364, 3192
Ensure Log Table (Case)	Data Table	1.1		2584, 3192
Case Study Row	Code	2		2804, 3192
Save Case Study	Data Table	1.1		3024, 3192
Case Study Log Row	Code	2		3244, 3192
Log Case Study	Data Table	1.1		3464, 3192
Spawn Case Study Run	Code	2		3684, 3192
Start Case Study Run	Execute Workflow	1.2		3904, 3192
Case Study Delivery	Code	2		4124, 3192
Via Webhook (Case Study)?	IF	2.3		4344, 3192
Case Study Started	Form page	2.5		4564, 3312
Start AI Visibility	Execute Workflow	1.2		2804, 3492
Build Monitor Response	Code	2		3464, 3492
Monitor Start	Code	2		2364, 3612
AI Visibility Run?	IF	2.3		2584, 3612
Monitor Started	Code	2		3024, 3612
Via Webhook (Monitor)?	IF	2.3		3244, 3612
Start Backlink Monitor	Execute Workflow	1.2		2804, 3732
Monitor Check Started	Form page	2.5		3464, 3732

SEO Agent — API

Node	Type	Version	Credential	Position
Respond Accepted	Respond to Webhook	1.1		720, -100
Prepare Payload	Code	2		960, -100
Start SEO Run	Execute Workflow	1.2		1200, -100
Webhook Trigger	Webhook	2	SEO API Key	0, 0
Quick Validate	Code	2		240, 0
Valid?	IF	2.3		480, 0
Respond Bad Request	Respond to Webhook	1.1		720, 120

SEO Agent — Rank Tracker

Node	Type	Version	Credential	Position
Has Email (Tracker)?	IF	2.3		2660, -100
Send Progress Email	Send Email	2.1	Gmail SMTP (SEO Agent)	2880, -100
Weekly Schedule	Schedule Trigger	1.2		0, 0
SERP Check	HTTP Request	4.5	DataForSEO	1560, 0
Parse Positions	Code	2		1780, 0
Save History	Data Table	1.1		2000, 0
Tracker Report	Code	2		2220, 0
Has Report?	IF	2.3		2440, 0
Ensure Ladder Table	Data Table	1.1		240, 100
Ensure History Table	Data Table	1.1		460, 100
Load Ladders	Data Table	1.1		680, 100
Load History	Data Table	1.1		900, 100
Tracker Plan	Code	2		1120, 100
Any Checks?	IF	2.3		1340, 100
Has Callback (Tracker)?	IF	2.3		2660, 120
POST Progress	HTTP Request	4.5		2880, 120
Manual Run	Execute Workflow Trigger	1.2		0, 220
Nothing To Track	No Op	1		1560, 260

SEO Agent — Site Tracker

Node	Type	Version	Credential	Position
Weekly Schedule	Schedule Trigger	1.2		0, 0
Ensure Sites Table	Data Table	1.1		220, 0
Ensure Metrics Table	Data Table	1.1		440, 0
Ensure Query Table	Data Table	1.1		660, 0
Ensure History Table (Site)	Data Table	1.1		880, 0
Load Sites	Data Table	1.1		1100, 0
Load Ladders	Data Table	1.1		1320, 0
Load Rank History	Data Table	1.1		1540, 0
Load Site Metrics	Data Table	1.1		1760, 0
Load Query History	Data Table	1.1		1980, 0
Site Plan	Code	2		2200, 0
Any Sites?	IF	2.3		2420, 0

GSC Sites	HTTP Request	4.5	Google Service Account (SEO Agent)	2640, 0
Resolve Properties	Code	2		2860, 0
GSC Requests	Code	2		3080, 0
Any GSC?	IF	2.3		3300, 0
GSC Query	HTTP Request	4.5	Google Service Account (SEO Agent)	3520, 0
Parse GSC	Code	2		3740, 0
Manual Run	Execute Workflow Trigger	1.2		0, 220
Ensure Trends Table	Data Table	1.1		880, 220
Ensure Log Table (Site)	Data Table	1.1		1100, 220
Ensure Alerts Table (Site)	Data Table	1.1		1320, 220
Ensure Checkin Table (Site)	Data Table	1.1		1540, 220
Load Content Log	Data Table	1.1		1980, 220
Load Console Alerts	Data Table	1.1		2200, 220
Load Check-ins	Data Table	1.1		2420, 220
Nothing To Track (Site)	No Op	1		2640, 220
Ensure AI Visibility Table (Site)	Data Table	1.1		1760, 440
Ensure Snapshots Table (Site)	Data Table	1.1		1980, 440
Ensure Audits Table (Site)	Data Table	1.1		2200, 440
Need GA4 Detect?	IF	2.3		2640, 440
GA4 Accounts	HTTP Request	4.5	Google Service Account (SEO Agent)	2860, 440
Stream Requests	Code	2		3080, 440
Any Streams?	IF	2.3		3300, 440
GA4 Streams	HTTP Request	4.5	Google Service Account (SEO Agent)	3520, 440
GA4 Requests	Code	2		3740, 440
Any GA4?	IF	2.3		3960, 440
GA4 Report	HTTP Request	4.5	Google Service Account (SEO Agent)	4180, 440
Parse GA4	Code	2		4400, 440
Load AI Visibility (Site)	Data Table	1.1		1760, 660
Load Backlink Snapshots (Site)	Data Table	1.1		1980, 660
Load Audits (Site)	Data Table	1.1		2200, 660
Inspect Requests	Code	2		2640, 660
Any Inspections?	IF	2.3		2860, 660
Inspect URL	HTTP Request	4.5	Google Service Account (SEO Agent)	3080, 660
Parse Inspection	Code	2		3300, 660
Trends Requests	Code	2		3520, 660

Any Trends?	IF	2.3		3740, 660
Google Trends	HTTP Request	4.5	DataForSEO	3960, 660
Parse Trends	Code	2		4180, 660
Site SERP Requests	Code	2		2640, 880
Any SERP?	IF	2.3		2860, 880
SERP Check (Site)	HTTP Request	4.5	DataForSEO	3080, 880
Parse Site SERP	Code	2		3300, 880
Save Site Rank History	Data Table	1.1		3520, 880
Site Metrics	Code	2		3740, 880
Save Site Metrics	Data Table	1.1		3960, 880
Query Rows	Code	2		4180, 880
Any Query Rows?	IF	2.3		4400, 880
Save Query Rows	Data Table	1.1		4620, 880
Trend Rows	Code	2		4840, 880
Any Trend Rows?	IF	2.3		5060, 880
Save Trends	Data Table	1.1		5280, 880
Site Rows	Code	2		5500, 880
Save Sites	Data Table	1.1		5720, 880
Brief Input	Code	2		2640, 1100
Site Brief	AI Agent	3.1		2860, 1100
Site Report	Code	2		3080, 1100
Prepare PDF Site	Code	2		3300, 1100
Render PDF Site	HTTP Request	4.5		3520, 1100
Attach PDF Site	Code	2		3740, 1100
Has Email (Site)?	IF	2.3		3960, 1100
Send Site Report	Send Email	2.1	Gmail SMTP (SEO Agent)	4180, 1100
Claude — Site Brief	Anthropic model	1.6	Anthropic (SEO Agent)	2760, 1300
Parser — Site Brief	Output parser	1.3		2980, 1300
Build Site Callback	Code	2		3960, 1320
Has Callback (Site)?	IF	2.3		4180, 1320
POST Site Report	HTTP Request	4.5		4400, 1320

SEO Agent — Console Alerts

Node	Type	Version	Credential	Position
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Has Email (Alert)?	IF	2.3		2400, -200
Email Owner (Alert)	Send Email	2.1	Gmail SMTP (SEO Agent)	2640, -200
Alert Target	Code	2		2160, -100
Mailbox Trigger	n8n-nodes-base.emailReadImap	2.2	Gmail IMAP (SEO Agent)	0, 0
Classify Notification	Code	2		240, 0
Relevant?	IF	2.3		480, 0
Ensure Alerts Table	Data Table	1.1		720, 0
Load Sites (Alerts)	Data Table	1.1		960, 0
Load Ladders (Alerts)	Data Table	1.1		1200, 0
Alert Row	Code	2		1440, 0
Save Alert	Data Table	1.1		1680, 0
Urgent?	IF	2.3		1920, 0
Has Callback (Alert)?	IF	2.3		2400, 0
POST Alert	HTTP Request	4.5		2640, 0
Stored	No Op	1		2160, 160
Ignore	No Op	1		720, 200

SEO Agent — Content Cadence

Node	Type	Version	Credential	Position
Planned Only	No Op	1		2860, -220
Has Email (Cadence)?	IF	2.3		4400, -220
Send Cadence Note	Send Email	2.1	Gmail SMTP (SEO Agent)	4620, -220
Weekly Schedule	Schedule Trigger	1.2		0, 0
Ensure Log Table	Data Table	1.1		220, 0
Ensure Cadence Table	Data Table	1.1		440, 0
Ensure Trends Table (Cadence)	Data Table	1.1		660, 0
Load Sites	Data Table	1.1		880, 0
Load Cadence	Data Table	1.1		1100, 0
Load Ladders	Data Table	1.1		1320, 0
Load Query History	Data Table	1.1		1540, 0
Load Trends	Data Table	1.1		1760, 0
Load Content Log	Data Table	1.1		1980, 0
Cadence Plan	Code	2		2200, 0
Any Pages?	IF	2.3		2420, 0

Dry Run?	IF	2.3		2640, 0
Start Content Runs	Execute Workflow	1.2		2860, 0
Log Rows	Code	2		3080, 0
Save Content Log	Data Table	1.1		3300, 0
Ladder Marks	Code	2		3520, 0
Any Ladder Marks?	IF	2.3		3740, 0
Mark Ladder Writing	Data Table	1.1		3960, 0
Cadence Note	Code	2		4180, 0
Manual Run	Execute Workflow Trigger	1.2		0, 220
Nothing To Write	No Op	1		2640, 220
Has Callback (Cadence)?	IF	2.3		4400, 220
POST Cadence Note	HTTP Request	4.5		4620, 220

SEO Agent — Site Admin

Node	Type	Version	Credential	Position
Ensure AI Prompts Table (Admin)	Data Table	1.1		480, -640
Ensure Prospects Table (Admin)	Data Table	1.1		480, -480
Ensure Monitors Table (Admin)	Data Table	1.1		480, -320
Delete Site	Data Table	1.1		760, -240
Ensure Sites Table (Admin)	Data Table	1.1		480, -160
Set Site Status	Data Table	1.1		760, -80
Admin Run	Execute Workflow Trigger	1.2		0, 0
Ensure Cadence Table (Admin)	Data Table	1.1		160, 0
Ensure Log Table (Admin)	Data Table	1.1		320, 0
Admin Route	Switch	3.4		480, 0
Admin Action	Code	2		640, 0
Admin Result	Code	2		1240, 0
Cadence Row (Admin)	Code	2		760, 80
Set Cadence	Data Table	1.1		1000, 80
Unpublish Log	Data Table	1.1		760, 240
Unpublish Ladder	Data Table	1.1		1000, 240
Load Monitors (Admin)	Data Table	1.1		760, 400
Monitor Row (Admin)	Code	2		1000, 400
Set Monitors	Data Table	1.1		1240, 400

Load Link Prospects (Admin)	Data Table	1.1		760, 560
Prospect Rows (Admin)	Code	2		1000, 560
Set Prospects	Data Table	1.1		1240, 560
Load AI Prompts (Admin)	Data Table	1.1		760, 720
AI Prompt Rows (Admin)	Code	2		1000, 720
Set AI Prompts	Data Table	1.1		1240, 720

SEO Agent — AI Visibility Tracker

Node	Type	Version	Credential	Position
Read Site (AI)	HTTP Request	4.5	Jina Reader	2640, -660
Site Text Jobs	Code	2		2200, -440
Need Site Text?	IF	2.3		2420, -440
Prompt Jobs	Code	2		2860, -440
Prompt Writer	AI Agent	3.1		2640, -220
Save Prompts	Data Table	1.1		3300, -220
Run AI Requests	HTTP Request	4.5	DataForSEO	3960, -220
Save AI Answers	Data Table	1.1		4620, -220
Claude — Prompt Writer	Anthropic model	1.6	Anthropic (SEO Agent)	2540, -20
Parser — Prompt Writer	Output parser	1.3		2760, -20
Weekly Schedule	Schedule Trigger	1.2		0, 0
Ensure Monitors Table	Data Table	1.1		220, 0
Ensure AI Prompts Table	Data Table	1.1		440, 0
Ensure AI Answers Table	Data Table	1.1		660, 0
Ensure AI Visibility Table	Data Table	1.1		880, 0
Ensure Prospects Table	Data Table	1.1		1100, 0
Ensure Sites Table	Data Table	1.1		1320, 0
Ensure Profiles Table	Data Table	1.1		1540, 0
Any Prompt Jobs?	IF	2.3		2420, 0
Prompt Rows	Code	2		2860, 0
Any New Prompts?	IF	2.3		3080, 0
AI Requests	Code	2		3520, 0
Any AI Requests?	IF	2.3		3740, 0
Parse AI Answers	Code	2		4180, 0
Any Answer Rows?	IF	2.3		4400, 0

Manual Run	Execute Workflow Trigger	1.2		0, 220
Load Sites	Data Table	1.1		220, 220
Load Ladders	Data Table	1.1		440, 220
Load Monitors	Data Table	1.1		660, 220
Load Profiles	Data Table	1.1		880, 220
Load AI Prompts	Data Table	1.1		1100, 220
Load AI Visibility	Data Table	1.1		1320, 220
Load Link Prospects	Data Table	1.1		1540, 220
AI Plan	Code	2		1760, 220
Any Sites (AI)?	IF	2.3		1980, 220
Nothing To Do (AI)	No Op	1		2200, 440
Save AI Prospects	Data Table	1.1		3300, 440
AI Metrics	Code	2		2200, 660
AI Vis Rows	Code	2		2420, 660
Save AI Visibility	Data Table	1.1		2640, 660
AI Prospect Rows	Code	2		2860, 660
Any AI Prospects?	IF	2.3		3080, 660
AI Brief Input	Code	2		3520, 660
AI Brief	AI Agent	3.1		3740, 660
AI Report	Code	2		3960, 660
Prepare PDF AI	Code	2		4180, 660
Render PDF AI	HTTP Request	4.5		4400, 660
Attach PDF AI	Code	2		4620, 660
Has Email (AI)?	IF	2.3		4840, 660
Send AI Report	Send Email	2.1	Gmail SMTP (SEO Agent)	5060, 660
Claude — AI Brief	Anthropic model	1.6	Anthropic (SEO Agent)	3640, 860
Parser — AI Brief	Output parser	1.3		3860, 860
Has Callback (AI)?	IF	2.3		5060, 860
POST AI Report	HTTP Request	4.5		5280, 860
Build AI Callback	Code	2		4840, 880

SEO Agent — Backlink Monitor

Node	Type	Version	Credential	Position
Run Gap Requests	HTTP Request	4.5	DataForSEO	3300, -220

Outreach Writer	AI Agent	3.1		4180, -220
Save Prospects	Data Table	1.1		4840, -220
Claude — Outreach Writer	Anthropic model	1.6	Anthropic (SEO Agent)	4080, -20
Parser — Outreach Writer	Output parser	1.3		4300, -20
Weekly Schedule	Schedule Trigger	1.2		0, 0
Ensure Monitors Table	Data Table	1.1		220, 0
Ensure Snapshots Table	Data Table	1.1		440, 0
Ensure Prospects Table	Data Table	1.1		660, 0
Ensure Sites Table	Data Table	1.1		880, 0
Ensure Profiles Table	Data Table	1.1		1100, 0
Ensure Case Table	Data Table	1.1		1320, 0
Ensure Log Table	Data Table	1.1		1540, 0
BL Requests	Code	2		2420, 0
Run BL Requests	HTTP Request	4.5	DataForSEO	2640, 0
Gap Requests	Code	2		2860, 0
Any Gap?	IF	2.3		3080, 0
Parse Backlinks	Code	2		3520, 0
Outreach Jobs	Code	2		3740, 0
Any Outreach?	IF	2.3		3960, 0
Prospect Rows	Code	2		4400, 0
Any Prospect Rows?	IF	2.3		4620, 0
Manual Run	Execute Workflow Trigger	1.2		0, 220
Load Sites	Data Table	1.1		220, 220
Load Ladders	Data Table	1.1		440, 220
Load Monitors	Data Table	1.1		660, 220
Load Profiles	Data Table	1.1		880, 220
Load Backlink Snapshots	Data Table	1.1		1100, 220
Load Link Prospects	Data Table	1.1		1320, 220
Load Content Log	Data Table	1.1		1540, 220
Load Case Studies	Data Table	1.1		1760, 220
BL Plan	Code	2		1980, 220
Any Sites (BL)?	IF	2.3		2200, 220
Nothing To Do (BL)	No Op	1		2420, 440
BL Snapshot Rows	Code	2		2420, 660
Save BL Snapshot	Data Table	1.1		2640, 660

BL Report	Code	2		2860, 660
Prepare PDF BL	Code	2		3080, 660
Render PDF BL	HTTP Request	4.5		3300, 660
Attach PDF BL	Code	2		3520, 660
Has Email (BL)?	IF	2.3		3740, 660
Send BL Report	Send Email	2.1	Gmail SMTP (SEO Agent)	3960, 660
Has Callback (BL)?	IF	2.3		3960, 860
POST BL Report	HTTP Request	4.5		4180, 860
Build BL Callback	Code	2		3740, 880

SEO Agent — Audit Scheduler

Node	Type	Version	Credential	Position
Monthly Schedule	Schedule Trigger	1.2		0, 0
Ensure Monitors Table	Data Table	1.1		220, 0
Ensure Audits Table	Data Table	1.1		440, 0
Ensure Sites Table	Data Table	1.1		660, 0
Ensure Profiles Table	Data Table	1.1		880, 0
Start Scheduled Audits	Execute Workflow	1.2		1760, 0
Audits Started	Code	2		1980, 0
Manual Run	Execute Workflow Trigger	1.2		0, 220
Load Sites	Data Table	1.1		220, 220
Load Ladders	Data Table	1.1		440, 220
Load Monitors	Data Table	1.1		660, 220
Load Profiles	Data Table	1.1		880, 220
Load Audits	Data Table	1.1		1100, 220
Audit Schedule Plan	Code	2		1320, 220
Any Audits Due?	IF	2.3		1540, 220
Nothing Due	No Op	1		1760, 440

SEO Agent — Publish to WordPress

Node	Type	Version	Credential	Position
From Content Run	Execute Workflow Trigger	1.2		0, 0
Build WP Draft	Code	2		240, 0
Create WP Draft	HTTP Request	4.5	WordPress (SEO Agent)	480, 0

WP Result	Code	2		720, 0
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SEO Agent — Error Handler

Node	Type	Version	Credential	Position
Error Trigger	Error Trigger	1		0, 0
Format Alert	Code	2		240, 0
Email Ops	Send Email	2.1	Gmail SMTP (SEO Agent)	480, 0

SEO Agent — Test Callback Receiver

Node	Type	Version	Credential	Position
Callback Webhook	Webhook	2		0, 0
Summarise Callback	Code	2		240, 0

SEO Agent — Test Runner

Node	Type	Version	Credential	Position
Test Webhook	Webhook	2	SEO API Key (test)	0, 0
Route Test	Code	2		240, 0
Respond Started	Respond to Webhook	1.1		480, 0
Prepare Forward	Code	2		720, 0
Run Target	Execute Workflow	1.2		960, 0

12. Appendix B: the page generated in the live test

The ladder's first page, "uae e invoicing penalties" (rung 1, Blog Post), exactly as the content pipeline delivered it on 2026-10-01 (execution 46): Opus brief and copy, Sonnet critique, deterministic QA, one Opus editor pass. Content score 85/100; text in [square brackets] is a placeholder the client must replace; the two internal links to the ladder's planned top page and sibling are part of the ladder design. JSON-LD blocks are omitted here (they are in the report files under harness/sample-output/LIVE-ladder-page-uae-e-invoicing-penalties.*).

Title: UAE E-Invoicing Penalties: Fines, Examples & Deadlines

Meta Description: UAE e-invoicing penalties explained: the confirmed fines under Cabinet Decision No. 106, worked cost examples and a prevention plan. Book a check.

Slug: uae-e-invoicing-penalties

Primary Keyword: uae e invoicing penalties

Secondary Keywords: cabinet decision no. 106 of 2025, uae e-invoicing penalty examples, e-invoicing penalties for recipients, e-invoicing system failure penalty, accredited service provider uae, uae e-invoicing deadline, penalty for not issuing an e-invoice, fta e-invoicing administrative penalties

UAE E-Invoicing Penalties: The Confirmed Fines, What They Could Cost and How to Avoid Them

Last updated: [Date] · Reviewed by [Name, UAE tax credential] · Technical review: [Reviewer name, role at Tech&]

UAE e-invoicing penalties are recurring monthly fines, so the cost of a late start accumulates every month. This guide gives your finance team four things: the confirmed fines in plain English, costed exposure for three scenarios, your phase deadline, and a prevention plan with an owner for each control. Where we have not confirmed the official text, we say so.

Quick answer: Under Cabinet Decision No. 106 of 2025, the Federal Tax Authority (FTA) can fine VAT-registered businesses that miss e-invoicing obligations. Failing to implement the Electronic Invoicing System or appoint an Accredited Service Provider costs [Price] per month or part month. Each late electronic invoice or credit note costs [Price] Reported wording sets an [Price] monthly cap for each document type. Voluntary adopters are not penalised.

What are the UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025?

The Ministry of Finance published [Cabinet Decision No. 106 of 2025 on e-invoicing violations and penalties](#) in December 2025. It sits under Federal Decree-Law No. 28 of 2022 on Tax Procedures, and the FTA enforces it. Two administrative penalties are confirmed. A business with no e-invoicing system or ASP pays [Price] per month. Each late document costs [Price]

What is the penalty for not issuing an e-invoice?

Failing to issue and transmit an electronic invoice on time costs [Price] per invoice. [Commentary on the decision](#) reports a cap of [Price] per month. A PDF does not count. The invoice must pass through your ASP in the structured electronic format. If you have no system or ASP at all, the separate [Price] monthly penalty also applies.

The violations at a glance

Violation (paraphrased from the MoF decision)	Plain-English meaning	Example	Who is liable	Fine	Cap
Failure to implement the e-invoicing system, including appointing an ASP, by the deadline	You are not connected through an accredited provider on time	Your mandatory date passes with the ASP contract unsigned	Business in scope	[Price] per month or part month	None stated; recurs monthly
Failure to issue and transmit an electronic invoice on time	The invoice does not go through your ASP in the structured electronic format	A batch of sales invoices goes out as PDFs	Issuer	[Price] per invoice	[Price] per month (reported)
Failure to issue and transmit an electronic credit note on time	The same rule, applied to credit notes	Month-end returns are credited manually	Issuer	[Price] per credit note	[Price] per month (reported)
Failure to notify the FTA of a system failure within the set period	You must tell the FTA when a fault stops you issuing	An ASP outage goes unreported	Business in scope	Not confirmed here	Check the decision
Failure to notify the ASP of changes to your registered data	Your ASP's record of your business must stay current	A new branch is never notified to the ASP	Business in scope	Not confirmed here	Check the decision
Recipient failure to notify the issuer of a system failure	A buyer must tell its supplier when it cannot receive	The buyer's ASP is down and the supplier is not told	Recipient (as reported)	Not confirmed here	Check the decision

We have not confirmed the amounts, caps or notification windows for rows four to six. This guide therefore prices only the confirmed fines. Commentary groups the breaches into these six violations, so check the full list against the decision before using it in board papers.

The monthly fine applies for each month or part of a month. One day into a new month counts as a full month. The decision excludes voluntary adopters.

[Image: Infographic summarising the confirmed UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025, with fines in AED]

Is e-invoicing mandatory in the UAE?

Yes. It is mandatory for VAT-registered businesses in scope, on a phased timetable set by Ministerial Decision No. 244 of 2025. Voluntary adoption opens from 1 July 2026. Businesses with revenue at or above [[Price] million] (<https://worldcompanysetup.com/en/blog/uae-e-invoicing-mandate-dubai-timeline-penalties/>) must appoint an ASP first, by 31 July 2026. They must go live from 1 January 2027.

When do e-invoicing penalties start, and which phase are you in?

Penalties start only once your deadline passes. Ministerial Decision No. 244 of 2025 sets two dates for each phase: an ASP appointment deadline and a go-live date. Ministerial Decision No. 243 of 2025 defines scope and exclusions. Confirm the dates below against the official decision before planning.

Phase	Appoint an ASP by	Mandatory e-invoicing from
Revenue of [Price] million or more	31 July 2026	1 January 2027
Revenue below [Price] million	31 March 2027	1 July 2027
Government entities	31 March 2027	1 October 2027

Published guides often disagree because they mix up these two dates. Your ASP deadline comes first, so it is your nearest exposure. A [secondary guide to Ministerial Decision No. 244](#) explains the phases in more detail.

Are there penalties during the voluntary phase?

No. Cabinet Decision No. 106 of 2025 states that its provisions do not apply to businesses that issue, transmit or exchange e-invoices voluntarily. The safest time to test is therefore from 1 July 2026 until your mandatory date. Use that window to test your ERP, your ASP connection and your system-failure procedure without any fine exposure.

A three-question rule to find your deadline

1. Are you VAT-registered and outside the exclusions in Ministerial Decision No. 243? If not, the mandate may not apply. Confirm this with your adviser.
2. Is your revenue at or above [Price] million? If yes, the first-phase dates apply.
3. Have you appointed an ASP? If not, your nearest exposure is the ASP deadline, not go-live.

Is the [Price] cap per violation or in total?

Reported wording of Cabinet Decision No. 106 sets a separate [Price] monthly cap for invoices and another for credit notes. The [Price] monthly system fine sits on top of both. The safe budgeting assumption is that penalties stack, giving a worst month of [Price] Confirm the clause with your adviser until the FTA clarifies it.

Reading	How the cap works	Worst month, all three breaches
Per-category (stacking)	[Price] cap for invoices, a separate [Price] for credit notes, plus the [Price] system fine	[Price]
Overall cap	One [Price] ceiling across all penalties in the month	[Price]

Commentary describes the document fine as [capped at [Price] per month for each](https://www.genzone.com/uae-e-invoicing-2026-guide/) category, which supports the stacking reading. Read the clause in the [MoF decision](#) yourself.

How much could non-compliance cost? Worked penalty examples

A business with no ASP and regular late invoices faces up to [Price] a month. That is [Price] for the system plus the [Price] invoice cap. Add late credit notes and the worst month reaches [Price] Both figures use the stacking reading. The fines [recur for each month or part of a month](#).

The exposure formula

> Monthly exposure = 5,000 (if no system or ASP) + min(late invoices × 100, 5,000) + min(late credit notes × 100, 5,000). All figures in AED per month, per-category reading. Under the overall-cap reading, the monthly total cannot exceed 5,000.

Three scenarios, three totals

UAE e-invoicing penalty exposure: three worked scenarios

Scenario	Months	Violations each month	Monthly penalty AED (per-category)	Total AED: per-category	Total AED: overall cap
A: enterprise above [Price]m	6	No ASP; 80 late invoices (8,000, capped to 5,000)	5,000 + 5,000 = 10,000	10,000 × 6 = 60,000	5,000 × 6 = 30,000

B: system live, process gaps	3	30 late invoices (3,000); 10 late credit notes (1,000)	$3,000 + 1,000 = 4,000$	$4,000 \times 3 = 12,000$	$4,000 \times 3 = 12,000$
C: no system at all	12	No ASP; 60 late invoices (6,000, capped to 5,000); 55 late credit notes (5,500, capped to 5,000)	$5,000 + 5,000 + 5,000 = 15,000$	$15,000 \times 12 = 180,000$	$5,000 \times 12 = 60,000$

Scenario B is the same under both readings because its monthly total stays below [Price] Scenario C shows the full gap between the readings: [Price] over a year. The MoF text governs these figures, so take advice before you report them.

Costs that do not appear on the penalty notice

Remediation also consumes finance and IT time during close. For the wider analysis, see [the full cost of e-invoicing non-compliance and mitigation](#).

What counts as a system failure, and what if your ASP goes down?

A system failure is a technical fault in your own system or your ASP's that stops you issuing or transmitting e-invoices. The decision requires you to notify the FTA within a set period and fines you if you do not. We have not confirmed that window or the amount, so take both from the [MoF decision](#).

An ASP outage does not remove your own obligations. Do not assume your provider will notify the FTA for you. Check in the decision who must notify, then write that duty into your ASP contract.

System-failure incident checklist

1. Detect and timestamp the failure, noting whether it sits in the ERP or the ASP.
2. Raise a ticket with your ASP and record the reference.
3. Notify the Federal Tax Authority within the required window and record the channel you used.
4. Log the affected invoices, correspondence and screenshots, with detection, notification and resolution times.
5. Re-transmit the backlog once the system failure is resolved, then reconcile.

What is an accredited service provider (ASP), and why does it matter for penalties?

An accredited service provider (ASP) is a provider accredited by the UAE authorities to exchange structured electronic invoices between businesses. It also reports invoice data to the FTA. Under UAE electronic invoicing, you cannot issue compliant e-invoices without an ASP. Missing your ASP appointment deadline triggers the [Price] monthly fine.

When you choose an ASP, check its accreditation status, PINT AE support, ERP connectors and uptime commitments. See the [Peppol e-invoicing software features CFOs should look for](#).

Who is liable: the issuer, the recipient or the ASP?

As reported, the business carries the liability, not the service provider (ASP). Confirm the article for each violation in the MoF decision. UAE e-invoicing is commonly described as a [five-corner model](#) on Peppol and PINT AE. Data flows from the supplier to the supplier's ASP, then to the buyer's ASP and the buyer. The FTA is the fifth corner and receives the invoice data.

Issuers must issue and transmit each invoice and electronic credit note in the structured format on time. Recipients need an ASP and the ability to receive e-invoices. Commentary indicates they must also notify the issuer of failures. See [what the FTA requires under UAE e-invoicing](#) for the full obligations.

Your ASP is a service provider, so write SLAs, uptime targets and incident-reporting duties into the contract.

[Image: Five-corner Peppol model diagram showing issuer, recipient, Accredited Service Providers and the FTA]

How do e-invoicing fines relate to VAT penalties, and can you appeal?

They are separate schedules and can apply together. You can also request reconsideration.

Late VAT payment vs e-invoicing fines

E-invoicing fines sit in Cabinet Decision No. 106 of 2025. General VAT penalties, including those for late payment, sit under Cabinet Decision No. 40 of 2017 as amended. A late electronic invoice can therefore attract both an e-invoicing fine and a separate VAT penalty. Take current late-payment rates from the FTA's published penalty schedule.

Requesting reconsideration from the FTA

Federal Decree-Law No. 28 of 2022 lets you ask the FTA to reconsider a penalty within 40 business days of being notified. If you disagree with the outcome, you can object to the Tax Disputes Resolution Committee. Your incident logs and ASP correspondence are your evidence, so keep them from day one.

How to avoid UAE e-invoicing penalties: owners, controls and a 90-day plan

Each violation needs one owner and one documented control. Start with ASP selection, because [one compliance guide warns that delay risks bottlenecks and [Price] monthly fines](<https://farahsolutions.ae/blog/uae-e-invoicing-guidelines-v1-1-compliance-guide>).

Prevention matrix by owner

Violation	Owner	Control	Evidence to keep
No system or ASP	CFO (Finance)	ASP contract signed before your ASP deadline	Signed contract
Late electronic invoice	IT/ERP	Automated transmission with a daily exception report	Exception reports
Late electronic credit note	Finance (AR)	Credit-note workflow routed through the ASP	Approval trail
No FTA failure notice	Finance	Incident procedure with a named notifier	Incident log
ASP data not updated	Finance	Quarterly master-data review	Change confirmations
Recipient fails to notify issuer	ASP and Finance (AP)	Contractual alerting SLA	ASP alerts

A 90-day penalty-avoidance plan

1. Days 1–30: confirm your phase, scope your entities, shortlist ASPs and run an ERP gap check for PINT AE fields.
2. Days 31–60: appoint the ASP, map invoice data and configure credit-note flows.
3. Days 61–90: test end to end during the voluntary window, when no penalties apply. Rehearse the system-failure procedure and train your AP and AR teams.

Dynamics 365 Finance & Operations and Business Central both need ERP-side changes. See [upgrading your ERP for Peppol PINT AE compliance](#).

Tech& is a [Partner status] with [Number] years of experience in UAE finance systems. Its readiness check covers [scope] over [duration]. It confirms your phase, checks your ERP gaps and ASP-integration readiness, and estimates

your exposure. [Case study: UAE client, industry, e-invoicing readiness outcome]

Frequently Asked Questions

Q1: Is e-invoicing mandatory in the UAE?

A1: Yes, for VAT-registered businesses in scope, on a phased timetable under Ministerial Decision No. 244 of 2025. Voluntary adoption opens from 1 July 2026. Businesses with revenue of [Price] million or more must appoint an ASP by 31 July 2026 and go live from 1 January 2027. Our [complete guide to e-invoicing in the UAE](#) covers scope.

Q2: What is the penalty for not issuing an e-invoice?

A2: Failing to issue and transmit an electronic invoice on time carries a penalty of [Price] per invoice, capped at [Price] per month according to [published commentary](#). If you have not implemented the system or appointed an ASP at all, a separate [Price] monthly penalty can apply as well.

Q3: Who is exempted from e-invoicing?

A3: Ministerial Decision No. 243 of 2025 lists the excluded transactions and entities. Being VAT-registered does not by itself put every transaction in scope. Check the official exclusion list against each of your revenue streams. Confirm any borderline cases with your tax adviser before your phase begins, because exclusions apply to specific transactions rather than to whole businesses by default.

Q4: What is the penalty for late payment of VAT in the UAE?

A4: Late VAT payment penalties fall under Cabinet Decision No. 40 of 2017 as amended, not under the e-invoicing decision. Rates have changed over time, so take the current figures from the Federal Tax Authority's published penalty schedule. Both penalty regimes can apply to the same business in the same period.

Q5: Are there penalties during the voluntary phase?

A5: No. Cabinet Decision No. 106 of 2025 states that its provisions do not apply to businesses that issue, transmit or exchange e-invoices voluntarily. This makes the voluntary window the safest time to test your ERP and ASP connection before your mandatory date.

Q6: What penalties apply to notification failures?

A6: The decision covers three notification duties. You must tell the FTA about a system failure and tell your ASP about changes to your registered data. Recipients must tell issuers when they cannot receive. We have not confirmed the amounts or time windows, so read the exact clauses in the MoF decision before you budget for them.

Q7: Do e-invoicing penalties apply to credit notes?

A7: Yes. Each electronic credit note not issued and transmitted on time costs [Price] capped at [Price] per month ([source](#)). Reported wording suggests this cap sits separately from the invoice cap. Late credit notes could therefore add to invoice penalties in the same month.

Q8: Where can I download the UAE e-invoicing penalties PDF?

A8: The official text is the Ministry of Finance's [Cabinet Decision on e-invoicing violations and penalties](#), published as a PDF. The violations table on this page summarises it in plain English. Check the decision itself for any amounts we mark as not confirmed.

Book your free readiness and penalty exposure check

Book a free e-invoicing readiness and penalty exposure check with Tech& before UAE e-invoicing penalties apply to you. In [duration] we confirm your phase and deadline, run an ERP gap check for PINT AE and review your ASP readiness. We then estimate your AED exposure using the worked examples above.

[Customer review: CFO quote on readiness project]

[Book an e-invoicing readiness check](#) · Phone: [Phone] · Email: [Email]

Other live samples in the same folder: the Keyword Ladder Plan (PDF, Word, callback JSON), the form verdict download, the "Choose your keyword" and ladder form pages, and the rank-tracker progress e-mail.

13. Appendix C: everything the live test generated, and what it was used for

Every artefact below was produced on 2026-10-01 for techand.ai and the keyword "e invoicing in uae". For each run: the inputs, the intermediate data each node produced (with the downstream step that consumed it), and the delivered report exactly as the customer received it. Full JSON of every item is in docs/runs/full/; the delivered files are in harness/sample-output/LIVE-*.

Run 18 — Keyword discovery (API, techand.ai, goal leads)

Site description (Site Describer, Claude Haiku) → used by Keyword Seeds, the relevance screen and every later prompt as the business context

```
{
  "business_name": "Tech&",
  "one_line_summary": "A Microsoft-aligned enterprise technology partner delivering AI-driven digital transformation solutions across the GCC region.",
  "business_description": "Tech& is a trusted Microsoft partner providing enterprise technology solutions for digital transformation and business growth across the GCC region. With 20+ years of proven industry experience, a strong client base spanning multiple industries, and a global collaboration reach, Tech& specializes in delivering Microsoft solutions, AI tools, and secure enterprise technology. The company guides businesses through technology growth while addressing complex business challenges, offering 24/7 support and deep regional expertise in the UAE and GCC.",
  "industry": "Technology Consulting and Enterprise Solutions",
  "products_or_services": [
    "Microsoft Dynamics 365 Finance & Operations",
    "Microsoft 365 and Copilot solutions",
    "AI-driven business transformation",
    "E-invoicing and Peppol solutions",
    "Cloud migration and Azure services",
    "Enterprise digital transformation consulting",
    "VAT compliance solutions"
  ],
  "target_audience": [
    "Enterprise organizations in the GCC region",
    "UAE and Saudi Arabia CFOs and finance teams",
    "Businesses seeking digital transformation",
    "Companies needing Microsoft technology implementation"
  ],
  "unique_selling_points": [
    "20+ years of proven industry experience",
    "Trusted Microsoft Global Partner status",
    "Deep expertise in UAE and GCC regulatory requirements",
    "Specialized knowledge in Peppol E-invoicing and VAT compliance",
    "24/7 support availability",
    "Demonstrated track record across multiple industries"
  ],
  "location_served": "GCC region (United Arab Emirates, Saudi Arabia, and surrounding Middle East markets)",
  "tone_of_brand": "Professional, trustworthy, expert-driven, forward-thinking, and solutions-oriented",
}
```

```
"suggested_meta_title": "Tech& | Microsoft Enterprise Solutions for GCC",
"suggested_meta_description": "Transform your business with Tech&'s Microsoft-aligned
enterprise solutions. 20+ years expertise in AI, digital transformation, and cloud services
across UAE and GCC.",
"seed_key
... (384 more characters in docs/runs/full)
```

Seeds by buyer motive (Keyword Seeds, Claude Sonnet → Seed List) → used by Research Requests for the seven DataForSEO pulls

```
{
  "primary_seed": "erp implementation partner",
  "seed_groups": {
    "services": [
      "erp implementation services",
      "finance and operations erp",
      "e invoicing solution",
      "vat compliance software",
      "cloud migration services",
      "ai business solutions",
      "digital transformation consulting",
      "microsoft 365 implementation"
    ],
    "problems": [
      "automate finance processes",
      "comply with uae e invoicing",
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      "streamline vat reporting"
    ],
    "comparisons": [
      "best erp for uae",
      "top erp implementation companies",
      "best digital transformation consultants",
      "erp software comparison",
      "best e invoicing providers uae"
    ],
    "pricing": [
      "erp implementation cost",
      "erp pricing uae",
      "e invoicing solution cost",
      "digital transformation consulting rates"
    ],
    "local": [
      "erp partner uae",
      "e invoicing uae",
      "digital transformation dubai",
      "erp company saudi arabia",
      "vat software uae"
    ],
    "audience": [
      "erp for enterprises uae",
      "finance software for cfos",
      "erp for manufacturing companies",
      "technology solutions for large enterprises",
      "erp for retail gcc"
    ]
  }
}
```

```

},
"services": [
  "erp implementation",
  "e invoicing and vat compliance",
  "cloud migration",
  "ai business solutions",
  "digital transformation consulting",
  "productivity and collaboration solutions"
]
}

```

Research pool (Collect Research) → scored candidates for the AI relevance screen

Pool 624 keywords from 10 pulls; by source {"competitor": 17, "ideas": 599, "suggestions": 8, "related": 2}; failures: []. Top candidates before the AI screen:

Keyword	Searches/mo	KD	CPC	Intent	Page type	Opportunity
e invoicing	1900	9	7.949999809265137	commercial		1070.8
his	3600	8	0.27000001072883606	informational		741.4
erp	5400	29	8.979999542236328	informational		550.2
armoplast	590	0		informational		141.6
gulf technical equipment	320	0		commercial		120
what is e invoicing in uae	390	13	3.7100000381469727	informational		115
numero uno training and consulting llc reviews	260	0		commercial		82.9
sap for business	320	14	6.630000114440918	commercial		65.4
orill	210	0		informational		65
erp consultants	110	0	8.140000343322754	commercial		36.5
erpnext software	140	36	10.170000076293945	commercial		28.7
accounting solutions	20	30	17.46	commercial		6.9
business expenses list	30		2.58	informational		5.6
business expense list	30		2.58	informational		5.6
finance erp system	40			commercial		5.3
erp finance	40			commercial		5.3
erp financials	40			commercial		5.3
system of accounting	20		15.89	informational		5.3
uae accounting system	10		47.62	commercial		5
assets in business	40			informational		4.8

220 more rows not shown.

Keyword strategy (Rank Keywords + Build Keyword Strategy) → the report, the API callback and the form's choice step

126 relevant of 240 AI-reviewed; 46 clusters; goal leads. Priority keywords (live-checked on Google):

Keyword	Searches/mo	KD	CPC	Intent	Page type	Opportunity	Who ranks now
e invoicing	1900	9	7.949999809265137	commercial	Service Page	1070.8	tax.gov.ae, mof.gov.ae, edicomgroup.com
what is e invoicing in uae	390	13	3.7100000381469727	informational	Guide	115	tax.gov.ae, mof.gov.ae, grantthornton.ae
erp consultants	110	0	8.140000343322754	commercial	Service Page	36.5	goodfirms.co, linkedin.com, erpadvisorsgroup.com
finance erp system	40			commercial	Product Page	5.3	oracle.com, ecisolutions.com, pwc.de
erp finance	40			commercial	Product Page	5.3	navan.com, hyperbots.com, oracle.com
ai in accounts payable	20			commercial	Blog Post	3.8	rillion.com, ramp.com, sap.com

Content plan (clusters with tiers):

Tier	Topic	Primary keyword	Page type	Keywords	Searches/mo	KD
Now	erp implementation services	erp consultants	Service Page	3	130	0
Now	dynamics 365 finance erp	finance erp system	Product Page	9	180	
Now	dynamics 365 finance	integrated financial management system	Product Page	12	120	
Now	finance accounting solutions	solutions for accounting	Service Page	9	90	
Now	dynamics 365 finance operations	corporate accounting system	Product Page	8	80	
Next	financial management solution	enterprise financial management	Service Page	5	50	
Next	accounts receivable system	accounts receivable management system	Product Page	4	40	
Next	cash management erp	erp cash management	Product Page	4	40	
Next	cash flow management	cash flow management solution	Service Page	4	40	
Next	spend management	expense management solutions	Service Page	4	40	
Next	ai in finance	ai in accounts payable	Blog Post	1	20	
Next	erp costing module	erp cost accounting	Product Page	2	20	
Next	financial management software	financial management software examples	Guide	4	40	
Later	e-invoicing uae	e invoicing	Guide	3	2300	9

Later	what is erp	erp	Guide	2	5420	29
Later	financial compliance	financial compliance solutions	Guide	6	70	
Later	accounting software guide	accounting software feature	Guide	4	40	
Later	finance process transformation	accounts payable transformation	Service Page	3	30	
Later	finance erp software	accounting solutions	Service Page	1	20	30
Later	financial reporting solutions	financial reporting solutions	Service Page	2	20	
Later	project accounting system	project accounting system	Product Page	2	20	
Later	budget planning	budget planning services	Service Page	3	30	
Later	financial tracking system	financial tracking software	Service Page	3	30	
Later	cash flow software	cash flow projection software	Guide	3	30	
Later	erp implementation cost	erp implementation cost breakdown	Guide	1	10	

21 more rows not shown.

AI search demand (DataForSEO AI keyword data):

Keyword	AI searches/mo	Google searches/mo
erp	546	5400
e invoicing	64	1900
what is e invoicing in uae	9	390
erp consultants	3	110

Delivered: Keyword Strategy report (PDF 88 KB + Word) → posted to the callback (stage keyword_strategy)

DELIVERED REPORT: KEYWORD-IDEAS-TECHAND-AI-2026-10-01.DOC

Keyword strategy for techand.ai

United Arab Emirates · goal: enquiries and leads · 624 keywords researched from 10 data pulls · 240 screened for relevance by AI · 126 relevant

Where to start: "e invoicing" — 1,900 searches/mo, difficulty 9 · easy, commercial intent, best page: Service Page. low competition; high commercial value (CPC 7.949999809265137); buyers comparing providers; 1 competitor(s) already rank for it; AI Overview present — write a quotable answer block.

Build first: erp implementation services (Service Page, 130/mo across 3 keywords) · dynamics 365 finance erp (Product Page, 180/mo across 9 keywords) · dynamics 365 finance (Product Page, 120/mo across 12 keywords) · finance accounting solutions (Service Page, 90/mo across 9 keywords) · dynamics 365 finance operations (Product Page, 80/mo across 8 keywords)

Priority keywords (6) — live-checked on Google

1. e invoicing

1,900 searches/mo Difficulty 9 · easy Commercial Service Page CPC \$7.949999809265137
Traffic potential ~371/mo AI search volume 64/mo

low competition; high commercial value (CPC 7.949999809265137); buyers comparing providers; 1 competitor(s) already rank for it; AI Overview present — write a quotable answer block.

Topic: e-invoicing uae · **SERP features:** ai_overview, people_also_ask, images, related_searches · **Who ranks now:** tax.gov.ae, mof.gov.ae, edicomgroup.com, tallysolutions.com, europe.thomsonreuters.com · **Your position:** not in top 20

People also ask: What is meant by e-invoicing? · Is e-invoicing mandatory in the UAE? · What is an e-invoice and how does it work? · What is the rule for e-invoicing?

2. what is e invoicing in uae

390 searches/mo Difficulty 13 · easy Informational Guide CPC \$3.7100000381469727 Traffic potential ~76/mo
AI search volume 9/mo

low competition; research-stage traffic that builds authority; 1 competitor(s) already rank for it; AI Overview present — write a quotable answer block.

Topic: e-invoicing uae · **SERP features:** ai_overview, people_also_ask, paid, related_searches · **Who ranks now:** tax.gov.ae, mof.gov.ae, grantthornton.ae, cleartax.com, edicomgroup.com · **Your position:** not in top 20

People also ask: What is the concept of e-invoicing? · How does e-invoicing work? · Who is exempted from e-invoicing in the UAE? · Why do you need an e-invoice?

3. erp consultants

110 searches/mo Difficulty 0 · easy Commercial Service Page CPC \$8.140000343322754
Traffic potential ~21/mo AI search volume 3/mo

low competition; high commercial value (CPC 8.140000343322754); buyers comparing providers; 1 competitor(s) already rank for it; AI Overview present — write a quotable answer block.

Topic: erp implementation services · **SERP features:** ai_overview, people_also_ask, discussions_and_forums, related_searches · **Who ranks now:** goodfirms.co, linkedin.com, erpadvorsgroup.com, datixinc.com, ae.indeed.com · **Your position:** not in top 20

People also ask: What does an ERP consultant do? · Which ERP is mostly used in UAE? · What are the top 10 ERP consulting companies? · What does ERP mean in consulting?

4. finance erp system

40 searches/mo Difficulty — Commercial Product Page Traffic potential ~12/mo

buyers comparing providers.

Topic: dynamics 365 finance erp · **SERP features:** ai_overview, people_also_ask, video, related_searches · **Who ranks now:** oracle.com, ecisolutions.com, pwc.de, gartner.com, spendesk.com · **Your position:** not in top 20

People also ask: What are ERP systems in finance? · Which ERP is mostly used in UAE? · Which ERP is best for finance? · What are the top 5 ERP systems?

5. erp finance

40 searches/mo Difficulty — Commercial Product Page Traffic potential ~12/mo

buyers comparing providers.

Topic: dynamics 365 finance erp · **SERP features:** ai_overview, images, people_also_ask · **Who ranks now:** navan.com, hyperbots.com, oracle.com, withless.com, umbrex.com · **Your position:** not in top 20

People also ask: What does ERP stand for in finance? · What is an ERP example? · What are the four types of ERP? · Which ERP is best for finance?

6. ai in accounts payable

20 searches/mo

Difficulty —

Commercial

Blog Post

Traffic potential ~6/mo

buyers comparing providers.

Topic: ai in finance · **SERP features:** ai_overview, people_also_ask, video, related_searches · **Who ranks now:** rillion.com, ramp.com, sap.com, abbyy.com, precoro.com · **Your position:** not in top 20

People also ask: Can AI take over accounts payable? · What are some AI solutions for accounts payable? · Is there an AI course specifically for accounts payable? · How is AI currently used in accounting?

Content plan by topic

One strong page per topic. "Now" = highest opportunity and winnable difficulty; "Next" = plan for the following months; "Later" = long-term or hard.

Tier	Topic / page	Primary keyword	Page type	Keywords	Total searches/mo	Difficulty
Now	Erp implementation services erp implementation consulting services, implementing accounting system	erp consultants	Service Page	3	130	0 · easy
Now	Dynamics 365 finance erp erp finance, erp financials, uae accounting system, financial and accounting software, erp financial systems	finance erp system	Product Page	9	180	—
Now	Dynamics 365 finance finance and accounting system, inventory and accounting system, automated financial management system, financial management information system, financial erp solutions	integrated financial management system	Product Page	12	120	—
Now	Finance accounting solutions financial accounting solutions, finance management solutions, finance and accounting solutions, accounting and financial solutions, accounting and finance solutions	solutions for accounting	Service Page	9	90	—
Now	Dynamics 365 finance operations accounting inventory system, financial information system, accounting system demo, accounting system training, accounting system setup	corporate accounting system	Product Page	8	80	—
Next	Financial management solution financial management solution, solution financial management, business financial solution, financial planning solution	enterprise financial management	Service Page	5	50	—

Next	Accounts receivable system account receivable solutions, manage receivables system options, business receivables	accounts receivable management system	Product Page	4	40	—
Next	Cash management erp business cash flow management software, cash management implementation, liquidity management solution	erp cash management	Product Page	4	40	—
Next	Cash flow management cash forecast software, financial projection software, cashflow forecast software	cash flow management solution	Service Page	4	40	—
Next	Spend management spend management system, expense solutions, spend management solutions	expense management solutions	Service Page	4	40	—
Next	Ai in finance	ai in accounts payable	Blog Post	1	20	—
Next	Erp costing module erp costing module	erp cost accounting	Product Page	2	20	—
Next	Financial management software financial management software list, best financial tracking software, software project financial management	financial management software examples	Guide	4	40	—
Later	E-invoicing uae what is e invoicing in uae, invoicing best practices	e invoicing	Guide	3	2,300	9 · easy
Later	What is erp erp accounting definition	erp	Guide	2	5,420	29 · easy
Later	Financial compliance compliance with ifrs, accounting and tax compliance, financial compliance, financial compliance monitoring, financial data compliance	financial compliance solutions	Guide	6	70	—
Later	Accounting software guide accounting software advantages, accounting software benefits, financial software examples	accounting software feature	Guide	4	40	—
Later	Finance process transformation finance process transformation, shared accounting services	accounts payable transformation	Service Page	3	30	—
Later	Finance erp software	accounting solutions	Service Page	1	20	30 · medium
Later	Financial reporting solutions financial ratio software	financial reporting solutions	Service Page	2	20	—

Later	Project accounting system project accounting services	project accounting system	Product Page	2	20	—
Later	Budget planning best budgeting system, budget tracking system	budget planning services	Service Page	3	30	—
Later	Financial tracking system financial tracking system, finance tracking system	financial tracking software	Service Page	3	30	—
Later	Cash flow software cash flow management software, liquidity management software	cash flow projection software	Guide	3	30	—
Later	Erp implementation cost	erp implementation cost breakdown	Guide	1	10	—

Quick wins (difficulty under 30, relevant, real demand)

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Trend	SERP features	Opportunity
e invoicing	1,900	9 · easy	7.949999809265137	Commercial	+222%	AI Overview, PAA	1,070.8
what is e invoicing in uae	390	13 · easy	3.7100000381469727	Informational	+333%	AI Overview, PAA	115
erp consultants	110	0 · easy	8.140000343322754	Commercial	-29%	AI Overview, PAA	36.5

Questions your buyers ask

Each is a guide, FAQ entry or section with a direct 40-60 word answer — the format AI Overviews and assistants quote.

Keyword	Searches/mo	Difficulty	CPC \$	Intent
what is e invoicing in uae	390	13 · easy	3.7100000381469727	Informational

Keywords competitors rank for and you should too

Competitors checked: instalogic.ae, craftinteractive.ae, hamtinfotech.com.

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Competitors in top 20
e invoicing	1,900	9 · easy	7.949999809265137	Commercial	hamtinfotech.com #10
what is e invoicing in uae	390	13 · easy	3.7100000381469727	Informational	hamtinfotech.com #17
erp consultants	110	0 · easy	8.140000343322754	Commercial	hamtinfotech.com #8

Long-tail phrases (4+ words)

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Trend
---------	-------------	------------	--------	--------	-------

what is e invoicing in uae	390	13 · easy	3.7100000381469727	Informational	+333%
financial and accounting software	10	—	93.89	Commercial	-50%
ai in accounts payable	20	—	—	Commercial	—
erp implementation cost breakdown	10	—	—	Commercial	0%
financial business management system	10	—	—	Commercial	0%
dynamics 365 finance and operations system requirements	10	—	—	Informational	0%
cash flow management solution	10	—	—	Commercial	0%
finance and accounting solutions	10	—	—	Commercial	0%
accounting and financial solutions	10	—	—	Commercial	—
accounting and finance solutions	10	—	—	Commercial	—
integrated financial management system	10	—	—	Commercial	0%
finance and accounting system	10	—	—	Commercial	—
accounting system in dubai	10	—	—	Commercial	0%
inventory and accounting system	10	—	—	Commercial	0%
accounts receivable management system	10	—	—	Commercial	0%

AI search demand

How often these keywords are asked in AI assistants (DataForSEO AI keyword data) next to Google demand.

Keyword	AI searches/mo	Google searches/mo
erp	546	5,400
e invoicing	64	1,900
what is e invoicing in uae	9	390
erp consultants	3	110

Commercial keywords

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Trend	SERP features	Opportunity
e invoicing	1,900	9 · easy	7.949999809265137	Commercial	+222%	AI Overview, PAA	1,070.8
erp consultants	110	0 · easy	8.140000343322754	Commercial	-29%	AI Overview, PAA	36.5
finance erp system	40	—	—	Commercial	-75%	—	5.3
erp finance	40	—	—	Commercial	-75%	—	5.3
erp financials	40	—	—	Commercial	-75%	—	5.3

uae accounting system	10	—	47.62	Commercial	0%	—	5
financial and accounting software	10	—	93.89	Commercial	-50%	—	3.9
ai in accounts payable	20	—	—	Commercial	—	—	3.8
accounting solutions	20	30 · medium	17.46	Commercial	-67%	Local pack, PAA	3.1
erp financial systems	10	—	—	Commercial	0%	—	1.9

Transactional keywords

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Trend	SERP features	Opportunity
implementing accounting system	10	—	—	Transactional	-100%	—	1.4
accounting software purchase	10	—	—	Transactional	—	—	0.9
accounting system demo	10	—	—	Transactional	0%	—	0.9

Informational keywords

Keyword	Searches/mo	Difficulty	CPC \$	Intent	Trend	SERP features	Opportunity
erp	5,400	29 · easy	8.979999542236328	Informational	0%	AI Overview, Video, PAA	247.6
what is e invoicing in uae	390	13 · easy	3.7100000381469727	Informational	+333%	AI Overview, PAA	115
dynamics 365 finance and operations system requirements	10	—	—	Informational	0%	—	1.9
compliance with ifrs	20	—	2.46	Informational	0%	—	1.7
erp accounting definition	20	—	0.09	Informational	+50%	—	1.4
accounting and tax compliance	10	—	4.78	Informational	0%	—	0.9
deferred output vat journal entry	10	—	—	Informational	0%	—	0.9

invoicing best practices	10	—	—	Informational	0%	—	0.9
financial compliance	10	—	—	Informational	0%	—	0.9
financial data compliance	10	—	—	Informational	0%	—	0.9

How this was built

Seeds: erp implementation partner, erp implementation services, finance and operations erp, e invoicing solution, vat compliance software, cloud migration services, ai business solutions, digital transformation consulting, microsoft 365 implementation, enterprise cloud solutions. Sources: competitor 17, ideas 599, suggestions 8, related 2. Opportunity = traffic potential (volume after SERP-feature click loss) × winnability (difficulty and top-10 link strength) × commercial value (CPC) × intent weight for your goal × trend, then multiplied by AI relevance (2 = core, 1 = adjacent, 0 = dropped).

Prepared by Dev SEO · Search volumes, difficulty, CPC and SERP features from DataForSEO for United Arab Emirates.

Run 16 — Technical site audit (API, techand.ai)

Crawl, extras and probes → findings for the scorer

Probe measurements: {"canonical_host": "www.techand.ai", "cache_control": "private, no-cache, no-store, max-age=0, must-revalidate", "cache_status": ["MISS", "MISS"], "homepage_ttfb_ms": null, "server": "Vercel", "sitemap_urls": 43, "crawled_urls": 88}

Probe findings:

Severity	Category	Finding	Evidence	Fix
High	Performance	HTML cannot be cached by the CDN (no-store / private)	Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS	Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can
Low	Security	Missing security headers	Not found: Content-Security-Policy	Add the missing headers in your hosting or CDN configuration; start Content-Security-Policy in Report-Only mode.

What works (probes): One canonical host: https://techand.ai 301-redirects to https://www.techand.ai; HTTP redirects to HTTPS; robots.txt lists the sitemap; XML sitemap lists 43 URLs; Every live page is listed in the sitemap; llms.txt exists (10296 characters); Site serves Markdown to AI agents on request; All 11 search and AI crawlers tested (GPTBot, ClaudeBot, PerplexityBot, Googlebot and more) receive the full page

Scored audit (Build Site Issues) → the Site Audit report, the e-mail subject and the callback

Health score **70/100** (Needs Improvement); raw 81, cap 70; category scores {"Crawlability & Indexing": 41, "On-Page": 89, "Content": 97, "Performance": 76, "Security": 97, "Images": 100, "Technical": 100, "AI Search Readiness": 100}; issue counts {"Critical": 1, "High": 3, "Medium": 6, "Low": 10, "Info": 3}.

Severity	Category	Finding	Evidence	Fix
Critical	Crawlability & Indexing	Broken pages (4xx, 5xx or timeout)	3 of 88 crawled pages	Fix or 301-redirect broken URLs to the most relevant live page.

High	Crawlability & Indexing	www and non-www versions are not consolidated	Crawler canonicalization test failed	Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.
High	Performance	HTML cannot be cached by the CDN (no-store / private)	Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS	Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can
High	Crawlability & Indexing	Broken internal links	21 broken links across 88 pages	Export the broken links from the crawl and fix or remove each one; redirect deleted pages to the closest live page.
Medium	Crawlability & Indexing	Missing canonical tag	45 of 88 crawled pages	Add a self-referencing canonical tag to every indexable page.
Medium	Crawlability & Indexing	Canonical points to a redirect	42 of 88 crawled pages	Point canonicals directly to the final URL.
Medium	Performance	No compression (gzip/brotli)	42 of 88 crawled pages	Enable gzip or brotli compression on the server.
Medium	Performance	Large image payload (over 1MB of images per page)	33 of 88 crawled pages	Compress images, serve WebP/AVIF, and make sure image CDN optimisation (e.g. f_auto,q_auto) is on.
Medium	Schema	Structured data with errors	12 of 88 crawled pages	Validate each page in the Rich Results Test / Schema Markup Validator and fix the reported properties.
Medium	On-Page	Brand name repeated in the title	1 of 88 crawled pages	Fix the title template so the brand appears once.
Low	Security	Missing security headers	Not found: Content-Security-Policy	Add the missing headers in your hosting or CDN configuration; start Content-Security-Policy in Report-Only mode.
Low	On-Page	Missing Open Graph tags	45 of 88 crawled pages	Add og:title, og:description and og:image.
Low	On-Page	URLs are not SEO-friendly	43 of 88 crawled pages	Use short, lowercase, hyphenated, descriptive URLs.
Low	Crawlability & Indexing	Internal URLs that redirect	42 of 88 crawled pages	Link directly to the final URL.
Low	Crawlability & Indexing	Links pointing to redirects	42 of 88 crawled pages	Update links to point to the final URL.
Low	Content	Articles with no outbound links to sources	13 of 88 crawled pages	Link key facts and statistics to their primary sources.
Low	On-Page	Title shorter than 30 characters	12 of 88 crawled pages	Expand to 50-60 characters with the main keyword and benefit.
Low	On-Page	Long or messy URLs	9 of 88 crawled pages	Use short, lowercase, hyphenated URLs.
Low	Crawlability & Indexing	Few internal links pointing to the page	6 of 88 crawled pages	Add contextual links from related pages with descriptive anchor text.
Low	Content	Light content (300-500 words)	1 of 88 crawled pages	Expand with answers to the questions buyers ask before contacting you.

Info	On-Page	Missing Twitter/X card tags	45 of 88 crawled pages	Add twitter:card and twitter:image tags.
Info	Crawlability & Indexing	Canonical points to a different page	43 of 88 crawled pages	Confirm each of these is intentional.
Info	On-Page	H1 is identical to the title tag	2 of 88 crawled pages	Optional: vary the H1 wording from the title.

What works: Valid SSL certificate on the main domain (expires 2026-12-29); HTTP correctly redirects to HTTPS; robots.txt is present; HTTP/2 is enabled; Missing pages return a proper 404; No broken images, scripts or stylesheets; No duplicate page titles; No duplicate meta descriptions; No duplicate content across pages; No redirect loops; All crawled pages are indexable; Every page has an H1; Every page has a title tag; One canonical host: https://techand.ai 301-redirects to https://www.techand.ai; HTTP redirects to HTTPS; robots.txt lists the sitemap; XML sitemap lists 43 URLs; Every live page is listed in the sitemap; llms.txt exists (10296 characters); Site serves Markdown to AI agents on request; All 11 search and AI crawlers tested (GPTBot, ClaudeBot, PerplexityBot, Googlebot and more) receive the full page

Delivered: Site Audit report (PDF + Word) → e-mailed to kinngahmed@gmail.com (250 OK) and posted to the callback (stage site_audit)

DELIVERED REPORT: SITE-AUDIT-TECHAND-AI-2026-10-01.DOC

SITE AUDIT REPORT

techand.ai

88 pages crawled • 2026-10-01

70/100

Needs Improvement

Site Health Score (technical)

Prepared by Dev SEO

Contents

1. Executive Summary

2. Scores by Category

3. What Works Well

4. Issues Found

5. Action Plan

6. Not Assessed in This Audit

7. Methodology

1. Executive Summary

We crawled 88 pages of techand.ai and ran additional checks on security, caching, crawler access and the sitemap. We found:

Critical	High	Medium	Low	Info
1	3	6	10	3

Score limit applied: 1 critical issue found — score capped at 70 until fixed. Calculated score before the limit: 81/100.

Fix These First

- 1. Broken pages (4xx, 5xx or timeout)** (3 of 88 crawled pages)
Fix or 301-redirect broken URLs to the most relevant live page.
- 2. www and non-www versions are not consolidated** (Crawler canonicalization test failed)
Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.
- 3. HTML cannot be cached by the CDN (no-store / private)** (Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS)
Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can cache HTML. Verify: repeat requests should show a cache HIT.
- 4. Broken internal links** (21 broken links across 88 pages)
Export the broken links from the crawl and fix or remove each one; redirect deleted pages to the closest live page.

Key Measurements

Homepage server response	N/A
Cache-Control header	private, no-cache, no-store, max-age=0, must-revalidate
CDN cache status	MISS / MISS
URLs in sitemap	43
Pages found by crawling	88
Server	Vercel

2. Scores by Category

Category	Score	Weight	
Crawlability & Indexing	41	22%	
On-Page	89	18%	
Content	97	13%	
Performance	76	15%	
Security	97	10%	

Images	100	7%	
Technical	100	5%	
AI Search Readiness	100	10%	

3. What Works Well

- Valid SSL certificate on the main domain (expires 2026-12-29)
- HTTP correctly redirects to HTTPS
- robots.txt is present
- HTTP/2 is enabled
- Missing pages return a proper 404
- No broken images, scripts or stylesheets
- No duplicate page titles
- No duplicate meta descriptions
- No duplicate content across pages
- No redirect loops
- All crawled pages are indexable
- Every page has an H1
- Every page has a title tag
- One canonical host: <https://techand.ai> 301-redirects to <https://www.techand.ai>
- HTTP redirects to HTTPS
- robots.txt lists the sitemap
- XML sitemap lists 43 URLs
- Every live page is listed in the sitemap
- llms.txt exists (10296 characters)
- Site serves Markdown to AI agents on request
- All 11 search and AI crawlers tested (GPTBot, ClaudeBot, PerplexityBot, Googlebot and more) receive the full page

4. Issues Found

Critical Priority (1)

Critical Broken pages (4xx, 5xx or timeout)

Crawlability & Indexing • 3 of 88 pages (3%)

Evidence: 3 of 88 crawled pages

Why it matters: Pages that return errors waste crawl budget and lose any ranking signals pointing to them.

How to fix: Fix or 301-redirect broken URLs to the most relevant live page.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae>

High Priority (3)

High www and non-www versions are not consolidated

Crawlability & Indexing • Whole website

Evidence: Crawler canonicalization test failed

Why it matters: If both versions do not resolve to one canonical host, links and signals are split.

How to fix: Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.

High HTML cannot be cached by the CDN (no-store / private)

Performance • Whole website

Evidence: Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS

Why it matters: Every visit forces the server to rebuild the page from scratch, which slows the first byte for every user and hurts Core Web Vitals. This is sometimes intentional (carts, logged-in areas) — confirm before changing.

How to fix: Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can cache HTML. Verify: repeat requests should show a cache HIT.

High Broken internal links

Crawlability & Indexing • Whole website

Evidence: 21 broken links across 88 pages

Why it matters: Links to pages that no longer exist waste crawl budget and send users to dead ends.

How to fix: Export the broken links from the crawl and fix or remove each one; redirect deleted pages to the closest live page.

Medium Priority (6)

Medium Missing canonical tag

Crawlability & Indexing • 45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Without a canonical, duplicate URL versions can compete with each other.

How to fix: Add a self-referencing canonical tag to every indexable page.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — No canonical

<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — No canonical

<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — No canonical

<https://techand.ai/ai-copilot-advisory-uae> — No canonical

<https://techand.ai/business-central-implementation-uae> — No canonical

<https://techand.ai/business-central-post-go-live-support-uae> — No canonical

<https://techand.ai/capabilities> — No canonical

<https://techand.ai/capabilities/business-applications> — No canonical

<https://techand.ai/capabilities/cloud-services> — No canonical

<https://techand.ai/capabilities/data> — No canonical

...and 35 more

Medium Canonical points to a redirect

Crawlability & Indexing • 42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Mixed signals slow down consolidation of ranking signals.

How to fix: Point canonicals directly to the final URL.

Affected pages:

<https://www.techand.ai/ai-copilot-advisory-uae>
<https://www.techand.ai/business-central-implementation-uae>
<https://www.techand.ai/business-central-post-go-live-support-uae>
<https://www.techand.ai/capabilities>
<https://www.techand.ai/capabilities/business-applications>
<https://www.techand.ai/capabilities/cloud-services>
<https://www.techand.ai/capabilities/data>
<https://www.techand.ai/contact-us>
<https://www.techand.ai/cookies>
<https://www.techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Medium No compression (gzip/brotli)

Performance • 42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Uncompressed pages take longer to download.

How to fix: Enable gzip or brotli compression on the server.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>
<https://techand.ai/business-central-implementation-uae>
<https://techand.ai/business-central-post-go-live-support-uae>
<https://techand.ai/capabilities>
<https://techand.ai/capabilities/business-applications>
<https://techand.ai/capabilities/cloud-services>
<https://techand.ai/capabilities/data>
<https://techand.ai/contact-us>
<https://techand.ai/cookies>
<https://techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Medium Large image payload (over 1MB of images per page)

Performance • 33 of 88 pages (38%)

Evidence: 33 of 88 crawled pages

Why it matters: Large images are the most common cause of slow mobile pages. The figure is the total size of image files the page references (including responsive sizes), so the real download may be smaller, but it signals unoptimised images.

How to fix: Compress images, serve WebP/AVIF, and make sure image CDN optimisation (e.g. f_auto,q_auto) is on.

Affected pages:

<https://www.techand.ai/> — 3.5MB of image files referenced
<https://www.techand.ai/business-central-implementation-uae> — 3.4MB of image files referenced
<https://www.techand.ai/business-central-post-go-live-support-uae> — 1.1MB of image files referenced
<https://www.techand.ai/capabilities/business-applications> — 1.8MB of image files referenced
<https://www.techand.ai/d365-finance-operations-implementation-rollout-uae> — 31.8MB of image files referenced
<https://www.techand.ai/dynamics-365-business-central-uae> — 9.0MB of image files referenced
<https://www.techand.ai/dynamics-365-cloud-migration-uae> — 31.2MB of image files referenced
<https://www.techand.ai/dynamics-365-crm-uae> — 4.3MB of image files referenced
<https://www.techand.ai/dynamics-365-finance-and-operations-uae> — 40.0MB of image files referenced
<https://www.techand.ai/dynamics-365-managed-services-uae> — 15.7MB of image files referenced
...and 23 more

Medium Structured data with errors

Schema • 12 of 88 pages (14%)

Evidence: 12 of 88 crawled pages

Why it matters: Search engines ignore structured data blocks that fail validation, so the page loses rich-result eligibility.

How to fix: Validate each page in the Rich Results Test / Schema Markup Validator and fix the reported properties.

Affected pages:

<https://www.techand.ai/insights/agent-ai>
<https://www.techand.ai/insights/autonomous-ai-customer-service>
<https://www.techand.ai/insights/data-sovereign-asset>
<https://www.techand.ai/insights/how-microsoft-365-copilot-is-changing-business-productivity>
<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>
<https://www.techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap>
<https://www.techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs>
<https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel>
<https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for>
<https://www.techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide>
...and 2 more

Medium Brand name repeated in the title

On-Page • 1 of 88 pages (1%)

Evidence: 1 of 88 crawled pages

Why it matters: A doubled brand suffix wastes title space and usually means the template adds the brand twice.

How to fix: Fix the title template so the brand appears once.

Affected pages:

<https://www.techand.ai/power-platform-analytics-uae> — "Power Platform & Analytics Services in UAE | Tech& | Tech&"

Low Priority (10)

Low Missing security headers

Security • Whole website

Evidence: Not found: Content-Security-Policy

Why it matters: Security headers protect visitors. They are not a direct ranking factor but show a well-maintained site.

How to fix: Add the missing headers in your hosting or CDN configuration; start Content-Security-Policy in Report-Only mode.

Low Missing Open Graph tags

On-Page • 45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Without them, links shared on LinkedIn, WhatsApp or Facebook show a poor preview.

How to fix: Add og:title, og:description and og:image.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — Missing og:title, og:image
<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — Missing og:title, og:image
<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — Missing og:title, og:image
<https://techand.ai/ai-copilot-advisory-uae> — Missing og:title, og:image
<https://techand.ai/business-central-implementation-uae> — Missing og:title, og:image
<https://techand.ai/business-central-post-go-live-support-uae> — Missing og:title, og:image
<https://techand.ai/capabilities> — Missing og:title, og:image
<https://techand.ai/capabilities/business-applications> — Missing og:title, og:image
<https://techand.ai/capabilities/cloud-services> — Missing og:title, og:image
<https://techand.ai/capabilities/data> — Missing og:title, og:image
...and 35 more

Low URLs are not SEO-friendly

On-Page • 43 of 88 pages (49%)

Evidence: 43 of 88 crawled pages

Why it matters: Dynamic parameters, long paths or non-descriptive URLs are harder to read and share.

How to fix: Use short, lowercase, hyphenated, descriptive URLs.

Affected pages:

<https://www.techand.ai/>
<https://www.techand.ai/ai-copilot-advisory-uae>
<https://www.techand.ai/business-central-implementation-uae>
<https://www.techand.ai/business-central-post-go-live-support-uae>
<https://www.techand.ai/capabilities>
<https://www.techand.ai/capabilities/business-applications>
<https://www.techand.ai/capabilities/cloud-services>
<https://www.techand.ai/capabilities/data>
<https://www.techand.ai/contact-us>
<https://www.techand.ai/cookies>
...and 33 more

Low Internal URLs that redirect

Crawlability & Indexing • 42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Redirects add load time and dilute crawl efficiency.

How to fix: Link directly to the final URL.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>
<https://techand.ai/business-central-implementation-uae>
<https://techand.ai/business-central-post-go-live-support-uae>
<https://techand.ai/capabilities>
<https://techand.ai/capabilities/business-applications>
<https://techand.ai/capabilities/cloud-services>
<https://techand.ai/capabilities/data>
<https://techand.ai/contact-us>
<https://techand.ai/cookies>
<https://techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Low Links pointing to redirects

Crawlability & Indexing • 42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Extra hops for users and crawlers.

How to fix: Update links to point to the final URL.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>
<https://techand.ai/business-central-implementation-uae>
<https://techand.ai/business-central-post-go-live-support-uae>
<https://techand.ai/capabilities>
<https://techand.ai/capabilities/business-applications>
<https://techand.ai/capabilities/cloud-services>
<https://techand.ai/capabilities/data>
<https://techand.ai/contact-us>
<https://techand.ai/cookies>
<https://techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Low Articles with no outbound links to sources

Content • 13 of 88 pages (15%)

Evidence: 13 of 88 crawled pages

Why it matters: Articles that cite no sources are less trustworthy and less likely to be quoted by AI answers.

How to fix: Link key facts and statistics to their primary sources.

Affected pages:

<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — 0 external links

<https://techand.ai/insights/agentica-ai> — 0 external links

<https://techand.ai/insights/autonomous-ai-customer-service> — 0 external links

<https://techand.ai/insights/data-sovereign-asset> — 0 external links

<https://techand.ai/insights/how-microsoft-365-copilot-is-changing-business-productivity> — 0 external links

<https://techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance> — 0 external links

<https://techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap> — 0 external links

<https://techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs> — 0 external links

<https://techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel> — 0 external links

<https://techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for> — 0 external links

...and 3 more

Low Title shorter than 30 characters

On-Page • 12 of 88 pages (14%)

Evidence: 12 of 88 crawled pages

Why it matters: Short titles waste space to include keywords and a reason to click.

How to fix: Expand to 50-60 characters with the main keyword and benefit.

Affected pages:

<https://www.techand.ai/capabilities> — 20 characters: "Capabilities | Tech&"

<https://www.techand.ai/capabilities/business-applications> — 29 characters: "Business Applications | Tech&"

<https://www.techand.ai/capabilities/cloud-services> — 22 characters: "Cloud Services | Tech&"

<https://www.techand.ai/capabilities/data> — 24 characters: "Data & Analytics | Tech&"

<https://www.techand.ai/contact-us> — 18 characters: "Contact Us | Tech&"

<https://www.techand.ai/cookies> — 21 characters: "Cookie Policy | Tech&"

<https://www.techand.ai/industries> — 27 characters: "Industries We Serve | Tech&"

<https://www.techand.ai/insights> — 16 characters: "Insights | Tech&"

<https://www.techand.ai/privacy> — 22 characters: "Privacy Policy | Tech&"

<https://www.techand.ai/security> — 23 characters: "Security Policy | Tech&"

...and 2 more

Low Long or messy URLs

On-Page • 9 of 88 pages (10%)

Evidence: 9 of 88 crawled pages

Why it matters: Long URLs, capitals or underscores are harder to read, share and remember.

How to fix: Use short, lowercase, hyphenated URLs.

Affected pages:

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance> — 105 characters

<https://www.techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap> — 134 characters

<https://www.techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs> — 101 characters

<https://www.techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide> — 114 characters

<https://www.techand.ai/insights/your-current-pdf-invoices-will-be-illegal-in-the-uae-by-2027-and-how-to-fix-it> — 110 characters

<https://techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance> — 101 characters

<https://techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap> — 130 characters

<https://techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide> — 110 characters

<https://techand.ai/insights/your-current-pdf-invoices-will-be-illegal-in-the-uae-by-2027-and-how-to-fix-it> — 106 characters

Low Few internal links pointing to the page

Crawlability & Indexing • 6 of 88 pages (7%)

Evidence: 6 of 88 crawled pages

Why it matters: Pages with few internal links look less important to search engines.

How to fix: Add contextual links from related pages with descriptive anchor text.

Affected pages:

<https://www.techand.ai/whatwedo/ai-automation> — 1 internal links in
<https://www.techand.ai/whatwedo/assess-envision> — 1 internal links in
<https://www.techand.ai/whatwedo/build-implement> — 1 internal links in
<https://www.techand.ai/whatwedo/global-rollouts> — 1 internal links in
<https://www.techand.ai/whatwedo/managed-services> — 1 internal links in
<https://www.techand.ai/whatwedo/run-optimize> — 1 internal links in

Low Light content (300-500 words)

Content • 1 of 88 pages (1%)

Evidence: 1 of 88 crawled pages

Why it matters: Commercial pages usually need more depth to compete.

How to fix: Expand with answers to the questions buyers ask before contacting you.

Affected pages:

<https://www.techand.ai/contact-us> — 334 words

Info Priority (3)

Info Missing Twitter/X card tags

On-Page • 45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Shared links on X show a plain preview.

How to fix: Add twitter:card and twitter:image tags.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — No twitter:card
<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — No twitter:card
<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — No twitter:card
<https://techand.ai/ai-copilot-advisory-uae> — No twitter:card
<https://techand.ai/business-central-implementation-uae> — No twitter:card
<https://techand.ai/business-central-post-go-live-support-uae> — No twitter:card
<https://techand.ai/capabilities> — No twitter:card
<https://techand.ai/capabilities/business-applications> — No twitter:card
<https://techand.ai/capabilities/cloud-services> — No twitter:card
<https://techand.ai/capabilities/data> — No twitter:card
...and 35 more

Info Canonical points to a different page

Crawlability & Indexing • 43 of 88 pages (49%)

Evidence: 43 of 88 crawled pages

Why it matters: Fine if intentional, but this page will not rank on its own.

How to fix: Confirm each of these is intentional.

Affected pages:

<https://www.techand.ai/> — → <https://techand.ai/>
<https://www.techand.ai/ai-copilot-advisory-uae> — → <https://techand.ai/ai-copilot-advisory-uae>
<https://www.techand.ai/business-central-implementation-uae> — → <https://techand.ai/business-central-implementation-uae>
<https://www.techand.ai/business-central-post-go-live-support-uae> — → <https://techand.ai/business-central-post-go-live-support-uae>
<https://www.techand.ai/capabilities> — → <https://techand.ai/capabilities>
<https://www.techand.ai/capabilities/business-applications> — → <https://techand.ai/capabilities/business-applications>
<https://www.techand.ai/capabilities/cloud-services> — → <https://techand.ai/capabilities/cloud-services>
<https://www.techand.ai/capabilities/data> — → <https://techand.ai/capabilities/data>
<https://www.techand.ai/contact-us> — → <https://techand.ai/contact-us>

Info H1 is identical to the title tag

On-Page • 2 of 88 pages (2%)

Evidence: 2 of 88 crawled pages

Why it matters: A slightly different H1 lets you target a second keyword variation.

How to fix: Optional: vary the H1 wording from the title.

Affected pages:

<https://www.techand.ai/insights/autonomous-ai-customer-service> — Both: "How Customer Services Are Achieving Productivity with AI"

<https://www.techand.ai/insights/how-microsoft-365-copilot-is-changing-business-productivity> — Both: "How Microsoft 365 Copilot Is Changing Business Productivity"

5. Action Plan

Phase 1 — Urgent (this week)

- Broken pages (4xx, 5xx or timeout)
- www and non-www versions are not consolidated
- HTML cannot be cached by the CDN (no-store / private)
- Broken internal links

Phase 2 — Important (next 2-4 weeks)

- Missing canonical tag
- Canonical points to a redirect
- No compression (gzip/brotli)
- Large image payload (over 1MB of images per page)
- Structured data with errors
- Brand name repeated in the title

Phase 3 — Polish (when time allows)

- Missing security headers
- Missing Open Graph tags
- URLs are not SEO-friendly
- Internal URLs that redirect
- Links pointing to redirects
- Articles with no outbound links to sources
- Title shorter than 30 characters
- Long or messy URLs
- Few internal links pointing to the page
- Light content (300-500 words)
- Missing Twitter/X card tags
- Canonical points to a different page
- H1 is identical to the title tag

6. Not Assessed in This Audit

This is a technical site audit. The following areas are part of the Full SEO Report and are not included in this score:

- Structured data (schema) and HTML-level checks
- Competitor benchmark (organic keywords, rankings, traffic, domain age)
- Core Web Vitals (LCP, CLS, INP) from Google PageSpeed and real users
- Content quality and E-E-A-T review (authors, case studies, claims, pricing)
- Backlinks and referring domains
- Keyword rankings by search intent and keyword gap
- AI answer visibility (AI Overviews, ChatGPT brand mentions)

7. Methodology

Technical site health only. Each category starts at 100. Findings written by the AI content review, and notes about data that could not be collected, are listed but do not change the score. Each other finding deducts Critical 30, High 15, Medium 7, Low 3, Info 1 points, scaled by the share of pages affected (60%-100% of the penalty); site-wide issues take the full penalty. Within a category, each further finding of the same severity counts 70% of the previous one, so many small issues cannot zero a category on their own. The score is the weighted average of category scores, capped at 70 with one critical issue, 60 with two or more, and 85 with five or more high-priority issues.

Severity	Points deducted
Critical	30
High	15
Medium	7
Low	3
Info	1

Data Sources

Source	Used for
Full site crawl (DataForSEO On-Page)	Page-level checks on up to 200 pages, including sitemap URLs; duplicate tags, non-indexable pages and redirect chains reports
Direct HTTP checks	www/HTTPS, caching, security headers, robots.txt, sitemap, llms.txt
Crawler simulation	Access for 11 search and AI crawlers (user-agent based)

AI crawler checks use the crawler user-agent only; real crawlers are also verified by IP, so a pass here means no user-agent based blocking.
Some extra crawl reports were unavailable: duplicate_title: Task Not Found.; duplicate_description: Task Not Found.; non_indexable: Task Not Found.; redirect_chains: Task Not Found.

Run 21 — Full SEO report (API, techand.ai)

Competitor benchmark (Find Competitors / Fallback SERP / Domain Overview / WHOIS + RDAP → Competitor Analysis) → Authority findings and the report's benchmark table

Domain	Organic keywords	Top 10	Est. visits/mo	Domain age (y)	Method
techand.ai	0	0	0	0.6	

theknowledgeacademy.com	1358	381	9897	17.6	
youglish.com	13745	1350	28492	10.7	
boardroom-ai.ae	1	0	1		

Keywords (Ranked Keywords + Keyword Ideas → Analyze Keywords) → keyword gap, easy wins, striking distance

Domain	Ranked keywords	Top 10
techand.ai	0	0
theknowledgeacademy.com	300	115
youglish.com	300	22
boardroom-ai.ae	0	0

Keyword gap 25, easy wins 10, striking distance 0. Easy wins:

Keyword	Searches/mo	KD	Position
ai in cloud	27100		
ai on cloud	27100		
cloud's ai	27100		
blackbox ai	14800		
vertically	9900		
employees	8100		
face swap ai	6600		
typewriting centre near me	6600		
closes	6600		
magic light ai	5400		

Backlinks (Backlink Summary + Backlink Gap → Analyze Backlinks)

Available: True

Domain	Referring domains	Backlinks	You
techand.ai	13	13	yes
theknowledgeacademy.com	9917	218676	
youglish.com	7806	845464	
boardroom-ai.ae			

Link gap (domains linking to competitors, not to you):

none

PageSpeed (Google PageSpeed Insights → Analyze PageSpeed) → Performance findings

Domain	Performance score	LCP (field)	CLS (field)	INP (field)	Available
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techand.ai	89				True
theknowledgeacademy.com	64	1638	0.16	162	True
youglsh.com	71	1679	0.05	194	True
boardroom-ai.ae	41				True

E-E-A-T content review (Claude Sonnet, Content Reviewer → Parse Content Review) → listed findings (not scored)

Severity	Category	Finding	Evidence	Fix
High	Content	Unproven scale and financial claims on contact page	"We bring together a massive team of experts and significant financial backing to ensure your project succeeds on time."	Replace it with verifiable facts: team size, years operating, number of projects delivered, and named leadership. Add a link to proof.
High	Content	No named authors or expert bylines on insight articles	The articles on D365 F&O reporting and Peppol e-invoicing show only "Dynamics 365 7 min read", with no author name or credentials shown.	Add bylines with bios, roles, Microsoft certifications and LinkedIn links. Add reviewer and last-updated dates.
High	Content	Partner status and certifications not evidenced	Pages list many Dynamics 365, Copilot and Azure OpenAI services, but the text shows no Microsoft partner badge, competencies or certifications.	Show Microsoft partner badges, designations, MCT/consultant certifications and links to the Partner directory listing in the footer and on the Why Tech& page.
High	Content	Case study and testimonial proof is thin	The homepage teaser "How Customer Service Teams Are Achieving a 20% to 45% Increase in..." is cut off. No client names, quotes or reviews appear in the supplied t	Publish named case studies with the client, scope, timeline and measurable results. Add testimonials, logos and third-party reviews such as Clutch or Gartner Pe
Medium	Content	Service pages are generic and duplicated	Every /whatwedo/ page repeats the same block: "We guide your business through every step of technology growth to ensure you stay ahead of competitors." Each pag	Write unique content per service: scope, methodology, typical timelines, deliverables, SLAs, a UAE example and FAQs.
Medium	Content	Heavy repeated mega-menu text on every page	The same navigation text, from "Dynamics 365 Finance & Operations UAE" to "Predictive Models", precedes the content on every page.	Make the navigation less crawlable-noisy and the labels more natural. Make sure the main content comes first in the HTML.
Medium	Content	Formatting artifact and only high-commitment CTAs	"Let's discuss how we can help your business" shows a raw HTML entity. CTAs are "Talk to Sales", "Call Now" and "Request Call Back".	Fix the entity encoding. Add softer CTAs such as a readiness checklist, a sample scope or a downloadable guide.
Medium	Content	No pricing or engagement-model guidance, and no leadership info	No pricing, ranges, or engagement models appear on the service pages. No team or leadership names appear in the text supplied.	Add indicative engagement models, typical project durations, and a leadership/team page.

AI visibility (brand SERP, AI Overviews, live ChatGPT / Perplexity answers → AI Visibility) → AI Search Readiness findings

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  }
]

```

Scored report (Build Site Issues) → Full SEO Report, e-mail and callback

Health score **64/100** (Needs Improvement); category scores {"Crawlability & Indexing": 34, "On-Page": 88, "Content": 65, "Schema": 47, "Performance": 66, "Security": 97, "Images": 93, "Technical": 93, "AI Search Readiness": 60, "Authority": 44}; issue counts {"Critical": 1, "High": 22, "Medium": 18, "Low": 16, "Info": 3}; not assessed: [].

Severity	Category	Finding	Evidence	Fix
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Critical	Crawlability & Indexing	Broken pages (4xx, 5xx or timeout)	3 of 88 crawled pages	Fix or 301-redirect broken URLs to the most relevant live page.
High	Crawlability & Indexing	www and non-www versions are not consolidated	Crawler canonicalization test failed	Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.
High	Performance	HTML cannot be cached by the CDN (no-store / private)	Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS	Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can
High	Schema	Malformed sameAs profile link	Invalid: https://twitter.com/Tech&	Remove or correct the invalid entry and link only real, active profiles.
High	Schema	Structured contact details are thinner than what the site shows	Missing in schema: streetAddress, addressLocality, telephone, email	Add full PostalAddress (streetAddress, addressLocality, addressRegion, postalCode, addressCountry), telephone and email to Organization schema.
High	Content	Experience claim conflicts with the founding date	"20+ years" on https://www.techand.ai/ but schema foundingDate is 2020	Make the claims consistent (e.g. "leadership with 20+ years of experience") or correct the founding date.
High	Schema	Service pages have no Service schema	5 of 5 sampled service pages	Add Service JSON-LD (serviceType, provider → Organization @id, areaServed) on each service page.
High	Schema	No BreadcrumbList schema	9 of 9 sampled inner pages	Add BreadcrumbList JSON-LD to every inner page.
High	Authority	The site does not rank in Google's top 10 for any keyword	Top-10 rankings in United Arab Emirates: techand.ai 0 vs competitor median 381 (theknowledgeacademy.com, youglish.com, boardroom-ai.ae)	Build authority first (partner listings, directories, reviews, mentions) and target long-tail keywords where competitors are weak.
High	Authority	Organic traffic is under 10% of competitors	Estimated monthly organic visits: techand.ai 0 vs competitor median 9,897	Close the keyword and backlink gaps shown in this report.
High	Authority	Very young domain competing with long-established sites	techand.ai registered 2026-02-19 (0.6 years) vs competitor median 14.2 years	Invest in off-site entity building (partner directories, reviews, mentions) and target long-tail, lower-difficulty searches first.
High	Authority	Keyword gap: searches competitors win and you do not	25 keywords where competitors rank in the top 10 and you are not in the top 20. Top: "vertically" (9900/mo), "employees" (8100/mo), "typewriting centre near me"	Create or improve a dedicated page for the highest-volume, lowest-difficulty gap keywords first.

High	Authority	Far fewer referring domains than competitors	13 referring domains vs competitor median 8862	Earn links from partner directories, industry associations, suppliers, press and guest content.
High	Content	Unproven scale and financial claims on contact page	"We bring together a massive team of experts and significant financial backing to ensure your project succeeds on time."	Replace it with verifiable facts: team size, years operating, number of projects delivered, and named leadership. Add a link to proof.
High	Content	No named authors or expert bylines on insight articles	The articles on D365 F&O reporting and Peppol e-invoicing show only "Dynamics 365 7 min read", with no author name or credentials shown.	Add bylines with bios, roles, Microsoft certifications and LinkedIn links. Add reviewer and last-updated dates.
High	Content	Partner status and certifications not evidenced	Pages list many Dynamics 365, Copilot and Azure OpenAI services, but the text shows no Microsoft partner badge, competencies or certifications.	Show Microsoft partner badges, designations, MCT/consultant certifications and links to the Partner directory listing in the footer and on the Why Tech& page.
High	Content	Case study and testimonial proof is thin	The homepage teaser "How Customer Service Teams Are Achieving a 20% to 45% Increase in..." is cut off. No client names, quotes or reviews appear in the supplied t	Publish named case studies with the client, scope, timeline and measurable results. Add testimonials, logos and third-party reviews such as Clutch or Gartner Pe
High	AI Search Readiness	ChatGPT recommends other providers, not you	For "I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with..." ChatGPT listed: nxaara.com, nexa	Get listed where AI answers draw their sources: the directories and review sites cited (nxaara.com, nexait.ai, oracle.com, ae.scc.com), plus comparison and "bes
High	AI Search Readiness	Perplexity recommends other providers, not you	For "I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with..." Perplexity listed: g42.ai, core4	Get listed where AI answers draw their sources: the directories and review sites cited (f6s.com, lumitech.co, sortlist.com, revenuebase.ai), plus comparison and
High	AI Search Readiness	Site does not rank at the top for its own brand name	Search "tech&": your position not in top 20	Use one consistent brand name everywhere, add Organization schema with alternateName, and build branded mentions.
High	Crawlability & Indexing	Far fewer pages indexed than crawled	Google site: estimate 5 vs 88 crawlable pages	Check Search Console → Pages for "Crawled – currently not indexed" and "Discovered – currently not indexed", improve internal links and content depth on those U
High	Content	Articles have no named human author	1 of 2 sampled articles	Credit a named expert with an author page, job title and LinkedIn (author @type Person with url and sameAs).

High	Crawlability & Indexing	Broken internal links	21 broken links across 88 pages	Export the broken links from the crawl and fix or remove each one; redirect deleted pages to the closest live page.
Medium	Schema	Few verified profiles linked in sameAs	1 valid sameAs link(s): https://www.linkedin.com/company/techand.ai/ , https://twitter.com/Tech&	Add sameAs links to every real business profile.
Medium	Content	No freshness signal on articles	2 of 2 sampled articles have dateModified equal to datePublished	Show a visible "Last reviewed" date and update dateModified only after real edits.
Medium	Technical	Contact form is not in the server-rendered HTML	Raw HTML of https://www.techand.ai/contact-us has 0 <form> and 0 <input> elements	Server-render the form (fields, labels, submit) or add a <noscript> fallback form.
Medium	Images	Images without width and height	76 images on sampled pages lack explicit dimensions	Add width and height (or aspect-ratio) to every image.
Medium	Authority	No rankings for informational searches	You rank for 0 informational keywords vs competitor median 234	Publish helpful guides that answer the questions buyers research.
Medium	Authority	High backlink spam score	Spam score 63	Review the worst linking domains and disavow clearly manipulative ones.
Medium	Performance	Largest Contentful Paint needs improvement (2.5-4s)	3481ms (lab test, mobile)	Speed up server response and optimise the main image.
Medium	Content	Service pages are generic and duplicated	Every /whatwedo/ page repeats the same block: "We guide your business through every step of technology growth to ensure you stay ahead of competitors." Each pag	Write unique content per service: scope, methodology, typical timelines, deliverables, SLAs, a UAE example and FAQs.
Medium	Content	Heavy repeated mega-menu text on every page	The same navigation text, from "Dynamics 365 Finance & Operations UAE" to "Predictive Models", precedes the content on every page.	Make the navigation less crawlable-noisy and the labels more natural. Make sure the main content comes first in the HTML.
Medium	Content	Formatting artifact and only high-commitment CTAs	"Let's discuss how we can help your business" shows a raw HTML entity. CTAs are "Talk to Sales", "Call Now" and "Request Call Back".	Fix the entity encoding. Add softer CTAs such as a readiness checklist, a sample scope or a downloadable guide.
Medium	Content	No pricing or engagement-model guidance, and no leadership info	No pricing, ranges, or engagement models appear on the service pages. No team or leadership names appear in the text supplied.	Add indicative engagement models, typical project durations, and a leadership/team page.

Medium	AI Search Readiness	Google shows AI Overviews for your key searches and does not cite you	"ai in cloud" (cited: aicloud.core42.ai, asus.com, youtube.com, nebius.com)	Add a concise answer block, question-style headings and primary-source citations on the relevant pages.
Medium	Crawlability & Indexing	Missing canonical tag	45 of 88 crawled pages	Add a self-referencing canonical tag to every indexable page.
Medium	Crawlability & Indexing	Canonical points to a redirect	42 of 88 crawled pages	Point canonicals directly to the final URL.
Medium	Performance	No compression (gzip/brotli)	42 of 88 crawled pages	Enable gzip or brotli compression on the server.
Medium	Performance	Large image payload (over 1MB of images per page)	33 of 88 crawled pages	Compress images, serve WebP/AVIF, and make sure image CDN optimisation (e.g. f_auto,q_auto) is on.
Medium	Schema	Structured data with errors	12 of 88 crawled pages	Validate each page in the Rich Results Test / Schema Markup Validator and fix the reported properties.
Medium	On-Page	Brand name repeated in the title	1 of 88 crawled pages	Fix the title template so the brand appears once.
Low	Security	Missing security headers	Not found: Content-Security-Policy	Add the missing headers in your hosting or CDN configuration; start Content-Security-Policy in Report-Only mode.
Low	Schema	No alternateName in Organization schema	alternateName missing	Add alternateName with common variants of the brand name.
Low	Schema	Organization schema has no @id	@id missing	Add "@id": "https://techand.ai/#organization".
Low	On-Page	Homepage H1 does not name the business or what it does	Homepage H1: "Value Driven Innovation through Automation"	Rewrite the H1 to name the brand, the service category and the location.
Low	Content	Phone number in a link but not shown on the page	8006427676 (https://www.techand.ai/); 8006427676 (https://www.techand.ai/contact-us)	Confirm the number, display it, or remove the link.
Low	Performance	No speculation rules for instant navigation	No <script type="speculationrules"> found	Add conservative speculation rules for key pages (e.g. contact and top services).

Low	Authority	Low-difficulty keywords you can target now	"ai in cloud" (27100/mo, difficulty 12), "ai on cloud" (27100/mo, difficulty 10), "cloud's ai" (27100/mo, difficulty 28), "blackbox ai" (14800/mo, difficulty 15)	Build or optimise one focused page per keyword group, starting with the highest volume.
Low	Performance	Top speed-up opportunities	Avoid multiple page redirects (~780ms)	Work through these in order of estimated savings.
Low	On-Page	Missing Open Graph tags	45 of 88 crawled pages	Add og:title, og:description and og:image.
Low	On-Page	URLs are not SEO-friendly	43 of 88 crawled pages	Use short, lowercase, hyphenated, descriptive URLs.
Low	Crawlability & Indexing	Internal URLs that redirect	42 of 88 crawled pages	Link directly to the final URL.
Low	Crawlability & Indexing	Links pointing to redirects	42 of 88 crawled pages	Update links to point to the final URL.
Low	On-Page	Title shorter than 30 characters	12 of 88 crawled pages	Expand to 50-60 characters with the main keyword and benefit.
Low	On-Page	Long or messy URLs	9 of 88 crawled pages	Use short, lowercase, hyphenated URLs.
Low	Crawlability & Indexing	Few internal links pointing to the page	6 of 88 crawled pages	Add contextual links from related pages with descriptive anchor text.
Low	Content	Light content (300-500 words)	1 of 88 crawled pages	Expand with answers to the questions buyers ask before contacting you.
Info	On-Page	Missing Twitter/X card tags	45 of 88 crawled pages	Add twitter:card and twitter:image tags.
Info	Crawlability & Indexing	Canonical points to a different page	43 of 88 crawled pages	Confirm each of these is intentional.
Info	On-Page	H1 is identical to the title tag	2 of 88 crawled pages	Optional: vary the H1 wording from the title.

Delivered: Full SEO Report (PDF + Word, 443 KB e-mail) → e-mailed and posted to the callback (stage full_report)

DELIVERED REPORT: FULL-SEO-REPORT-TECHAND-AI-2026-10-01.DOC

FULL SEO REPORT

techand.ai

United Arab Emirates • 88 pages crawled • 2026-10-01

64/100

Needs Improvement
SEO Health Score

Contents

1. Executive Summary
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1. Executive Summary

About the site: Tech& is a Dubai-based (DAFZA) Microsoft Dynamics 365, Business Central, Power Platform and AI/Copilot consultancy serving GCC enterprises and banks, the public sector, retail, hospitality and other industries.

The main thing holding this site back: The site makes broad capability and scale claims but gives no verifiable proof, such as named experts, Microsoft credentials, named client case studies or reviews, so buyers have little reason to trust it.

Critical	High	Medium	Low	Info
1	22	18	16	3

Score limit applied: 1 critical issue found — score capped at 70 until fixed. Calculated score before the limit: 64/100.

Key Numbers: You vs Competitors

Metric	You	Competitor median
Keywords ranking (top 30)	0	1,358
Keywords in top 10	0	381
Estimated monthly organic visits	0	9,897
Referring domains	13	8,861.5
Mobile PageSpeed	89	64

Domain age (years)	0.6	14.2
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Fix These First

- 1. Broken pages (4xx, 5xx or timeout)** (3 of 88 crawled pages)
Fix or 301-redirect broken URLs to the most relevant live page.
- 2. www and non-www versions are not consolidated** (Crawler canonicalization test failed)
Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.
- 3. HTML cannot be cached by the CDN (no-store / private)** (Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS)
Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can cache HTML. Verify: repeat requests should show a cache HIT.
- 4. Malformed sameAs profile link** (Invalid: <https://twitter.com/Tech&>)
Remove or correct the invalid entry and link only real, active profiles.
- 5. Structured contact details are thinner than what the site shows** (Missing in schema: streetAddress, addressLocality, telephone, email)
Add full PostalAddress (streetAddress, addressLocality, addressRegion, postalCode, addressCountry), telephone and email to Organization schema.
- 6. Experience claim conflicts with the founding date** ("20+ years" on <https://www.techand.ai/> but schema foundingDate is 2020)
Make the claims consistent (e.g. "leadership with 20+ years of experience") or correct the founding date.
- 7. Service pages have no Service schema** (5 of 5 sampled service pages)
Add Service JSON-LD (serviceType, provider → Organization @id, areaServed) on each service page.

2. Scorecard

Category	Score	Weight	
Crawlability & Indexing	34	15%	
On-Page	88	14%	
Content	65	16%	
Schema	47	12%	
Performance	66	13%	
Security	97	7%	

Images	93	7%	
Technical	93	6%	
AI Search Readiness	60	10%	
Authority	44	12%	

3. Competitor Benchmark

Competitors: auto-detected from Google results for: ai & automation, assess & envision, build & implement. Country: United Arab Emirates.

Domain	Keywords	Top 10	Est. visits/mo	Traffic value \$	Ref. domains	Domain rank	PageSpeed	Age (yrs)
techand.ai (you)	0	0	0	0	13	0	89	0.6
theknowledgeacademy.com	300	115	9,897	31,618	9,917	377	64	17.6
youghish.com	300	22	28,492	16,964	7,806	591	71	10.7
boardroom-ai.ae	0	0	1	0	N/A	N/A	41	N/A

4. Keywords by Search Intent

Keyword research based on your services: ai & automation, assess & envision, build & implement

Current Rankings per Intent (top 30): You vs Competitors

Intent	techand.ai (you)	theknowledgeacademy.com	youghish.com	boardroom-ai.ae
Informational	0	234	295	0
Navigational	0	29	1	0
Commercial	0	32	0	0
Transactional	0	5	4	0

Your site does not rank in the top 30 for any keyword in United Arab Emirates yet. The opportunities below show which searches to target, grouped by search intent.

4a. Keyword Opportunities by Search Intent

Informational Keywords

People researching or learning (how, what, guide)

Keyword	Searches/mo	Difficulty	CPC \$	Your position	Competitors in top 10
ai	165,000	72 (very hard)	0.49	—	—
character ai	110,000	42 (medium)	0.07	—	—
ai mode	49,500	42 (medium)	0.48	—	—
humanize ai	40,500	53 (hard)	0.48	—	—
google ai	40,500	79 (very hard)	1.44	—	—
cloud's ai	27,100	28 (easy)	3.1	—	—
meta ai	22,200	52 (hard)	0.89	—	—
ai humanizer	18,100	48 (medium)	0.57	—	—
blackbox ai	14,800	15 (easy)	1.65	—	—
vertically	9,900	4 (easy)	N/A	—	youghish.com #9

Navigational Keywords

People looking for a specific brand or website

Keyword	Searches/mo	Difficulty	CPC \$	Your position	Competitors in top 10
console google cloud platform	4,400	51 (hard)	0.36000001430511475	—	theknowledgeacademy.com #9
runway ai	1,900	41 (medium)	2.71	—	—
teacher training institute ajman	1,900	0 (easy)	1.4800000190734863	—	theknowledgeacademy.com #10
the knowledge academy	1,300	8 (easy)	2.119999885559082	—	theknowledgeacademy.com #1
rockwell automation	590	52 (hard)	2.5	—	—
ai academy	590	23 (easy)	4	—	—
ai clothes changer	590	17 (easy)	0.72	—	—
brand of makeup cosmetics	590	0 (easy)	0.5699999928474426	—	theknowledgeacademy.com #10
near me english language course	590	7 (easy)	2.8299999237060547	—	theknowledgeacademy.com #9
trading center near me	590	0 (easy)	0.20000000298023224	—	theknowledgeacademy.com #1

Commercial Keywords

People comparing providers before deciding (best, services, company, vs)

Keyword	Searches/mo	Difficulty	CPC \$	Your position	Competitors in top 10
ai in cloud	27,100	12 (easy)	3.1	—	—

ai on cloud	27,100	10 (easy)	3.1	—	—
flow ai	27,100	35 (medium)	0.92	—	—
typewriting centre near me	6,600	0 (easy)	0.4000000059604645	—	theknowledgeacademy.com #6
presentation ai	2,900	58 (hard)	3.23	—	—
ai tools	2,400	58 (hard)	7.77	—	—
microsoft ai	1,900	49 (medium)	2.75	—	—
ai agent	1,900	51 (hard)	8.22	—	—
building automation systems	1,300	26 (easy)	0.57	—	—
articulate course	1,000	0 (easy)	4.739999771118164	—	theknowledgeacademy.com #5

Transactional Keywords

People ready to act (price, cost, hire, buy, quote)

Keyword	Searches/mo	Difficulty	CPC \$	Your position	Competitors in top 10
ai image generator	12,100	66 (hard)	0.62	—	—
ai video generator free	6,600	47 (medium)	0.71	—	—
face swap ai	6,600	26 (easy)	0.63	—	—
magic light ai	5,400	8 (easy)	0.17	—	—
ai photo generator	4,400	49 (medium)	0.69	—	—
ai image generator free	2,400	66 (hard)	0.51	—	—
ai image editor	2,400	46 (medium)	0.75	—	—
free ai image generator	2,400	52 (hard)	0.66	—	—
ai picture generator	1,900	61 (hard)	0.62	—	—
powerpoint ai	1,300	30 (medium)	3.02	—	—

Easy Wins (difficulty under 30)

Keyword	Intent	Searches/mo	Difficulty	Competitors in top 10
ai in cloud	Commercial	27,100	12	—
ai on cloud	Commercial	27,100	10	—
cloud's ai	Informational	27,100	28	—
blackbox ai	Informational	14,800	15	—
vertically	Informational	9,900	4	youglish.com #9
employees	Informational	8,100	12	theknowledgeacademy.com #6
face swap ai	Transactional	6,600	26	—

typewriting centre near me	Commercial	6,600	0	theknowledgeacademy.com #6
closes	Informational	6,600	14	youglish.com #9
magic light ai	Transactional	5,400	8	—

Keyword Gap: Competitors Rank, You Don't

Keyword	Searches/mo	Difficulty	Intent	Competitors ranking
vertically	9,900	4 (easy)	Informational	youglish.com #9
employees	8,100	12 (easy)	Informational	theknowledgeacademy.com #6
typewriting centre near me	6,600	0 (easy)	Commercial	theknowledgeacademy.com #6
closes	6,600	14 (easy)	Informational	youglish.com #9
console google cloud platform	4,400	51 (hard)	Navigational	theknowledgeacademy.com #9
networks	4,400	44 (medium)	Informational	theknowledgeacademy.com #9
pmp certification	4,400	70 (very hard)	Informational	theknowledgeacademy.com #9
pronunciate	4,400	0 (easy)	Informational	youglish.com #9
pronunciation	4,400	37 (medium)	Informational	youglish.com #6
pmp	3,600	11 (easy)	Informational	theknowledgeacademy.com #6
training	3,600	18 (easy)	Informational	theknowledgeacademy.com #4
rumbled	3,600	0 (easy)	Informational	youglish.com #8
worked	3,600	3 (easy)	Informational	youglish.com #9
supervisors	2,900	7 (easy)	Informational	theknowledgeacademy.com #8
trapezium area	2,900	0 (easy)	Informational	theknowledgeacademy.com #7
consciously	2,900	2 (easy)	Informational	youglish.com #6
exploring	2,900	17 (easy)	Informational	youglish.com #10
what doing	2,900	0 (easy)	Informational	youglish.com #9
curriculums	2,400	4 (easy)	Informational	theknowledgeacademy.com #5
differentiating	2,400	9 (easy)	Informational	youglish.com #8

5. Authority & Backlinks

Domain	Domain rank	Backlinks	Referring domains	Followed %	Spam score
techand.ai (you)	0	13	13	100%	63
theknowledgeacademy.com	377	218,676	9,917	64%	58
youglish.com	591	845,464	7,806	39%	5
boardroom-ai.ae	N/A	N/A	N/A	N/A	N/A

Findings

High The site does not rank in Google's top 10 for any keyword

Whole website

Evidence: Top-10 rankings in United Arab Emirates: techand.ai 0 vs competitor median 381 (theknowledgeacademy.com, youglish.com, boardroom-ai.ae)

Why it matters: Without top-10 rankings the site gets almost no organic traffic, however good the pages are.

How to fix: Build authority first (partner listings, directories, reviews, mentions) and target long-tail keywords where competitors are weak.

High Organic traffic is under 10% of competitors

Whole website

Evidence: Estimated monthly organic visits: techand.ai 0 vs competitor median 9,897

Why it matters: Most searchers in this market are going to competitors.

How to fix: Close the keyword and backlink gaps shown in this report.

High Very young domain competing with long-established sites

Whole website

Evidence: techand.ai registered 2026-02-19 (0.6 years) vs competitor median 14.2 years

Why it matters: New domains have little accumulated trust; on-page fixes alone rarely move competitive terms in the first year.

How to fix: Invest in off-site entity building (partner directories, reviews, mentions) and target long-tail, lower-difficulty searches first.

High Keyword gap: searches competitors win and you do not

Whole website

Evidence: 25 keywords where competitors rank in the top 10 and you are not in the top 20. Top: "vertically" (9900/mo), "employees" (8100/mo), "typewriting centre near me" (6600/mo), "closes" (6600/mo), "console google cloud platform" (4400/mo)

Why it matters: Each of these is proven demand in your market that is currently going to competitors.

How to fix: Create or improve a dedicated page for the highest-volume, lowest-difficulty gap keywords first.

High Far fewer referring domains than competitors

Whole website

Evidence: 13 referring domains vs competitor median 8862

Why it matters: Links from other websites are one of the strongest ranking signals; competitors are far ahead.

How to fix: Earn links from partner directories, industry associations, suppliers, press and guest content.

Medium No rankings for informational searches

Whole website

Evidence: You rank for 0 informational keywords vs competitor median 234

Why it matters: You are missing a whole stage of the buyer journey that competitors capture.

How to fix: Publish helpful guides that answer the questions buyers research.

Medium High backlink spam score

Whole website

Evidence: Spam score 63

Why it matters: A high share of low-quality links can hold rankings back.

How to fix: Review the worst linking domains and disavow clearly manipulative ones.

Low Low-difficulty keywords you can target now

Whole website

Evidence: "ai in cloud" (27100/mo, difficulty 12), "ai on cloud" (27100/mo, difficulty 10), "cloud's ai" (27100/mo, difficulty 28), "blackbox ai" (14800/mo, difficulty 15), "vertically" (9900/mo, difficulty 4)

Why it matters: These searches have real demand and weak competition, the best starting point for a young site.

How to fix: Build or optimise one focused page per keyword group, starting with the highest volume.

6. Content Quality & E-E-A-T

Trust Signals on Your Site

named authors	no
case studies	partial
testimonials reviews	no
credentials certifications	no
pricing transparency	no
team or leadership info	no
clear contact details	yes

Competitors' Trust Signals

Competitor	Authors	Case studies	Testimonials	Pricing	Depth	Note
theknowledgeacademy.com	unclear	no	unclear	unclear	moderate	Gives concrete delivery-format detail and stats (90-day access, trainers with 10+ years' experience), but it is a UK training business and not a direct competitor.
youghish.com	no	no	no	unclear	thin	An unrelated pronunciation tool, not a relevant competitor. It is useful only as a reminder of how clear its single-purpose messaging is.
boardroom-ai.ae	no	no	no	unclear	moderate	A UAE AI vendor with a free 7-day trial and a low-friction CTA, specific metrics, a step-by-step process and a compliance disclaimer. Its CTAs and process clarity beat Tech&'s, though it also lacks named proof.

Findings

High Experience claim conflicts with the founding date
Whole website

Evidence: "20+ years" on <https://www.techand.ai/> but schema foundingDate is 2020

Why it matters: Contradictory facts damage trust with buyers and with search quality systems.

How to fix: Make the claims consistent (e.g. "leadership with 20+ years of experience") or correct the founding date.

Affected pages:

<https://www.techand.ai/>

High Unproven scale and financial claims on contact page

Whole website

Evidence: "We bring together a massive team of experts and significant financial backing to ensure your project succeeds on time."

Why it matters: The claim is vague and unverifiable. It gives no headcount, funding source or delivery record. Readers and quality raters read this as filler, and it weakens trust.

How to fix: Replace it with verifiable facts: team size, years operating, number of projects delivered, and named leadership. Add a link to proof.

Affected pages:

<https://www.techand.ai/contact-us>

High No named authors or expert bylines on insight articles

Whole website

Evidence: The articles on D365 F&O reporting and Peppol e-invoicing show only "Dynamics 365 7 min read", with no author name or credentials shown.

Why it matters: Technical and finance-compliance content (UAE e-invoicing, consolidation) is YMYL-adjacent. Without a named expert, the experience and expertise signals are weak.

How to fix: Add bylines with bios, roles, Microsoft certifications and LinkedIn links. Add reviewer and last-updated dates.

Affected pages:

<https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel>

High Partner status and certifications not evidenced

Whole website

Evidence: Pages list many Dynamics 365, Copilot and Azure OpenAI services, but the text shows no Microsoft partner badge, competencies or certifications.

Why it matters: Claims to deliver Microsoft products need proof such as Solutions Partner designation, to be credible to enterprise and banking buyers.

How to fix: Show Microsoft partner badges, designations, MCT/consultant certifications and links to the Partner directory listing in the footer and on the Why Tech& page.

Affected pages:

<https://www.techand.ai/>

High Case study and testimonial proof is thin

Whole website

Evidence: The homepage teaser "How Customer Service Teams Are Achieving a 20% to 45% Increase in..." is cut off. No client names, quotes or reviews appear in the supplied text.

Why it matters: A percentage without a named client or source reads as unsubstantiated. Enterprise buyers in the UAE look for reference clients.

How to fix: Publish named case studies with the client, scope, timeline and measurable results. Add testimonials, logos and third-party reviews such as Clutch or Gartner Peer Insights.

Affected pages:

<https://www.techand.ai/>

High Articles have no named human author

1 of 88 pages (50%)

Evidence: 1 of 2 sampled articles

Why it matters: Expertise signals (E-E-A-T) depend on a real, credentialed author, especially on advice or compliance topics.

How to fix: Credit a named expert with an author page, job title and LinkedIn (author @type Person with url and sameAs).

Affected pages:

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>

Medium No freshness signal on articles

2 of 88 pages (100%)

Evidence: 2 of 2 sampled articles have dateModified equal to datePublished

Why it matters: Time-sensitive content looks unmaintained without a genuine update date.

How to fix: Show a visible "Last reviewed" date and update dateModified only after real edits.

Affected pages:

<https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for>

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>

Medium Service pages are generic and duplicated

Whole website

Evidence: Every /whatwedo/ page repeats the same block: "We guide your business through every step of technology growth to ensure you stay ahead of competitors." Each page also carries the same "Assess & Envision" copy. Managed Services is described as "Expert IT management for your daily needs".

Why it matters: Duplicated boilerplate and generic descriptions give each page little unique value. The pages lack timelines, deliverables and UAE specifics.

How to fix: Write unique content per service: scope, methodology, typical timelines, deliverables, SLAs, a UAE example and FAQs.

Affected pages:

<https://www.techand.ai/whatwedo/managed-services>

Medium Heavy repeated mega-menu text on every page

Whole website

Evidence: The same navigation text, from "Dynamics 365 Finance & Operations UAE" to "Predictive Models", precedes the content on every page.

Why it matters: It dilutes the unique content, and the menu shows up as page text. Menu labels also look keyword-stuffed ("...UAE" repeated).

How to fix: Make the navigation less crawlable-noisy and the labels more natural. Make sure the main content comes first in the HTML.

Affected pages:

<https://www.techand.ai/contact-us>

Medium Formatting artifact and only high-commitment CTAs

Whole website

Evidence: "Let's discuss how we can help your business" shows a raw HTML entity. CTAs are "Talk to Sales", "Call Now" and "Request Call Back".

Why it matters: The artifact looks unpolished. There is no low-friction option such as a guide, assessment or pricing guidance.

How to fix: Fix the entity encoding. Add softer CTAs such as a readiness checklist, a sample scope or a downloadable guide.

Affected pages:

<https://www.techand.ai/contact-us>

Medium No pricing or engagement-model guidance, and no leadership info

Whole website

Evidence: No pricing, ranges, or engagement models appear on the service pages. No team or leadership names appear in the text supplied.

Why it matters: Buyers can't qualify the cost. Anonymity lowers trust.

How to fix: Add indicative engagement models, typical project durations, and a leadership/team page.

Affected pages:

<https://www.techand.ai/whatwedo/build-implement>

Low Phone number in a link but not shown on the page

Whole website

Evidence: 8006427676 (<https://www.techand.ai/>); 8006427676 (<https://www.techand.ai/contact-us>)

Why it matters: Hidden or unlabeled numbers can be picked up by directories and AI as conflicting contact details.

How to fix: Confirm the number, display it, or remove the link.

Low Light content (300-500 words)

1 of 88 pages (1%)

Evidence: 1 of 88 crawled pages

Why it matters: Commercial pages usually need more depth to compete.

How to fix: Expand with answers to the questions buyers ask before contacting you.

Affected pages:

<https://www.techand.ai/contact-us> — 334 words

7. Performance & Core Web Vitals

Domain	Mobile score	LCP	CLS	INP (real users)	TBT	Server response
techand.ai (you)	89	3.48s	0.00	N/A	58ms	168ms
theknowledgeacademy.com	64	1.64s	0.16	162ms	11ms	3ms
youghish.com	71	1.68s	0.05	194ms	442ms	2ms
boardroom-ai.ae	41	18.41s	0.00	N/A	635ms	9ms

Biggest Speed Savings (your homepage)

- Avoid multiple page redirects — about 780ms

Findings

High HTML cannot be cached by the CDN (no-store / private)

Whole website

Evidence: Cache-Control: private, no-cache, no-store, max-age=0, must-revalidate; CDN cache status on two requests: MISS / MISS

Why it matters: Every visit forces the server to rebuild the page from scratch, which slows the first byte for every user and hurts Core Web Vitals. This is sometimes intentional (carts, logged-in areas) — confirm before changing.

How to fix: Remove the setting that forces dynamic rendering (for Next.js: force-dynamic, or cookies()/headers() in server components) or enable revalidation so the CDN can cache HTML. Verify: repeat requests should show a cache HIT.

Medium Largest Contentful Paint needs improvement (2.5-4s)

Whole website

Evidence: 3481ms (lab test, mobile)

Why it matters: Google rates LCP above 2.5 seconds as Needs Improvement.

How to fix: Speed up server response and optimise the main image.

Affected pages:

<https://techand.ai/>

Medium No compression (gzip/brotli)

42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Uncompressed pages take longer to download.

How to fix: Enable gzip or brotli compression on the server.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>

<https://techand.ai/business-central-implementation-uae>

<https://techand.ai/business-central-post-go-live-support-uae>

<https://techand.ai/capabilities>

<https://techand.ai/capabilities/business-applications>

<https://techand.ai/capabilities/cloud-services>

<https://techand.ai/capabilities/data>

<https://techand.ai/contact-us>

<https://techand.ai/cookies>

<https://techand.ai/d365-finance-operations-implementation-rollout-uae>

...and 32 more

Medium Large image payload (over 1MB of images per page)

33 of 88 pages (38%)

Evidence: 33 of 88 crawled pages

Why it matters: Large images are the most common cause of slow mobile pages. The figure is the total size of image files the page references (including responsive sizes), so the real download may be smaller, but it signals unoptimised images.

How to fix: Compress images, serve WebP/AVIF, and make sure image CDN optimisation (e.g. f_auto,q_auto) is on.

Affected pages:

<https://www.techand.ai/> — 3.5MB of image files referenced

<https://www.techand.ai/business-central-implementation-uae> — 3.4MB of image files referenced

<https://www.techand.ai/business-central-post-go-live-support-uae> — 1.1MB of image files referenced

<https://www.techand.ai/capabilities/business-applications> — 1.8MB of image files referenced

<https://www.techand.ai/d365-finance-operations-implementation-rollout-uae> — 31.8MB of image files referenced

<https://www.techand.ai/dynamics-365-business-central-uae> — 9.0MB of image files referenced

<https://www.techand.ai/dynamics-365-cloud-migration-uae> — 31.2MB of image files referenced

<https://www.techand.ai/dynamics-365-crm-uae> — 4.3MB of image files referenced

<https://www.techand.ai/dynamics-365-finance-and-operations-uae> — 40.0MB of image files referenced

<https://www.techand.ai/dynamics-365-managed-services-uae> — 15.7MB of image files referenced

...and 23 more

Low No speculation rules for instant navigation

Whole website

Evidence: No <script type="speculationrules"> found

Why it matters: Prefetching likely next pages makes navigation feel instant.

How to fix: Add conservative speculation rules for key pages (e.g. contact and top services).

Low Top speed-up opportunities

Whole website

Evidence: Avoid multiple page redirects (~780ms)

Why it matters: Google lists these as the biggest savings available.

How to fix: Work through these in order of estimated savings.

Affected pages:

<https://techand.ai/>

8. AI Search Visibility

What AI Assistants Say (live answers)

Assistant	Question	Mentions you	Providers named	Sources cited
ChatGPT (web search on)	What is Tech& (techand.ai)? What does the company do, where is it based and who is it for? Answer in 3 sentences. If you cannot find it, say exactly: I could not find this company.	Yes	techand.ai	techand.ai
ChatGPT (web search on)	I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with their website domains, best first, one line each.	No	nxaara.com, nexait.ai, oracle.com, ae.scc.com, cloud.netcity.ae, themiddleeast.cloud, itpro.com	nxaara.com, nexait.ai, oracle.com, ae.scc.com, cloud.netcity.ae, themiddleeast.cloud
Perplexity	I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with their website domains, best first, one line each.	No	g42.ai, core42.ai, presight.ai, aiqintelligence.ae, ai71.ai, nodeshift.cloud, cloudstation.ai, smartuniverse.ai	f6s.com, lumitech.co, sortlist.com, revenuebase.ai

ChatGPT (web search on) — about the company:

Tech& is a technology consulting firm based in Dubai, United Arab Emirates, specializing in enterprise automation and digital transformation solutions. They offer services such as AI and automation, data analytics, cloud services, and cybersecurity to organizations across various industries, including banking, retail, and public sector. Their mission is to empower enterprises in the GCC region to achieve measurable business impact through cutting-edge technology. ([techand.ai])(https://techand.ai/?utm_source=openai)

ChatGPT (web search on) — recommendation:

Here are eight leading AI cloud service providers in the United Arab Emirates, listed in order of prominence: 1. ****NXAARA AI Cloud****: Offers GPU compute, model fine-tuning, and inference services, with a focus on Arabic and multilingual workloads. ([nxaara.com])(https://www.nxaara.com/?utm_source=openai) 2. ****NexaAI****: Provides private AI, GPU compute, and AI infrastructure solutions, emphasizing data sovereignty and compliance with UAE regulations. ([nexait.ai])(https://nexait.ai/?utm_source=openai) 3. ****du****: A major telecom operator that has deployed Oracle Alloy to offer sovereign cloud capabilities, including AI services, to the UAE public sector. ([oracle.com])(<https://www.oracle.com...>)

Perplexity — recommendation:

1. ****G42**** — Sovereign AI infrastructure, cloud, and sector solutions; ****g42.ai****. [1][2] 2. ****Core42**** — UAE-based sovereign cloud, AI infrastructure, and GPU computing; ****core42.ai****. [4] 3. ****Presight**** — Enterprise and government AI analytics platforms; ****presight.ai****. [4] 4. ****AIQ**** — AI solutions for energy and industrial operations; ****aiqintelligence.ae****. [4] 5. ****AI71**** — Enterprise generative-AI platforms and applications; ****ai71.ai****. [4] 6. ****NodeShift**** — Sovereign cloud and AI infrastructure for regulated organizations;

nodeshift.cloud.^{[6][8]} 7. **CloudStation** — AI-agent and application deployment across cloud environments; **cloudstation.ai**.^[1] 8. **Smart Uni...**

Google index estimate (site: search): 5 URLs vs 88 pages crawled. Confirm the exact figure in Search Console.

Google Results & AI Overviews

Search	AI Overview	You cited	Competitors cited	Your position	Competitor positions
tech& (brand)	Yes	No	—	Not in top 20	theknowledgeacademy.com: — youglish.com: — boardroom-ai.ae: —
best ai in cloud company united arab emirates (category)	No	No	—	Not in top 20	theknowledgeacademy.com: — youglish.com: — boardroom-ai.ae: —
ai in cloud (top keyword)	Yes	No	—	Not in top 20	theknowledgeacademy.com: — youglish.com: — boardroom-ai.ae: —

AI assistant checks are live answers from ChatGPT (web search on) and Perplexity via the DataForSEO LLM Responses API; each run may differ slightly. Google AI Overview data comes from a live search.

Findings

High ChatGPT recommends other providers, not you

Whole website

Evidence: For "I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with..." ChatGPT listed: nxaara.com, nexait.ai, oracle.com, ae.scc.com, cloud.netcity.ae, themiddleeast.cloud

Why it matters: Buyers increasingly ask AI assistants for shortlists; not being named means not being considered.

How to fix: Get listed where AI answers draw their sources: the directories and review sites cited (nxaara.com, nexait.ai, oracle.com, ae.scc.com), plus comparison and "best X" pages that name you.

High Perplexity recommends other providers, not you

Whole website

Evidence: For "I need to choose a provider. Which are the best ai in cloud company United Arab Emirates? Give me up to 8 companies with..." Perplexity listed: g42.ai, core42.ai, presight.ai, aiqintelligence.ae, ai71.ai, nodeshift.cloud

Why it matters: Buyers increasingly ask AI assistants for shortlists; not being named means not being considered.

How to fix: Get listed where AI answers draw their sources: the directories and review sites cited (f6s.com, lumitech.co, sortlist.com, revenuebase.ai), plus comparison and "best X" pages that name you.

High Site does not rank at the top for its own brand name

Whole website

Evidence: Search "tech&": your position not in top 20

Why it matters: If people cannot find you by name, every other channel (ads, referrals, AI) leaks visitors.

How to fix: Use one consistent brand name everywhere, add Organization schema with alternateName, and build branded mentions.

Medium Google shows AI Overviews for your key searches and does not cite you

Whole website

Evidence: "ai in cloud" (cited: aicloud.core42.ai, asus.com, youtube.com, nebius.com)

Why it matters: These queries already have an AI answer; pages that are not cited lose visibility.

How to fix: Add a concise answer block, question-style headings and primary-source citations on the relevant pages.

9. Structured Data (Schema)

Score: 47/100

High Malformed sameAs profile link

Whole website

Evidence: Invalid: <https://twitter.com/Tech&>

Why it matters: A broken identity link is worse than none: it confuses entity matching in Google and AI systems.

How to fix: Remove or correct the invalid entry and link only real, active profiles.

High Structured contact details are thinner than what the site shows

Whole website

Evidence: Missing in schema: streetAddress, addressLocality, telephone, email

Why it matters: Search engines cannot match your business listing to your site when the address and phone in schema are incomplete.

How to fix: Add full PostalAddress (streetAddress, addressLocality, addressRegion, postalCode, addressCountry), telephone and email to Organization schema.

High Service pages have no Service schema

5 of 88 pages (100%)

Evidence: 5 of 5 sampled service pages

Why it matters: Service schema tells search engines exactly what you offer, where and by whom, and separates similar service pages.

How to fix: Add Service JSON-LD (serviceType, provider → Organization @id, areaServed) on each service page.

Affected pages:

<https://www.techand.ai/whatwedo/ai-automation>

<https://www.techand.ai/whatwedo/assess-envision>

<https://www.techand.ai/whatwedo/build-implement>

<https://www.techand.ai/whatwedo/global-rollouts>

<https://www.techand.ai/whatwedo/managed-services>

High No BreadcrumbList schema

9 of 88 pages (100%)

Evidence: 9 of 9 sampled inner pages

Why it matters: Breadcrumbs show site hierarchy in search results and help search engines understand structure.

How to fix: Add BreadcrumbList JSON-LD to every inner page.

Affected pages:

<https://www.techand.ai/contact-us>

<https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel>

<https://www.techand.ai/whatwedo/ai-automation>

<https://www.techand.ai/whatwedo/assess-envision>

<https://www.techand.ai/whatwedo/build-implement>

<https://www.techand.ai/whatwedo/global-rollouts>

<https://www.techand.ai/whatwedo/managed-services>

<https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for>

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>

Medium Few verified profiles linked in sameAs

Whole website

Evidence: 1 valid sameAs link(s): <https://www.linkedin.com/company/techand.ai/>, <https://twitter.com/Tech&>

Why it matters: Entity recognition works best with 4-6 independent profiles (LinkedIn, directories, Wikidata, Crunchbase, review sites).

How to fix: Add sameAs links to every real business profile.

Medium Structured data with errors

12 of 88 pages (14%)

Evidence: 12 of 88 crawled pages

Why it matters: Search engines ignore structured data blocks that fail validation, so the page loses rich-result eligibility.

How to fix: Validate each page in the Rich Results Test / Schema Markup Validator and fix the reported properties.

Affected pages:

<https://www.techand.ai/insights/agentic-ai>

<https://www.techand.ai/insights/autonomous-ai-customer-service>

<https://www.techand.ai/insights/data-sovereign-asset>

<https://www.techand.ai/insights/how-microsoft-365-copilot-is-changing-business-productivity>

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>

<https://www.techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap>

<https://www.techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs>

<https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel>

<https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for>

<https://www.techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide>

...and 2 more

Low No alternateName in Organization schema

Whole website

Evidence: alternateName missing

Why it matters: Helps search engines and AI map spelling variants of the brand to one entity.

How to fix: Add alternateName with common variants of the brand name.

Low Organization schema has no @id

Whole website

Evidence: @id missing

Why it matters: An @id lets other schema (Service, Article) reference the same organisation cleanly.

How to fix: Add "@id": "<https://techand.ai/#organization>".

Schema Found on Sampled Pages

Page	Type	Schema types
https://www.techand.ai/	home	Organization
https://www.techand.ai/contact-us	contact	Organization
https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel	about	Organization, BlogPosting
https://www.techand.ai/whatwedo/ai-automation	service	Organization
https://www.techand.ai/whatwedo/assess-envision	service	Organization
https://www.techand.ai/whatwedo/build-implement	service	Organization

https://www.techand.ai/whatwedo/global-rollouts	service	Organization
https://www.techand.ai/whatwedo/managed-services	service	Organization
https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for	article	Organization, BlogPosting, FAQPage
https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance	article	Organization, BlogPosting, FAQPage

10. Crawlability & Indexing

Score: 34/100

Critical Broken pages (4xx, 5xx or timeout)

3 of 88 pages (3%)

Evidence: 3 of 88 crawled pages

Why it matters: Pages that return errors waste crawl budget and lose any ranking signals pointing to them.

How to fix: Fix or 301-redirect broken URLs to the most relevant live page.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae>

<https://www.techand.ai/insights/dynamics-365-project-operations-uae>

<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae>

High www and non-www versions are not consolidated

Whole website

Evidence: Crawler canonicalization test failed

Why it matters: If both versions do not resolve to one canonical host, links and signals are split.

How to fix: Make the www version redirect (301) to the main domain with a valid certificate for both hostnames.

High Far fewer pages indexed than crawled

Whole website

Evidence: Google site: estimate 5 vs 88 crawlable pages

Why it matters: Pages that are not indexed cannot rank. This is an estimate; confirm in Search Console.

How to fix: Check Search Console → Pages for "Crawled – currently not indexed" and "Discovered – currently not indexed", improve internal links and content depth on those URLs.

High Broken internal links

Whole website

Evidence: 21 broken links across 88 pages

Why it matters: Links to pages that no longer exist waste crawl budget and send users to dead ends.

How to fix: Export the broken links from the crawl and fix or remove each one; redirect deleted pages to the closest live page.

Medium Missing canonical tag

45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Without a canonical, duplicate URL versions can compete with each other.

How to fix: Add a self-referencing canonical tag to every indexable page.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — No canonical

<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — No canonical
<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — No canonical
<https://techand.ai/ai-copilot-advisory-uae> — No canonical
<https://techand.ai/business-central-implementation-uae> — No canonical
<https://techand.ai/business-central-post-go-live-support-uae> — No canonical
<https://techand.ai/capabilities> — No canonical
<https://techand.ai/capabilities/business-applications> — No canonical
<https://techand.ai/capabilities/cloud-services> — No canonical
<https://techand.ai/capabilities/data> — No canonical
...and 35 more

Medium Canonical points to a redirect

42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Mixed signals slow down consolidation of ranking signals.

How to fix: Point canonicals directly to the final URL.

Affected pages:

<https://www.techand.ai/ai-copilot-advisory-uae>
<https://www.techand.ai/business-central-implementation-uae>
<https://www.techand.ai/business-central-post-go-live-support-uae>
<https://www.techand.ai/capabilities>
<https://www.techand.ai/capabilities/business-applications>
<https://www.techand.ai/capabilities/cloud-services>
<https://www.techand.ai/capabilities/data>
<https://www.techand.ai/contact-us>
<https://www.techand.ai/cookies>
<https://www.techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Low Internal URLs that redirect

42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Redirects add load time and dilute crawl efficiency.

How to fix: Link directly to the final URL.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>
<https://techand.ai/business-central-implementation-uae>
<https://techand.ai/business-central-post-go-live-support-uae>
<https://techand.ai/capabilities>
<https://techand.ai/capabilities/business-applications>
<https://techand.ai/capabilities/cloud-services>
<https://techand.ai/capabilities/data>
<https://techand.ai/contact-us>
<https://techand.ai/cookies>
<https://techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Low Links pointing to redirects

42 of 88 pages (48%)

Evidence: 42 of 88 crawled pages

Why it matters: Extra hops for users and crawlers.

How to fix: Update links to point to the final URL.

Affected pages:

<https://techand.ai/ai-copilot-advisory-uae>
<https://techand.ai/business-central-implementation-uae>

<https://techand.ai/business-central-post-go-live-support-uae>
<https://techand.ai/capabilities>
<https://techand.ai/capabilities/business-applications>
<https://techand.ai/capabilities/cloud-services>
<https://techand.ai/capabilities/data>
<https://techand.ai/contact-us>
<https://techand.ai/cookies>
<https://techand.ai/d365-finance-operations-implementation-rollout-uae>
...and 32 more

Low Few internal links pointing to the page
6 of 88 pages (7%)

Evidence: 6 of 88 crawled pages

Why it matters: Pages with few internal links look less important to search engines.

How to fix: Add contextual links from related pages with descriptive anchor text.

Affected pages:

<https://www.techand.ai/whatwedo/ai-automation> — 1 internal links in
<https://www.techand.ai/whatwedo/assess-envision> — 1 internal links in
<https://www.techand.ai/whatwedo/build-implement> — 1 internal links in
<https://www.techand.ai/whatwedo/global-rollouts> — 1 internal links in
<https://www.techand.ai/whatwedo/managed-services> — 1 internal links in
<https://www.techand.ai/whatwedo/run-optimize> — 1 internal links in

Info Canonical points to a different page
43 of 88 pages (49%)

Evidence: 43 of 88 crawled pages

Why it matters: Fine if intentional, but this page will not rank on its own.

How to fix: Confirm each of these is intentional.

Affected pages:

<https://www.techand.ai/> — → <https://techand.ai/>
<https://www.techand.ai/ai-copilot-advisory-uae> — → <https://techand.ai/ai-copilot-advisory-uae>
<https://www.techand.ai/business-central-implementation-uae> — → <https://techand.ai/business-central-implementation-uae>
<https://www.techand.ai/business-central-post-go-live-support-uae> — → <https://techand.ai/business-central-post-go-live-support-uae>
<https://www.techand.ai/capabilities> — → <https://techand.ai/capabilities>
<https://www.techand.ai/capabilities/business-applications> — → <https://techand.ai/capabilities/business-applications>
<https://www.techand.ai/capabilities/cloud-services> — → <https://techand.ai/capabilities/cloud-services>
<https://www.techand.ai/capabilities/data> — → <https://techand.ai/capabilities/data>
<https://www.techand.ai/contact-us> — → <https://techand.ai/contact-us>
<https://www.techand.ai/cookies> — → <https://techand.ai/cookies>
...and 33 more

11. On-Page SEO

Score: 88/100

Medium Brand name repeated in the title
1 of 88 pages (1%)

Evidence: 1 of 88 crawled pages

Why it matters: A doubled brand suffix wastes title space and usually means the template adds the brand twice.

How to fix: Fix the title template so the brand appears once.

Affected pages:

<https://www.techand.ai/power-platform-analytics-uae> — "Power Platform & Analytics Services in UAE | Tech& | Tech&"

Low Homepage H1 does not name the business or what it does

Whole website

Evidence: Homepage H1: "Value Driven Innovation through Automation"

Why it matters: The homepage H1 is the strongest on-page signal of who you are, what you sell and where.

How to fix: Rewrite the H1 to name the brand, the service category and the location.

Affected pages:

<https://www.techand.ai/>

Low Missing Open Graph tags

45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Without them, links shared on LinkedIn, WhatsApp or Facebook show a poor preview.

How to fix: Add og:title, og:description and og:image.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — Missing og:title, og:image

<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — Missing og:title, og:image

<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — Missing og:title, og:image

<https://techand.ai/ai-copilot-advisory-uae> — Missing og:title, og:image

<https://techand.ai/business-central-implementation-uae> — Missing og:title, og:image

<https://techand.ai/business-central-post-go-live-support-uae> — Missing og:title, og:image

<https://techand.ai/capabilities> — Missing og:title, og:image

<https://techand.ai/capabilities/business-applications> — Missing og:title, og:image

<https://techand.ai/capabilities/cloud-services> — Missing og:title, og:image

<https://techand.ai/capabilities/data> — Missing og:title, og:image

...and 35 more

Low URLs are not SEO-friendly

43 of 88 pages (49%)

Evidence: 43 of 88 crawled pages

Why it matters: Dynamic parameters, long paths or non-descriptive URLs are harder to read and share.

How to fix: Use short, lowercase, hyphenated, descriptive URLs.

Affected pages:

<https://www.techand.ai/>

<https://www.techand.ai/ai-copilot-advisory-uae>

<https://www.techand.ai/business-central-implementation-uae>

<https://www.techand.ai/business-central-post-go-live-support-uae>

<https://www.techand.ai/capabilities>

<https://www.techand.ai/capabilities/business-applications>

<https://www.techand.ai/capabilities/cloud-services>

<https://www.techand.ai/capabilities/data>

<https://www.techand.ai/contact-us>

<https://www.techand.ai/cookies>

...and 33 more

Low Title shorter than 30 characters

12 of 88 pages (14%)

Evidence: 12 of 88 crawled pages

Why it matters: Short titles waste space to include keywords and a reason to click.

How to fix: Expand to 50-60 characters with the main keyword and benefit.

Affected pages:

<https://www.techand.ai/capabilities> — 20 characters: "Capabilities | Tech&"

<https://www.techand.ai/capabilities/business-applications> — 29 characters: "Business Applications | Tech&"

<https://www.techand.ai/capabilities/cloud-services> — 22 characters: "Cloud Services | Tech&"

<https://www.techand.ai/capabilities/data> — 24 characters: "Data & Analytics | Tech&"

<https://www.techand.ai/contact-us> — 18 characters: "Contact Us | Tech&"
<https://www.techand.ai/cookies> — 21 characters: "Cookie Policy | Tech&"
<https://www.techand.ai/industries> — 27 characters: "Industries We Serve | Tech&"
<https://www.techand.ai/insights> — 16 characters: "Insights | Tech&"
<https://www.techand.ai/privacy> — 22 characters: "Privacy Policy | Tech&"
<https://www.techand.ai/security> — 23 characters: "Security Policy | Tech&"
...and 2 more

Low Long or messy URLs

9 of 88 pages (10%)

Evidence: 9 of 88 crawled pages

Why it matters: Long URLs, capitals or underscores are harder to read, share and remember.

How to fix: Use short, lowercase, hyphenated URLs.

Affected pages:

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance> — 105 characters
<https://www.techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap> — 134 characters
<https://www.techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs> — 101 characters
<https://www.techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide> — 114 characters
<https://www.techand.ai/insights/your-current-pdf-invoices-will-be-illegal-in-the-uae-by-2027-and-how-to-fix-it> — 110 characters
<https://techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance> — 101 characters
<https://techand.ai/insights/microsofts-saudi-arabia-east-datacenter-goes-live-in-november-2026-what-it-means-for-your-d365-roadmap> — 130 characters
<https://techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide> — 110 characters
<https://techand.ai/insights/your-current-pdf-invoices-will-be-illegal-in-the-uae-by-2027-and-how-to-fix-it> — 106 characters

Info Missing Twitter/X card tags

45 of 88 pages (51%)

Evidence: 45 of 88 crawled pages

Why it matters: Shared links on X show a plain preview.

How to fix: Add twitter:card and twitter:image tags.

Affected pages:

<https://www.techand.ai/capabilities/dynamics-365-project-operations-uae> — No twitter:card
<https://www.techand.ai/insights/dynamics-365-project-operations-uae> — No twitter:card
<https://www.techand.ai/whatwedo/dynamics-365-project-operations-uae> — No twitter:card
<https://techand.ai/ai-copilot-advisory-uae> — No twitter:card
<https://techand.ai/business-central-implementation-uae> — No twitter:card
<https://techand.ai/business-central-post-go-live-support-uae> — No twitter:card
<https://techand.ai/capabilities> — No twitter:card
<https://techand.ai/capabilities/business-applications> — No twitter:card
<https://techand.ai/capabilities/cloud-services> — No twitter:card
<https://techand.ai/capabilities/data> — No twitter:card
...and 35 more

Info H1 is identical to the title tag

2 of 88 pages (2%)

Evidence: 2 of 88 crawled pages

Why it matters: A slightly different H1 lets you target a second keyword variation.

How to fix: Optional: vary the H1 wording from the title.

Affected pages:

<https://www.techand.ai/insights/autonomous-ai-customer-service> — Both: "How Customer Services Are Achieving Productivity with AI"
<https://www.techand.ai/insights/how-microsoft-365-copilot-is-changing-business-productivity> — Both: "How Microsoft 365 Copilot Is Changing Business Productivity"

12. Images, Security & Technical

Images — 93/100

Medium Images without width and height

Whole website

Evidence: 76 images on sampled pages lack explicit dimensions

Why it matters: Images without reserved space can shift the layout while loading (CLS).

How to fix: Add width and height (or aspect-ratio) to every image.

Affected pages:

<https://www.techand.ai/>

<https://www.techand.ai/contact-us>

<https://www.techand.ai/insights/the-financial-reporting-trap-why-d365-fo-users-still-reach-for-excel>

<https://www.techand.ai/whatwedo/ai-automation>

<https://www.techand.ai/whatwedo/assess-envision>

<https://www.techand.ai/whatwedo/build-implement>

<https://www.techand.ai/whatwedo/global-rollouts>

<https://www.techand.ai/whatwedo/managed-services>

<https://www.techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for>

<https://www.techand.ai/insights/how-microsoft-dynamics-365-finance-operations-supports-uae-vat-compliance>

Security — 97/100

Low Missing security headers

Whole website

Evidence: Not found: Content-Security-Policy

Why it matters: Security headers protect visitors. They are not a direct ranking factor but show a well-maintained site.

How to fix: Add the missing headers in your hosting or CDN configuration; start Content-Security-Policy in Report-Only mode.

Technical — 93/100

Medium Contact form is not in the server-rendered HTML

Whole website

Evidence: Raw HTML of <https://www.techand.ai/contact-us> has 0 <form> and 0 <input> elements

Why it matters: Crawlers and AI agents that do not run JavaScript see no way to contact you.

How to fix: Server-render the form (fields, labels, submit) or add a <noscript> fallback form.

Affected pages:

<https://www.techand.ai/contact-us>

13. What Works Well

- Valid SSL certificate on the main domain (expires 2026-12-29)
- HTTP correctly redirects to HTTPS
- robots.txt is present
- HTTP/2 is enabled
- Missing pages return a proper 404
- No broken images, scripts or stylesheets
- No duplicate page titles
- No duplicate meta descriptions

- No duplicate content across pages
- No redirect loops
- All crawled pages are indexable
- Every page has an H1
- Every page has a title tag
- One canonical host: <https://techand.ai> 301-redirects to <https://www.techand.ai>
- HTTP redirects to HTTPS
- robots.txt lists the sitemap
- XML sitemap lists 43 URLs
- Every live page is listed in the sitemap
- llms.txt exists (10296 characters)
- Site serves Markdown to AI agents on request
- All 11 search and AI crawlers tested (GPTBot, ClaudeBot, PerplexityBot, Googlebot and more) receive the full page
- Structured data present on the homepage (Organization)
- Organization schema found (Organization)
- Sampled articles link to external sources
- Stable layout (CLS 0.00)
- Clear contact details: a UAE phone number, a DAFZA Dubai address, an email and business hours (UTC+4).
- Strong UAE and GCC positioning, with UAE-specific service naming and industry pages including banking and public sector.
- Timely, relevant insight content such as Peppol e-invoicing for UAE CFOs and D365 reporting, with a table of contents and read time.
- A broad, well-organised Microsoft stack offering covering D365 F&O, Business Central, Power Platform and Copilot/AI.
- ChatGPT (with web search) correctly identifies the company

14. Action Plan

Phase 1 — Fix the foundations (Week 1)

- Broken pages (4xx, 5xx or timeout) [Critical]
- www and non-www versions are not consolidated [High]
- HTML cannot be cached by the CDN (no-store / private) [High]
- Malformed sameAs profile link [High]
- Structured contact details are thinner than what the site shows [High]
- Service pages have no Service schema [High]
- No BreadcrumbList schema [High]
- Far fewer pages indexed than crawled [High]
- Broken internal links [High]

Phase 2 — Content and on-page (Weeks 2-5)

- Experience claim conflicts with the founding date [High]
- Unproven scale and financial claims on contact page [High]
- No named authors or expert bylines on insight articles [High]
- Partner status and certifications not evidenced [High]

- Case study and testimonial proof is thin [High]
- Articles have no named human author [High]
- Few verified profiles linked in sameAs [Medium]
- No freshness signal on articles [Medium]
- Images without width and height [Medium]
- Service pages are generic and duplicated [Medium]
- Heavy repeated mega-menu text on every page [Medium]
- Formatting artifact and only high-commitment CTAs [Medium]
- No pricing or engagement-model guidance, and no leadership info [Medium]
- Structured data with errors [Medium]
- Brand name repeated in the title [Medium]

Phase 3 — Authority, keywords and AI visibility (run in parallel) (Weeks 2-12)

- The site does not rank in Google's top 10 for any keyword [High]
- Organic traffic is under 10% of competitors [High]
- Very young domain competing with long-established sites [High]
- Keyword gap: searches competitors win and you do not [High]
- Far fewer referring domains than competitors [High]
- ChatGPT recommends other providers, not you [High]
- Perplexity recommends other providers, not you [High]
- Site does not rank at the top for its own brand name [High]
- No rankings for informational searches [Medium]
- High backlink spam score [Medium]
- Google shows AI Overviews for your key searches and does not cite you [Medium]
- Low-difficulty keywords you can target now [Low]

Phase 4 — Polish (Months 2-3)

- Contact form is not in the server-rendered HTML [Medium]
- Largest Contentful Paint needs improvement (2.5-4s) [Medium]
- Missing canonical tag [Medium]
- Canonical points to a redirect [Medium]
- No compression (gzip/brotli) [Medium]
- Large image payload (over 1MB of images per page) [Medium]
- Missing security headers [Low]
- No alternateName in Organization schema [Low]
- Organization schema has no @id [Low]
- Homepage H1 does not name the business or what it does [Low]
- Phone number in a link but not shown on the page [Low]
- No speculation rules for instant navigation [Low]
- Top speed-up opportunities [Low]
- Missing Open Graph tags [Low]
- URLs are not SEO-friendly [Low]
- Internal URLs that redirect [Low]
- Links pointing to redirects [Low]
- Title shorter than 30 characters [Low]

- Long or messy URLs [Low]
- Few internal links pointing to the page [Low]
- Light content (300-500 words) [Low]

15. Leading Indicators to Track

- SEO Health Score on the next audit (now 64/100)
- Homepage CDN cache status shows HIT and server response under 200ms
- Keywords in Google top 10 (now 0, competitor median 381)
- Rankings for the easy-win keywords listed in section 4
- Referring domains (now 13, competitor median 8,861.5)
- Mobile PageSpeed score (now 89)
- Whether ChatGPT and Perplexity name you for category queries, and whether Google AI Overviews cite you
- Position for your brand name in Google

16. Methodology & Data Sources

Full SEO health. Each category starts at 100. Findings written by the AI content review, and notes about data that could not be collected, are listed but do not change the score. Each other finding deducts Critical 30, High 15, Medium 7, Low 3, Info 1 points, scaled by the share of pages affected (60%-100% of the penalty); site-wide issues take the full penalty. Within a category, each further finding of the same severity counts 70% of the previous one, so many small issues cannot zero a category on their own. The score is the weighted average of category scores, capped at 70 with one critical issue, 60 with two or more, and 85 with five or more high-priority issues.

Severity	Points deducted
Critical	30
High	15
Medium	7
Low	3
Info	1

Source	Used for
DataForSEO On-Page crawl	Page-level technical and on-page checks (up to 200 pages incl. sitemap)
Direct HTTP and HTML checks	www/HTTPS, caching, headers, robots.txt, sitemap, llms.txt, schema, forms, images
DataForSEO Labs	Competitors, rankings, traffic estimates, keyword ideas by intent, keyword gap
DataForSEO Backlinks	Backlinks, referring domains, link gap
Google PageSpeed Insights	Lighthouse lab data and real-user Core Web Vitals (mobile)
Google search results (live)	Brand and category rankings, AI Overview citations, competitor discovery
AI content review (Claude)	Content quality and E-E-A-T review of sampled pages (listed, not scored)
DataForSEO LLM Responses	Live ChatGPT (web search) and Perplexity answers about the brand and category
DataForSEO On-Page reports	Duplicate titles/descriptions, non-indexable pages, redirect chains

Domain registry data	Domain registration date
----------------------	--------------------------

AI crawler checks use the crawler user-agent only; real crawlers are also verified by IP, so a pass here means no user-agent based blocking.

Some extra crawl reports were unavailable: duplicate_title: Task Not Found.; duplicate_description: Task Not Found.; non_indexable: Task Not Found.; redirect_chains: Task Not Found.

Run 27 — Keyword ladder for "e invoicing in uae" (API, techand.ai, pages_now 1)

Head-term research (sitemap, client pages, SERP, competitor pages, facts, Competitor Analyzer, keyword data, site authority) → Verdict

```
{
  "keyword_data": {
    "search_volume": 880,
    "cpc": 6.85,
    "ads_competition": "MEDIUM",
    "keyword_difficulty": 2,
    "difficulty_label": "Easy",
    "google_intent": "informational",
    "trend_yearly_pct": 85,
    "trend_quarterly_pct": -18,
    "monthly_trend": [
      {
        "month": "2026-08",
        "searches": 720
      },
      {
        "month": "2026-07",
        "searches": 880
      },
      {
        "month": "2026-06",
        "searches": 880
      },
      {
        "month": "2026-05",
        "searches": 1000
      },
      {
        "month": "2026-04",
        "searches": 720
      },
      {
        "month": "2026-03",
        "searches": 480
      },
      {
        "month": "2026-02",
        "searches": 720
      },
      {
        "month": "2026-01",
        "searches": 880
      },
      {

```

```

    "month": "2025-12",
    "searches": 1000
  },
  {
    "month": "2025-11",
    "searches": 880
  },
  {
    "month": "2025-10",
    "searches": 880
  },
  {
    "month": "2025-09",
    "searches": 590
  }
],
"competitor_backlink_strength": {
  "avg_referring_domains": 5,
  "avg_backlinks": 10,
  "avg_dofollow": 7
},
"site_authority": null,
"competitor_analysis (excerpt)": {
  "search_intent": "commercial",
  "ranking_page_types": "The SERP mixes government sources (mof.gov.ae, u.ae), Big-4 and advisory explainers (PwC PDF, Maros VAT), and vendor or ASP guides (ClearTax, EDICOM). Most pages are informational guides, and the vendor ones add a soft sales pitch. The AI Overview cites pwc.com, grantthornton.ae, hawksford.com, marosavat.com and zoho.com. No featured snippet is present.",
  "recommended_word_count": 1800,
  "content_gaps": [
    "No competitor gives a decision tool for which wave applies to a business and its own dates, including how revenue is measured",
    "Practical guidance on choosing an ASP is thin: mof.gov.ae only lists 'considerations' as a heading, and ClearTax and EDICOM mostly pitch themselves",
    "No competitor provides a table, since all six pages show 0 tables, and none gives a side-by-side comparison of Wave 1 and Wave 2 or of penalties",
    "Few answer the PAA question 'Who is eligible for an e-invoice?' directly in a short answer block",
    "ERP and accounting-software readiness (data dictionary mapping, TIN/endpoint ID capture) is covered only in outline",
    "Little on the free zone and mainland angle, or on the 10 May 2026 deadline extension, outside ClearTax",
    "Related searches (sample, template, PDF, list of ASPs) are not served with compliant example layouts",
    "No implementation timeline or cost and effort breakdown for onboarding"
  ],
  "unique_angle": "A 'which wave are you and what do you do by when' service page: a wave-finder table, a backwards-planned onboarding timeline and an ASP selection scorecard, ending in a free e-invoicing readiness assessment for UAE businesses.",
  "feasibility_note": "This is a hard but partly winnable target. Government (mof.gov.ae, u.ae), PwC and established advisory or vendor sites such as ClearTax and EDICOM hold the top positions, and the AI Overview a
... (406 more characters in docs/runs/full)

```

Verdict on the head term (Claude Sonnet) → feasibility of the ladder, alternatives, the top page type

```

{
  "verdict": "GO_WITH_CHANGES",
  "verdict_score": 72,
  "verdict_reasons": [
    "Volume of 880/month with a CPC of $6.85 shows real commercial value. The mandate is close: Wave 1 must appoint an ASP by 30 Oct 2026 and go live on 1 Jan 2027, and Wave 1 is exactly Tech&'s enterprise audience (revenue ≥ AED 50m).",
    "Keyword difficulty is 2 (Easy) and competitors average only 5 referring domains and 10 backlinks. Link barriers are low, even though the site's authority is unknown.",
    "The content gaps are clear. All six ranking pages have 0 tables, and none offers a wave-finder, an ASP selection scorecard or a backwards-planned onboarding timeline. A structured page with a lead path can beat ClearTax (2,628 words) and EDICOM (1,888 words), and MoF (713 words, no lead path) is a weak competitor on conversion.",
    "The SERP is informational and mixed. It has government, Big-4 and vendor pages, an AI Overview and no featured snippet. A pure service page would mismatch that intent, so the page needs to be a guide-style service page."
  ],
  "verdict_changes": [
    "Build a hybrid service page: open with a 40-70 word answer on what e-invoicing is and the mandatory dates, then give the detailed guide content, then the Dynamics 365 / Microsoft e-invoicing service offer.",
    "Add a Wave 1 vs Wave 2 wave-finder table, a backwards-planned timeline from 30 Oct 2026 and 31 Mar 2027, and an ASP selection scorecard labelled as Tech&'s own framework.",
    "Add question-style H2s and FAQ schema matching the People Also Ask questions, plus a sample invoice field table and a readiness checklist, with a free readiness assessment CTA.",
    "Add trust signals: a named expert author, a last-updated date, citations to MoF, Ministerial Decisions 243, 244 and 64, and Cabinet Decision 106, ASP partner or accreditation status, a UAE contact, and case studies.",
    "Support the page with spoke content on deadline, requirements, ASP list and ERP readiness (D365, SAP), and build internal links and a few UAE finance backlinks."
  ],
  "recommended_page": "Service Page",
  "secondary_keywords": [
    "uae e-invoicing requirements",
    "e invoicing uae deadline",
    "e invoicing uae 2026",
    "list of asp for e invoicing in uae",
    "uae e-invoicing accredited service provider",
    "e invoicing uae fta",
    "peppol e-invoicing uae",
    "dynamics 365 e-invoicing uae",
    "e invoicing uae penalties",
    "pint-ae"
  ],
  "expected_visits_top3": 300,
  "time_to_rank_months": 4
}

```

Ladder research pool (Ladder Requests → Run Ladder Research → Ladder Pool) → the AI relevance screen

Pool 19 keywords; by source {"suggestions": 10, "related": 12}; site rankings found 0; head tokens ["invoicing", "uae"].

Keyword	Searches/mo	KD	CPC	Intent	Page type	Opportunity	Your position
uae e invoicing penalties	1600	0		transactional		405.6	
e invoicing uae fta	260	9	4.18	transactional		134.5	

accredited service provider for e invoicing in uae	110	2	7.04	commercial		77	
uae e invoicing requirements	140	10	3.28	commercial		59.6	
uae e invoicing timeline	110	0	3.4	transactional		54.7	
uae e invoicing accredited service provider	110	2	9.01	transactional		53.5	
e invoicing uae 2026	110		6	transactional		51.5	
list of asp for e invoicing in uae	140		4.62	transactional		47.8	
uae e invoicing accredited service provider list	50		6.62	commercial		17.6	
is e invoicing mandatory in uae	50		2.38	informational		11.9	
is e-invoicing mandatory in uae	50		2.38	informational		11.9	
accredited service providers in uae	20		6.07	commercial		6.9	
uae e invoicing guidelines pdf	20		3.98	informational		4.1	
e invoicing in uae timeline	10	0		transactional		3.9	
e invoicing service providers in uae	20			commercial		3.8	
when is e invoicing applicable in uae	10		2.67	informational		1.9	

Ladder plan (Ladder Plan) → the plan report, the Data Table rows, the page runs and the tracker

Rung 1 (Rung 1 — long-tail pages (difficulty up to 25), months [1, 2], 6 keywords available):

#	Page (primary keyword)	Supporting	Type	Searches/mo	KD	You today	URL	Status
1	uae e invoicing penalties		Blog Post	1600	0		https://techand.ai/uae-e-invoicing-penalties/	new
2	e invoicing uae fta		Guide	260	9		https://techand.ai/e-invoicing-uae-fta/	new
3	accredited service provider for e invoicing in uae	uae e invoicing accredited service provider	Service Page	220	2		https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	new
4	uae e invoicing requirements		Guide	140	10		https://techand.ai/uae-e-invoicing-requirements/	new

Rung 2 (Rung 2 — mid-difficulty pages (26-45), months [3, 5], 10 keywords available):

#	Page (primary keyword)	Supporting	Type	Searches/mo	KD	You today	URL	Status
5	e invoicing uae 2026	e invoicing in uae timeline, when is e invoicing applicable in uae	Blog Post	130			https://techand.ai/e-invoicing-uae-2026/	new

6	list of asp for e invoicing in uae	uae e invoicing accredited service provider list, accredited service providers in uae	Comparison Page	210			https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	new
7	is e invoicing mandatory in uae	is e-invoicing mandatory in uae	FAQ Page	100			https://techand.ai/is-e-invoicing-mandatory-in-uae/	new
8	e invoicing service providers in uae		Service Page	20			https://techand.ai/e-invoicing-service-providers-in-uae/	new

Rung 3 (Rung 3 — strong pages (46-60), months , 0 keywords available):

none

Top page: {"keyword": "e invoicing in uae", "page_type": "Service Page", "volume": 880, "kd": 2, "target_url": "https://techand.ai/e-invoicing-in-uae/", "exists": false, "months": [6, 9]}

Internal link map:

From	To	Type
https://techand.ai/uae-e-invoicing-penalties/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/uae-e-invoicing-penalties/	https://techand.ai/e-invoicing-uae-fta/	sideways
https://techand.ai/e-invoicing-uae-fta/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-uae-fta/	https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	sideways
https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-requirements/	sideways
https://techand.ai/uae-e-invoicing-requirements/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/uae-e-invoicing-requirements/	https://techand.ai/uae-e-invoicing-penalties/	sideways
https://techand.ai/e-invoicing-uae-2026/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-uae-2026/	https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	sideways
https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	https://techand.ai/is-e-invoicing-mandatory-in-uae/	sideways
https://techand.ai/is-e-invoicing-mandatory-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/is-e-invoicing-mandatory-in-uae/	https://techand.ai/e-invoicing-service-providers-in-uae/	sideways
https://techand.ai/e-invoicing-service-providers-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-service-providers-in-uae/	https://techand.ai/e-invoicing-uae-2026/	sideways
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-penalties/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-uae-fta/	down

https://techand.ai/e-invoicing-in-uae/	https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-requirements/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-uae-2026/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/is-e-invoicing-mandatory-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-service-providers-in-uae/	down

Requirements:

Item	Detail
Links to the top page	Top-10 pages for "e invoicing in uae" average 5 referring domains. Plan 5+ relevant links (partners, directories, PR, guest articles) to the top page over the ladder period.
Local trust signals	The head term is local (uae): a complete Google Business Profile, consistent name/address/phone and recent reviews support every rung.
Entity and trust pages	Organization schema, an about page with real credentials, author bylines and case studies; the content QA already adds schema to every generated page.
Publish on schedule and interlink	Each rung page links up to the top page and sideways to one sibling; the top page links down to every rung page. Ranking only starts once the pages are live.

Stats: {"pool_size": 19, "ai_reviewed": 16, "relevant": 16, "pages_total": 9, "rung_pages": 8, "keywords_covered": 21, "traffic_potential_total": 844, "site_rankings_found": 0, "research_failures": []}

Delivered: Keyword Ladder Plan (PDF 72 KB + Word) → e-mailed (123 KB, 250 OK), posted to the callback (stage ladder_plan), 9 rows written to seo_ladders, first page run spawned

DELIVERED REPORT: KEYWORD-LADDER-E-INVOICING-IN-UAE-2026-10-01.DOC

KEYWORD LADDER PLAN

e invoicing in uae

techand.ai • United Arab Emirates • 2026-10-01

Prepared by Dev SEO

1. The plan in one page

Destination keyword	e invoicing in uae — 880 searches/mo, difficulty 2 · easy, CPC 6.85
Is it realistic?	Winnable (verdict GO_WITH_CHANGES, 72/100)
Where you rank today	not in the top 60 for the destination keyword
The ladder	9 pages: 4 on rung 1, 4 on rung 2 and the top page
Timeline	Rung 1: months 1-2 · Rung 2: months 3-5 · Top — e invoicing in uae: months 6-9
Traffic potential	~844 visits/month across the ladder once the pages rank in the top 3

Written now	"uae e invoicing penalties" (rung 1) — delivered separately as page content
Tracking	Positions checked weekly; progress e-mail with the next rung; stops when the top page holds the top 3 for 4 checks

No keywords fell into rung 3; the ladder skips it.

2. Is the destination realistic?

Google ranks the head term for the site with the most complete, trusted coverage of the topic. The ladder builds that coverage from the bottom: specific pages that are winnable now, linking up to one strong page for the head term.

Why

- Volume of 880/month with a CPC of \$6.85 shows real commercial value. The mandate is close: Wave 1 must appoint an ASP by 30 Oct 2026 and go live on 1 Jan 2027, and Wave 1 is exactly Tech&'s enterprise audience (revenue ≥ AED 50m).
- Keyword difficulty is 2 (Easy) and competitors average only 5 referring domains and 10 backlinks. Link barriers are low, even though the site's authority is unknown.
- The content gaps are clear. All six ranking pages have 0 tables, and none offers a wave-finder, an ASP selection scorecard or a backwards-planned onboarding timeline. A structured page with a lead path can beat ClearTax (2,628 words) and EDICOM (1,888 words), and MoF (713 words, no lead path) is a weak competitor on conversion.
- The SERP is informational and mixed. It has government, Big-4 and vendor pages, an AI Overview and no featured snippet. A pure service page would mismatch that intent, so the page needs to be a guide-style service page.

What must change first

- Build a hybrid service page: open with a 40-70 word answer on what e-invoicing is and the mandatory dates, then give the detailed guide content, then the Dynamics 365 / Microsoft e-invoicing service offer.
- Add a Wave 1 vs Wave 2 wave-finder table, a backwards-planned timeline from 30 Oct 2026 and 31 Mar 2027, and an ASP selection scorecard labelled as Tech&'s own framework.
- Add question-style H2s and FAQ schema matching the People Also Ask questions, plus a sample invoice field table and a readiness checklist, with a free readiness assessment CTA.
- Add trust signals: a named expert author, a last-updated date, citations to MoF, Ministerial Decisions 243, 244 and 64, and Cabinet Decision 106, ASP partner or accreditation status, a UAE contact, and case studies.
- Support the page with spoke content on deadline, requirements, ASP list and ERP readiness (D365, SAP), and build internal links and a few UAE finance backlinks.

Expected visits/month at #1-3	300
Time to rank (verdict estimate)	4 months

3. The ladder

One page per topic. Lower rungs are specific and winnable; each rung links up to the top page. Pages marked "improve" already exist on your site.

Rung 1 — long-tail pages (difficulty up to 25) — months 1-2

#	Page (primary keyword)	Supporting keywords	Page type	Searches/mo	Difficulty	You today	Page on site
1	uae e invoicing penalties	—	Blog Post	1,600	0 · easy	not in top 60	new https://techand.ai/uae-e-invoicing-penalties/
2	e invoicing uae fta	—	Guide	260	9 · easy	not in top 60	new https://techand.ai/e-invoicing-uae-fta/
3	accredited service provider for e invoicing in uae	uae e invoicing accredited service provider	Service Page	220	2 · easy	not in top 60	new https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/
4	uae e invoicing requirements	—	Guide	140	10 · easy	not in top 60	new https://techand.ai/uae-e-invoicing-requirements/

Rung 2 — mid-difficulty pages (26-45) — months 3-5

#	Page (primary keyword)	Supporting keywords	Page type	Searches/mo	Difficulty	You today	Page on site
5	e invoicing uae 2026	e invoicing in uae timeline, when is e invoicing applicable in uae	Blog Post	130	N/A	not in top 60	new https://techand.ai/e-invoicing-uae-2026/
6	list of asp for e invoicing in uae	uae e invoicing accredited service provider list, accredited service providers in uae	Comparison Page	210	N/A	not in top 60	new https://techand.ai/list-of-asp-for-e-invoicing-in-uae/
7	is e invoicing mandatory in uae	is e-invoicing mandatory in uae	FAQ Page	100	N/A	not in top 60	new https://techand.ai/is-e-invoicing-mandatory-in-uae/
8	e invoicing service providers in uae	—	Service Page	20	N/A	not in top 60	new https://techand.ai/e-invoicing-service-providers-in-uae/

Top — e invoicing in uae — months 6-9

Page	new https://techand.ai/e-invoicing-in-uae/
Page type	Service Page
Supporting keywords	uae e-invoicing requirements, e invoicing uae deadline, e invoicing uae 2026, list of asp for e invoicing in uae, uae e-invoicing accredited service provider, e invoicing uae fta
Searches/mo · difficulty	880 · 2 · easy
You today	not in top 60

4. Internal link map

Every rung page links up to the top page and sideways to one sibling; the top page links down to every rung page.

From	Links to	Type
https://techand.ai/uae-e-invoicing-penalties/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/uae-e-invoicing-penalties/	https://techand.ai/e-invoicing-uae-fta/	sideways
https://techand.ai/e-invoicing-uae-fta/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-uae-fta/	https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	sideways
https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-requirements/	sideways
https://techand.ai/uae-e-invoicing-requirements/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/uae-e-invoicing-requirements/	https://techand.ai/uae-e-invoicing-penalties/	sideways
https://techand.ai/e-invoicing-uae-2026/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-uae-2026/	https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	sideways
https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	https://techand.ai/is-e-invoicing-mandatory-in-uae/	sideways
https://techand.ai/is-e-invoicing-mandatory-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/is-e-invoicing-mandatory-in-uae/	https://techand.ai/e-invoicing-service-providers-in-uae/	sideways
https://techand.ai/e-invoicing-service-providers-in-uae/	https://techand.ai/e-invoicing-in-uae/	up
https://techand.ai/e-invoicing-service-providers-in-uae/	https://techand.ai/e-invoicing-uae-2026/	sideways
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-penalties/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-uae-fta/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/accredited-service-provider-for-e-invoicing-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/uae-e-invoicing-requirements/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-uae-2026/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/list-of-asp-for-e-invoicing-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/is-e-invoicing-mandatory-in-uae/	down
https://techand.ai/e-invoicing-in-uae/	https://techand.ai/e-invoicing-service-providers-in-uae/	down

5. Pages written now

#	Keyword	Page type	URL	Links to
---	---------	-----------	-----	----------

1	uae e invoicing penalties	Blog Post	https://techand.ai/uae-e-invoicing-penalties/ (new)	e invoicing in uae, e invoicing uae fta
---	---------------------------	-----------	---	---

Each page runs through the full content pipeline (competitor analysis, verified facts, brief, copywriter, critic, QA, editor) and arrives as its own PDF and Word file at kinngahmed@gmail.com. Publish it at the URL above, then the weekly tracker measures the climb.

6. What else is needed

Item	Detail
Links to the top page	Top-10 pages for "e invoicing in uae" average 5 referring domains. Plan 5+ relevant links (partners, directories, PR, guest articles) to the top page over the ladder period.
Local trust signals	The head term is local (uae): a complete Google Business Profile, consistent name/address/phone and recent reviews support every rung.
Entity and trust pages	Organization schema, an about page with real credentials, author bylines and case studies; the content QA already adds schema to every generated page.
Publish on schedule and interlink	Each rung page links up to the top page and sideways to one sibling; the top page links down to every rung page. Ranking only starts once the pages are live.

7. Tracking and next rungs

The ladder is registered for weekly position checks (Google top 100 for every page's primary keyword). The progress e-mail shows position changes, which rungs are in the top 10 / top 3, and the recommendation for the next rung with a ready-to-send request. Pages for the next rung are written when you confirm (recommend-only mode). Tracking stops when top rung in the top 3 for 4 consecutive checks.

8. How this was built

19 keywords were pulled around the head term (long-tail suggestions, related searches, keyword ideas for the head term and your services, and the keywords your site already ranks for). 16 were screened by AI for topical relevance to the head term; 16 survived. Rungs follow difficulty: rung 1 up to 25 (long-tail, 20+ searches), rung 2 26-45, rung 3 46-60, the head term on top. One page per topic cluster, 2-4 pages per rung. Traffic potential = searches after SERP-feature click loss × a top-3 click share. Positions and metrics from DataForSEO for United Arab Emirates.

Run 35 — Rank Tracker (manual run)

Positions (SERP Check → Parse Positions → `seo_rank_history`) and the recommendation (Tracker Report) → progress e-mail and callback (`stage_rank_tracker`)

Rung	Keyword	Position now	Previous	Delta	URL
1	uae e invoicing penalties	0			
1	e invoicing uae fta	0			
1	accredited service provider for e invoicing in uae	0			
1	uae e invoicing requirements	0			
2	e invoicing uae 2026	0			

2	list of asp for e invoicing in uae	0			
2	is e invoicing mandatory in uae	0			
2	e invoicing service providers in uae	0			
4	e invoicing in uae	0			

Rungs: [{"rung": 1, "pages": 4, "top10": 0, "top3": 0, "reached": false}, {"rung": 2, "pages": 4, "top10": 0, "top3": 0, "reached": false}]

Next step: Work on rung 1: none of its pages is in the top 10 yet. Write or improve: "uae e invoicing penalties" (already written — publish it), "e invoicing uae fta", "accredited service provider for e invoicing in uae", "uae e invoicing requirements".

Ready-to-send API body for the next page:

```
{
  "mode": "keyword",
  "keyword": "e invoicing uae fta",
  "page_type": "Guide",
  "country": "United Arab Emirates",
  "domain": "techand.ai",
  "existing_page_url": "",
  "email": "kinngahmed@gmail.com",
  "ladder_id": "lad_mupia2l2c0tp",
  "ladder_rung": 1,
  "ladder_head": "e invoicing in uae",
  "force_content": true
}
```

Delivered: progress e-mail (250 OK)

Hi,

Position check for your keyword ladder towards **e invoicing in uae** on **techand.ai** (9 keywords, Google top 100).

Rung	Keyword	Now	Before	Ranking URL
1	uae e invoicing penalties	not in top 100	first check	—
1	e invoicing uae fta	not in top 100	first check	—
1	accredited service provider for e invoicing in uae	not in top 100	first check	—
1	uae e invoicing requirements	not in top 100	first check	—
2	e invoicing uae 2026	not in top 100	first check	—
2	list of asp for e invoicing in uae	not in top 100	first check	—
2	is e invoicing mandatory in uae	not in top 100	first check	—
2	e invoicing service providers in uae	not in top 100	first check	—
Top	e invoicing in uae	not in top 100	first check	—

Rungs: Rung 1: 0/4 in top 10, 0 in top 3 · Rung 2: 0/4 in top 10, 0 in top 3

Next step: Work on rung 1: none of its pages is in the top 10 yet. Write or improve: "uae e invoicing penalties" (already written — publish it), "e invoicing uae fta", "accredited service provider for e invoicing in uae", "uae e invoicing requirements".

Start the next page: open the [SEO Assistant form](#) ("I know my keyword", keyword **e invoicing uae fta**, domain techand.ai), or send this to the API (<http://localhost:5678/webhook/seo-keyword-check>):

```
{
  "mode": "keyword",
  "keyword": "e invoicing uae fta",
  "page_type": "Guide",
  "country": "United Arab Emirates",
  "domain": "techand.ai",
  "existing_page_url": "",
  "email": "kinngahmed@gmail.com",
  "ladder_id": "lad_mupia2l2c0tp",
  "ladder_rung": 1,
  "ladder_head": "e invoicing in uae",
  "force_content": true
}
```

Checks run weekly; tracking stops once the top page holds the top 3 for four checks. Positions come from DataForSEO live SERPs.

Run 46 — The ladder's first page: "uae e invoicing penalties" (content run spawned by the ladder)

Normalised input (Normalize Input) → every later node

```
{
  "mode": "keyword",
  "keyword": "uae e invoicing penalties",
  "page_type": "Blog Post",
  "domain": "techand.ai",
  "country": "United Arab Emirates",
  "business": "Tech& is a trusted Microsoft partner specializing in enterprise digital transformation and AI-driven business solutions for finance leaders in the UAE and Saudi Arabia, including Dynamics 365 and e-invoicing readiness.",
  "audience": "Enterprise CFOs and finance leaders in the GCC",
  "goal": "leads",
  "include_content": true,
  "include_seo_report": true,
  "force_content": true,
  "ladder_id": "lad_mupia2l2c0tp",
  "ladder_rung": 1,
  "ladder_head": "e invoicing in uae",
  "ladder_links": [
    {
      "url": "https://techand.ai/e-invoicing-in-uae/",
      "path": "/e-invoicing-in-uae/",
      "role": "top",
      "keyword": "e invoicing in uae",
      "planned": true
    },
    {
      "url": "https://techand.ai/e-invoicing-uae-fta/",
      "path": "/e-invoicing-uae-fta/",
      "role": "sibling",
      "keyword": "e invoicing uae fta",
      "planned": true
    }
  ]
}
```

```
}
],
"via_webhook": true,
"callback_url": "http://localhost:5678/webhook/seo-callback"
}
```

Site pages (Fetch Sitemaps → Collect Site URLs) → internal-link candidates for the brief and the QA

43 URLs in the sitemap; existing page: null. Candidates (ladder pages first):

URL	Ladder role	Planned
https://techand.ai/		
https://techand.ai/e-invoicing-in-uae/	top	yes
https://techand.ai/e-invoicing-uae-fta/	sibling	yes
https://techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs		
https://techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for		
https://techand.ai/ai-copilot-advisory-uae		
https://techand.ai/business-central-implementation-uae		
https://techand.ai/business-central-post-go-live-support-uae		
https://techand.ai/d365-finance-operations-implementation-rollout-uae		
https://techand.ai/dynamics-365-business-central-uae		
https://techand.ai/dynamics-365-cloud-migration-uae		
https://techand.ai/dynamics-365-crm-uae		

30 more rows not shown.

Client pages (Read Client Pages → Collect Client Pages) → real facts and trust signals for the brief

2 pages read; signals: {"phone_shown": true, "email_shown": true, "years_claims": ["20+ Years"], "credentials": [], "numbers": ["365 Project"], "locations": []}

Competitor pages (SERP Top 10 → Pick Top 6 → Read Competitor Pages → Analyze Competitor Pages) → term model, questions and statistics for the analyzer and the QA coverage check

6 competitor pages read (avg 1476 words); SERP features: ["ai_overview", "people_also_ask", "related_searches"]; AI Overview cites ["hawksford.com", "marmin.ai", "zoho.com", "uae-asp.ae", "youtube.com"].

Rank	Domain	Title	Words
1	mof.gov.ae	Cabinet Decision No. 106 of 2025	672
2	hawksford.com	UAE e-invoicing: timeline, business actions and penalties	2074
3	wafeq.com	UAE E-Invoicing Penalties: What Businesses Need to ...	2925
4	krestonmenon.com	UAE E-Invoicing Update: Violations & Penalties Explained	804
5	tallysolutions.com	UAE E-Invoicing Penalties and System Failure Reporting	1078
6	cleartax.com	e-Invoicing Penalties Under Cabinet Decision 106/2025	1305

Term model (top shared terms): cabinet decision (6), accredited service provider (6), uae e-invoicing (5), e-invoicing system (5), system failure (5), federal tax authority (5), credit notes (5), electronic invoicing (4), penalty aed (4), administrative penalties (4), system failures (4), tax authority fta (4), service provider asp (4), structured electronic (4), service provider within (4), electronic invoicing system (3), federal decree-law (3), electronic invoice (3), invoice data (3), electronic invoices (3), aed million (3), ministry finance (3), electronic credit note (3), issue and transmit (3), aed per month (3)

Competitor questions: What is e-invoicing? · Why is the UAE implementing e-invoicing? · What VAT-registered businesses must do to prepare for UAE e-invoicing? · How we can help · Is e-invoicing mandatory in the UAE? · Is a UAE e-invoice the same as a PDF tax invoice? · Are there additional considerations for large enterprises in the e-invoicing procedure? · What happens if my business is not ready for UAE e-invoicing mandate deadlines? · What to Consider When Choosing an ASP: · What to Do Right Now

Verified facts (Facts SERP → Collect Facts) → the brief's evidence plan; only these numbers may appear with a source

Fact	Source	Authority	URL
	abdelhamidcpa.com	medium	https://abdelhamidcpa.com/2026/05/29/uae-e-invoicing-system-2026/
	growacross.com	medium	https://growacross.com/insights/uae-e-invoicing-2026
	kaidolab.com	medium	https://kaidolab.com/blog/uae-e-invoicing-asp-deadline-2026/
	genzone.com	medium	https://www.genzone.com/uae-e-invoicing-2026-guide/
	farahsolutions.ae	medium	https://farahsolutions.ae/blog/uae-e-invoicing-guidelines-v1-1-compliance-guide
	worldcompanysetup.com	medium	https://worldcompanysetup.com/en/blog/uae-e-invoicing-mandate-dubai-timeline-penalties/
	daxin-global.ae	medium	https://daxin-global.ae/blog/uae-e-invoicing-guide-2026/
	daxin-global.ae	medium	https://daxin-global.ae/blog/uae-e-invoicing-guide-2026/
	covoro.ai	medium	https://covoro.ai/uae/fta-announces-penalties-for-non-compliance-with-uae-e-invoicing/
	qeemahcloud.com	medium	https://qeemahcloud.com/en/blog/uae-e-invoicing-deadlines-2026/?srsId=AU7gw4WNyQVaepjxnAO0IBhnccG4lwbsJEKQue8sg3o-iwSYmvpmKNG9

Competitor analysis (Claude Sonnet, Competitor Analyzer → Parse Analysis) → verdict and brief

```
{
  "search_intent": "informational",
  "intent_explanation": "Searchers want the exact fine amounts under Cabinet Decision No. 106 of 2025, when the fines start, and how to avoid them. Leads come from readers who are unsure about their phase deadline or ASP choice, so a readiness check or consultation offer fits.",
  "ranking_page_types": "A mix of the primary source (the MoF Cabinet Decision PDF), advisory firm explainers (Hawksford, Kreston Menon), and software vendor or ASP guides (Wafeq, Tally, ClearTax). Most vendor pages target one angle, such as system failure or the penalty table. The AI Overview cites hawksford.com, marmin.ai, zoho.com, uae-asp.ae and youtube.com, and there is no featured snippet.",
  "recommended_word_count": 1800,
  "common_topics": [
    "Cabinet Decision No. 106 of 2025 and what it covers",
    "The six violations and fine amounts: AED 5,000 per month for no system or ASP, AED 100 per invoice or credit note capped at AED 5,000 per month, AED 1,000 per day for system failure notification",
    "Issuer vs recipient obligations",
    "Accredited Service Provider (ASP) appointment",
  ]
}
```

```

"System failure definition and FTA notification",
"Phased timeline: voluntary from July 2026, mandatory for AED 50m+ businesses from January 2027",
"Voluntary adopters not penalised",
"Federal Tax Authority as enforcer",
"Credit notes and structured electronic format (not PDF)"
],
"common_heading_patterns": [
  "What is Cabinet Decision No. 106 of 2025 about?",
  "Who does the decision apply to?",
  "Penalty table / key violations and penalties",
  "System failure notification to the FTA",
  "What to do now / compliance guidelines",
  "FAQs: Is e-invoicing mandatory in the UAE?"
],
"content_gaps": [
  "Only a few pages (the MoF PDF, Kreston Menon, ClearTax, Wafeq) list all six violations, and none cover the legal wording and the plain-English meaning side by side with scenarios",
  "No page works out total exposure over 3, 6 or 12 months for different business types",
  "The competitor data shows conflicting phase dates (Wafeq: ASP deadline 31 July 2026 for AED 50m+ and 31 March 2027 for the rest; ClearTax: all VAT-registered by July 2027; Hawksford: mandatory from January 2027). Readers get no reconciliation or advice to check against the primary source",
  "The relationship to existing VAT penalties under Cabinet Decision No. 40 of 2017 is not covered, though 'late payment of VAT' appears in People Also Ask",
  "How to contest or appeal an e-invoicing penalty is not covered in the data",
  "Related searches ask for a penalty calculator, PDF and ASP list, and no ranking page offers these",
  "Recipient-side penalties are covered thinly, with limited practical guidance",
  "Whether penalties stack, and whether the AED 5,000 cap applies per category or overall, is stated differently across pages (Wafeq says stacking, ClearTax says per category)"
],
"competitor_strengths": [
  {
    "domain": "mof.gov.ae",
    "strength": "The primary source, with the official wording and the full annexed penalty table"
  },
  {
    "domain": "hawksford.com",
    "strength": "Thorough on timeline, five-corner model and exclusions; cited in the AI Overview"
  },
  {
    "domain": "wafeq.com",
    "strength": "The most detailed penalty-focused page (about 2,900 words), covering stacking, deadlines and the voluntary window"
  },
  {
    "domain": "krestonmenon.com",
    "strength": "Clear split of issuer and recipient penalties"
  },
  {
    "domain": "tallysolutions.com",
    "strength": "Deep coverage of system failure and the AED 1,000 daily penalty, with
... (5,232 more characters in docs/runs/full)

```

Keyword data and site authority (Keyword Data, Site Authority → Merge Keyword Data) → verdict

```
{
  "keyword_data": {
    "search_volume": 1600,
    "cpc": null,
    "ads_competition": "LOW",
    "keyword_difficulty": 0,
    "difficulty_label": "Easy",
    "google_intent": "transactional",
    "trend_yearly_pct": null,
    "trend_quarterly_pct": 0,
    "monthly_trend": [
      {
        "month": "2026-08",
        "searches": 10
      },
      {
        "month": "2026-07",
        "searches": 40
      },
      {
        "month": "2026-06",
        "searches": 10
      },
      {
        "month": "2026-05",
        "searches": 10
      },
      {
        "month": "2026-04",
        "searches": 10
      },
      {
        "month": "2026-03",
        "searches": 10
      },
      {
        "month": "2026-02",
        "searches": 30
      },
      {
        "month": "2026-01",
        "searches": 50
      },
      {
        "month": "2025-12",
        "searches": 18100
      },
      {
        "month": "2025-11",
        "searches": 0
      },
      {
        "month": "2025-10",
        "searches": 0
      },
      {
        "month": "2025-09",
        "searches": 0
      }
    ]
  }
}
```

```

],
"competitor_backlink_strength": {
  "avg_referring_domains": 4,
  "avg_backlinks": 11,
  "avg_dofollow": 7
},
"site_authority": null
}

```

Verdict (Claude Sonnet) → brief; force_content keeps the page even on AVOID

```

{
  "verdict": "GO_WITH_CHANGES",
  "verdict_score": 62,
  "verdict_reasons": [
    "The volume data is unreliable. The 1,600 headline figure sits next to a single 18,100 spike in Dec 2025, when Cabinet Decision No. 106 was published. Every month from Feb to Aug 2026 shows only 10-50 searches. Real demand is likely a few hundred a month, rising around regulatory deadlines (July 2026 voluntary start, January 2027 mandatory for AED 50m+).",
    "Ranking looks feasible. Competitors average only 4 referring domains and 11 backlinks, and keyword difficulty is rated Easy. Kreston Menon is thin at about 804 words, and Hawksford covers penalties only in an FAQ.",
    "Clear content gaps exist: no page has a penalty calculator, worked exposure examples, a reconciliation of the conflicting phase dates, or an explanation of whether the AED 5,000 cap is per category or overall. A 1,800+ word guide can beat Wafeq's 2,925-word vendor page on usefulness.",
    "The fit with the business goal is good. Readers worried about fines are the same people who need an ASP and readiness help, and Tech& sells e-invoicing readiness to enterprise CFOs. The audience is narrow but high-value, so a few leads would justify the page. The keyword is informational, despite the 'transactional' label, and fits a blog post."
  ],
  "verdict_changes": [
    "Build it as a practical penalty-exposure guide, not a generic explainer. Open with a 40-70 word answer block, then add a penalty table (violation, amount, cap, who is liable).",
    "Add a calculator-style table with worked examples, for instance 6 months without an ASP plus 80 missed invoices a month. Offer a downloadable penalty summary PDF as the lead magnet.",
    "Reconcile the conflicting phase dates and the cap and stacking question, citing mof.gov.ae and FTA sources, with a visible updated date and a named author with UAE tax credentials.",
    "Close with a phase-readiness check and a consultation CTA aimed at CFOs of AED 50m+ businesses. Add FAQs using the People Also Ask wording.",
    "Support the page with cluster content: penalties calculator, recipient-side penalties, system failure checklist, and ASP selection. Link these to a Dynamics 365 e-invoicing service page."
  ],
  "recommended_page": "Blog Post",
  "secondary_keywords": [
    "uae e-invoicing penalties calculator",
    "uae e-invoicing penalties pdf",
    "cabinet decision 106 of 2025",
    "e-invoicing system failure penalty uae",
    "uae e-invoicing deadline 2027",
    "e-invoicing penalties for recipients uae",
    "accredited service provider uae e-invoicing",
    "uae e-invoicing AED 50 million threshold",
    "FTA e-invoicing fines"
  ],
  "expected_visits_top3": 250,

```

```
"time_to_rank_months": 5
}
```

Content brief (Claude Opus, Strategy Brief → Parse Brief) → the only instruction the copywriter gets

```
{
  "meta_title": "UAE E-Invoicing Penalties: Fines, Examples & Calculator",
  "meta_description": "UAE e-invoicing penalties explained: every fine under Cabinet Decision No. 106, worked cost examples and a prevention plan. Book a readiness check.",
  "slug": "uae-e-invoicing-penalties",
  "h1": "UAE E-Invoicing Penalties: Every Fine, What It Could Cost and How to Avoid It",
  "primary_keyword": "uae e invoicing penalties",
  "secondary_keywords": [
    "cabinet decision no. 106 of 2025",
    "uae e-invoicing penalties calculator",
    "e-invoicing penalties for recipients",
    "e-invoicing system failure penalty",
    "accredited service provider uae",
    "uae e-invoicing deadline",
    "penalty for not issuing an e-invoice",
    "fta e-invoicing administrative penalties"
  ],
  "target_audience": "CFOs, finance directors and tax or finance-systems leads at VAT-registered UAE enterprises, especially those at or above the AED 50 million revenue threshold, who need to know their penalty exposure and which deadline applies to them.",
  "search_intent": "Informational with commercial follow-through. The reader wants exact fine amounts, start dates and their own exposure, then a way to avoid the fines (an ASP and readiness help).",
  "tone": "Professional and direct. Write like a senior adviser briefing a CFO: precise, calm, no scare tactics, figures first.",
  "target_word_count": 1800,
  "answer_block": "Under Cabinet Decision No. 106 of 2025, the Federal Tax Authority (FTA) can fine businesses that miss their e-invoicing obligations. Failing to implement the Electronic Invoicing System or appoint an Accredited Service Provider costs AED 5,000 per month or part of a month. Each electronic invoice or credit note not issued and transmitted on time costs AED 100, capped at AED 5,000 per month. Voluntary adopters are not penalised.",
  "outline": [
    {
      "h2": "What are the UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025?",
      "purpose": "Win the AI Overview and snippet. Give the reader the full penalty picture in one table straight away, so they can stop searching.",
      "key_points": [
        "Open with the answer block verbatim (40-70 words), directly under the H1 and a visible 'Last updated: [date]' line plus 'Reviewed by [Name, UAE tax credential]'.",
        "One line of context: the Ministry of Finance published the decision in December 2025. It sits under Federal Decree-Law No. 28 of 2022 (Tax Procedures) and is enforced by the FTA.",
        "An HTML penalty table with all six violations. Columns: Violation | Plain-English meaning | Who is liable (Issuer/Recipient) | Fine | Cap. Use verified figures for the AED 5,000/month and AED 100-per-document rows. For the system-failure notification fine (reported as AED 1,000 per day) and the remaining rows, take the wording and amounts directly from the MoF PDF and link to it. Do not publish any amount you have not checked against that PDF.",
        "Say clearly that penalties recur. The AED 5,000 applies for each month or part of a month of delay.",
        "Note that Article 2 excludes voluntary adopters from these penalties.",
        "Offer the downloadable one-page 'UAE e-invoicing penalties PDF summary' as a lead magnet, with an email gate. This covers the 'pdf' related search."
      ]
    },
    {
      "h3s": [
        "The six violations at a glance",

```

```

    "Download the penalty summary (PDF)"
  ],
  "target_words": 250,
  "format": "table",
  "evidence": [
    "AED 5,000 per month for failing to implement the Electronic Invoicing System (including failure to appoint an ASP): covoro.ai",
    "AED 100 per document, capped at AED 5,000 per month: genzone.com",
    "Penalties are recurring, not one-off; AED 5,000 for each month of delay or part of a month: kaidolab.com",
    "Primary source to link: https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf"
  ]
},
{
  "h2": "When do e-invoicing penalties start, and which phase are you in?",
  "purpose": "Clear up the date confusion that every competitor leaves unresolved, so the reader can identify their own deadline.",
  "key_points": [
    "Explain the phased approach as published under Ministerial Decisions No. 243 and 244 of 2025: a voluntary start from 1 July 2026, then mandatory phases led by businesses at or above the AED 50 million threshold. Do not state any date as fact unless it appears in the Ministerial Decisions or on the FTA site. Link to the official text.",
    "Add a reconciliation table. Columns: Source | AED 50m+ ASP deadline | AED 50m+ go-live | Everyone else. Rows: Wafeq (ASP by 31 July 2026; others by 31 March 2027), Hawksford (mandatory from January 2027), ClearTax (all VAT-registered by July 2027), Official MoF/FTA text [writer to fill from the primary source]. Label the competitor rows 'as reported by'.",
    "Explain the difference between the ASP appointment deadline and the go-live date. Most of the confusion comes from mixing the two.",
    "Give a three-question decision rule: (1) Are you VAT-registered and not on the exclusion list? (2) Is your revenue at or above AED 50m? (3) Have you appointed an ASP? Each answer points to the date that applies.",
    "State that penalties only bite once your mandatory date passes. Before then, voluntary issuance carries no penalty."
  ],
  "h3s": [
    "Why different sources quote different dates",
    "A three-question rule to find your deadline"
  ],
  "target_words": 230,
  "format": "table",
  "evidence": [
    "Penalties once mandatory: AED 5,000 per month plus AED 100 per missing invoice: growacross.com",
    "A separate Cabinet Decision sets out the violations and administrative penalties specific to e-invoicing: abdelhamidcpa.com",
    "Ministerial Decision No. 244 and the 5-corner model: qeemahcloud.com",
    "placeholder: official dates from the Ministerial Decisions 243/244 and the FTA"
  ]
},
{
  "h2": "Is the AED 5,000 monthly cap per violation or in total?",
  "purpose": "Answer the question competitors contradict each other on, and set honest expectations about how the cap may be applied.",
  "key_points": [
    "State that sources disagree. Wafeq describes the penalties as stacking; ClearTax treats the cap as per category. Reported wording indicates the AED 5,000 cap applies separately to invoices and to credit notes.",
    "Explain both readings with a short example: under the per-category reading, a business late on invoices AND credit notes in one month could face up to AED 10,000, plus AED 5,000 if

```

```

no system is in place.",
  "Advise readers to plan budgets on the higher (per-category, stacking) reading until the
  FTA publishes clarification, and to confirm with a tax adviser.",
  "Point readers to the exact clause in the MoF PDF so they can read it themselves."
],
"h3s": [
  "How the penalties can stack in a single month"
],
"target_words": 180,
"format": "comparison",
"evidence": [
  "AED 100 per document, capped a
... (23,031 more characters in docs/runs/full)

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Editorial critique of the first draft (Claude Sonnet, Critic → Parse Critique) → revision instructions for the editor

```

{
  "score": 58,
  "verdict": "revise",
  "strengths": [
    "Strong information gain: a source-conflict table for deadlines, a per-category vs overall
    cap reading, a stacking example, and an exposure formula with three costed scenarios.",
    "Honest handling of unverified facts: unknown amounts are placeholdered and the AED 1,000
    daily figure is flagged in the FAQ instead of being stated.",
    "Practical, specific tools: an incident checklist, incident log, prevention matrix with
    owners, 90-day plan, and a three-question deadline rule.",
    "Structure is clear: comparison tables are present, the CTA comes last, and the FAQ uses the
    Q:/A: format."
  ],
  "problems": [
    {
      "severity": "high",
      "location": "Table rows four to six: [Amount per MoF decision] / [Cap per MoF decision]",
      "issue": "Three of the six core penalties have no amounts, even though the primary MoF PDF
      is linked. The page's main promise, 'every fine', is not met. The H1 and meta also say 'every
      fine'. The 'six violations' framing is unverified.",
      "fix": "Open the MoF PDF, fill in the amounts, caps and notification windows, and cite the
      clause. If they cannot be confirmed, change 'every fine' to 'the confirmed fines' in the H1,
      intro and meta."
    },
    {
      "severity": "high",
      "location": "Why different sources quote different dates table: Official MoF/FTA text row",
      "issue": "The official dates are all placeholders, and the 'Everyone else' column is thin.
      The user's most urgent question, 'when do I get hit', is left unanswered. Wafeq, Hawksford and
      ClearTax appear in the table with no links.",
      "fix": "Fill in the dates from Ministerial Decision No. 244 of 2025. Link each cited
      source. Otherwise, drop the unlinked sources."
    },
    {
      "severity": "high",
      "location": "Scenario table: Scenario A 'Total: overall cap reading' 30,000 and Scenario
      C",
      "issue": "The numbers are inconsistent with the stated readings. Under the overall cap of
      AED 5,000 per month, Scenario A over 6 months should be 30,000, which is consistent, but
      Scenario C's 'before invoice-level fines' mixes in an undefined total. The opening claim 'up
      to AED 10,000 a month' contradicts the stacking rule, where the maximum is AED 15,000. The

```

```

title promises a 'Calculator' but there is only a formula and a table.",
  "fix": "Recompute all the totals and show the arithmetic. Reconcile the 10,000 and 15,000
claims. Either build a real interactive calculator or remove 'Calculator' from the title and
the calculator references."
},
{
  "severity": "high",
  "location": "Ministry of Finance published the decision in December 2025. It is a [separate
Cabinet Decision...](https://abdelhamidcpa.com/...)",
  "issue": "A primary-law claim is sourced to a third-party blog (abdelhamidcpa) with a date
slug of 2026/05/29. Other key numbers are sourced to marketing blogs (covoro, genzone,
kaidolab, growacross). The anchor text implies authority it does not have, and the sources are
weak.",
  "fix": "Cite the MoF PDF directly for the decision, the dates and the amounts. Use
secondary sources only for context, with honest anchor text."
},
{
  "severity": "medium",
  "location": "Is the AED 5,000 monthly cap
... (5,364 more characters in docs/runs/full)

```

Content QA after the editor pass (Content QA) → report's quality section, qa_summary in e-mail/callback

```

{
  "content_score": 85,
  "score_breakdown": {
    "structure": 25,
    "coverage": 16,
    "readability": 15,
    "specificity": 10,
    "clean_language": 10,
    "length": 10
  },
  "passed": false,
  "main_content_words": 2458,
  "target_word_count": 1800,
  "coverage_pct": 63,
  "keyword_mentions": 8,
  "keyword_density_pct": 0.27,
  "h2_count": 12,
  "faq_count": 8,
  "internal_links": 6,
  "external_links": 11,
  "readability": {
    "flesch": 61.3,
    "avg_sentence_words": 12.9,
    "long_sentences": 3,
    "passive_pct": 2
  },
  "banned_phrases": [],
  "warnings": [
    "Main content is too long: 2458 words vs target 1800 – cut repetition and secondary detail,
do not add anything",
    "Snippet questions from the brief that have no matching heading: Is the AED 5,000 cap per
violation or in total?"
  ],
  "auto_fixes": [
    "Replaced 43 invented price(s) with [Price]",
    "Removed 1 line(s) of notes that sat between the meta lines and the H1",

```

```
"Generated Article + BreadcrumbList + FAQPage schema"
],
"schema_types": [
  "Article",
  "BreadcrumbList",
  "FAQPage"
],
"critic_score": 58,
"critic_verdict": "revise"
}
```

Delivered: keyword report + page content (PDF 187 KB + Word 52 KB) → posted to the callback (stage content, with ladder_id / rung / head). The page itself is also in Appendix B.

DELIVERED REPORT: SEO-REPORT-AND-CONTENT-UAE-E-INVOICING-PENALTIES-2026-10-01.DOC

SEO REPORT + PAGE CONTENT

uae e invoicing penalties

Blog Post • United Arab Emirates • techand.ai • 2026-10-01

Prepared by Dev SEO

1. Keyword Verdict

Verdict	GO, WITH CHANGES
Score	62/100
Recommended Page Type	Blog Post
Expected visits/month at #1-3	250
Time to rank	5 months

Why

- The volume data is unreliable. The 1,600 headline figure sits next to a single 18,100 spike in Dec 2025, when Cabinet Decision No. 106 was published. Every month from Feb to Aug 2026 shows only 10-50 searches. Real demand is likely a few hundred a month, rising around regulatory deadlines (July 2026 voluntary start, January 2027 mandatory for AED 50m+).
- Ranking looks feasible. Competitors average only 4 referring domains and 11 backlinks, and keyword difficulty is rated Easy. Kreston Menon is thin at about 804 words, and Hawksford covers penalties only in an FAQ.
- Clear content gaps exist: no page has a penalty calculator, worked exposure examples, a reconciliation of the conflicting phase dates, or an explanation of whether the AED 5,000 cap is per category or overall. A 1,800+ word guide can beat Wafeq's 2,925-word vendor page on usefulness.
- The fit with the business goal is good. Readers worried about fines are the same people who need an ASP and readiness help, and Tech& sells e-invoicing readiness to enterprise CFOs. The audience is narrow but high-value, so a few leads would justify the page. The keyword is informational, despite the 'transactional' label, and fits a blog post.

What must change first

- Build it as a practical penalty-exposure guide, not a generic explainer. Open with a 40-70 word answer block, then add a penalty table (violation, amount, cap, who is liable).
- Add a calculator-style table with worked examples, for instance 6 months without an ASP plus 80 missed invoices a month. Offer a downloadable penalty summary PDF as the lead magnet.
- Reconcile the conflicting phase dates and the cap and stacking question, citing mof.gov.ae and FTA sources, with a visible updated date and a named author with UAE tax credentials.
- Close with a phase-readiness check and a consultation CTA aimed at CFOs of AED 50m+ businesses. Add FAQs using the People Also Ask wording.
- Support the page with cluster content: penalties calculator, recipient-side penalties, system failure checklist, and ASP selection. Link these to a Dynamics 365 e-invoicing service page.

Risks / Watch-outs

- The AI Overview already answers the question (AED 100 to AED 5,000), citing hawksford.com and zoho.com, and the MoF PDF ranks #1. Click-through will be low, and a top-3 position may bring only 100-300 visits a month.
- The site's authority is unknown, and the topic is crowded with vendors (Wafeq, Zoho, ClearTax, Tally). Penalty and phase details are changing and conflicting, so outdated or wrong figures would damage credibility with CFOs.
- Many searchers are SMEs and accountants looking for a PDF or calculator, not enterprise buyers, so lead quality may be mixed.

Secondary Keywords Worth Targeting

- uae e-invoicing penalties calculator
- uae e-invoicing penalties pdf
- cabinet decision 106 of 2025
- e-invoicing system failure penalty uae
- uae e-invoicing deadline 2027
- e-invoicing penalties for recipients uae
- accredited service provider uae e-invoicing
- uae e-invoicing AED 50 million threshold
- FTA e-invoicing fines

2. Keyword Overview

Monthly Searches	1,600
Keyword Difficulty	0/100 — Easy
Cost Per Click	N/A
Ads Competition	LOW
Search Intent	informational
Search Trend	N/A
Competitor Link Strength	Avg 4 referring domains, 11 backlinks

3. What the Results Page Looks Like

Featured snippet	None
AI Overview	Shown — cites hawksford.com, marmin.ai, zoho.com, uae-asp.ae, youtube.com
People Also Ask	Is e-invoicing mandatory in the UAE? • What is the penalty for not issuing an e-invoice? • Who is exempted from e-invoicing? • What is the penalty for late payment of VAT in the UAE?
Related searches	Uae e invoicing penalties pdf • Uae e invoicing penalties calculator • UAE e Invoicing Guidelines • E invoicing UAE 2026 • UAE e Invoicing Guidelines PDF • UAE e invoicing data Dictionary • List of ASP for E invoicing in UAE • E invoicing UAE FTA
Other features	ai_overview, people_also_ask, related_searches
Results for this query	129

How to win the features: The AI Overview cites the AED 100 to AED 5,000 range, so open with a 40-70 word answer that names the decision and states all three fine types. Follow it with a clean HTML penalty table and a Q&A-style FAQ that uses the People Also Ask wording. Add a calculator-style table and a downloadable PDF to cover the 'pdf' and 'calculator' related searches. There is no featured snippet, so the table and the answer block are the best candidates. Cite mof.gov.ae and add a visible updated date.

4. Competitor Content Analysis

Searchers want the exact fine amounts under Cabinet Decision No. 106 of 2025, when the fines start, and how to avoid them. Leads come from readers who are unsure about their phase deadline or ASP choice, so a readiness check or consultation offer fits.

Top Ranking Pages

Rank	Domain	Page Title	Words
#1	mof.gov.ae	Cabinet Decision No. 106 of 2025	672
#2	hawksford.com	UAE e-invoicing: timeline, business actions and penalties	2,074
#3	wafeq.com	UAE E-Invoicing Penalties: What Businesses Need to ...	2,925
#4	krestonmenon.com	UAE E-Invoicing Update: Violations & Penalties Explained	804
#5	tallysolutions.com	UAE E-Invoicing Penalties and System Failure Reporting	1,078
#6	cleartax.com	e-Invoicing Penalties Under Cabinet Decision 106/2025	1,305

What Competitors Do Well

- **mof.gov.ae:** The primary source, with the official wording and the full annexed penalty table
- **hawksford.com:** Thorough on timeline, five-corner model and exclusions; cited in the AI Overview
- **wafeq.com:** The most detailed penalty-focused page (about 2,900 words), covering stacking, deadlines and the voluntary window
- **krestonmenon.com:** Clear split of issuer and recipient penalties
- **tallysolutions.com:** Deep coverage of system failure and the AED 1,000 daily penalty, with reporting steps
- **cleartax.com:** Penalty table, enforcement mechanics and a phase table

Where Competitors Are Weak

- **mof.gov.ae:** A legal PDF with no explanation, examples or guidance

- **hawksford.com:** Penalties are covered only in an FAQ; the page focuses on timeline and framework
- **wafeq.com:** Vendor-promotional, with a discount pop-up and a sales slant; it has no tables
- **krestonmenon.com:** Short (about 800 words), has no named author, and lacks worked examples or tables
- **tallysolutions.com:** Narrow focus on system failure, with a product pitch
- **cleartax.com:** Its phase dates differ from other pages, and it has no worked examples

Content Gaps to Fill

- Only a few pages (the MoF PDF, Kreston Menon, ClearTax, Wafeq) list all six violations, and none cover the legal wording and the plain-English meaning side by side with scenarios
- No page works out total exposure over 3, 6 or 12 months for different business types
- The competitor data shows conflicting phase dates (Wafeq: ASP deadline 31 July 2026 for AED 50m+ and 31 March 2027 for the rest; ClearTax: all VAT-registered by July 2027; Hawksford: mandatory from January 2027). Readers get no reconciliation or advice to check against the primary source
- The relationship to existing VAT penalties under Cabinet Decision No. 40 of 2017 is not covered, though 'late payment of VAT' appears in People Also Ask
- How to contest or appeal an e-invoicing penalty is not covered in the data
- Related searches ask for a penalty calculator, PDF and ASP list, and no ranking page offers these
- Recipient-side penalties are covered thinly, with limited practical guidance
- Whether penalties stack, and whether the AED 5,000 cap applies per category or overall, is stated differently across pages (Wafeq says stacking, ClearTax says per category)

Recommended Unique Angle: The practical penalty exposure guide for CFOs: every fine in plain English, a cost calculator with worked examples, an honest reconciliation of conflicting dates and cap readings, and a prevention checklist by owner. It ends with a phase-readiness check that leads to a Tech& consultation.

Ranking Feasibility: This is moderately difficult. The MoF PDF, Big-4-style advisers (Hawksford) and software vendors with deep penalty pages (Wafeq, Tally, ClearTax) hold the top positions. A well-structured blog post can still compete, because several ranking pages are thin (Kreston Menon at about 800 words) and none offers worked examples or a calculator. Expect to need strong depth, fresh dates and backlinks. Supporting long-tail targets such as 'uae e-invoicing penalties calculator' and 'e-invoicing penalties for recipients' are easier wins.

5. Content Quality Check

Status	Needs review
Main Content Words	2,458 (target 1,800)
FAQ Words	478
Keyword Mentions	8 (density 0.27%)
Sections (H2)	12
FAQs	8
Meta title / description length	54 / 146 characters
Internal links to existing pages	6
Quick answer block	65 words
Editing rounds	2

Schema generated	Article, BreadcrumbList, FAQPage
Content score	85/100 (structure 25/25, topic coverage 16/25, readability 15/15, specificity 10/10, clean language 10/10, length 10/15)
Topic coverage	63% of the terms top-ranking pages share
Readability	Flesch 61.3 · 12.9 words per sentence · passive voice 2%
Filler phrases left	none
Editorial review	58/100 — revise

Warnings (please review before publishing)

- Main content is too long: 2458 words vs target 1800 — cut repetition and secondary detail, do not add anything
- Snippet questions from the brief that have no matching heading: Is the AED 5,000 cap per violation or in total?

Automatic fixes applied

- Replaced 43 invented price(s) with [Price]
- Removed 1 line(s) of notes that sat between the meta lines and the H1
- Generated Article + BreadcrumbList + FAQPage schema

Text in [square brackets] must be replaced with your real business details before publishing.

6. Page Content

SEO Meta Tags

Title	UAE E-Invoicing Penalties: Fines, Examples & Deadlines
Meta Description	UAE e-invoicing penalties explained: the confirmed fines under Cabinet Decision No. 106, worked cost examples and a prevention plan. Book a check.
Slug	uae-e-invoicing-penalties
Primary Keyword	uae e invoicing penalties
Secondary Keywords	cabinet decision no. 106 of 2025, uae e-invoicing penalty examples, e-invoicing penalties for recipients, e-invoicing system failure penalty, accredited service provider uae, uae e-invoicing deadline, penalty for not issuing an e-invoice, fta e-invoicing administrative penalties

UAE E-Invoicing Penalties: The Confirmed Fines, What They Could Cost and How to Avoid Them

Last updated: [Date] · Reviewed by [Name, UAE tax credential] · Technical review: [Reviewer name, role at Tech&]

UAE e-invoicing penalties are recurring monthly fines, so the cost of a late start accumulates every month. This guide gives your finance team four things: the confirmed fines in plain English, costed exposure for three scenarios, your phase deadline, and a prevention plan with an owner for each control. Where we have not confirmed the official text, we say so.

Quick answer: Under Cabinet Decision No. 106 of 2025, the Federal Tax Authority (FTA) can fine VAT-registered businesses that miss e-invoicing obligations. Failing to implement the Electronic Invoicing System or appoint an Accredited Service Provider costs [Price] per month or part month. Each late electronic invoice or credit note costs [Price] Reported wording sets an [Price] monthly cap for each document type. Voluntary adopters are not penalised.

What are the UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025?

The Ministry of Finance published [Cabinet Decision No. 106 of 2025 on e-invoicing violations and penalties](#) in December 2025. It sits under Federal Decree-Law No. 28 of 2022 on Tax Procedures, and the FTA enforces it. Two administrative penalties are confirmed. A business with no e-invoicing system or ASP pays [Price] per month. Each late document costs [Price]

What is the penalty for not issuing an e-invoice?

Failing to issue and transmit an electronic invoice on time costs [Price] per invoice. [Commentary on the decision](#) reports a cap of [Price] per month. A PDF does not count. The invoice must pass through your ASP in the structured electronic format. If you have no system or ASP at all, the separate [Price] monthly penalty also applies.

The violations at a glance

Violation (paraphrased from the MoF decision)	Plain-English meaning	Example	Who is liable	Fine	Cap
Failure to implement the e-invoicing system, including appointing an ASP, by the deadline	You are not connected through an accredited provider on time	Your mandatory date passes with the ASP contract unsigned	Business in scope	[Price] per month or part month	None stated; recurs monthly
Failure to issue and transmit an electronic invoice on time	The invoice does not go through your ASP in the structured electronic format	A batch of sales invoices goes out as PDFs	Issuer	[Price] per invoice	[Price] per month (reported)
Failure to issue and transmit an electronic credit note on time	The same rule, applied to credit notes	Month-end returns are credited manually	Issuer	[Price] per credit note	[Price] per month (reported)
Failure to notify the FTA of a system failure within the set period	You must tell the FTA when a fault stops you issuing	An ASP outage goes unreported	Business in scope	Not confirmed here	Check the decision
Failure to notify the ASP of changes to your registered data	Your ASP's record of your business must stay current	A new branch is never notified to the ASP	Business in scope	Not confirmed here	Check the decision
Recipient failure to notify the issuer of a system failure	A buyer must tell its supplier when it cannot receive	The buyer's ASP is down and the supplier is not told	Recipient (as reported)	Not confirmed here	Check the decision

We have not confirmed the amounts, caps or notification windows for rows four to six. This guide therefore prices only the confirmed fines. Commentary groups the breaches into these six violations, so check the full list against the decision before using it in board papers.

The monthly fine applies for each month or part of a month. One day into a new month counts as a full month. The decision excludes voluntary adopters.

[Image: Infographic summarising the confirmed UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025, with fines in AED]

Is e-invoicing mandatory in the UAE?

Yes. It is mandatory for VAT-registered businesses in scope, on a phased timetable set by Ministerial Decision No. 244 of 2025. Voluntary adoption opens from 1 July 2026. Businesses with revenue at or above [Price] million (<https://worldcompanysetup.com/en/blog/uae-e-invoicing-mandate-dubai-timeline-penalties/>) must appoint an ASP first, by 31 July 2026. They must go live from 1 January 2027.

When do e-invoicing penalties start, and which phase are you in?

Penalties start only once your deadline passes. Ministerial Decision No. 244 of 2025 sets two dates for each phase: an ASP appointment deadline and a go-live date. Ministerial Decision No. 243 of 2025 defines scope and exclusions. Confirm the dates below against the official decision before planning.

Phase	Appoint an ASP by	Mandatory e-invoicing from
Revenue of [Price] million or more	31 July 2026	1 January 2027
Revenue below [Price] million	31 March 2027	1 July 2027
Government entities	31 March 2027	1 October 2027

Published guides often disagree because they mix up these two dates. Your ASP deadline comes first, so it is your nearest exposure. A [secondary guide to Ministerial Decision No. 244](#) explains the phases in more detail.

Are there penalties during the voluntary phase?

No. Cabinet Decision No. 106 of 2025 states that its provisions do not apply to businesses that issue, transmit or exchange e-invoices voluntarily. The safest time to test is therefore from 1 July 2026 until your mandatory date. Use that window to test your ERP, your ASP connection and your system-failure procedure without any fine exposure.

A three-question rule to find your deadline

1. Are you VAT-registered and outside the exclusions in Ministerial Decision No. 243? If not, the mandate may not apply. Confirm this with your adviser.
2. Is your revenue at or above [Price] million? If yes, the first-phase dates apply.
3. Have you appointed an ASP? If not, your nearest exposure is the ASP deadline, not go-live.

Is the [Price] cap per violation or in total?

Reported wording of Cabinet Decision No. 106 sets a separate [Price] monthly cap for invoices and another for credit notes. The [Price] monthly system fine sits on top of both. The safe budgeting assumption is that penalties stack, giving a worst month of [Price]. Confirm the clause with your adviser until the FTA clarifies it.

Reading	How the cap works	Worst month, all three breaches
Per-category (stacking)	[Price] cap for invoices, a separate [Price] for credit notes, plus the [Price] system fine	[Price]
Overall cap	One [Price] ceiling across all penalties in the month	[Price]

Commentary describes the document fine as [capped at [Price] per month for each](<https://www.genzone.com/uae-e-invoicing-2026-guide/>) category, which supports the stacking reading. Read the clause in the [MoF decision](#) yourself.

How much could non-compliance cost? Worked penalty examples

A business with no ASP and regular late invoices faces up to **[Price]** a month. That is **[Price]** for the system plus the **[Price]** invoice cap. Add late credit notes and the worst month reaches **[Price]** Both figures use the stacking reading. The fines [recur for each month or part of a month](#).

The exposure formula

Monthly exposure = 5,000 (if no system or ASP) + min(late invoices × 100, 5,000) + min(late credit notes × 100, 5,000). All figures in AED per month, per-category reading. Under the overall-cap reading, the monthly total cannot exceed 5,000.

Three scenarios, three totals

UAE e-invoicing penalty exposure: three worked scenarios

Scenario	Months	Violations each month	Monthly penalty AED (per-category)	Total AED: per-category	Total AED: overall cap
A: enterprise above [Price] m	6	No ASP; 80 late invoices (8,000, capped to 5,000)	5,000 + 5,000 = 10,000	10,000 × 6 = 60,000	5,000 × 6 = 30,000
B: system live, process gaps	3	30 late invoices (3,000); 10 late credit notes (1,000)	3,000 + 1,000 = 4,000	4,000 × 3 = 12,000	4,000 × 3 = 12,000
C: no system at all	12	No ASP; 60 late invoices (6,000, capped to 5,000); 55 late credit notes (5,500, capped to 5,000)	5,000 + 5,000 + 5,000 = 15,000	15,000 × 12 = 180,000	5,000 × 12 = 60,000

Scenario B is the same under both readings because its monthly total stays below **[Price]** Scenario C shows the full gap between the readings: **[Price]** over a year. The MoF text governs these figures, so take advice before you report them.

Costs that do not appear on the penalty notice

Remediation also consumes finance and IT time during close. For the wider analysis, see [the full cost of e-invoicing non-compliance and mitigation](#).

What counts as a system failure, and what if your ASP goes down?

A system failure is a technical fault in your own system or your ASP's that stops you issuing or transmitting e-invoices. The decision requires you to notify the FTA within a set period and fines you if you do not. We have not confirmed that window or the amount, so take both from the [MoF decision](#).

An ASP outage does not remove your own obligations. Do not assume your provider will notify the FTA for you. Check in the decision who must notify, then write that duty into your ASP contract.

System-failure incident checklist

1. Detect and timestamp the failure, noting whether it sits in the ERP or the ASP.
2. Raise a ticket with your ASP and record the reference.
3. Notify the Federal Tax Authority within the required window and record the channel you used.
4. Log the affected invoices, correspondence and screenshots, with detection, notification and resolution times.
5. Re-transmit the backlog once the system failure is resolved, then reconcile.

What is an accredited service provider (ASP), and why does it matter for penalties?

An accredited service provider (ASP) is a provider accredited by the UAE authorities to exchange structured electronic invoices between businesses. It also reports invoice data to the FTA. Under UAE electronic invoicing, you cannot issue compliant e-invoices without an ASP. Missing your ASP appointment deadline triggers the **[Price]** monthly fine.

When you choose an ASP, check its accreditation status, PINT AE support, ERP connectors and uptime commitments. See the [Peppol e-invoicing software features CFOs should look for](#).

Who is liable: the issuer, the recipient or the ASP?

As reported, the business carries the liability, not the service provider (ASP). Confirm the article for each violation in the MoF decision. UAE e-invoicing is commonly described as a [five-corner model](#) on Peppol and PINT AE. Data flows from the supplier to the supplier's ASP, then to the buyer's ASP and the buyer. The FTA is the fifth corner and receives the invoice data.

Issuers must issue and transmit each invoice and electronic credit note in the structured format on time. Recipients need an ASP and the ability to receive e-invoices. Commentary indicates they must also notify the issuer of failures. See [what the FTA requires under UAE e-invoicing](#) for the full obligations.

Your ASP is a service provider, so write SLAs, uptime targets and incident-reporting duties into the contract.

[Image: Five-corner Peppol model diagram showing issuer, recipient, Accredited Service Providers and the FTA]

How do e-invoicing fines relate to VAT penalties, and can you appeal?

They are separate schedules and can apply together. You can also request reconsideration.

Late VAT payment vs e-invoicing fines

E-invoicing fines sit in Cabinet Decision No. 106 of 2025. General VAT penalties, including those for late payment, sit under Cabinet Decision No. 40 of 2017 as amended. A late electronic invoice can therefore attract both an e-invoicing fine and a separate VAT penalty. Take current late-payment rates from the FTA's published penalty schedule.

Requesting reconsideration from the FTA

Federal Decree-Law No. 28 of 2022 lets you ask the FTA to reconsider a penalty within 40 business days of being notified. If you disagree with the outcome, you can object to the Tax Disputes Resolution Committee. Your incident logs and ASP correspondence are your evidence, so keep them from day one.

How to avoid UAE e-invoicing penalties: owners, controls and a 90-day plan

Each violation needs one owner and one documented control. Start with ASP selection, because [one compliance guide warns that delay risks bottlenecks and **[Price]** monthly fines](<https://farahsolutions.ae/blog/uae-e-invoicing-guidelines-v1-1-compliance-guide>).

Prevention matrix by owner

Violation	Owner	Control	Evidence to keep
No system or ASP	CFO (Finance)	ASP contract signed before your ASP deadline	Signed contract
Late electronic invoice	IT/ERP	Automated transmission with a daily exception report	Exception reports
Late electronic credit note	Finance (AR)	Credit-note workflow routed through the ASP	Approval trail
No FTA failure notice	Finance	Incident procedure with a named notifier	Incident log

ASP data not updated	Finance	Quarterly master-data review	Change confirmations
Recipient fails to notify issuer	ASP and Finance (AP)	Contractual alerting SLA	ASP alerts

A 90-day penalty-avoidance plan

1. Days 1–30: confirm your phase, scope your entities, shortlist ASPs and run an ERP gap check for PINT AE fields.
2. Days 31–60: appoint the ASP, map invoice data and configure credit-note flows.
3. Days 61–90: test end to end during the voluntary window, when no penalties apply. Rehearse the system-failure procedure and train your AP and AR teams.

Dynamics 365 Finance & Operations and Business Central both need ERP-side changes. See [upgrading your ERP for Peppol PINT AE compliance](#).

Tech& is a [Partner status] with [Number] years of experience in UAE finance systems. Its readiness check covers [scope] over [duration]. It confirms your phase, checks your ERP gaps and ASP-integration readiness, and estimates your exposure. [Case study: UAE client, industry, e-invoicing readiness outcome]

Frequently Asked Questions

Is e-invoicing mandatory in the UAE?

Yes, for VAT-registered businesses in scope, on a phased timetable under Ministerial Decision No. 244 of 2025. Voluntary adoption opens from 1 July 2026. Businesses with revenue of [Price] million or more must appoint an ASP by 31 July 2026 and go live from 1 January 2027. Our [complete guide to e-invoicing in the UAE](#) covers scope.

What is the penalty for not issuing an e-invoice?

Failing to issue and transmit an electronic invoice on time carries a penalty of [Price] per invoice, capped at [Price] per month according to [published commentary](#). If you have not implemented the system or appointed an ASP at all, a separate [Price] monthly penalty can apply as well.

Who is exempted from e-invoicing?

Ministerial Decision No. 243 of 2025 lists the excluded transactions and entities. Being VAT-registered does not by itself put every transaction in scope. Check the official exclusion list against each of your revenue streams. Confirm any borderline cases with your tax adviser before your phase begins, because exclusions apply to specific transactions rather than to whole businesses by default.

What is the penalty for late payment of VAT in the UAE?

Late VAT payment penalties fall under Cabinet Decision No. 40 of 2017 as amended, not under the e-invoicing decision. Rates have changed over time, so take the current figures from the Federal Tax Authority's published penalty schedule. Both penalty regimes can apply to the same business in the same period.

Are there penalties during the voluntary phase?

No. Cabinet Decision No. 106 of 2025 states that its provisions do not apply to businesses that issue, transmit or exchange e-invoices voluntarily. This makes the voluntary window the safest time to test your ERP and ASP connection before your mandatory date.

What penalties apply to notification failures?

The decision covers three notification duties. You must tell the FTA about a system failure and tell your ASP about changes to your registered data. Recipients must tell issuers when they cannot receive. We have not confirmed the amounts or time windows, so read the exact clauses in the MoF decision before you budget for them.

Do e-invoicing penalties apply to credit notes?

Yes. Each electronic credit note not issued and transmitted on time costs [Price] capped at [Price] per month (source). Reported wording suggests this cap sits separately from the invoice cap. Late credit notes could therefore add to invoice penalties in the same month.

Where can I download the UAE e-invoicing penalties PDF?

The official text is the Ministry of Finance's [Cabinet Decision on e-invoicing violations and penalties](#), published as a PDF. The violations table on this page summarises it in plain English. Check the decision itself for any amounts we mark as not confirmed.

Book your free readiness and penalty exposure check

Book a free e-invoicing readiness and penalty exposure check with Tech& before UAE e-invoicing penalties apply to you. In [duration] we confirm your phase and deadline, run an ERP gap check for PINT AE and review your ASP readiness. We then estimate your AED exposure using the worked examples above.

[Customer review: CFO quote on readiness project]

[Book an e-invoicing readiness check](#) · Phone: [Phone] · Email: [Email]

7. Internal Links Used

Anchor text	Links to (existing page)
complete guide to e-invoicing in the UAE	https://techand.ai/e-invoicing-in-uae/
what the FTA requires under UAE e-invoicing	https://techand.ai/e-invoicing-uae-fta/
the full cost of e-invoicing non-compliance and mitigation	https://techand.ai/insights/the-cost-of-non-compliance-uae-e-invoicing-penalties-mitigation-costs
upgrading your ERP for Peppol PINT AE compliance	https://techand.ai/insights/upgrading-your-erp-for-uae-peppol-pint-ae-compliance-a-technical-integration-guide
Peppol e-invoicing software features CFOs should look for	https://techand.ai/insights/top-peppol-e-invoicing-software-features-every-uae-cfo-must-look-for
book an e-invoicing readiness check	https://techand.ai/contact-us

8. Image Suggestions

Placement	Alt text
After 'What are the UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025?'	Infographic summarising the six UAE e-invoicing penalties under Cabinet Decision No. 106 of 2025 with fines in AED
After 'When do e-invoicing penalties start, and which phase are you in?'	UAE e-invoicing timeline showing the 1 July 2026 voluntary start and mandatory phases for businesses above AED 50 million
After 'How much could non-compliance cost? Worked penalty examples'	Bar chart comparing UAE e-invoicing penalty exposure over 3, 6 and 12 months for three business scenarios
After 'Who is liable: the issuer, the recipient or the ASP?'	DCTCE five-corner Peppol model diagram showing issuer, recipient, Accredited Service Providers and the FTA

9. Topic Coverage vs Top-Ranking Pages

Terms the top pages share and how often this page uses them.

Term	Target	Found	
cabinet decision	5	11	ok
accredited service provider	4	4	ok
uae e-invoicing	6	13	ok
e-invoicing system	4	1	add
system failure	4	7	ok
federal tax authority	3	3	ok
credit notes	3	5	ok
electronic invoicing	3	2	add
administrative penalties	2	1	add
structured electronic	1	2	ok
federal decree-law	2	2	ok
ministry of finance	2	2	ok
electronic credit note	2	2	ok
issue and transmit	2	3	ok
aed per month	2	1	add
vat-registered businesses	2	3	ok
electronic invoice	3	5	ok
invoice data	1	3	ok
penalty aed	1	0	add
system failures	1	0	add
tax authority fta	1	0	add
service provider asp	1	0	add
service provider within	1	0	add
electronic invoicing system	1	1	ok
electronic invoices	1	1	ok
aed million	1	0	add
ministry finance	1	2	ok

10. Sources Cited

Anchor	URL
--------	-----

Cabinet Decision No. 106 of 2025 on e-invoicing violations and penalties	https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf
Commentary on the decision	https://www.genzone.com/uae-e-invoicing-2026-guide/
MoF decision	https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf
secondary guide to Ministerial Decision No. 244	https://qeemahcloud.com/en/blog/uae-e-invoicing-deadlines-2026/?srsltid=AU7gw4WNyQVaepjxnAO0IBhncG4lwbSJEKQue8sg3o-iwSYmvpmKNG9
MoF decision	https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf
recur for each month or part of a month	https://kaidolab.com/blog/uae-e-invoicing-asp-deadline-2026/
MoF decision	https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf
five-corner model	https://qeemahcloud.com/en/blog/uae-e-invoicing-deadlines-2026/?srsltid=AU7gw4WNyQVaepjxnAO0IBhncG4lwbSJEKQue8sg3o-iwSYmvpmKNG9
published commentary	https://www.genzone.com/uae-e-invoicing-2026-guide/
source	https://www.genzone.com/uae-e-invoicing-2026-guide/
Cabinet Decision on e-invoicing violations and penalties	https://mof.gov.ae/wp-content/uploads/2025/12/Cabinet-Decision-Violations-and-Penalties-eInvoicing-final-version-en-8.12.25.pdf

11. Title & Meta Alternatives

Titles

- UAE E-Invoicing Penalties Explained: What Each Fine Could Cost You
- Cabinet Decision No. 106 of 2025: UAE E-Invoicing Fines and How to Avoid Them

Meta descriptions

- How much could UAE e-invoicing non-compliance cost you? See every fine, the AED 5,000 monthly cap and 3- to 12-month examples. Get a free penalty exposure review.

12. Editorial Review

Score	58/100 — revise
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Strengths

- Strong information gain: a source-conflict table for deadlines, a per-category vs overall cap reading, a stacking example, and an exposure formula with three costed scenarios.
- Honest handling of unverified facts: unknown amounts are placeholdered and the AED 1,000 daily figure is flagged in the FAQ instead of being stated.
- Practical, specific tools: an incident checklist, incident log, prevention matrix with owners, 90-day plan, and a three-question deadline rule.
- Structure is clear: comparison tables are present, the CTA comes last, and the FAQ uses the Q:/A: format.

Issues raised (addressed by the editor)

- [high] Three of the six core penalties have no amounts, even though the primary MoF PDF is linked. The page's main promise, 'every fine', is not met. The H1 and meta also say 'every fine'. The 'six violations' framing is unverified.
- [high] The official dates are all placeholders, and the 'Everyone else' column is thin. The user's most urgent question, 'when do I get hit', is left unanswered. Wafeq, Hawksford and ClearTax appear in the table with no links.
- [high] The numbers are inconsistent with the stated readings. Under the overall cap of AED 5,000 per month, Scenario A over 6 months should be 30,000, which is consistent, but Scenario C's 'before invoice-level fines' mixes in an undefined total. The opening claim 'up to AED 10,000 a month' contradicts the stacking rule, where the maximum is AED 15,000. The title promises a 'Calculator' but there is only a formula and a table.
- [high] A primary-law claim is sourced to a third-party blog (abdelhamidcpa) with a date slug of 2026/05/29. Other key numbers are sourced to marketing blogs (covoro, genzone, kaidolab, growacross). The anchor text implies authority it does not have, and the sources are weak.
- [medium] The answer is hedged and not a direct 40-60 word snippet. The Quick answer says 'capped at AED 5,000 per month' with no hedge, while the section says the cap is disputed.
- [medium] Both are asserted without a source. The liability claim and the five-corner model rely on a vendor blog. The statement 'If you assume the provider will report the outage, the notification fine falls on you' is unsupported.
- [medium] It says 'four things' but lists three or four vaguely. The opening claim that 'a late start compounds quickly' is loose, since the fines are flat and do not compound.
- [medium] There are too many unfilled placeholders for a client-ready draft. Reconsideration deadlines and the exclusions in the FAQ are also unfilled, which weakens the appeal and exemption answers.

13. Structured Data (JSON-LD)

Paste each block in the page's <head> or before </body>. Replace every [square-bracket] value first.

Article

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FAQPage

Google stopped showing FAQ rich results in May 2026; keep for other engines and AI assistants, or drop it.

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Ministerial Decision No. 244 of 2025. Voluntary adoption opens from 1 July 2026. Businesses
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Run 38 — Form path: "Check if my keyword is right" for "e invoicing in uae"

Verdict (Claude Sonnet) → report, download page and e-mail

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    "Ranking is realistic: KD is 2 (Easy), and top-10 competitors average only 5 referring domains and 10 backlinks. Thin pages rank, such as ajmsglobal.com at 664 words and youngandright.ae at about 122 words. An unknown-authority site can compete on content quality.",  
    "Intent does not match a pure service page. Google labels it informational, and the SERP is guides from advisory firms and vendors (roedl.com, auditfirmsdubai.ae, wafeq.com). No pure service page ranks. A guide-style service page with lead CTAs fits the pattern.",  
    "Content gaps are clear: competitors disagree on dates (30 Oct 2026 vs 31 July 2026), cover penalties weakly, and give no ASP selection criteria, cost guidance or sector detail. Tech& can win with a dated, cited guide that includes a phase table and a readiness scorecard."  
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    "Add a dated 'Which phase am I in?' table with the AED 50M threshold, ASP deadline, go-live and penalty dates, each cited to tax.gov.ae or the MoF. Reconcile the conflicting dates and show a last-updated date.",  
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}
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DELIVERED REPORT: KEYWORD-VERDICT-E-INVOICING-IN-UAE-2026-10-01.DOC

KEYWORD VERDICT

e invoicing in uae

Service Page • United Arab Emirates • techand.ai • 2026-10-01

Prepared by Dev SEO

1. Keyword Verdict

Verdict	GO, WITH CHANGES
Score	72/100
Recommended Page Type	Guide-style Service Page (e-invoicing readiness and implementation guide with lead CTAs)
Expected visits/month at #1-3	250
Time to rank	4 months

Why

- Demand is real and rising: 880 monthly searches, +85% year on year, and a CPC of \$6.85 that shows commercial value. The -18% quarterly dip and the 720 in Aug 2026 look like seasonality, not decline. Mandate deadlines should keep interest high into 2027.
- Ranking is realistic: KD is 2 (Easy), and top-10 competitors average only 5 referring domains and 10 backlinks. Thin pages rank, such as ajmsglobal.com at 664 words and youngandright.ae at about 122 words. An unknown-authority site can compete on content quality.
- Intent does not match a pure service page. Google labels it informational, and the SERP is guides from advisory firms and vendors (roedl.com, auditfirmsdubai.ae, wafeq.com). No pure service page ranks. A guide-style service page with lead CTAs fits the pattern.
- Content gaps are clear: competitors disagree on dates (30 Oct 2026 vs 31 July 2026), cover penalties weakly, and give no ASP selection criteria, cost guidance or sector detail. Tech& can win with a dated, cited guide that includes a phase table and a readiness scorecard.

What must change first

- Open with a 40-70 word direct answer, then use sections that match the PAA questions: mandatory?, who is eligible?, how it works, how to do it. Add FAQ schema.
- Add a dated 'Which phase am I in?' table with the AED 50M threshold, ASP deadline, go-live and penalty dates, each cited to tax.gov.ae or the MoF. Reconcile the conflicting dates and show a last-updated date.
- Position Tech& around Dynamics 365 and ERP integration with an ASP. Include an ASP selection scorecard, a readiness checklist, a 30/60/90-day plan and a free readiness assessment CTA.
- Add a named expert author, plus supporting pages and internal links for 'UAE e-invoicing ASP', 'e-invoicing implementation UAE' and 'e-invoicing penalties UAE'.

Risks / Watch-outs

- The AI Overview cites tax.gov.ae, PwC and Grant Thornton and will absorb many clicks. Real organic traffic may be well below the 880 volume.

- Many searchers are researching rather than buying. Lead conversion will be low unless the CTAs are strong. Tech& is also a Dynamics 365 partner, not a tax advisory firm, so E-E-A-T on tax topics is weaker.
- Deadlines and rules are shifting, so the page needs frequent updates to stay accurate and cited.

Secondary Keywords Worth Targeting

- e-invoicing UAE implementation
- UAE e-invoicing ASP
- accredited service provider UAE e-invoicing
- UAE e-invoicing penalties
- UAE e-invoicing deadline 2026
- Peppol UAE e-invoicing
- e-invoicing Dynamics 365 UAE
- is e-invoicing mandatory in UAE
- UAE e-invoicing requirements
- PINT-AE

Run 41 — Form path: discovery with the "Choose your keyword" step

Business text given on the form (no site read). Choice options shown on the form page:

- Let the system choose for my goal
- 1. electronic invoicing system · 40 searches/mo · difficulty ? · Commercial · Service Page
- 2. digital invoice · 30 searches/mo · difficulty ? · Commercial · Service Page
- 3. einvoice implementation · 20 searches/mo · difficulty ? · Commercial · Service Page
- 4. e invoicing implementation · 20 searches/mo · difficulty ? · Commercial · Service Page
- 5. automation of accounts payable · 30 searches/mo · difficulty ? · Commercial · Service Page
- 6. top accounts payable automation software · 20 searches/mo · difficulty ? · Commercial · Comparison Page
- 7. dynamics 365 implementation · 20 searches/mo · difficulty ? · Commercial · Service Page

User's pick: **e invoicing implementation** (chosen_by user); it replaced the system suggestion **electronic invoicing system** as the pipeline keyword and in the report. Priority keywords of this run:

Keyword	Searches/mo	KD	CPC	Intent	Page type	Opportunity
electronic invoicing system	40		11.17	commercial	Service Page	15.6
digital invoice	30		12.86	commercial	Service Page	8.4
einvoice implementation	20		2.6	commercial	Service Page	5.8
e invoicing implementation	20		2.6	commercial	Service Page	5.8
automation of accounts payable	30			commercial	Service Page	5.6
top accounts payable automation software	20			commercial	Comparison Page	4.9
dynamics 365 implementation	20			commercial	Service Page	2.4

The keyword-ideas download page was served; the chained report did not run in this execution (completion ended it) — the redesigned flow spawns it as its own run.

Run 43 — Form path: technical audit started from the form (spawned execution)

Same report type as run 16; health score **70/100** (Needs Improvement), issue counts {"Critical": 1, "High": 3, "Medium": 6, "Low": 10, "Info": 3}; top issues: Broken pages (4xx, 5xx or timeout) · www and non-www versions are not consolidated · HTML cannot be cached by the CDN (no-store / private) · Broken internal links. E-mailed to kinngahmed@gmail.com (208 KB, 250 OK); saved as LIVE-form-audit-techand.doc.html.

Run 29/30 — Error handler check

A deliberate API request without callback or e-mail made *API Reject* throw; the Error Handler's *Format Alert* produced the ops e-mail below (sent, 250 OK).

Subject: [SEO Agent] Run failed at "API Reject" — SEO Agent v4 — Audit, Keywords, Verdict & Content

Workflow: SEO Agent v4 — Audit, Keywords, Verdict & Content

Execution: 29 (integrated) — [open](#)

Failed node: API Reject

When: 2026-10-01T12:23:57.977Z

API request rejected (no callback_url to notify): API requests need a callback_url (https) or an email address so the result can be delivered. [line 4]

Typical causes: DataForSEO/Jina credentials or credits, a site that blocks readers, an AI response that was not valid JSON, or a crawl that never finished.

If a customer is waiting for an emailed report, re-run the execution from the failed node or tell them what happened.